

CS1231S Midterm Tome

Susanna S. Epp · Discrete Mathematics with Applications, 5th edition

248 pages selected from 1057, plus a 4-page supplement · the sections underlying Lectures 1 to 6.

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SPEAKING MATHEMATICALLY

*Therefore O students study mathematics and do not build
without foundations.* —Leonardo da Vinci (1452–1519)

The aim of this book is to introduce you to a mathematical way of thinking that can serve you in a wide variety of situations. Often when you start work on a mathematical problem, you may have only a vague sense of how to proceed. You may begin by looking at examples, drawing pictures, playing around with notation, rereading the problem to focus on more of its details, and so forth. The closer you get to a solution, however, the more your thinking has to crystallize. And the more you need to understand, the more you need language that expresses mathematical ideas clearly, precisely, and unambiguously.

This chapter will introduce you to some of the special language that is a foundation for much mathematical thought, the language of variables, sets, relations, and functions. Think of the chapter like the exercises you would do before an important sporting event. Its goal is to warm up your mental muscles so that you can do your best.

1.1 Variables

A variable is sometimes thought of as a mathematical “John Doe” because you can use it as a placeholder when you want to talk about something but either (1) you imagine that it has one or more values but you don’t know what they are, or (2) you want whatever you say about it to be equally true for all elements in a given set, and so you don’t want to be restricted to considering only a particular, concrete value for it. To illustrate the first use, consider asking

Is there a number with the following property: doubling it and adding 3 gives the same result as squaring it?

In this sentence you can introduce a variable to replace the potentially ambiguous word “it”:

Is there a number x with the property that $2x + 3 = x^2$?

The advantage of using a variable is that it allows you to give a temporary name to what you are seeking so that you can perform concrete computations with it to help discover its possible values. To emphasize the role of the variable as a placeholder, you might write the following:

Is there a number $\square$ with the property that $2 \cdot \square + 3 = \square^2$?

The emptiness of the box can help you imagine filling it in with a variety of different values, some of which might make the two sides equal and others of which might not.

In this sense, a variable in a computer program is similar to a mathematical variable because it creates a location in computer memory (either actual or virtual) into which values can be placed.

To illustrate the second use of variables, consider the statement

No matter what number might be chosen, if it is greater than 2, then its square is greater than 4.

In this case introducing a variable to give a temporary name to the (arbitrary) number you might choose enables you to maintain the generality of the statement, and replacing all instances of the word “it” by the name of the variable ensures that possible ambiguity is avoided:

No matter what number n might be chosen, if n is greater than 2, then n^2 is greater than 4.

Example 1.1.1 Writing Sentences Using Variables

Use variables to rewrite the following sentences more formally.

- Are there numbers with the property that the sum of their squares equals the square of their sum?
- Given any real number, its square is nonnegative.

Solution

- Are there numbers a and b with the property that $a^2 + b^2 = (a + b)^2$?
Or: Are there numbers a and b such that $a^2 + b^2 = (a + b)^2$?
Or: Do there exist any numbers a and b such that $a^2 + b^2 = (a + b)^2$?
- Given any real number r , r^2 is nonnegative.
Or: For any real number r , $r^2 \geq 0$.
Or: For every real number r , $r^2 \geq 0$.

Note In part (a) the answer is yes. For instance, $a = 1$ and $b = 0$ would work. Can you think of other numbers that would also work?

Some Important Kinds of Mathematical Statements

Three of the most important kinds of sentences in mathematics are universal statements, conditional statements, and existential statements:

A **universal statement** says that a certain property is true for all elements in a set. (For example: *All positive numbers are greater than zero.*)

A **conditional statement** says that if one thing is true then some other thing also has to be true. (For example: *If 378 is divisible by 18, then 378 is divisible by 6.*)

Given a property that may or may not be true, an **existential statement** says that there is at least one thing for which the property is true. (For example: *There is a prime number that is even.*)

In later sections we will define each kind of statement carefully and discuss all of them in detail. The aim here is for you to realize that combinations of these statements can be expressed in a variety of different ways. One way uses ordinary, everyday language and another expresses the statement using one or more variables. The exercises are designed to help you start becoming comfortable in translating from one way to another.

Universal Conditional Statements

Universal statements contain some variation of the words “for every” and conditional statements contain versions of the words “if-then.” A **universal conditional statement** is a statement that is both universal and conditional. Here is an example:

For every animal a , if a is a dog, then a is a mammal.

One of the most important facts about universal conditional statements is that they can be rewritten in ways that make them appear to be purely universal or purely conditional. For example, the previous statement can be rewritten in a way that makes its conditional nature explicit but its universal nature implicit:

If a is a dog, then a is a mammal.

Or: If an animal is a dog, then the animal is a mammal.

The statement can also be expressed so as to make its universal nature explicit and its conditional nature implicit:

For every dog a , a is a mammal.

Or: All dogs are mammals.

The crucial point is that the ability to translate among various ways of expressing universal conditional statements is enormously useful for doing mathematics and many parts of computer science.

Example 1.1.2

Rewriting a Universal Conditional Statement

Fill in the blanks to rewrite the following statement:

For every real number x , if x is nonzero then x^2 is positive.

- If a real number is nonzero, then its square _____.
- For every nonzero real number x , _____.
- If x _____, then _____.
- The square of any nonzero real number is _____.
- All nonzero real numbers have _____.

Note If you introduce x in the first part of the sentence, be sure to include it in the second part of the sentence.

Solution

- is positive
- x^2 is positive
- is a nonzero real number; x^2 is positive
- positive
- positive squares (*or:* squares that are positive)

Universal Existential Statements

A **universal existential statement** is a statement that is universal because its first part says that a certain property is true for all objects of a given type, and it is existential because its second part asserts the existence of something. For example:

Every real number has an additive inverse.

Note For a number b to be an additive inverse for a number a means that $a + b = 0$.

In this statement the property “has an additive inverse” applies universally to all real numbers. “Has an additive inverse” asserts the existence of something—an additive inverse—for each real number. However, the nature of the additive inverse depends on the real number; different real numbers have different additive inverses. Knowing that an additive inverse is a real number, you can rewrite this statement in several ways, some less formal and some more formal:*

All real numbers have additive inverses.

Or: For every real number r , there is an additive inverse for r .

Or: For every real number r , there is a real number s such that s is an additive inverse for r .

Introducing names for the variables simplifies references in further discussion. For instance, after the third version of the statement you might go on to write: When r is positive, s is negative, when r is negative, s is positive, and when r is zero, s is also zero.

One of the most important reasons for using variables in mathematics is that it gives you the ability to refer to quantities unambiguously throughout a lengthy mathematical argument, while not restricting you to consider only specific values for them.

Example 1.1.3

Rewriting a Universal Existential Statement

Fill in the blanks to rewrite the following statement: Every pot has a lid.

- All pots _____.
- For every pot P , there is _____.
- For every pot P , there is a lid L such that _____.

Solution

- have lids
- a lid for P
- L is a lid for P

Existential Universal Statements

An **existential universal statement** is a statement that is existential because its first part asserts that a certain object exists and is universal because its second part says that the object satisfies a certain property for all things of a certain kind. For example:

There is a positive integer that is less than or equal to every positive integer.

This statement is true because the number one is a positive integer, and it satisfies the property of being less than or equal to every positive integer. We can rewrite the statement in several ways, some less formal and some more formal:

Some positive integer is less than or equal to every positive integer.

Or: There is a positive integer m that is less than or equal to every positive integer.

Or: There is a positive integer m such that every positive integer is greater than or equal to m .

Or: There is a positive integer m with the property that for every positive integer n , $m \leq n$.

*A conditional could be used to help express this statement, but we postpone the additional complexity to a later chapter.

Example 1.1.4 Rewriting an Existential Universal Statement

Fill in the blanks to rewrite the following statement in three different ways:

There is a person in my class who is at least as old as every person in my class.

- Some _____ is at least as old as _____.
- There is a person p in my class such that p is _____.
- There is a person p in my class with the property that for every person q in my class, p is _____.

Solution

- person in my class; every person in my class
- at least as old as every person in my class
- at least as old as q

Some of the most important mathematical concepts, such as the definition of limit of a sequence, can only be defined using phrases that are universal, existential, and conditional, and they require the use of all three phrases “for every,” “there is,” and “if-then.” For example, if $a_1, a_2, a_3, \dots$ is a sequence of real numbers, saying that

the limit of a_n as n approaches infinity is L

means that

for every positive real number ε , **there is** an integer N such that
for every integer n , **if** $n > N$ **then** $-\varepsilon < a_n - L < \varepsilon$.

TEST YOURSELF

Answers to Test Yourself questions are located at the end of each section.

- A universal statement asserts that a certain property is _____ for _____.
- A conditional statement asserts that if one thing _____ then some other thing _____.
- Given a property that may or may not be true, an existential statement asserts that _____ for which the property is true.

EXERCISE SET 1.1

Appendix B contains either full or partial solutions to all exercises with blue numbers. When the solution is not complete, the exercise number has an “H” next to it. A “*” next to an exercise number signals that the exercise is more challenging than usual. Be careful not to get into the habit of turning to the solutions too quickly. Make every effort to work exercises on your own before checking your answers. See the Preface for additional sources of assistance and further study.

In each of 1–6, fill in the blanks using a variable or variables to rewrite the given statement.

- Is there a real number whose square is -1 ?
 - Is there a real number x such that _____?
 - Does there exist _____ such that $x^2 = -1$?
- Is there an integer that has a remainder of 2 when it is divided by 5 and a remainder of 3 when it is divided by 6?
 - Is there an integer n such that n has _____?
 - Does there exist _____ such that if n is divided by 5 the remainder is 2 and if _____?

Note: There are integers with this property. Can you think of one?
- Given any two distinct real numbers, there is a real number in between them.

- a. Given any two distinct real numbers a and b , there is a real number c such that c is ____.
 - b. For any two ____, ____ such that c is between a and b .
4. Given any real number, there is a real number that is greater.
 - a. Given any real number r , there is ____ s such that s is ____.
 - b. For any ____, ____ such that $s > r$.
 5. The reciprocal of any positive real number is positive.
 - a. Given any positive real number r , the reciprocal of ____.
 - b. For any real number r , if r is ____, then ____.
 - c. If a real number r ____, then ____.
 6. The cube root of any negative real number is negative.
 - a. Given any negative real number s , the cube root of ____.
 - b. For any real number s , if s is ____, then ____.
 - c. If a real number s ____, then ____.
 7. Rewrite the following statements less formally, without using variables. Determine, as best as you can, whether the statements are true or false.
 - a. There are real numbers u and v with the property that $u + v < u - v$.
 - b. There is a real number x such that $x^2 < x$.
 - c. For every positive integer n , $n^2 \geq n$.
 - d. For all real numbers a and b , $|a + b| \leq |a| + |b|$.
- In each of 8–13, fill in the blanks to rewrite the given statement.
8. For every object J , if J is a square then J has four sides.
 - a. All squares ____.
 - b. Every square ____.
 - c. If an object is a square, then it ____.
 - d. If J ____, then J ____.
 - e. For every square J , ____.
 9. For every equation E , if E is quadratic then E has at most two real solutions.
 - a. All quadratic equations ____.
 - b. Every quadratic equation ____.
 - c. If an equation is quadratic, then it ____.
 - d. If E ____, then E ____.
 - e. For every quadratic equation E , ____.
 10. Every nonzero real number has a reciprocal.
 - a. All nonzero real numbers ____.
 - b. For every nonzero real number r , there is ____ for r .
 - c. For every nonzero real number r , there is a real number s such that ____.
 11. Every positive number has a positive square root.
 - a. All positive numbers ____.
 - b. For every positive number e , there is ____ for e .
 - c. For every positive number e , there is a positive number r such that ____.
 12. There is a real number whose product with every number leaves the number unchanged.
 - a. Some ____ has the property that its ____.
 - b. There is a real number r such that the product of r ____.
 - c. There is a real number r with the property that for every real number s , ____.
 13. There is a real number whose product with every real number equals zero.
 - a. Some ____ has the property that its ____.
 - b. There is a real number a such that the product of a ____.
 - c. There is a real number a with the property that for every real number b , ____.

ANSWERS FOR TEST YOURSELF

1. true; all elements of a set 2. is true; also has to be true 3. there is at least one thing

1.2 The Language of Sets

... when we attempt to express in mathematical symbols a condition proposed in words. First, we must understand thoroughly the condition. Second, we must be familiar with the forms of mathematical expression. —George Polyá (1887–1985)

Use of the word *set* as a formal mathematical term was introduced in 1879 by Georg Cantor (1845–1918). For most mathematical purposes we can think of a set intuitively, as

Cantor did, simply as a collection of elements. For instance, if C is the set of all countries that are currently in the United Nations, then the United States is an element of C , and if I is the set of all integers from 1 to 100, then the number 57 is an element of I .

Set-Roster Notation

If S is a set, the notation $x \in S$ means that x is an element of S . The notation $x \notin S$ means that x is not an element of S . A set may be specified using the **set-roster notation** by writing all of its elements between braces. For example, $\{1, 2, 3\}$ denotes the set whose elements are 1, 2, and 3. A variation of the notation is sometimes used to describe a very large set, as when we write $\{1, 2, 3, \dots, 100\}$ to refer to the set of all integers from 1 to 100. A similar notation can also describe an infinite set, as when we write $\{1, 2, 3, \dots\}$ to refer to the set of all positive integers. (The symbol $\dots$ is called an **ellipsis** and is read “and so forth.”)

The **axiom of extension** says that a set is completely determined by what its elements are—not the order in which they might be listed or the fact that some elements might be listed more than once.

Example 1.2.1 Using the Set-Roster Notation

- a. Let $A = \{1, 2, 3\}$, $B = \{3, 1, 2\}$, and $C = \{1, 1, 2, 3, 3, 3\}$. What are the elements of A , B , and C ? How are A , B , and C related?
- b. Is $\{0\} = 0$?
- c. How many elements are in the set $\{1, \{1\}\}$?
- d. For each nonnegative integer n , let $U_n = \{n, -n\}$. Find U_1 , U_2 , and U_0 .

- Solution**
- a. A , B , and C have exactly the same three elements: 1, 2, and 3. Therefore, A , B , and C are simply different ways to represent the same set.
 - b. $\{0\} \neq 0$ because $\{0\}$ is a set with one element, namely 0, whereas 0 is just the symbol that represents the number zero.
 - c. The set $\{1, \{1\}\}$ has two elements: 1 and the set whose only element is 1.
 - d. $U_1 = \{1, -1\}$, $U_2 = \{2, -2\}$, $U_0 = \{0, -0\} = \{0, 0\} = \{0\}$. ■

Certain sets of numbers are so frequently referred to that they are given special symbolic names. These are summarized in the following table.

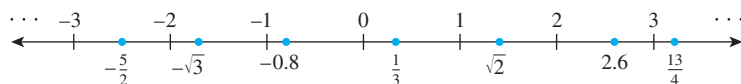
Note The $\mathbf{Z}$ is the first letter of the German word for integers, *Zahlen*. It stands for the *set* of all integers and should not be used as a shorthand for the word *integer*.
When the Symbols $\mathbf{R}$, $\mathbf{Q}$, and $\mathbf{Z}$ are handwritten, they appear as $\mathbb{R}$, $\mathbb{Q}$, and $\mathbb{Z}$.

Symbol	Set
$\mathbf{R}$	the set of all real numbers
$\mathbf{Z}$	the set of all integers
$\mathbf{Q}$	the set of all rational numbers, or quotients of integers

Addition of a superscript $+$ or $-$ or the letters *nonneg* indicates that only the positive or negative or nonnegative elements of the set, respectively, are to be included. Thus $\mathbf{R}^+$ denotes the set of positive real numbers, and $\mathbf{Z}^{\text{nonneg}}$ refers to the set of nonnegative integers: 0, 1, 2, 3, 4, and so forth. Some authors refer to the set of nonnegative integers as the set of **natural numbers** and denote it as $\mathbf{N}$. Other authors call only the positive

integers natural numbers. To prevent confusion, we simply avoid using the phrase *natural numbers* in this book.

The set of real numbers is usually pictured as the set of all points on a line, as shown below. The number 0 corresponds to a middle point, called the *origin*. A unit of distance is marked off, and each point to the right of the origin corresponds to a positive real number found by computing its distance from the origin. Each point to the left of the origin corresponds to a negative real number, which is denoted by computing its distance from the origin and putting a minus sign in front of the resulting number. The set of real numbers is therefore divided into three parts: the set of positive real numbers, the set of negative real numbers, and the number 0. *Note that 0 is neither positive nor negative.* Labels are given for a few real numbers corresponding to points on the line shown below.



The real number line is called *continuous* because it is imagined to have no holes. The set of integers corresponds to a collection of points located at fixed intervals along the real number line. Thus every integer is a real number, and because the integers are all separated from each other, the set of integers is called *discrete*. The name *discrete mathematics* comes from the distinction between continuous and discrete mathematical objects.

Another way to specify a set uses what is called the *set-builder notation*.

Note We read the left-hand brace as “the set of all” and the vertical line as “such that.” In all other mathematical contexts, however, we do not use a vertical line to denote the words “such that”; we abbreviate “such that” as “s. t.” or “s. th.” or “ $\therefore$.”

Set-Builder Notation

Let S denote a set and let $P(x)$ be a property that elements of S may or may not satisfy. We may define a new set to be **the set of all elements x in S such that $P(x)$ is true.** We denote this set as follows:

$$\{x \in S \mid P(x)\}$$

the set of all
such that

Occasionally we will write $\{x \mid P(x)\}$ without being specific about where the element x comes from. It turns out that unrestricted use of this notation can lead to genuine contradictions in set theory. We will discuss one of these in Section 6.4 and will be careful to use this notation purely as a convenience in cases where the set S could be specified if necessary.

Example 1.2.2

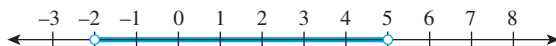
Using the Set-Builder Notation

Given that $\mathbf{R}$ denotes the set of all real numbers, $\mathbf{Z}$ the set of all integers, and $\mathbf{Z}^+$ the set of all positive integers, describe each of the following sets.

- $\{x \in \mathbf{R} \mid -2 < x < 5\}$
- $\{x \in \mathbf{Z} \mid -2 < x < 5\}$
- $\{x \in \mathbf{Z}^+ \mid -2 < x < 5\}$

Solution

- $\{x \in \mathbf{R} \mid -2 < x < 5\}$ is the open interval of real numbers (strictly) between -2 and 5 . It is pictured as follows:



- b. $\{x \in \mathbf{Z} \mid -2 < x < 5\}$ is the set of all integers (strictly) between -2 and 5 . It is equal to the set $\{-1, 0, 1, 2, 3, 4\}$.
- c. Since all the integers in $\mathbf{Z}^+$ are positive, $\{x \in \mathbf{Z}^+ \mid -2 < x < 5\} = \{1, 2, 3, 4\}$. ■

Subsets

A basic relation between sets is that of subset.

Definition

If A and B are sets, then A is called a **subset** of B , written $A \subseteq B$, if, and only if, every element of A is also an element of B .

Symbolically:

$A \subseteq B$ means that for every element x , if $x \in A$ then $x \in B$.

The phrases A is contained in B and B contains A are alternative ways of saying that A is a subset of B .

It follows from the definition of subset that for a set A not to be a subset of a set B means that there is at least one element of A that is not an element of B . Symbolically:

$A \not\subseteq B$ means that there is at least one element x such that $x \in A$ and $x \notin B$.

Definition

Let A and B be sets. A is a **proper subset** of B if, and only if, every element of A is in B but there is at least one element of B that is not in A .

Example 1.2.3

Subsets

Let $A = \mathbf{Z}^+$, $B = \{n \in \mathbf{Z} \mid 0 \leq n \leq 100\}$, and $C = \{100, 200, 300, 400, 500\}$. Evaluate the truth and falsity of each of the following statements.

- $B \subseteq A$
- C is a proper subset of A
- C and B have at least one element in common
- $C \subseteq B$
- $C \subseteq C$

Solution

- False. Zero is not a positive integer. Thus zero is in B but zero is not in A , and so $B \not\subseteq A$.
- True. Each element in C is a positive integer and, hence, is in A , but there are elements in A that are not in C . For instance, 1 is in A and not in C .
- True. For example, 100 is in both C and B .
- False. For example, 200 is in C but not in B .
- True. Every element in C is in C . In general, the definition of subset implies that all sets are subsets of themselves. ■

Example 1.2.4 Distinction between $\in$ and $\subseteq$

Which of the following are true statements?

- a. $2 \in \{1, 2, 3\}$ b. $\{2\} \in \{1, 2, 3\}$ c. $2 \subseteq \{1, 2, 3\}$
 d. $\{2\} \subseteq \{1, 2, 3\}$ e. $\{2\} \subseteq \{\{1\}, \{2\}\}$ f. $\{2\} \in \{\{1\}, \{2\}\}$

Solution Only (a), (d), and (f) are true.

For (b) to be true, the set $\{1, 2, 3\}$ would have to contain the element $\{2\}$. But the only elements of $\{1, 2, 3\}$ are 1, 2, and 3, and 2 is not equal to $\{2\}$. Hence (b) is false.

For (c) to be true, the number 2 would have to be a set and every element in the set 2 would have to be an element of $\{1, 2, 3\}$. This is not the case, so (c) is false.

For (e) to be true, every element in the set containing only the number 2 would have to be an element of the set whose elements are $\{1\}$ and $\{2\}$. But 2 is not equal to either $\{1\}$ or $\{2\}$, and so (e) is false. ■

Cartesian Products

With the introduction of Georg Cantor's set theory in the late nineteenth century, it began to seem possible to put mathematics on a firm logical foundation by developing all of its various branches from set theory and logic alone. A major stumbling block was how to use sets to define an ordered pair because the definition of a set is unaffected by the order in which its elements are listed. For example, $\{a, b\}$ and $\{b, a\}$ represent the same set, whereas in an ordered pair we want to be able to indicate which element comes first.

In 1914 crucial breakthroughs were made by Norbert Wiener (1894–1964), a young American who had recently received his Ph.D. from Harvard, and the German mathematician Felix Hausdorff (1868–1942). Both gave definitions showing that an ordered pair can be defined as a certain type of set, but both definitions were somewhat awkward. Finally, in 1921, the Polish mathematician Kazimierz Kuratowski (1896–1980) published the following definition, which has since become standard. It says that an ordered pair is a set of the form

$$\{\{a\}, \{a, b\}\}.$$

This set has elements, $\{a\}$ and $\{a, b\}$. If $a \neq b$, then the two sets are distinct and a is in both sets whereas b is not. This allows us to distinguish between a and b and say that a is the first element of the ordered pair and b is the second element of the pair. If $a = b$, then we can simply say that a is both the first and the second element of the pair. In this case the set that defines the ordered pair becomes $\{\{a\}, \{a, a\}\}$, which equals $\{\{a\}\}$.

However, it was only long after ordered pairs had been used extensively in mathematics that mathematicians realized that it was possible to define them entirely in terms of sets, and, in any case, the set notation would be cumbersome to use on a regular basis. The usual notation for ordered pairs refers to $\{\{a\}, \{a, b\}\}$ more simply as (a, b) .

Notation

Given elements a and b , the symbol (a, b) denotes the **ordered pair** consisting of a and b together with the specification that a is the first element of the pair and b is the second element. Two ordered pairs (a, b) and (c, d) are equal if, and only if, $a = c$ and $b = d$. Symbolically:

$$(a, b) = (c, d) \text{ means that } a = c \text{ and } b = d.$$



ArchivePL/Alamy Stock Photo

Kazimierz Kuratowski
(1896–1980)

Example 1.2.5 Ordered Pairs

- Is $(1, 2) = (2, 1)$?
- Is $(3, \frac{5}{10}) = (\sqrt{9}, \frac{1}{2})$?
- What is the first element of $(1, 1)$?

Solution

- No. By definition of equality of ordered pairs,

$$(1, 2) = (2, 1) \text{ if, and only if, } 1 = 2 \text{ and } 2 = 1.$$

But $1 \neq 2$, and so the ordered pairs are not equal.

- Yes. By definition of equality of ordered pairs,

$$\left(3, \frac{5}{10}\right) = \left(\sqrt{9}, \frac{1}{2}\right) \text{ if, and only if, } 3 = \sqrt{9} \text{ and } \frac{5}{10} = \frac{1}{2}.$$

Because these equations are both true, the ordered pairs are equal.

- In the ordered pair $(1, 1)$, the first and the second elements are both 1. ■

The notation for an *ordered n -tuple* generalizes the notation for an ordered pair to a set with any finite number of elements. It also takes both order and multiplicity into account.

Definition

Let n be a positive integer and let $x_1, x_2, \dots, x_n$ be (not necessarily distinct) elements. The **ordered n -tuple**, $(x_1, x_2, \dots, x_n)$, consists of $x_1, x_2, \dots, x_n$ together with the ordering: first x_1 , then x_2 , and so forth up to x_n . An ordered 2-tuple is called an **ordered pair**, and an ordered 3-tuple is called an **ordered triple**.

Two ordered n -tuples $(x_1, x_2, \dots, x_n)$ and $(y_1, y_2, \dots, y_n)$ are **equal** if, and only if, $x_1 = y_1, x_2 = y_2, \dots$, and $x_n = y_n$.

Symbolically:

$$(x_1, x_2, \dots, x_n) = (y_1, y_2, \dots, y_n) \Leftrightarrow x_1 = y_1, x_2 = y_2, \dots, x_n = y_n.$$

Example 1.2.6 Ordered n -tuples

- Is $(1, 2, 3, 4) = (1, 2, 4, 3)$?
- Is $(3, (-2)^2, \frac{1}{2}) = (\sqrt{9}, 4, \frac{3}{6})$?

Solution

- No. By definition of equality of ordered 4-tuples,

$$(1, 2, 3, 4) = (1, 2, 4, 3) \Leftrightarrow 1 = 1, 2 = 2, 3 = 4, \text{ and } 4 = 3$$

But $3 \neq 4$, and so the ordered 4-tuples are not equal.

- Yes. By definition of equality of ordered triples,

$$\left(3, (-2)^2, \frac{1}{2}\right) = \left(\sqrt{9}, 4, \frac{3}{6}\right) \Leftrightarrow 3 = \sqrt{9} \text{ and } (-2)^2 = 4 \text{ and } \frac{1}{2} = \frac{3}{6}.$$

Because these equations are all true, the two ordered triples are equal. ■

Definition

Given sets $A_1, A_2, \dots, A_n$, the **Cartesian product** of $A_1, A_2, \dots, A_n$, denoted $A_1 \times A_2 \times \cdots \times A_n$, is the set of all ordered n -tuples $(a_1, a_2, \dots, a_n)$ where $a_1 \in A_1$, $a_2 \in A_2, \dots, a_n \in A_n$.

Symbolically:

$$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_1 \in A_1, a_2 \in A_2, \dots, a_n \in A_n\}.$$

In particular,

$$A_1 \times A_2 = \{(a_1, a_2) \mid a_1 \in A_1 \text{ and } a_2 \in A_2\}$$

is the Cartesian product of A_1 and A_2 .

Example 1.2.7 Cartesian Products

Let $A = \{x, y\}$, $B = \{1, 2, 3\}$, and $C = \{a, b\}$.

- Find $A \times B$.
- Find $B \times A$.
- Find $A \times A$.
- How many elements are in $A \times B$, $B \times A$, and $A \times A$?
- Find $(A \times B) \times C$.
- Find $A \times B \times C$.
- Let $\mathbf{R}$ denote the set of all real numbers. Describe $\mathbf{R} \times \mathbf{R}$.

Solution

- $A \times B = \{(x, 1), (y, 1), (x, 2), (y, 2), (x, 3), (y, 3)\}$
- $B \times A = \{(1, x), (1, y), (2, x), (2, y), (3, x), (3, y)\}$
- $A \times A = \{(x, x), (x, y), (y, x), (y, y)\}$
- $A \times B$ has 6 elements. Note that this is the number of elements in A times the number of elements in B . $B \times A$ has 6 elements, the number of elements in B times the number of elements in A . $A \times A$ has 4 elements, the number of elements in A times the number of elements in A .
- The Cartesian product of A and B is a set, so it may be used as one of the sets making up another Cartesian product. This is the case for $(A \times B) \times C$.

$$\begin{aligned} (A \times B) \times C &= \{(u, v) \mid u \in A \times B \text{ and } v \in C\} \quad \text{by definition of Cartesian product} \\ &= \{((x, 1), a), ((x, 2), a), ((x, 3), a), ((y, 1), a), \\ &\quad ((y, 2), a), ((y, 3), a), ((x, 1), b), ((x, 2), b), ((x, 3), b), \\ &\quad ((y, 1), b), ((y, 2), b), ((y, 3), b)\} \end{aligned}$$

- The Cartesian product $A \times B \times C$ is superficially similar to but is not quite the same mathematical object as $(A \times B) \times C$. $(A \times B) \times C$ is a set of ordered pairs of which one element is itself an ordered pair, whereas $A \times B \times C$ is a set of ordered triples. By definition of Cartesian product,

Note This is why it makes sense to call a Cartesian product a product!

$$\begin{aligned}
 A \times B \times C &= \{(u, v, w) \mid u \in A, v \in B, \text{ and } w \in C\} \\
 &= \{(x, 1, a), (x, 2, a), (x, 3, a), (y, 1, a), (y, 2, a), (y, 3, a), (x, 1, b), \\
 &\quad (x, 2, b), (x, 3, b), (y, 1, b), (y, 2, b), (y, 3, b)\}.
 \end{aligned}$$

- g. $\mathbf{R} \times \mathbf{R}$ is the set of all ordered pairs (x, y) where both x and y are real numbers. If horizontal and vertical axes are drawn on a plane and a unit length is marked off, then each ordered pair in $\mathbf{R} \times \mathbf{R}$ corresponds to a unique point in the plane, with the first and second elements of the pair indicating, respectively, the horizontal and vertical positions of the point. The term **Cartesian plane** is often used to refer to a plane with this coordinate system, as illustrated in Figure 1.2.1.

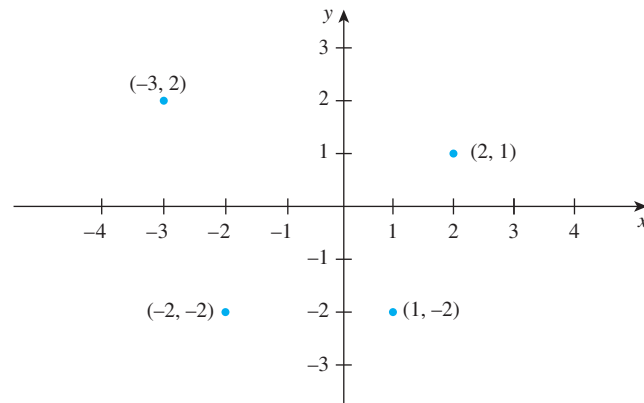


FIGURE 1.2.1 A Cartesian Plane

Another notation, which is important in both mathematics and computer science, denotes objects called *strings*.*

Definition

Let n be a positive integer. Given a finite set A , a **string of length n over A** is an ordered n -tuple of elements of A written without parentheses or commas. The elements of A are called the **characters** of the string. The **null string** over A is defined to be the “string” with no characters. It is often denoted λ and is said to have length 0. If $A = \{0, 1\}$, then a string over A is called a **bit string**.

Example 1.2.8 Strings

Let $A = \{a, b\}$. List all the strings of length 3 over A with at least two characters that are the same.

Solution

$$aab, aba, baa, aaa, bba, bab, abb, bbb$$

In computer programming it is important to distinguish among different kinds of data structures and to respect the notations that are used for them. Similarly in mathematics, it is important to distinguish among, say, $\{a, b, c\}$, $\{\{a, b\}, c\}$, (a, b, c) , $(a, (b, c))$, abc , and so forth, because these are all significantly different objects.

*A more formal definition of string, using recursion, is given in Section 5.9.

TEST YOURSELF

- When the elements of a set are given using the set-roster notation, the order in which they are listed _____.
- The symbol $\mathbf{R}$ denotes _____.
- The symbol $\mathbf{Z}$ denotes _____.
- The symbol $\mathbf{Q}$ denotes _____.
- The notation $\{x \mid P(x)\}$ is read _____.
- For a set A to be a subset of a set B means that _____.
- Given sets A and B , the Cartesian product $A \times B$ is _____.
- Given sets A , B , and C , the Cartesian product $A \times B \times C$ is _____.
- A string of length n over a set S is an ordered n -tuple of elements of S , written without _____ or _____.

EXERCISE SET 1.2

- Which of the following sets are equal?
 $A = \{a, b, c, d\}$ $B = \{d, e, a, c\}$
 $C = \{d, b, a, c\}$ $D = \{a, a, d, e, c, e\}$
- Write in words how to read each of the following out loud.
 - $\{x \in \mathbf{R}^+ \mid 0 < x < 1\}$
 - $\{x \in \mathbf{R} \mid x \leq 0 \text{ or } x \geq 1\}$
 - $\{n \in \mathbf{Z} \mid n \text{ is a factor of } 6\}$
 - $\{n \in \mathbf{Z}^+ \mid n \text{ is a factor of } 6\}$
- Is $4 = \{4\}$?
 - How many elements are in the set $\{3, 4, 3, 5\}$?
 - How many elements are in the set $\{1, \{1\}, \{1, \{1\}\}\}$?
- Is $2 \in \{2\}$?
 - How many elements are in the set $\{2, 2, 2, 2\}$?
 - How many elements are in the set $\{0, \{0\}\}$?
 - Is $\{0\} \in \{\{0\}, \{1\}\}$?
 - Is $0 \in \{\{0\}, \{1\}\}$?
- Which of the following sets are equal?
 $A = \{0, 1, 2\}$
 $B = \{x \in \mathbf{R} \mid -1 \leq x < 3\}$
 $C = \{x \in \mathbf{R} \mid -1 < x < 3\}$
 $D = \{x \in \mathbf{Z} \mid -1 < x < 3\}$
 $E = \{x \in \mathbf{Z}^+ \mid -1 < x < 3\}$
- For each integer n , let $T_n = \{n, n^2\}$. How many elements are in each of T_2 , T_{-3} , T_1 , and T_0 ? Justify your answers.
- Use the set-roster notation to indicate the elements in each of the following sets.
 - $S = \{n \in \mathbf{Z} \mid n = (-1)^k, \text{ for some integer } k\}$.
 - $T = \{m \in \mathbf{Z} \mid m = 1 + (-1)^i, \text{ for some integer } i\}$.
 - $U = \{r \in \mathbf{Z} \mid 2 \leq r \leq -2\}$
 - $V = \{s \in \mathbf{Z} \mid s > 2 \text{ or } s < 3\}$
 - $W = \{t \in \mathbf{Z} \mid 1 < t < -3\}$
 - $X = \{u \in \mathbf{Z} \mid u \leq 4 \text{ or } u \geq 1\}$
- Let $A = \{c, d, f, g\}$, $B = \{f, j\}$, and $C = \{d, g\}$. Answer each of the following questions. Give reasons for your answers.
 - Is $B \subseteq A$?
 - Is $C \subseteq A$?
 - Is $C \subseteq C$?
 - Is C a proper subset of A ?
- Is $3 \in \{1, 2, 3\}$?
 - Is $1 \subseteq \{1\}$?
 - Is $\{2\} \in \{1, 2\}$?
 - Is $\{3\} \in \{1, \{2\}, \{3\}\}$?
 - Is $1 \in \{1\}$?
 - Is $\{2\} \subseteq \{1, \{2\}, \{3\}\}$?
 - Is $\{1\} \subseteq \{1, 2\}$?
 - Is $1 \in \{\{1\}, 2\}$?
 - Is $\{1\} \subseteq \{1, \{2\}\}$?
 - Is $\{1\} \subseteq \{1\}$?
- Is $((-2)^2, -2^2) = (-2^2, (-2)^2)$?
 - Is $(5, -5) = (-5, 5)$?
 - Is $(8 - 9, \sqrt[3]{-1}) = (-1, -1)$?
 - Is $(\frac{-2}{4}, (-2)^3) = (\frac{3}{6}, -8)$?
- Let $A = \{w, x, y, z\}$ and $B = \{a, b\}$. Use the set-roster notation to write each of the following sets, and indicate the number of elements that are in each set.
 - $A \times B$
 - $B \times A$
 - $A \times A$
 - $B \times B$

12. Let $S = \{2, 4, 6\}$ and $T = \{1, 3, 5\}$. Use the set-roster notation to write each of the following sets, and indicate the number of elements that are in each set.
- $S \times T$
 - $T \times S$
 - $S \times S$
 - $T \times T$
13. Let $A = \{1, 2, 3\}$, $B = \{u\}$, and $C = \{m, n\}$. Find each of the following sets.
- $A \times (B \times C)$
 - $(A \times B) \times C$
 - $A \times B \times C$
14. Let $R = \{a\}$, $S = \{x, y\}$, and $T = \{p, q, r\}$. Find each of the following sets.
- $R \times (S \times T)$
 - $(R \times S) \times T$
 - $R \times S \times T$
15. Let $S = \{0, 1\}$. List all the strings of length 4 over S that contain three or more 0's.
16. Let $T = \{x, y\}$. List all the strings of length 5 over T that have exactly one y .

ANSWERS FOR TEST YOURSELF

1. does not matter 2. the set of all real numbers 3. the set of all integers 4. the set of all rational numbers 5. the set of all x such that $P(x)$ 6. every element in A is an element

in B 7. the set of all ordered pairs (a, b) where a is in A and b is in B 8. the set of ordered triples of the form (a, b, c) where $a \in A$, $b \in B$, and $c \in C$ 9. parentheses; commas

1.3 The Language of Relations and Functions

Mathematics is a language. —Josiah Willard Gibbs (1839–1903)

There are many kinds of relationships in the world. For instance, we say that two people are related by blood if they share a common ancestor and that they are related by marriage if one shares a common ancestor with the spouse of the other. We also speak of the relationship between student and teacher, between people who work for the same employer, and between people who share a common ethnic background.

Similarly, the objects of mathematics may be related in various ways. A set A may be said to be related to a set B if A is a subset of B , or if A is not a subset of B , or if A and B have at least one element in common. A number x may be said to be related to a number y if $x < y$, or if x is a factor of y , or if $x^2 + y^2 = 1$. Two identifiers in a computer program may be said to be related if they have the same first eight characters, or if the same memory location is used to store their values when the program is executed. And the list could go on!

Let $A = \{0, 1, 2\}$ and $B = \{1, 2, 3\}$ and let us say that an element x in A is related to an element y in B if, and only if, x is less than y . Let us use the notation $x R y$ as a shorthand for the sentence “ x is related to y .” Then

$0 R 1$	since	$0 < 1$,	
$0 R 2$	since	$0 < 2$,	
$0 R 3$	since	$0 < 3$,	
$1 R 2$	since	$1 < 2$,	
$1 R 3$	since	$1 < 3$,	and
$2 R 3$	since	$2 < 3$.	

On the other hand, if the notation $x \not R y$ represents the sentence “ x is not related to y ,” then

$$1 \not R 1 \quad \text{since} \quad 1 \not\leq 1,$$

$$2 \not R 1 \quad \text{since} \quad 2 \not\leq 1, \quad \text{and}$$

$$2 \not R 2 \quad \text{since} \quad 2 \not\leq 2.$$

Recall that the Cartesian product of A and B , $A \times B$, consists of all ordered pairs whose first element is in A and whose second element is in B :

$$A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}.$$

In this case,

$$A \times B = \{(0, 1), (0, 2), (0, 3), (1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)\}.$$

The elements of some ordered pairs in $A \times B$ are related, whereas the elements of other ordered pairs are not. Consider the set of all ordered pairs in $A \times B$ whose elements are related

$$\{(0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (2, 3)\}.$$

Observe that knowing which ordered pairs lie in this set is equivalent to knowing which elements are related to which. The relation itself can therefore be thought of as the totality of ordered pairs whose elements are related by the given condition. The formal mathematical definition of relation, based on this idea, was introduced by the American mathematician and logician C. S. Peirce in the nineteenth century.

Definition

Let A and B be sets. A **relation R from A to B** is a subset of $A \times B$. Given an ordered pair (x, y) in $A \times B$, x is **related to y by R** , written $x R y$, if, and only if, (x, y) is in R . The set A is called the **domain** of R and the set B is called its **co-domain**.

The notation for a relation R may be written symbolically as follows:

$$x R y \quad \text{means that} \quad (x, y) \in R.$$

The notation $x \not R y$ means that x is not related to y by R :

$$x \not R y \quad \text{means that} \quad (x, y) \notin R.$$

Example 1.3.1 A Relation as a Subset

Let $A = \{1, 2\}$ and $B = \{1, 2, 3\}$ and define a relation R from A to B as follows: Given any $(x, y) \in A \times B$,

$$(x, y) \in R \quad \text{means that} \quad \frac{x - y}{2} \text{ is an integer.}$$

- State explicitly which ordered pairs are in $A \times B$ and which are in R .
- Is $1 R 3$? Is $2 R 3$? Is $2 R 2$?
- What are the domain and co-domain of R ?

Solution

- $A \times B = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)\}$. To determine explicitly the composition of R , examine each ordered pair in $A \times B$ to see whether its elements satisfy the defining condition for R .

$$\begin{aligned}
 (1, 1) &\in R \text{ because } \frac{1-1}{2} = \frac{0}{2} = 0, \text{ which is an integer.} \\
 (1, 2) &\notin R \text{ because } \frac{1-2}{2} = \frac{-1}{2}, \text{ which is not an integer.} \\
 (1, 3) &\in R \text{ because } \frac{1-3}{2} = \frac{-2}{2} = -1, \text{ which is an integer.} \\
 (2, 1) &\notin R \text{ because } \frac{2-1}{2} = \frac{1}{2}, \text{ which is not an integer.} \\
 (2, 2) &\in R \text{ because } \frac{2-2}{2} = \frac{0}{2} = 0, \text{ which is an integer.} \\
 (2, 3) &\notin R \text{ because } \frac{2-3}{2} = \frac{-1}{2}, \text{ which is not an integer.}
 \end{aligned}$$

Thus

$$R = \{(1, 1), (1, 3), (2, 2)\}$$

- b. Yes, $1 R 3$ because $(1, 3) \in R$.
 No, $2 R 3$ because $(2, 3) \notin R$.
 Yes, $2 R 2$ because $(2, 2) \in R$.
- c. The domain of R is $\{1, 2\}$ and the co-domain is $\{1, 2, 3\}$.

Example 1.3.2

The Circle Relation

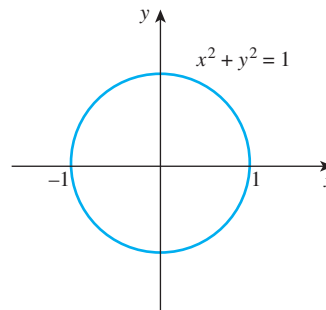
Define a relation C from $\mathbf{R}$ to $\mathbf{R}$ as follows: For any $(x, y) \in \mathbf{R} \times \mathbf{R}$,

$$(x, y) \in C \text{ means that } x^2 + y^2 = 1.$$

- a. Is $(1, 0) \in C$? Is $(0, 0) \in C$? Is $(-\frac{1}{2}, \frac{\sqrt{3}}{2}) \in C$? Is $-2 C 0$? Is $0 C (-1)$? Is $1 C 1$?
- b. What are the domain and co-domain of C ?
- c. Draw a graph for C by plotting the points of C in the Cartesian plane.

Solution

- a. Yes, $(1, 0) \in C$ because $1^2 + 0^2 = 1$.
 No, $(0, 0) \notin C$ because $0^2 + 0^2 = 0 \neq 1$.
 Yes, $(-\frac{1}{2}, \frac{\sqrt{3}}{2}) \in C$ because $(-\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2 = \frac{1}{4} + \frac{3}{4} = 1$.
 No, $-2 \not C 0$ because $(-2)^2 + 0^2 = 4 \neq 1$.
 Yes, $0 C (-1)$ because $0^2 + (-1)^2 = 1$.
 No, $1 \not C 1$ because $1^2 + 1^2 = 2 \neq 1$.
- b. The domain and co-domain of C are both $\mathbf{R}$, the set of all real numbers.
- c.



Arrow Diagram of a Relation

Suppose R is a relation from a set A to a set B . The **arrow diagram for R** is obtained as follows:

1. Represent the elements of A as points in one region and the elements of B as points in another region.
2. For each x in A and y in B , draw an arrow from x to y if, and only if, x is related to y by R . Symbolically:

Draw an arrow from x to y

if, and only if, $x R y$

if, and only if, $(x, y) \in R$.

Example 1.3.3 Arrow Diagrams of Relations

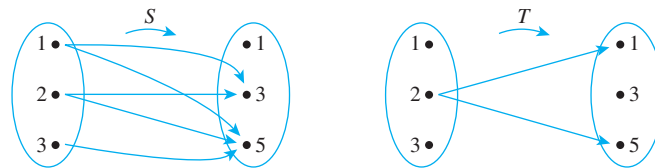
Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3\}$ and define relations S and T from A to B as follows:
For every $(x, y) \in A \times B$,

$(x, y) \in S$ means that $x < y$ S is a “less than” relation.

$T = \{(2, 1), (2, 5)\}$.

Draw arrow diagrams for S and T .

Solution



These example relations illustrate that it is possible to have several arrows coming out of the same element of A pointing in different directions. Also, it is quite possible to have an element of A that does not have an arrow coming out of it. ■

Functions

In Section 1.2 we showed that ordered pairs can be defined in terms of sets and we defined Cartesian products in terms of ordered pairs. In this section we introduced relations as subsets of Cartesian products. Thus we can now define functions in a way that depends only on the concept of set. Although this definition is not obviously related to the way we usually work with functions in mathematics, it is satisfying from a theoretical point of view, and computer scientists like it because it is particularly well suited for operating with functions on a computer.

Definition

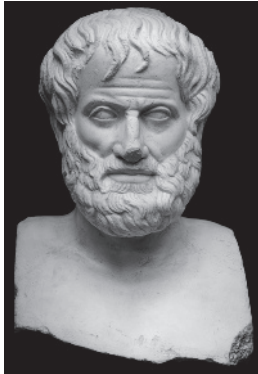
A **function F from a set A to a set B** is a relation with domain A and co-domain B that satisfies the following two properties:

1. For every element x in A , there is an element y in B such that $(x, y) \in F$.
2. For all elements x in A and y and z in B ,

if $(x, y) \in F$ and $(x, z) \in F$, then $y = z$.

CHAPTER 2

THE LOGIC OF COMPOUND STATEMENTS



Aristotle
(384 B.C.–322 B.C.)

Mohamed Osama/Alamy Stock Photo

The first great treatises on logic were written by the Greek philosopher Aristotle. They were a collection of rules for deductive reasoning that were intended to serve as a basis for the study of every branch of knowledge. In the seventeenth century, the German philosopher and mathematician Gottfried Leibniz conceived the idea of using symbols to mechanize the process of deductive reasoning in much the same way that algebraic notation had mechanized the process of reasoning about numbers and their relationships. Leibniz's idea was realized in the nineteenth century by the English mathematicians George Boole and Augustus De Morgan, who founded the modern subject of symbolic logic. With research continuing to the present day, symbolic logic has provided, among other things, the theoretical basis for many areas of computer science such as digital logic circuit design (see Sections 2.4 and 2.5), relational database theory (see Section 8.1), automata theory and computability (see Section 7.4 and Chapter 12), and artificial intelligence (see Sections 3.3, 10.1, and 10.5).

2.1 Logical Form and Logical Equivalence

Logic is a science of the necessary laws of thought, without which no employment of the understanding and the reason takes place. —Immanuel Kant, 1785

An argument is a sequence of statements aimed at demonstrating the truth of an assertion. The assertion at the end of the sequence is called the *conclusion*, and the preceding statements are called *premises*. To have confidence in the conclusion that you draw from an argument, you must be sure that the premises are acceptable on their own merits or follow from other statements that are known to be true.

In logic, the form of an argument is distinguished from its content. Logical analysis won't help you determine the intrinsic merit of an argument's content, but it will help you analyze an argument's form to determine whether the truth of the conclusion follows *necessarily* from the truth of the premises. For this reason logic is sometimes defined as the science of necessary inference or the science of reasoning.

Consider the following two arguments. They have very different content but their logical form is the same. To help make this clear, we use letters like p , q , and r to represent component sentences; we let the expression “not p ” refer to the sentence “It is not the case that p ”; and we let the symbol $\therefore$ stand for the word “therefore.”

Argument 1

p	q	r
{	{	{
If the bell rings or the flag drops,	then the race is over.	

not r	not p	not q
{	{	{
$\therefore$ If the race is not over,	then the bell hasn't rung and	the flag hasn't dropped.

Argument 2

$$\begin{array}{c}
 \begin{array}{ccc}
 \overbrace{x=2}^p & \text{or} & \overbrace{x=-2}^q, \text{ then } \overbrace{x^2=4}^r. \\
 \underbrace{\text{not } r} & & \underbrace{\text{not } p} & & \underbrace{\text{not } q} \\
 \therefore \text{ If } x^2 \neq 4, \text{ then } x \neq 2 \text{ and } x \neq -2.
 \end{array}
 \end{array}$$

The common form of the arguments is

If p or q , then r .

$\therefore$ If not r , then not p and not q .

In exercise 10 in Section 2.3 you will show that this form of argument is *valid* in the sense that *if* its assumptions are true, then its conclusion must also be true.

Example 2.1.1 Identifying Logical Form

Fill in the blanks below so that argument (b) has the same form as argument (a). Then represent the common form of the arguments using letters to stand for component sentences.

- a. If Jane is a math major or Jane is a computer science major, then Jane will take Math 150.
Jane is a computer science major.
Therefore, Jane will take Math 150.
- b. If logic is easy or (1), then (2).
I will study hard.
Therefore, I will get an A in this course.

Solution

- I (will) study hard.
- I will get an A in this course.

Common form: If p or q , then r .

q .
Therefore, r .

Statements

Most of the definitions of formal logic have been developed so that they agree with the natural or intuitive logic used by people who have been educated to think clearly and use language carefully. The differences that exist between formal and intuitive logic are necessary to avoid ambiguity and obtain consistency.

In any mathematical theory, new terms are defined by using those that have been previously defined. However, this process has to start somewhere. A few initial terms necessarily remain undefined. In logic, the words *sentence*, *true*, and *false* are the initial undefined terms.

Definition

A **statement** (or **proposition**) is a sentence that is true or false but not both.

For example, “Two plus two equals four” and “Two plus two equals five” are both statements, the first because it is true and the second because it is false. On the other hand, the truth or falsity of

$$x^2 + 2 = 11$$

depends on the value of x . For some values of x , it is true ($x = 3$ and $x = -3$), whereas for other values it is false. Similarly, the truth or falsity of

$$x + y > 0$$

depends on the values of x and y . For instance, when $x = -1$ and $y = 2$ it is true, whereas when $x = -1$ and $y = 1$ it is false. In Section 3.1 we will discuss ways to transform sentences of these forms into statements.

Compound Statements

Note $\sim p$ means “not p ”
 $p \wedge q$ means “ p and q ”
 $p \vee q$ means “ p or q ”

We now introduce three symbols that are used to build more complicated logical expressions out of simpler ones. The symbol $\sim$ denotes *not*, $\wedge$ denotes *and*, and $\vee$ denotes *or*. Given a statement p , the sentence “ $\sim p$ ” is read “not p ” or “It is not the case that p .” In some computer languages the symbol $\neg$ is used in place of $\sim$. Given another statement q , the sentence “ $p \wedge q$ ” is read “ p and q .” The sentence “ $p \vee q$ ” is read “ p or q .”

In expressions that include the symbol $\sim$ as well as $\wedge$ or $\vee$, the **order of operations** specifies that $\sim$ is performed first. For instance, $\sim p \wedge q = (\sim p) \wedge q$. In logical expressions, as in ordinary algebraic expressions, the order of operations can be overridden through the use of parentheses. Thus $\sim(p \wedge q)$ represents the negation of the conjunction of p and q . In this, as in most treatments of logic, the symbols $\wedge$ and $\vee$ are considered coequal in order of operation, and an expression such as $p \wedge q \vee r$ is considered ambiguous. This expression must be written as either $(p \wedge q) \vee r$ or $p \wedge (q \vee r)$ to have meaning.

A variety of English words translate into logic as $\wedge$, $\vee$, or $\sim$. For instance, the word *but* translates the same as *and* when it links two independent clauses, as in “Jim is tall but he is not heavy.” Generally, the word *but* is used in place of *and* when the part of the sentence that follows is, in some way, unexpected. Another example involves the words *neither-nor*. When Shakespeare wrote, “Neither a borrower nor a lender be,” he meant, “Do not be a borrower and do not be a lender.” So if p and q are statements, then

p but q	means	p and q
neither p nor q	means	$\sim p$ and $\sim q$.

Example 2.1.2 Translating from English to Symbols: *But* and *Neither-Nor*

Write each of the following sentences symbolically, letting h = “It is hot” and s = “It is sunny.”

- It is not hot but it is sunny.
- It is neither hot nor sunny.

Solution

- The given sentence is equivalent to “It is not hot and it is sunny,” which can be written symbolically as $\sim h \wedge s$.
- To say it is neither hot nor sunny means that it is not hot and it is not sunny. Therefore, the given sentence can be written symbolically as $\sim h \wedge \sim s$. ■

The notation for inequalities involves *and* and *or* statements. For instance, if x , a , and b are particular real numbers, then

Note The point of specifying x , a , and b to be *particular* real numbers is to ensure that sentences such as “ $x < a$ ” and “ $x \geq b$ ” are either true or false and hence that they are statements.

$$\begin{array}{llll} x \leq a & \text{means} & x < a & \text{or} & x = a \\ a \leq x \leq b & \text{means} & a \leq x & \text{and} & x \leq b. \end{array}$$

Note that the inequality $2 \leq x \leq 1$ is not satisfied by any real numbers because

$$2 \leq x \leq 1 \quad \text{means} \quad 2 \leq x \quad \text{and} \quad x \leq 1,$$

and this is false no matter what number x happens to be.

Example 2.1.3 And, Or, and Inequalities

Suppose x is a particular real number. Let p , q , and r symbolize “ $0 < x$,” “ $x < 3$,” and “ $x = 3$,” respectively. Write the following inequalities symbolically:

- a. $x \leq 3$ b. $0 < x < 3$ c. $0 < x \leq 3$

Solution

- a. $q \vee r$ b. $p \wedge q$ c. $p \wedge (q \vee r)$ ■

Truth Values

In Examples 2.1.2 and 2.1.3 we built compound sentences out of component statements and the terms *not*, *and*, and *or*. If such sentences are to be statements, however, they must have well-defined **truth values**—they must be either true or false. We now define such compound sentences as statements by specifying their truth values in terms of the statements that compose them.

Note Think of negation like this:

The negation of a statement is a statement that exactly expresses what it would mean for the statement to be false.

Definition

If p is a statement variable, the **negation** of p is “not p ” or “It is not the case that p ” and is denoted $\sim p$. It has opposite truth value from p : if p is true, $\sim p$ is false; if p is false, $\sim p$ is true.

The truth values for negation are summarized in a *truth table*.

Truth Table for $\sim p$

p	$\sim p$
T	F
F	T

In ordinary language the sentence “It is hot and it is sunny” is understood to be true when both conditions—being hot and being sunny—are satisfied. If it is hot but not sunny, or sunny but not hot, or neither hot nor sunny, the sentence is understood to be

false. The formal definition of truth values for an *and* statement agrees with this general understanding.

Definition

If p and q are statement variables, the **conjunction** of p and q is “ p and q ,” denoted $p \wedge q$. It is true when, and only when, both p and q are true. If either p or q is false, or if both are false, $p \wedge q$ is false.

The truth values for conjunction can also be summarized in a truth table. The table is obtained by considering the four possible combinations of truth values for p and q . Each combination is displayed in one row of the table; the corresponding truth value for the whole statement is placed in the right-most column of that row. Note that the only row containing a T is the first one because an *and* statement is true only when both components are true.

Note The only way for an *and* statement to be true is for both components to be true. So in the truth table for an *and* statement the first row is the only row with a T.

Truth Table for $p \wedge q$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

The order of truth values for p and q in the table above is TT, TF, FT, FF. It is not absolutely necessary to write the truth values in this order, although it is customary to do so. So please use this order for all truth tables involving two statement variables. Example 2.1.5 shows the standard order for truth tables that involve three statement variables.

In the case of disjunction—statements of the form “ p or q ”—intuitive logic offers two alternative interpretations. In ordinary language *or* is sometimes used in an exclusive sense (p or q but not both) and sometimes in an inclusive sense (p or q or both). A waiter who says you may have “coffee, tea, or milk” uses the word *or* in an exclusive sense: Extra payment is generally required if you want more than one beverage. On the other hand, a waiter who offers “cream or sugar” uses the word *or* in an inclusive sense: You are entitled to both cream and sugar if you wish to have them.

Mathematicians and logicians avoid possible ambiguity about the meaning of the word *or* by understanding it to mean the inclusive “and/or.” The symbol $\vee$ comes from the Latin word *vel*, which means *or* in its inclusive sense. To express the exclusive *or*, the phrase *p or q but not both* is used.

Note The statement “ $2 \leq 2$ ” means that 2 is less than 2 or 2 equals 2. It is true because $2 = 2$.

Definition

If p and q are statement variables, the **disjunction** of p and q is “ p or q ,” denoted $p \vee q$. It is true when either p is true, or q is true, or both p and q are true; it is false only when both p and q are false.

Here is the truth table for disjunction:

Note The only way for an *or* statement to be false is for both components to be false. So in the truth table for an *or* statement the last row is the only row with an *F*.

Truth Table for $p \vee q$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Evaluating the Truth of More General Compound Statements

Note Java, C, and C++ use the following notations:

~	!
^	&&
∨	

Now that truth values have been assigned to $\sim p$, $p \wedge q$, and $p \vee q$, consider the question of assigning truth values to more complicated expressions such as $\sim p \vee q$, $(p \vee q) \wedge \sim(p \wedge q)$, and $(p \wedge q) \vee r$. Such expressions are called *statement forms* (or *propositional forms*). The close relationship between statement forms and *Boolean expressions* is discussed in Section 2.4.

Definition

A **statement form** (or **propositional form**) is an expression made up of statement variables (such as p , q , and r) and logical connectives (such as $\sim$, $\wedge$, and $\vee$) that becomes a statement when actual statements are substituted for the component statement variables. The **truth table** for a given statement form displays the truth values that correspond to all possible combinations of truth values for its component statement variables.

To compute the truth values for a statement form, follow rules similar to those used to evaluate algebraic expressions. For each combination of truth values for the statement variables, first evaluate the expressions within the innermost parentheses, then evaluate the expressions within the next innermost set of parentheses, and so forth, until you have the truth values for the complete expression.

Example 2.1.4 Truth Table for Exclusive Or

Note Exclusive *or* is often symbolized as $p \oplus q$ or $p \text{ XOR } q$.

Construct the truth table for the statement form $(p \vee q) \wedge \sim(p \wedge q)$. Note that when *or* is used in its exclusive sense, the statement “ p or q ” means “ p or q but not both” or “ p or q and not both p and q ,” which translates into symbols as $(p \vee q) \wedge \sim(p \wedge q)$.

Solution Set up columns labeled p , q , $p \vee q$, $p \wedge q$, $\sim(p \wedge q)$, and $(p \vee q) \wedge \sim(p \wedge q)$. Fill in the p and q columns with all the logically possible combinations of T’s and F’s. Then use the truth tables for $\vee$ and $\wedge$ to fill in the $p \vee q$ and $p \wedge q$ columns with the appropriate truth values. Next fill in the $\sim(p \wedge q)$ column by taking the opposites of the truth values for $p \wedge q$. For example, the entry for $\sim(p \wedge q)$ in the first row is F because in the first row the truth value of $p \wedge q$ is T. Finally, fill in the $(p \vee q) \wedge \sim(p \wedge q)$ column by considering the truth values for an *and* statement together with the truth values for $p \vee q$ and $\sim(p \wedge q)$. Since an *and* statement is true only when both components are true and since rows 2 and 3 are the only two rows where both $p \vee q$ and $\sim(p \wedge q)$ are true, put T in rows 2 and 3 and F in the remaining rows.

Note To fill out a truth table for an *and* statement, first put a T in each row where both components are true; then put an F in each of the remaining rows.

Truth Table for *Exclusive Or*: $(p \vee q) \wedge \sim(p \wedge q)$

p	q	$p \vee q$	$p \wedge q$	$\sim(p \wedge q)$	$(p \vee q) \wedge \sim(p \wedge q)$
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F

Example 2.1.5

Truth Table for $(p \wedge q) \vee \sim r$

Note To fill out a truth table for an *or* statement, first put an F in each row where both components are false; then put a T in each of the remaining rows.

Construct a truth table for the statement form $(p \wedge q) \vee \sim r$.

Solution Make columns headed p , q , r , $p \wedge q$, $\sim r$, and $(p \wedge q) \vee \sim r$. Enter the eight logically possible combinations of truth values for p , q , and r in the three left-most columns. Then fill in the truth values for $p \wedge q$ and for $\sim r$. Complete the table by considering the truth values for $(p \wedge q)$ and for $\sim r$ and the definition of an *or* statement. Since an *or* statement is false only when both components are false, the only rows in which the entry is F are the third, fifth, and seventh rows because those are the only rows in which the expressions $p \wedge q$ and $\sim r$ are both false. The entry for all the other rows is T.

p	q	r	$p \wedge q$	$\sim r$	$(p \wedge q) \vee \sim r$
T	T	T	T	F	T
T	T	F	T	T	T
T	F	T	F	F	F
T	F	F	F	T	T
F	T	T	F	F	F
F	T	F	F	T	T
F	F	T	F	F	F
F	F	F	F	T	T

The essential point about assigning truth values to compound statements is that it allows you—using logic alone—to judge the truth of a compound statement on the basis of your knowledge of the truth of its component parts. Logic does not help you determine the truth or falsity of the component statements. Rather, logic helps link these separate pieces of information together into a coherent whole.

Logical Equivalence

The statements

6 is greater than 2 and 2 is less than 6

are two different ways of saying the same thing. Why? Because of the definition of the phrases *greater than* and *less than*. By contrast, although the statements

(1) Dogs bark and cats meow and (2) Cats meow and dogs bark

are also two different ways of saying the same thing, the reason has nothing to do with the definition of the words. It has to do with the logical form of the statements. Any two statements whose logical forms are related in the same way as (1) and (2) would either both be true or both be false. You can see this by examining the following truth table, where the statement variables p and q are substituted for the component statements “Dogs bark” and “Cats meow,” respectively. The table shows that for each combination of truth values for p and q , $p \wedge q$ is true when, and only when, $q \wedge p$ is true. In such a case, the statement forms are called *logically equivalent*, and we say that (1) and (2) are *logically equivalent statements*.

p	q	$p \wedge q$	$q \wedge p$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	F	F


 $p \wedge q$ and $q \wedge p$ always
have the same truth
values, so they are
logically equivalent

Definition

Two *statement forms* are called **logically equivalent** if, and only if, they have identical truth values for each possible substitution of statements for their statement variables. The logical equivalence of statement forms P and Q is denoted by writing $P \equiv Q$.

Two *statements* are called **logically equivalent** if, and only if, they have logically equivalent forms when identical component statement variables are used to replace identical component statements.

Testing Whether Two Statement Forms P and Q Are Logically Equivalent

1. Construct a truth table with one column for the truth values of P and another column for the truth values of Q .
2. Check each combination of truth values of the statement variables to see whether the truth value of P is the same as the truth value of Q .
 - a. If in each row the truth value of P is the same as the truth value of Q , then P and Q are logically equivalent.
 - b. If in some row P has a different truth value from Q , then P and Q are not logically equivalent.

Example 2.1.6 Double Negative Property: $\sim(\sim p) \equiv p$

Construct a truth table to show that the negation of the negation of a statement is logically equivalent to the statement, annotating the table with a sentence of explanation.

Solution

p	$\sim p$	$\sim(\sim p)$
T	F	T
F	T	F

p and $\sim(\sim p)$ always have the same truth values, so they are logically equivalent

There are two ways to show that statement forms P and Q are *not* logically equivalent. As indicated previously, one is to use a truth table to find rows for which their truth values differ. The other way is to find concrete statements for each of the two forms, one of which is true and the other of which is false. The next example illustrates both of these ways.

Example 2.1.7 Showing Nonequivalence

Show that the statement forms $\sim(p \wedge q)$ and $\sim p \wedge \sim q$ are not logically equivalent.

Solution

a. This method uses a truth table annotated with a sentence of explanation.

p	q	$\sim p$	$\sim q$	$p \wedge q$	$\sim(p \wedge q)$	$\sim p \wedge \sim q$
T	T	F	F	T	F	F
T	F	F	T	F	T	F
F	T	T	F	F	T	F
F	F	T	T	F	T	T

$\sim(p \wedge q)$ and $\sim p \wedge \sim q$ have different truth values in rows 2 and 3, so they are not logically equivalent

b. This method uses an example to show that $\sim(p \wedge q)$ and $\sim p \wedge \sim q$ are not logically equivalent. Let p be the statement “ $0 < 1$ ” and let q be the statement “ $1 < 0$.” Then

$\sim(p \wedge q)$ is “It is not the case that both $0 < 1$ and $1 < 0$,”

which is true. On the other hand,

$\sim p \wedge \sim q$ is “ $0 \not< 1$ and $1 \not< 0$,”

which is false. This example shows that there are concrete statements you can substitute for p and q to make one of the statement forms true and the other false. Therefore, the statement forms are not logically equivalent.

Example 2.1.8 Negations of And and Or: De Morgan's Laws

For the statement “John is tall and Jim is redheaded” to be true, both components must be true. So for the statement to be false, one or both components must be false. Thus the negation can be written as “John is not tall or Jim is not redheaded.” In general, the negation

of the conjunction of two statements is logically equivalent to the disjunction of their negations. That is, statements of the forms $\sim(p \wedge q)$ and $\sim p \vee \sim q$ are logically equivalent. Check this using truth tables.

Solution

p	q	$\sim p$	$\sim q$	$p \wedge q$	$\sim(p \wedge q)$	$\sim p \vee \sim q$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

$\sim(p \wedge q)$ and $\sim p \vee \sim q$ always have the same truth values, so they are logically equivalent

Symbolically,

$$\sim(p \wedge q) \equiv \sim p \vee \sim q.$$

In the exercises at the end of this section you are asked to show the analogous law that the negation of the disjunction of two statements is logically equivalent to the conjunction of their negations:

$$\sim(p \vee q) \equiv \sim p \wedge \sim q.$$



Augustus De Morgan
(1806–1871)

Paul Fearn/Alamy Stock Photo

The two logical equivalences of Example 2.1.8 are known as **De Morgan’s laws** of logic in honor of Augustus De Morgan, who was the first to state them in formal mathematical terms.

De Morgan’s Laws

The negation of an *and* statement is logically equivalent to the *or* statement in which each component is negated.
The negation of an *or* statement is logically equivalent to the *and* statement in which each component is negated.

Example 2.1.9 Applying De Morgan’s Laws

Write negations for each of the following statements:

- a. John is 6 feet tall and he weighs at least 200 pounds.
- b. The bus was late or Tom’s watch was slow.

Solution

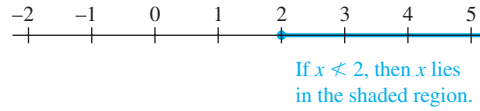
- a. John is not 6 feet tall or he weighs less than 200 pounds.
- b. The bus was not late and Tom’s watch was not slow.

Since the statement “neither p nor q ” means the same as “ $\sim p$ and $\sim q$,” an alternative answer for (b) is “Neither was the bus late nor was Tom’s watch slow.”

If x is a particular real number, saying that x is not less than 2 ($x \nless 2$) means that x does not lie to the left of 2 on the number line. This is equivalent to saying that either $x = 2$ or x lies to the right of 2 on the number line ($x = 2$ or $x > 2$). Hence,

$$x \nless 2 \text{ is equivalent to } x \geq 2.$$

Pictorially,



Similarly,

$$\begin{aligned} x \ngtr 2 & \text{ is equivalent to } x \leq 2, \\ x \neq 2 & \text{ is equivalent to } x > 2, \text{ and} \\ x \ngtr 2 & \text{ is equivalent to } x < 2. \end{aligned}$$

Example 2.1.10 Inequalities and De Morgan's Laws

Use De Morgan's laws to write the negation of $-1 < x \leq 4$.

Solution The given statement is equivalent to

$$-1 < x \text{ and } x \leq 4.$$

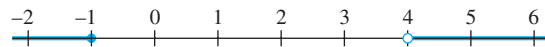
By De Morgan's laws, the negation is

$$-1 \nless x \text{ or } x \neq 4,$$

which is equivalent to

$$-1 \geq x \text{ or } x > 4.$$

Pictorially, if $-1 \geq x$ or $x > 4$, then x lies in the shaded region of the number line, as shown below.



Caution! The negation of $-1 < x \leq 4$ is *not* $-1 \nless x \neq 4$. It is also *not* $-1 \geq x > 4$.

De Morgan's laws are frequently used in writing computer programs. For instance, suppose you want your program to delete all files modified outside a certain range of dates, say from date 1 through date 2 inclusive. You would use the fact that

$$\sim(\text{date1} \leq \text{file_modification_date} \leq \text{date2})$$

is equivalent to

$$(\text{file_modification_date} < \text{date1}) \text{ or } (\text{date2} < \text{file_modification_date}).$$

Example 2.1.11 A Cautionary Example

According to De Morgan's laws, the negation of

$$p: \text{Jim is tall and Jim is thin}$$

is

$$\sim p: \text{Jim is not tall or Jim is not thin}$$

because the negation of an *and* statement is the *or* statement in which the two components are negated.

Unfortunately, a potentially confusing aspect of the English language can arise when you are taking negations of this kind. Note that statement p can be written more compactly as

p' : Jim is tall and thin.

When it is so written, another way to negate it is

$\sim(p')$: Jim is not tall and thin.

But in this form the negation looks like an *and* statement. Doesn't that violate De Morgan's laws?

Actually no violation occurs. The reason is that in formal logic the words *and* and *or* are allowed only between complete statements, not between sentence fragments. So when you apply De Morgan's laws, you must have complete statements on either side of each *and* and on either side of each *or*.



Caution! Although the laws of logic are extremely useful, they should be used as an *aid* to thinking, not as a mechanical substitute for it.

Tautologies and Contradictions

It has been said that all of mathematics reduces to tautologies. Although this is formally true, most working mathematicians think of their subject as having substance as well as form. Nonetheless, an intuitive grasp of basic logical tautologies is part of the equipment of anyone who reasons with mathematics.

Definition

A **tautology** is a statement form that is always true regardless of the truth values of the individual statements substituted for its statement variables. A statement whose form is a tautology is a **tautological statement**.

A **contradiction** is a statement form that is always false regardless of the truth values of the individual statements substituted for its statement variables. A statement whose form is a contradiction is a **contradictory statement**.

According to this definition, the truth of a tautological statement and the falsity of a contradictory statement are due to the logical structure of the statements themselves and are independent of the meanings of the statements.

Example 2.1.12

Tautologies and Contradictions

Show that the statement form $p \vee \sim p$ is a tautology and that the statement form $p \wedge \sim p$ is a contradiction.

Solution

p	$\sim p$	$p \vee \sim p$	$p \wedge \sim p$
T	F	T	F
F	T	T	F

↑

all T's, so
 $p \vee \sim p$ is
a tautology

↑

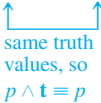
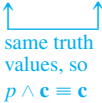
all F's, so
 $p \wedge \sim p$ is a
contradiction

Example 2.1.13**Logical Equivalence Involving Tautologies and Contradictions**

If **t** is a tautology and **c** is a contradiction, show that $p \wedge \mathbf{t} \equiv p$ and $p \wedge \mathbf{c} \equiv \mathbf{c}$.

Solution

p	t	$p \wedge \mathbf{t}$	p	c	$p \wedge \mathbf{c}$
T	T	T	T	F	F
F	T	F	F	F	F

Summary of Logical Equivalences

Knowledge of logically equivalent statements is very useful for constructing arguments. It often happens that it is difficult to see how a conclusion follows from one form of a statement, whereas it is easy to see how it follows from a logically equivalent form of the statement. A number of logical equivalences are summarized in Theorem 2.1.1 for future reference.

Theorem 2.1.1 Logical Equivalences

Given any statement variables p , q , and r , a tautology **t** and a contradiction **c**, the following logical equivalences hold.

- | | | |
|--|---|---|
| 1. <i>Commutative laws:</i> | $p \wedge q \equiv q \wedge p$ | $p \vee q \equiv q \vee p$ |
| 2. <i>Associative laws:</i> | $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ | $(p \vee q) \vee r \equiv p \vee (q \vee r)$ |
| 3. <i>Distributive laws:</i> | $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ | $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ |
| 4. <i>Identity laws:</i> | $p \wedge \mathbf{t} \equiv p$ | $p \vee \mathbf{c} \equiv p$ |
| 5. <i>Negation laws:</i> | $p \vee \sim p \equiv \mathbf{t}$ | $p \wedge \sim p \equiv \mathbf{c}$ |
| 6. <i>Double negative law:</i> | $\sim(\sim p) \equiv p$ | |
| 7. <i>Idempotent laws:</i> | $p \wedge p \equiv p$ | $p \vee p \equiv p$ |
| 8. <i>Universal bound laws:</i> | $p \vee \mathbf{t} \equiv \mathbf{t}$ | $p \wedge \mathbf{c} \equiv \mathbf{c}$ |
| 9. <i>De Morgan's laws:</i> | $\sim(p \wedge q) \equiv \sim p \vee \sim q$ | $\sim(p \vee q) \equiv \sim p \wedge \sim q$ |
| 10. <i>Absorption laws:</i> | $p \vee (p \wedge q) \equiv p$ | $p \wedge (p \vee q) \equiv p$ |
| 11. <i>Negations of t and c:</i> | $\sim \mathbf{t} \equiv \mathbf{c}$ | $\sim \mathbf{c} \equiv \mathbf{t}$ |

The proofs of laws 4 and 6, the first parts of laws 1 and 5, and the second part of law 9 have already been given as examples in the text. Proofs of the other parts of the theorem are left as exercises. In fact, it can be shown that the first five laws of Theorem 2.1.1 form a core from which the other laws can be derived. The first five laws are the axioms for a mathematical structure known as a Boolean algebra, which is discussed in Section 6.4.

The equivalences of Theorem 2.1.1 are general laws of thought that occur in all areas of human endeavor. They can also be used in a formal way to rewrite complicated statement forms more simply.

Example 2.1.14 Simplifying Statement Forms

Use Theorem 2.1.1 to verify the logical equivalence

$$\sim(\sim p \wedge q) \wedge (p \vee q) \equiv p.$$

Solution Use the laws of Theorem 2.1.1 to replace sections of the statement form on the left by logically equivalent expressions. Each time you do this, you obtain a logically equivalent statement form. Continue making replacements until you obtain the statement form on the right.

$$\begin{aligned} \sim(\sim p \wedge q) \wedge (p \vee q) &\equiv (\sim(\sim p) \vee \sim q) \wedge (p \vee q) && \text{by De Morgan's laws} \\ &\equiv (p \vee \sim q) \wedge (p \vee q) && \text{by the double negative law} \\ &\equiv (p \vee (\sim q \wedge q)) && \text{by the distributive law} \\ &\equiv p \vee (q \wedge \sim q) && \text{by the commutative law for } \wedge \\ &\equiv p \vee \mathbf{c} && \text{by the negation law} \\ &\equiv p && \text{by the identity law} \end{aligned}$$

Skill in simplifying statement forms is useful in constructing logically efficient computer programs and in designing digital logic circuits.

Although the properties in Theorem 2.1.1 can be used to prove the logical equivalence of two statement forms, they cannot be used to prove that statement forms are not logically equivalent. On the other hand, truth tables can always be used to determine both equivalence and nonequivalence, and truth tables are easy to program on a computer. When truth tables are used, however, checking for equivalence always requires 2^n steps, where n is the number of variables. Sometimes you can quickly see that two statement forms are equivalent by Theorem 2.1.1, whereas it would take quite a bit of calculating to show their equivalence using truth tables. For instance, it follows immediately from the associative law for $\wedge$ that $p \wedge (\sim q \wedge \sim r) \equiv (p \wedge \sim q) \wedge \sim r$, whereas a truth table verification requires constructing a table with eight rows.

TEST YOURSELF

Answers to Test Yourself questions are located at the end of each section.

1. An *and* statement is true when, and only when, both components are _____.
2. An *or* statement is false when, and only when, both components are _____.
3. Two statement forms are logically equivalent when, and only when, they always have _____.
4. De Morgan's laws say (1) that the negation of an *and* statement is logically equivalent to the _____ statement in which each component is _____, and (2) that the negation of an *or* statement is logically equivalent to the _____ statement in which each component is _____.
5. A tautology is a statement that is always _____.
6. A contradiction is a statement that is always _____.

EXERCISE SET 2.1*

In each of 1–4 represent the common form of each argument using letters to stand for component sentences, and fill in the blanks so that the argument in part (b) has the same logical form as the argument in part (a).

1. **a.** If all integers are rational, then the number 1 is rational.
All integers are rational.
Therefore, the number 1 is rational.
- b.** If all algebraic expressions can be written in prefix notation, then _____.

Therefore, $(a + 2b)(a^2 - b)$ can be written in prefix notation.
2. **a.** If all computer programs contain errors, then this program contains an error.
This program does not contain an error.
Therefore, it is not the case that all computer programs contain errors.
- b.** If _____, then _____.
2 is not odd.
Therefore, it is not the case that all prime numbers are odd.
3. **a.** This number is even or this number is odd.
This number is not even.
Therefore, this number is odd.
- b.** _____ or logic is confusing.
My mind is not shot.
Therefore, _____.
4. **a.** If the program syntax is faulty, then the computer will generate an error message.
If the computer generates an error message, then the program will not run.
Therefore, if the program syntax is faulty, then the program will not run.
- b.** If this simple graph _____, then it is complete.
If this graph _____, then any two of its vertices can be joined by a path.
Therefore, if this simple graph has 4 vertices and 6 edges, then _____.
5. Indicate which of the following sentences are statements.
 - a.** 1,024 is the smallest four-digit number that is a perfect square.
 - b.** She is a mathematics major.

- c.** $128 = 2^6$
- d.** $x = 2^6$

Write the statements in 6–9 in symbolic form using the symbols $\sim$, $\vee$, and $\wedge$ and the indicated letters to represent component statements.

6. Let s = “stocks are increasing” and i = “interest rates are steady.”
 - a.** Stocks are increasing but interest rates are steady.
 - b.** Neither are stocks increasing nor are interest rates steady.
7. Juan is a math major but not a computer science major. (m = “Juan is a math major,” c = “Juan is a computer science major”)
8. Let h = “John is healthy,” w = “John is wealthy,” and s = “John is wise.”
 - a.** John is healthy and wealthy but not wise.
 - b.** John is not wealthy but he is healthy and wise.
 - c.** John is neither healthy, wealthy, nor wise.
 - d.** John is neither wealthy nor wise, but he is healthy.
 - e.** John is wealthy, but he is not both healthy and wise.
9. Let p = “ $x > 5$,” q = “ $x = 5$,” and r = “ $10 > x$.”
 - a.** $x \geq 5$
 - b.** $10 > x > 5$
 - c.** $10 > x \geq 5$
10. Let p be the statement “DATAENDFLAG is off,” q the statement “ERROR equals 0,” and r the statement “SUM is less than 1,000.” Express the following sentences in symbolic notation.
 - a.** DATAENDFLAG is off, ERROR equals 0, and SUM is less than 1,000.
 - b.** DATAENDFLAG is off but ERROR is not equal to 0.
 - c.** DATAENDFLAG is off; however, ERROR is not 0 or SUM is greater than or equal to 1,000.
 - d.** DATAENDFLAG is on and ERROR equals 0 but SUM is greater than or equal to 1,000.
 - e.** Either DATAENDFLAG is on or it is the case that both ERROR equals 0 and SUM is less than 1,000.
11. In the following sentence, is the word *or* used in its inclusive or exclusive sense? A team wins the playoffs if it wins two games in a row or a total of three games.

*For exercises with blue numbers or letters, solutions are given in Appendix B. The symbol **H** indicates that only a hint or a partial solution is given. The symbol * signals that an exercise is more challenging than usual.

- *46. Let the symbol $\oplus$ denote *exclusive or*; so $p \oplus q \equiv (p \vee q) \wedge \sim(p \wedge q)$. Hence the truth table for $p \oplus q$ is as follows:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

- a. Find simpler statement forms that are logically equivalent to $p \oplus p$ and $(p \oplus p) \oplus p$.
- b. Is $(p \oplus q) \oplus r \equiv p \oplus (q \oplus r)$? Justify your answer.
- c. Is $(p \oplus q) \wedge r \equiv (p \wedge r) \oplus (q \wedge r)$? Justify your answer.
- *47. In logic and in standard English, a double negative is equivalent to a positive. There is one fairly common English usage in which a “double positive” is equivalent to a negative. What is it? Can you think of others?

In 48 and 49 below, a logical equivalence is derived from Theorem 2.1.1. Supply a reason for each step.

$$\begin{aligned}
 48. \quad (p \wedge \sim q) \vee (p \wedge q) &\equiv p \wedge (\sim q \vee q) && \text{by (a)} \\
 &\equiv p \wedge (q \vee \sim q) && \text{by (b)} \\
 &\equiv p \wedge \mathbf{t} && \text{by (c)} \\
 &\equiv p && \text{by (d)}
 \end{aligned}$$

Therefore, $(p \wedge \sim q) \vee (p \wedge q) \equiv p$.

$$\begin{aligned}
 49. \quad (p \vee \sim q) \wedge (\sim p \vee \sim q) & \\
 &\equiv (\sim q \vee p) \wedge (\sim q \vee \sim p) && \text{by (a)} \\
 &\equiv \sim q \vee (p \wedge \sim p) && \text{by (b)} \\
 &\equiv \sim q \vee \mathbf{c} && \text{by (c)} \\
 &\equiv \sim q && \text{by (d)}
 \end{aligned}$$

Therefore, $(p \vee \sim q) \wedge (\sim p \vee \sim q) \equiv \sim q$.

Use Theorem 2.1.1 to verify the logical equivalences in 50–54. Supply a reason for each step.

$$50. (p \wedge \sim q) \vee p \equiv p \qquad 51. p \wedge (\sim q \vee p) \equiv p$$

$$52. \sim(p \vee \sim q) \vee (\sim p \wedge \sim q) \equiv \sim p$$

$$53. \sim((\sim p \wedge q) \vee (\sim p \wedge \sim q)) \vee (p \wedge q) \equiv p$$

$$54. (p \wedge (\sim(\sim p \vee q))) \vee (p \wedge q) \equiv p$$

ANSWERS FOR TEST YOURSELF

1. true 2. false 3. the same truth values 4. *or*; negated; *and*; negated 5. true 6. false

2.2 Conditional Statements

... *hypothetical reasoning implies the subordination of the real to the realm of the possible* ... —Jean Piaget, 1972

When you make a logical inference or deduction, you reason *from* a hypothesis *to* a conclusion. Your aim is to be able to say, “If such and such is known, *then* something or other must be the case.”

Let p and q be statements. A sentence of the form “If p then q ” is denoted symbolically by “ $p \rightarrow q$ ”; p is called the *hypothesis* and q is called the *conclusion*. For instance, consider the following statement:

If $\underbrace{4,686 \text{ is divisible by } 6}_{\text{hypothesis}}$, then $\underbrace{4,686 \text{ is divisible by } 3}_{\text{conclusion}}$

Such a sentence is called *conditional* because the truth of statement q is conditioned on the truth of statement p .

The notation $p \rightarrow q$ indicates that $\rightarrow$ is a connective, like $\wedge$ or $\vee$, which can be used to join statements to create new statements. To define $p \rightarrow q$ as a statement, therefore, we must specify the truth values for $p \rightarrow q$ as we specified truth values for $p \wedge q$ and for $p \vee q$. As is the case with the other connectives, the formal definition of truth values for $\rightarrow$ (if-then) is based on its everyday, intuitive meaning. Consider an example.

Suppose you go to interview for a job at a store and the owner of the store makes you the following promise:

If you show up for work Monday morning, then you will get the job.

Under what circumstances are you justified in saying the owner spoke falsely? That is, under what circumstances is the above sentence false? The answer is: You *do* show up for work Monday morning and you do *not* get the job.

After all, the owner's promise only says you will get the job *if* a certain condition (showing up for work Monday morning) is met; it says nothing about what will happen if the condition is *not* met. So if the condition is not met, you cannot in fairness say the promise is false regardless of whether or not you get the job.

The above example was intended to convince you that *the only combination of circumstances in which you would call a conditional sentence false occurs when the hypothesis is true and the conclusion is false*. In all other cases, you would not call the sentence false. This implies that the only row of the truth table for $p \rightarrow q$ that should be filled in with an F is the row where p is T and q is F. No other row should contain an F. But each row of a truth table must be filled in with either a T or an F. Thus all other rows of the truth table for $p \rightarrow q$ must be filled in with T's.

Truth Table for $p \rightarrow q$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Definition

If p and q are statement variables, the **conditional** of q by p is “If p then q ” or “ p implies q ” and is denoted $p \rightarrow q$. It is false when p is true and q is false; otherwise it is true. We call p the **hypothesis** (or **antecedent**) of the conditional and q the **conclusion** (or **consequent**).

A conditional statement that is true by virtue of the fact that its hypothesis is false is often called **vacuously true** or **true by default**. Thus the statement “If you show up for work Monday morning, then you will get the job” is vacuously true if you do not show up for work Monday morning. In general, when the “if” part of an if-then statement is false, the statement as a whole is said to be true, regardless of whether the conclusion is true or false.

Example 2.2.1

A Conditional Statement with a False Hypothesis

Consider the statement

If $0 = 1$ then $1 = 2$.

As strange as it may seem, since the hypothesis of this statement is false, the statement as a whole is true. ■

Note For example, if you hypothesize that $0 = 1$, then, by adding 1 to both sides of the equation, you can deduce that $1 = 2$.

The philosopher Willard Van Orman Quine advises against using the phrase “ p implies q ” to mean “ $p \rightarrow q$ ” because the word *implies* suggests that q can be logically deduced from p and this is often not the case. Nonetheless, the phrase is used by many people, probably because it is a convenient replacement for the $\rightarrow$ symbol. And, of course, in many cases a conclusion can be deduced from a hypothesis, even when the hypothesis is false.

In expressions that include $\rightarrow$ as well as other logical operators such as $\wedge$, $\vee$, and $\sim$, the **order of operations** is that $\rightarrow$ is performed last. Thus, according to the specification of order of operations in Section 2.1, $\sim$ is performed first, then $\wedge$ and $\vee$, and finally $\rightarrow$.

Example 2.2.2 Truth Table for $p \vee \sim q \rightarrow \sim p$

Construct a truth table for the statement form $p \vee \sim q \rightarrow \sim p$.

Solution By the order of operations given above, the following two expressions are equivalent: $p \vee \sim q \rightarrow \sim p$ and $(p \vee (\sim q)) \rightarrow (\sim p)$, and this order governs the construction of the truth table. First fill in the four possible combinations of truth values for p and q , and then enter the truth values for $\sim p$ and $\sim q$ using the definition of negation. Next fill in the $p \vee \sim q$ column using the definition of $\vee$. Finally, fill in the $p \vee \sim q \rightarrow \sim p$ column using the definition of $\rightarrow$.

Note The only rows in which the hypothesis $p \vee \sim q$ is true and the conclusion $\sim p$ is false are the first and second rows. So you put F's in those two rows and T's in the other two rows.

		conclusion		hypothesis	
p	q	$\sim p$	$\sim q$	$p \vee \sim q$	$p \vee \sim q \rightarrow \sim p$
T	T	F	F	T	F
T	F	F	T	T	F
F	T	T	F	F	T
F	F	T	T	T	T

Logical Equivalences Involving $\rightarrow$

Imagine that you are trying to solve a problem involving three statements: p , q , and r . Suppose you know that the truth of r follows from the truth of p and also that the truth of r follows from the truth of q . Then no matter whether p or q is the case, the truth of r must follow. The division-into-cases method of analysis is based on this idea.

Example 2.2.3 Division into Cases: Showing That $p \vee q \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$

Use truth tables to show the logical equivalence of the statement forms $p \vee q \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$. Annotate the table with a sentence of explanation.

Solution First fill in the eight possible combinations of truth values for p , q , and r . Then fill in the columns for $p \vee q$, $p \rightarrow r$, and $q \rightarrow r$ using the definitions of *or* and *if-then*. For instance, the $p \rightarrow r$ column has F's in the second and fourth rows because these are the rows in which p is true and r is false. Next fill in the $p \vee q \rightarrow r$ column using the definition of *if-then*. The rows in which the hypothesis $p \vee q$ is true and the conclusion r is false are the second, fourth, and sixth. So F's go in these rows and T's in all the others. The complete table shows that $p \vee q \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ have the same truth values for each combination of truth values of p , q , and r . Hence the two statement forms are logically equivalent.

p	q	r	$p \vee q$	$p \rightarrow r$	$q \rightarrow r$	$p \vee q \rightarrow r$	$(p \rightarrow r) \wedge (q \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	T	T	T	T	T
T	F	F	T	F	T	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	F
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T


 $p \vee q \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$
 always have the same truth values,
 so they are logically equivalent

Representation of If-Then as Or

In exercise 13(a) at the end of this section you are asked to use truth tables to show that

$$p \rightarrow q \equiv \sim p \vee q.$$

The logical equivalence of “if p then q ” and “not p or q ” is occasionally used in everyday speech. Here is one instance.

Example 2.2.4

Application of the Equivalence between $\sim p \vee q$ and $p \rightarrow q$

Rewrite the following statement in if-then form.

Either you get to work on time or you are fired.

Solution Let $\sim p$ be

You get to work on time.

and q be

You are fired.

Then the given statement is $\sim p \vee q$. Also p is

You do not get to work on time.

So the equivalent if-then version, $p \rightarrow q$, is

If you do not get to work on time, then you are fired. ■

The Negation of a Conditional Statement

By definition, $p \rightarrow q$ is false if, and only if, its hypothesis, p , is true and its conclusion, q , is false. It follows that

The negation of “if p then q ” is logically equivalent to “ p and not q .”

This can be restated symbolically as follows:

$$\sim(p \rightarrow q) \equiv p \wedge \sim q$$

To obtain this result you can also start from the logical equivalence $p \rightarrow q \equiv \sim p \vee q$. Take the negation of both sides to obtain

$$\begin{aligned}\sim(p \rightarrow q) &\equiv \sim(\sim p \vee q) \\ &\equiv \sim(\sim p) \wedge (\sim q) && \text{by De Morgan's laws} \\ &\equiv p \wedge \sim q && \text{by the double negative law.}\end{aligned}$$

Yet another way to derive this result is to construct truth tables for $\sim(p \rightarrow q)$ and for $p \wedge \sim q$ and to check that they have the same truth values. (See exercise 13(b) at the end of this section.)

Example 2.2.5

Negations of If-Then Statements

Write negations for each of the following statements:

- If my car is in the repair shop, then I cannot get to class.
- If Sara lives in Athens, then she lives in Greece.

Solution

- My car is in the repair shop and I can get to class.
- Sara lives in Athens and she does not live in Greece. (Sara might live in Athens, Georgia; Athens, Ohio; or Athens, Wisconsin.)



Caution! Remember that the negation of an if-then statement does not start with the word *if*.

It is tempting to write the negation of an if-then statement as another if-then statement. Please resist that temptation!

The Contrapositive of a Conditional Statement

One of the most fundamental laws of logic is the equivalence between a conditional statement and its contrapositive.

Definition

The **contrapositive** of a conditional statement of the form “If p then q ” is

If $\sim q$ then $\sim p$.

Symbolically,

The contrapositive of $p \rightarrow q$ is $\sim q \rightarrow \sim p$.

The fact is that

A conditional statement is logically equivalent to its contrapositive.

You are asked to establish this equivalence in exercise 26 at the end of this section.

Example 2.2.6 Writing the Contrapositive

Write each of the following statements in its equivalent contrapositive form:

- If Howard can swim across the lake, then Howard can swim to the island.
- If today is Easter, then tomorrow is Monday.

Solution

- If Howard cannot swim to the island, then Howard cannot swim across the lake.
- If tomorrow is not Monday, then today is not Easter. ■

When you are trying to solve certain problems, you may find that the contrapositive form of a conditional statement is easier to work with than the original statement. Replacing a statement by its contrapositive may give the extra push that helps you over the top in your search for a solution. This logical equivalence is also the basis for one of the most important laws of deduction, modus tollens (to be explained in Section 2.3), and for the contrapositive method of proof (to be explained in Section 4.7).

The Converse and Inverse of a Conditional Statement

The fact that a conditional statement and its contrapositive are logically equivalent is very important and has wide application. Two other variants of a conditional statement are *not* logically equivalent to the statement.

Definition

Suppose a conditional statement of the form “If p then q ” is given.

- The **converse** is “If q then p .”
- The **inverse** is “If $\sim p$ then $\sim q$.”

Symbolically,

The converse of $p \rightarrow q$ is $q \rightarrow p$,

and

The inverse of $p \rightarrow q$ is $\sim p \rightarrow \sim q$.

Example 2.2.7 Writing the Converse and the Inverse

Write the converse and inverse of each of the following statements:

- If Howard can swim across the lake, then Howard can swim to the island.
- If today is Easter, then tomorrow is Monday.

Solution

- Converse:** If Howard can swim to the island, then Howard can swim across the lake.

Inverse: If Howard cannot swim across the lake, then Howard cannot swim to the island.

- Converse:** If tomorrow is Monday, then today is Easter.

Inverse: If today is not Easter, then tomorrow is not Monday. ■



Caution! Many people believe that if a conditional statement is true, then its converse and inverse must also be true. This is not correct! The converse might be true, but it does not have to be true.

Note that while the statement “If today is Easter, then tomorrow is Monday” is always true, both its converse and inverse are false on every Sunday except Easter.

1. A conditional statement and its converse are *not* logically equivalent.
2. A conditional statement and its inverse are *not* logically equivalent.
3. The converse and the inverse of a conditional statement are logically equivalent to each other.

In exercises 24, 25, and 27 at the end of this section, you are asked to use truth tables to verify the statements in the box above. Note that the truth of statement 3 also follows from the observation that the inverse of a conditional statement is the contrapositive of its converse.

Only If and the Biconditional

To say “ p only if q ” means that p can take place *only* if q takes place also. That is, if q does not take place, then p cannot take place. Another way to say this is that if p occurs, then q must also occur (by the logical equivalence between a statement and its contrapositive).

Definition

If p and q are statements,

p only if q means “if not q then not p ,”

or, equivalently,

“if p then q .”

Example 2.2.8

Converting Only If to If-Then

Rewrite the following statement in if-then form in two ways, one of which is the contrapositive of the other.

John will break the world’s record for the mile run only if he runs the mile in under four minutes.

Solution *Version 1:* If John does not run the mile in under four minutes, then he will not break the world’s record.

Version 2: If John breaks the world’s record, then he will have run the mile in under four minutes. ■



Caution! “ p only if q ” does *not* mean “ p if q .”

Note that it is possible for “ p only if q ” to be true at the same time that “ p if q ” is false. For instance, to say that John will break the world’s record only if he runs the mile in under four minutes does not mean that John will break the world’s record if he runs the mile in under four minutes. His time could be under four minutes but still not be fast enough to break the record.

Definition

Given statement variables p and q , the **biconditional of p and q** is “ p if, and only if, q ” and is denoted $p \leftrightarrow q$. It is true if both p and q have the same truth values and is false if p and q have opposite truth values. The words *if and only if* are sometimes abbreviated **iff**.

The biconditional has the following truth table:

Truth Table for $p \leftrightarrow q$

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

In order of operations $\leftrightarrow$ is coequal with $\rightarrow$. As with $\wedge$ and $\vee$, the only way to indicate precedence between them is to use parentheses. The full hierarchy of operations for the five logical operators is shown below.

Order of Operations for Logical Operators

1. $\sim$ Evaluate negations first.
2. $\wedge, \vee$ Evaluate $\wedge$ and $\vee$ second. When both are present, parentheses may be needed.
3. $\rightarrow, \leftrightarrow$ Evaluate $\rightarrow$ and $\leftrightarrow$ third. When both are present, parentheses may be needed.

According to the separate definitions of *if* and *only if*, saying “ p if, and only if, q ” should mean the same as saying both “ p if q ” and “ p only if q .” The following annotated truth table shows that this is the case:

Truth Table Showing That $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

p	q	$p \rightarrow q$	$q \rightarrow p$	$p \leftrightarrow q$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

$p \leftrightarrow q$ and $(p \rightarrow q) \wedge (q \rightarrow p)$
always have the same truth values,
so they are logically equivalent

Example 2.2.9 *If and Only If*

Rewrite the following statement as a conjunction of two if-then statements:

This computer program is correct if, and only if, it produces correct answers for all possible sets of input data.

Solution If this program is correct, then it produces the correct answers for all possible sets of input data; and if this program produces the correct answers for all possible sets of input data, then it is correct. ■

Necessary and Sufficient Conditions

The phrases *necessary condition* and *sufficient condition*, as used in formal English, correspond exactly to their definitions in logic.

Definition

If r and s are statements:

r is a **sufficient condition** for s means “if r then s .”

r is a **necessary condition** for s means “if not r then not s .”

In other words, to say “ r is a sufficient condition for s ” means that the occurrence of r is *sufficient* to guarantee the occurrence of s . On the other hand, to say “ r is a necessary condition for s ” means that if r does not occur, then s cannot occur either:

The occurrence of r is *necessary* to obtain the occurrence of s . Note that because of the equivalence between a statement and its contrapositive,

r is a necessary condition for s also means “if s then r .”

Consequently,

r is a necessary and sufficient condition for s means “ r if, and only if, s .”

Example 2.2.10 *Interpreting Necessary and Sufficient Conditions*

Consider the statement “If John is eligible to vote, then he is at least 18 years old.” The truth of the condition “John is eligible to vote” is *sufficient* to ensure the truth of the condition “John is at least 18 years old.” In addition, the condition “John is at least 18 years old” is *necessary* for the condition “John is eligible to vote” to be true. If John were younger than 18, then he would not be eligible to vote. ■

Example 2.2.11 *Converting a Sufficient Condition to If-Then Form*

Rewrite the following statement in the form “If A then B ”:

Pia’s birth on U.S. soil is a sufficient condition for her to be a U.S. citizen.

Solution If Pia was born on U.S. soil, then she is a U.S. citizen. ■

Example 2.2.12 Converting a Necessary Condition to If-Then Form

Use the contrapositive to rewrite the following statement in two ways:

George's attaining age 35 is a necessary condition for his being president of the United States.

Solution *Version 1:* If George has not attained the age of 35, then he cannot be president of the United States.

Version 2: If George can be president of the United States, then he has attained the age of 35. ■

Remarks

1. *In logic, a hypothesis and conclusion are not required to have related subject matters.*

In ordinary speech we never say things like “If computers are machines, then Babe Ruth was a baseball player” or “If $2 + 2 = 5$, then Mickey Mouse is president of the United States.” We formulate a sentence like “If p then q ” only if there is some connection of content between p and q .

In logic, however, the two parts of a conditional statement need not have related meanings. The reason? If there were such a requirement, who would enforce it? What one person perceives as two unrelated clauses may seem related to someone else. There would have to be a central arbiter to check each conditional sentence before anyone could use it, to be sure its clauses were in proper relation. This is impractical, to say the least!

Thus a statement like “if computers are machines, then Babe Ruth was a baseball player” is allowed, and it is even called true because both its hypothesis and its conclusion are true. Similarly, the statement “If $2 + 2 = 5$, then Mickey Mouse is president of the United States” is allowed and is called true because its hypothesis is false, even though doing so may seem ridiculous.

In mathematics it often happens that a carefully formulated definition that successfully covers the situations for which it was primarily intended is later seen to be satisfied by some extreme cases that the formulator did not have in mind. But those are the breaks, and it is important to get into the habit of exploring definitions fully to seek out and understand *all* their instances, even the unusual ones.

2. *In informal language, simple conditionals are often used to mean biconditionals.*

The formal statement “ p if, and only if, q ” is seldom used in ordinary language. Frequently, when people intend the biconditional they leave out either the *and only if* or the *if and*. That is, they say either “ p if q ” or “ p only if q ” when they really mean “ p if, and only if, q .” For example, consider the statement “You will get dessert if, and only if, you eat your dinner.” Logically, this is equivalent to the conjunction of the following two statements.

Statement 1: If you eat your dinner, then you will get dessert.

Statement 2: You will get dessert only if you eat your dinner.

or

If you do not eat your dinner, then you will not get dessert.

Now how many parents in the history of the world have said to their children “You will get dessert if, and only if, you eat your dinner”? Not many! Most say either “If you eat your dinner, you will get dessert” (these take the positive approach—they emphasize the reward) or “You will get dessert only if you eat your dinner” (these take the

negative approach—they emphasize the punishment). Yet the parents who promise the reward intend to suggest the punishment as well, and those who threaten the punishment will certainly give the reward if it is earned. Both sets of parents expect that their conditional statements will be interpreted as biconditionals.

Since we often (correctly) interpret conditional statements as biconditionals, it is not surprising that we may come to believe (mistakenly) that conditional statements are always logically equivalent to their inverses and converses. In formal settings, however, statements must have unambiguous interpretations. If-then statements can't sometimes mean "if-then" and other times mean "if and only if." When using language in mathematics, science, or other situations where precision is important, it is essential to interpret if-then statements according to the formal definition and not to confuse them with their converses and inverses.

TEST YOURSELF

1. An *if-then* statement is false if, and only if, the hypothesis is _____ and the conclusion is _____.
2. The negation of "if p then q " is _____.
3. The converse of "if p then q " is _____.
4. The contrapositive of "if p then q " is _____.
5. The inverse of "if p then q " is _____.
6. A conditional statement and its contrapositive are _____.
7. A conditional statement and its converse are not _____.
8. " R is a sufficient condition for S " means "if _____ then _____."
9. " R is a necessary condition for S " means "if _____ then _____."
10. " R only if S " means "if _____ then _____."

EXERCISE SET 2.2

Rewrite the statements in 1–4 in *if-then* form.

1. This loop will repeat exactly N times if it does not contain a **stop** or a **go to**.
2. I am on time for work if I catch the 8:05 bus.
3. Freeze or I'll shoot.
4. Fix my ceiling or I won't pay my rent.

Construct truth tables for the statement forms in 5–11.

5. $\sim p \vee q \rightarrow \sim q$
6. $(p \vee q) \vee (\sim p \wedge q) \rightarrow q$
7. $p \wedge \sim q \rightarrow r$
8. $\sim p \vee q \rightarrow r$
9. $p \wedge \sim r \leftrightarrow q \vee r$
10. $(p \rightarrow r) \leftrightarrow (q \rightarrow r)$
11. $(p \rightarrow (q \rightarrow r)) \leftrightarrow ((p \wedge q) \rightarrow r)$
12. Use the logical equivalence established in Example 2.2.3, $p \vee q \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$,

to rewrite the following statement. (Assume that x represents a fixed real number.)

If $x > 2$ or $x < -2$, then $x^2 > 4$.

13. Use truth tables to verify the following logical equivalences. Include a few words of explanation with your answers.

- a. $p \rightarrow q \equiv \sim p \vee q$
- b. $\sim(p \rightarrow q) \equiv p \wedge \sim q$.

- H 14. a. Show that the following statement forms are all logically equivalent:

$p \rightarrow q \vee r$, $p \wedge \sim q \rightarrow r$, and $p \wedge \sim r \rightarrow q$

- b. Use the logical equivalences established in part (a) to rewrite the following sentence in two different ways. (Assume that n represents a fixed integer.)

If n is prime, then n is odd or n is 2.

ANSWERS FOR TEST YOURSELF

1. true; false 2. $p \wedge \sim q$ 3. if q then p 4. if $\sim q$ then $\sim p$ 5. if $\sim p$ then $\sim q$ 6. logically equivalent 7. logically equivalent
 8. $R; S$ 9. $S; R$ 10. $R; S$

2.3 Valid and Invalid Arguments

“Contrariwise,” continued Tweedledee, “if it was so, it might be; and if it were so, it would be; but as it isn’t, it ain’t. That’s logic.” —Lewis Carroll, *Through the Looking Glass*

In mathematics and logic an argument is not a dispute. It is simply a sequence of statements ending in a conclusion. In this section we show how to determine whether an argument is valid—that is, whether the conclusion follows *necessarily* from the preceding statements. We will show that this determination depends only on the form of an argument, not on its content.

It was shown in Section 2.1 that the logical form of an argument can be abstracted from its content. For example, the argument

If Socrates is a man, then Socrates is mortal.
 Socrates is a man.
 $\therefore$ Socrates is mortal.

has the abstract form

If p then q
 p
 $\therefore q$

When considering the abstract form of an argument, think of p and q as variables for which statements may be substituted. An argument form is called *valid* if, and only if, whenever statements are substituted that make all the premises true, the conclusion is also true.

Definition

An **argument** is a sequence of statements, and an **argument form** is a sequence of statement forms. All statements in an argument and all statement forms in an argument form, except for the final one, are called **premises** (or **assumptions** or **hypotheses**). The final statement or statement form is called the **conclusion**. The symbol $\therefore$, which is read “therefore,” is normally placed just before the conclusion.

To say that an *argument form* is **valid** means that no matter what particular statements are substituted for the statement variables in its premises, if the resulting premises are all true, then the conclusion is also true. To say that an *argument* is **valid** means that its form is valid.

The crucial fact about a valid argument is that the truth of its conclusion follows *necessarily* or *inescapably* or *by logical form alone* from the truth of its premises. It is impossible to have a valid argument with all true premises and a false conclusion. When an argument is valid and its premises are true, the truth of the conclusion is said to be *inferred*

or *deduced* from the truth of the premises. If a conclusion “ain’t necessarily so,” then it isn’t a valid deduction.

Testing an Argument Form for Validity

1. Identify the premises and conclusion of the argument form.
2. Construct a truth table showing the truth values of all the premises and the conclusion.
3. A row of the truth table in which all the premises are true is called a **critical row**. If there is a critical row in which the conclusion is false, then it is possible for an argument of the given form to have true premises and a false conclusion, and so the argument form is invalid. If the conclusion in *every* critical row is true, then the argument form is valid.

Example 2.3.1

Determining Validity or Invalidity

Determine whether the following argument form is valid or invalid by drawing a truth table, indicating which columns represent the premises and which represent the conclusion, and annotating the table with a sentence of explanation. When you fill in the table, you only need to indicate the truth values for the conclusion in the rows where all the premises are true (the critical rows) because the truth values of the conclusion in the other rows are irrelevant to the validity or invalidity of the argument.



Caution! If at least one premise of an argument is false, then we have no information about the conclusion: It might be true or it might be false.

$$p \rightarrow q \vee \sim r$$

$$q \rightarrow p \wedge r$$

$$\therefore p \rightarrow r$$

Solution The truth table shows that even though there are several situations in which the premises and the conclusion are all true (rows 1, 7, and 8), there is one situation (row 4) where the premises are true and the conclusion is false.

premises						conclusion		
p	q	r	$\sim r$	$q \vee \sim r$	$p \wedge r$	$p \rightarrow q \vee \sim r$	$q \rightarrow p \wedge r$	$p \rightarrow r$
T	T	T	F	T	T	T	T	T
T	T	F	T	T	F	T	F	
T	F	T	F	F	T	F	T	
T	F	F	T	T	F	T	T	F
F	T	T	F	T	F	T	F	
F	T	F	T	T	F	T	F	
F	F	T	F	F	F	T	T	T
F	F	F	T	T	F	T	T	T

This row shows that an argument of this form can have true premises and a false conclusion. Hence this form of argument is invalid.

Modus Ponens and Modus Tollens

An argument form consisting of two premises and a conclusion is called a **syllogism**. The first and second premises are called the **major premise** and **minor premise**, respectively. The most famous form of syllogism in logic is called **modus ponens**. It has the following form:

If p then q .
 p
 $\therefore q$

Here is an argument of this form:

If the sum of the digits of 371,487 is divisible by 3,
then 371,487 is divisible by 3.
The sum of the digits of 371,487 is divisible by 3.
 $\therefore$ 371,487 is divisible by 3.

The term *modus ponens* is Latin meaning “method of affirming” (the conclusion is an affirmation). Long before you saw your first truth table, you were undoubtedly being convinced by arguments of this form. Nevertheless, it is instructive to prove that modus ponens is a valid form of argument, if for no other reason than to confirm the agreement between the formal definition of validity and the intuitive concept. To do so, we construct a truth table for the premises and conclusion.

premises			conclusion	
p	q	$p \rightarrow q$	p	q
T	T	T	T	T
T	F	F	T	
F	T	T	F	
F	F	T	F	

← critical row

The first row is the only one in which both premises are true, and the conclusion in that row is also true. Hence the argument form is valid.

Now consider another valid argument form called **modus tollens**. It has the following form:

If p then q .
 $\sim q$
 $\therefore \sim p$

Here is an example of modus tollens:

If Zeus is human, then Zeus is mortal.
Zeus is not mortal.
 $\therefore$ Zeus is not human.

An intuitive explanation for the validity of modus tollens uses proof by contradiction. It goes like this:

Suppose

(1) If Zeus is human, then Zeus is mortal; and

(2) Zeus is not mortal.

Must Zeus necessarily be nonhuman?

Yes!

Because, if Zeus were human, then by (1) he would be mortal.

But by (2) he is not mortal.

Hence, Zeus cannot be human.

Modus tollens is Latin meaning “method of denying” (the conclusion is a denial). The validity of modus tollens can be shown to follow from modus ponens together with the fact that a conditional statement is logically equivalent to its contrapositive. Or it can be established formally by using a truth table. (See exercise 13.)

Studies by cognitive psychologists have shown that although nearly 100% of college students have a solid, intuitive understanding of modus ponens, less than 60% are able to apply modus tollens correctly.* Yet in mathematical reasoning, modus tollens is used almost as often as modus ponens. Thus it is important to study the form of modus tollens carefully to learn to use it effectively.

Example 2.3.2 Recognizing Modus Ponens and Modus Tollens

Use modus ponens or modus tollens to fill in the blanks of the following arguments so that they become valid inferences.

- a. If there are more pigeons than there are pigeonholes, then at least two pigeons roost in the same hole.

There are more pigeons than there are pigeonholes.

∴ _____

- b. If 870,232 is divisible by 6, then it is divisible by 3.

870,232 is not divisible by 3.

∴ _____

Solution

- a. At least two pigeons roost in the same hole. by modus ponens

- b. 870,232 is not divisible by 6. by modus tollens

Additional Valid Argument Forms: Rules of Inference

A **rule of inference** is a form of argument that is valid. Thus modus ponens and modus tollens are both rules of inference. The following are additional examples of rules of inference that are frequently used in deductive reasoning.

Example 2.3.3 Generalization

The following argument forms are valid:

- | | |
|---|---|
| <p>a. p</p> <p>∴ $p \vee q$</p> | <p>b. q</p> <p>∴ $p \vee q$</p> |
|---|---|

**Cognitive Psychology and Its Implications*, 3d ed. by John R. Anderson (New York: Freeman, 1990), pp. 292–297.

These argument forms are used for making generalizations. For instance, according to the first, if p is true, then, more generally, “ p or q ” is true for *any* other statement q . As an example, suppose you are given the job of counting the upperclassmen at your school. You ask what class Anton is in and are told he is a junior.

You reason as follows:

Anton is a junior.
 $\therefore$ (more generally) Anton is a junior or Anton is a senior.

Knowing that upperclassman means junior or senior, you add Anton to your list. ■

Example 2.3.4 Specialization

The following argument forms are valid:

$$\begin{array}{ll} \text{a.} & p \wedge q \\ & \therefore p \\ \text{b.} & p \wedge q \\ & \therefore q \end{array}$$

These argument forms are used for specializing. When classifying objects according to some property, you often know much more about them than whether they do or do not have that property. When this happens, you discard extraneous information as you concentrate on the particular property of interest.

For instance, suppose you are looking for a person who knows graph algorithms to work with you on a project. You discover that Ana knows both numerical analysis and graph algorithms. You reason as follows:

Ana knows numerical analysis and Ana knows graph algorithms.
 $\therefore$ (in particular) Ana knows graph algorithms.

Accordingly, you invite her to work with you on your project. ■

Both generalization and specialization are used frequently in mathematics to tailor facts to fit into hypotheses of known theorems in order to draw further conclusions. Elimination, transitivity, and proof by division into cases are also widely used tools.

Example 2.3.5 Elimination

The following argument forms are valid:

$$\begin{array}{ll} \text{a.} & p \vee q \\ & \sim q \\ & \therefore p \\ \text{b.} & p \vee q \\ & \sim p \\ & \therefore q \end{array}$$

These argument forms say that when you have only two possibilities and you can rule one out, the other must be the case. For instance, suppose you know that for a particular number x ,

$$x - 3 = 0 \quad \text{or} \quad x + 2 = 0.$$

If you also know that x is not negative, then $x \neq -2$, so

$$x + 2 \neq 0.$$

By elimination, you can then conclude that

$$\therefore x - 3 = 0. \quad \blacksquare$$

Example 2.3.6 Transitivity

The following argument form is valid:

$$\begin{aligned} p &\rightarrow q \\ q &\rightarrow r \\ \therefore p &\rightarrow r \end{aligned}$$

Many arguments in mathematics contain chains of if-then statements. From the fact that one statement implies a second and the second implies a third, you can conclude that the first statement implies the third. In the example below suppose n is a particular integer.

If n is divisible by 18, then n is divisible by 9.

If n is divisible by 9, then the sum of the digits of n is divisible by 9.

$\therefore$ If n is divisible by 18, then the sum of the digits of n is divisible by 9. ■

Example 2.3.7 Proof by Division into Cases

The following argument form is valid:

$$\begin{aligned} p &\vee q \\ p &\rightarrow r \\ q &\rightarrow r \\ \therefore r \end{aligned}$$

It often happens that you know one thing or another is true. If you can show that in either case a certain conclusion follows, then this conclusion must also be true. For instance, suppose you know that x is a particular nonzero real number that is not zero. The trichotomy property of the real numbers says that any real number is positive, negative, or zero. Thus (by elimination) you know that x is positive or x is negative. You can deduce that $x^2 > 0$ by arguing as follows:

$$\begin{aligned} &x \text{ is positive or } x \text{ is negative.} \\ &\text{If } x \text{ is positive, then } x^2 > 0. \\ &\text{If } x \text{ is negative, then } x^2 > 0. \\ \therefore &x^2 > 0. \end{aligned}$$

The rules of valid inference are used constantly in problem solving. Here is an example from everyday life.

Example 2.3.8 Application: A More Complex Deduction

You are about to leave for class in the morning and discover that you don't have your glasses. You know the following statements are true:

- If I was reading my class notes in the kitchen, then my glasses are on the kitchen table.
- If my glasses are on the kitchen table, then I saw them at breakfast.
- I did not see my glasses at breakfast.
- I was reading my class notes in the living room or I was reading my class notes in the kitchen.
- If I was reading my class notes in the living room then my glasses are on the coffee table.

Where are the glasses?

Solution Let RK = I was reading my class notes in the kitchen.
 GK = My glasses are on the kitchen table.
 SB = I saw my glasses at breakfast.
 RL = I was reading my class notes in the living room.
 GC = My glasses are on the coffee table.

Here is a sequence of steps you might use to reach the answer, together with the rules of inference that allow you to draw the conclusion of each step:

1. $RK \rightarrow GK$ by (a)
 $GK \rightarrow SB$ by (b)
 $\therefore RK \rightarrow SB$ by transitivity
2. $RK \rightarrow SB$ by the conclusion of (1)
 $\sim SB$ by (c)
 $\therefore \sim RK$ by modus tollens
3. $RL \vee RK$ by (d)
 $\sim RK$ by the conclusion of (2)
 $\therefore RL$ by elimination
4. $RL \rightarrow GC$ by (e)
 RL by the conclusion of (3)
 $\therefore GC$ by modus ponens

Thus the glasses are on the coffee table. ■

Fallacies

A **fallacy** is an error in reasoning that results in an invalid argument. Three common fallacies are **using ambiguous premises**, and treating them as if they were unambiguous, **circular reasoning** (assuming what is to be proved without having derived it from the premises), and **jumping to a conclusion** (without adequate grounds). In this section we discuss two other fallacies, called *converse error* and *inverse error*, which give rise to arguments that superficially resemble those that are valid by modus ponens and modus tollens but are not, in fact, valid.

As in previous examples, you can show that an argument is invalid by constructing a truth table for the argument form and finding at least one critical row in which all the premises are true but the conclusion is false. Another way is to find an argument of the same form with true premises and a false conclusion.

For an argument to be valid, every argument of the same form whose premises are all true must have a true conclusion. It follows that for an argument to be invalid means that there is an argument of that form whose premises are all true and whose conclusion is false.

Example 2.3.9 Converse Error

Show that the following argument is invalid:

If Zeke is a cheater, then Zeke sits in the back row.
 Zeke sits in the back row.
 $\therefore$ Zeke is a cheater.

Solution Many people recognize the invalidity of the above argument intuitively, reasoning something like this: The first premise gives information about Zeke *if* it is known he is a cheater. It doesn't give any information about him if it is not already known that he is a cheater. One can certainly imagine a person who is not a cheater but happens to sit in the back row. Then if that person's name is substituted for Zeke, the first premise is true by default and the second premise is also true but the conclusion is false.

The general form of the previous argument is as follows:

$$\begin{array}{l} p \rightarrow q \\ q \\ \therefore p \end{array}$$

In exercise 12(a) at the end of this section you are asked to use a truth table to show that this form of argument is invalid. ■

The fallacy underlying this invalid argument form is called the **converse error** because the conclusion of the argument would follow from the premises if the premise $p \rightarrow q$ were replaced by its converse. Such a replacement is not allowed, however, because a conditional statement is not logically equivalent to its converse. Converse error is also known as the *fallacy of affirming the consequent*.

A related common reasoning error is shown in the next example.

Example 2.3.10 Inverse Error

Consider the following argument:

If these two vertices are adjacent, then they do not have the same color.
 These two vertices are not adjacent.
 $\therefore$ These two vertices have the same color.

Note that this argument has the following form:

$$\begin{array}{l} p \rightarrow q \\ \sim p \\ \therefore \sim q \end{array}$$



Caution! In logic, the words *true* and *valid* have very different meanings. A valid argument may have a false conclusion, and an invalid argument may have a true conclusion.

You are asked to give a truth table verification of the invalidity of this argument form in exercise 12(b) at the end of this section.

The fallacy underlying this invalid argument form is called the **inverse error** because the conclusion of the argument would follow from the premises if the premise $p \rightarrow q$ were replaced by its inverse. Such a replacement is not allowed, however, because a conditional statement is not logically equivalent to its inverse. Inverse error is also known as the *fallacy of denying the antecedent*. ■

Sometimes people lump together the ideas of validity and truth. If an argument seems valid, they accept the conclusion as true. And if an argument seems fishy (really a slang expression for invalid), they think the conclusion must be false. This is not correct!

Example 2.3.11 A Valid Argument with a False Premise and a False Conclusion

The argument below is valid by modus ponens. But its major premise is false, and so is its conclusion.

If Canada is north of the United States, then temperatures in Canada never rise above freezing.

Canada is north of the United States.

∴ Temperatures in Canada never rise above freezing. ■

Example 2.3.12 An Invalid Argument with True Premises and a True Conclusion

The argument below is invalid by the converse error, but it has a true conclusion.

If New York is a big city, then New York has tall buildings.

New York has tall buildings.

∴ New York is a big city. ■

Definition

An argument is called **sound** if, and only if, it is valid *and* all its premises are true. An argument that is not sound is called **unsound**.

The important thing to note is that validity is a property of argument *forms*: If an argument is valid, then so is every other argument that has the same form. Similarly, if an argument is invalid, then so is every other argument that has the same form. What characterizes a valid argument is that no argument whose form is valid can have all true premises and a false conclusion. For each valid argument, there are arguments of that form with all true premises and a true conclusion, with at least one false premise and a true conclusion, and with at least one false premise and a false conclusion. On the other hand, for each invalid argument, there are arguments of that form with every combination of truth values for the premises and conclusion, including all true premises and a false conclusion. The bottom line is that we can only be sure that the conclusion of an argument is true when we know that the argument is sound, that is, when we know both that the argument is valid and that it has all true premises.

Contradictions and Valid Arguments

The concept of logical contradiction can be used to make inferences through a technique of reasoning called the *contradiction rule*. Suppose p is some statement whose truth you wish to deduce.

Contradiction Rule

If you can show that the supposition that statement p is false leads logically to a contradiction, then you can conclude that p is true.

Example 2.3.13**Contradiction Rule**

Show that the following argument form is valid:

$$\begin{array}{l} \sim p \rightarrow \mathbf{c}, \text{ where } \mathbf{c} \text{ is a contradiction} \\ \therefore p \end{array}$$

Solution Construct a truth table for the premise and the conclusion of this argument.

premises			conclusion
p	$\sim p$	$\mathbf{c}$	$\sim p \rightarrow \mathbf{c}$
T	F	F	T
F	T	F	F

There is only one critical row in which the premise is true, and in this row the conclusion is also true. Hence this form of argument is valid.

The contradiction rule is the logical heart of the method of proof by contradiction. A slight variation also provides the basis for solving many logical puzzles by eliminating contradictory answers: *If an assumption leads to a contradiction, then that assumption must be false.*

Example 2.3.14**Knights and Knaves**

The logician Raymond Smullyan describes an island containing two types of people: knights who always tell the truth and knaves who always lie.* You visit the island and are approached by two natives who speak to you as follows:

A says: B is a knight.

B says: A and I are of opposite type.

What are A and B ?

Solution A and B are both knaves. To see this, reason as follows:

Suppose A is a knight.

$\therefore$ What A says is true. by definition of knight

$\therefore B$ is also a knight. That's what A said.

$\therefore$ What B says is true. by definition of knight

$\therefore A$ and B are of opposite types. That's what B said.

$\therefore$ We have arrived at the following contradiction: A and B are both knights and A and B are of opposite type.

$\therefore$ The supposition is false. by the contradiction rule

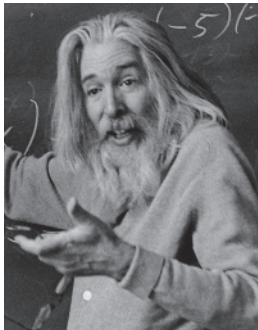
$\therefore A$ is not a knight. negation of supposition

$\therefore A$ is a knave. by elimination: It's given that all inhabitants are knights or knaves, so since A is not a knight, A is a knave.

$\therefore$ What A says is false.

$\therefore B$ is not a knight.

$\therefore B$ is also a knave. by elimination



Raymond Smullyan
(1919–2017)

*Raymond Smullyan has written a delightful series of whimsical yet profound books of logical puzzles starting with *What Is the Name of This Book?* (Englewood Cliffs, New Jersey: Prentice-Hall, 1978). Other good sources of logical puzzles are the many excellent books of Martin Gardner, such as *Aha! Insight* and *Aha! Gotcha* (New York: W. H. Freeman, 1978, 1982).

This reasoning shows that if the problem has a solution at all, then A and B must both be knaves. It is conceivable, however, that the problem has no solution. The problem statement could be inherently contradictory. If you look back at the solution, though, you can see that it does work out for both A and B to be knaves.

Summary of Rules of Inference

Table 2.3.1 summarizes some of the most important rules of inference.

TABLE 2.3.1 Valid Argument Forms

Modus Ponens	$ \begin{array}{l} p \rightarrow q \\ p \\ \therefore q \end{array} $	Elimination	a. $\begin{array}{l} p \vee q \\ \sim q \\ \therefore p \end{array}$ b. $\begin{array}{l} p \vee q \\ \sim p \\ \therefore q \end{array}$
Modus Tollens	$ \begin{array}{l} p \rightarrow q \\ \sim q \\ \therefore \sim p \end{array} $	Transitivity	$ \begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \therefore p \rightarrow r \end{array} $
Generalization	a. $\begin{array}{l} p \\ \therefore p \vee q \end{array}$ b. $\begin{array}{l} q \\ \therefore p \vee q \end{array}$	Proof by Division into Cases	$ \begin{array}{l} p \vee q \\ p \rightarrow r \\ q \rightarrow r \\ \therefore r \end{array} $
Specialization	a. $\begin{array}{l} p \wedge q \\ \therefore p \end{array}$ b. $\begin{array}{l} p \wedge q \\ \therefore q \end{array}$		
Conjunction	$ \begin{array}{l} p \\ q \\ \therefore p \wedge q \end{array} $	Contradiction Rule	$ \begin{array}{l} \sim p \rightarrow c \\ \therefore p \end{array} $

TEST YOURSELF

1. For an argument to be valid means that every argument of the same form whose premises _____ has a _____ conclusion.
2. For an argument to be invalid means that there is an argument of the same form whose premises _____ and whose conclusion _____.
3. For an argument to be sound means that it is _____ and its premises _____. In this case we can be sure that its conclusion _____.

EXERCISE SET 2.3

Use modus ponens or modus tollens to fill in the blanks in the arguments of 1–5 so as to produce valid inferences.

1. If $\sqrt{2}$ is rational, then $\sqrt{2} = a/b$ for some integers a and b .
 It is not true that $\sqrt{2} = a/b$ for some integers a and b .
 $\therefore$ _____
2. If $1 - 0.99999 \dots$ is less than every positive real number, then it equals zero.

 $\therefore$ The number $1 - 0.99999 \dots$ equals zero.
3. If logic is easy, then I am a monkey’s uncle.
 I am not a monkey’s uncle.
 $\therefore$ _____
4. If this graph can be colored with three colors, then it can colored with four colors.
 This graph cannot be colored with four colors.
 $\therefore$ _____
5. If they were unsure of the address, then they would have telephoned.

 $\therefore$ They were sure of the address.

THE LOGIC OF QUANTIFIED STATEMENTS

In Chapter 2 we discussed the logical analysis of compound statements—those made of simple statements joined by the connectives $\sim$, $\wedge$, $\vee$, $\rightarrow$, and $\leftrightarrow$. Such analysis casts light on many aspects of human reasoning, but it cannot be used to determine validity in the majority of everyday and mathematical situations. For example, the argument

All men are mortal.
Socrates is a man.
 $\therefore$ Socrates is mortal.

is intuitively perceived as correct. Yet its validity cannot be derived using the methods outlined in Section 2.3. To determine validity in examples like this, it is necessary to separate the statements into parts in much the same way that you separate declarative sentences into subjects and predicates. And you must analyze and understand the special role played by words that denote quantities such as “all” or “some.” The symbolic analysis of predicates and quantified statements is called the **predicate calculus**. The symbolic analysis of ordinary compound statements (as outlined in Sections 2.1–2.3) is called the **statement calculus** (or the **propositional calculus**).

3.1 Predicates and Quantified Statements I

... it was not till within the last few years that it has been realized how fundamental any and some are to the very nature of mathematics. —A. N. Whitehead (1861–1947)

As noted in Section 2.1, the sentence “ $x^2 + 2 = 11$ ” is not a statement because it may be either true or false depending on the value of x . Similarly, the sentence “ $x + y > 0$ ” is not a statement because its truth value depends on the values of the variables x and y .

In grammar, the word *predicate* refers to the part of a sentence that gives information about the subject. In the sentence “James is a student at Bedford College,” the word *James* is the subject and the phrase *is a student at Bedford College* is the predicate. The predicate is the part of the sentence from which the subject has been removed.

In logic, predicates can be obtained by removing some or all of the nouns from a statement. For instance, let P stand for “is a student at Bedford College” and let Q stand for “is a student at.” Then both P and Q are *predicate symbols*. The sentences “ x is a student at Bedford College” and “ x is a student at y ” are symbolized as $P(x)$ and as $Q(x, y)$, respectively, where x and y are *predicate variables* that take values in appropriate sets. When concrete values are substituted in place of predicate variables, a statement results. For simplicity, we define a *predicate* to be a predicate symbol together with suitable predicate variables. In some other treatments of logic, such objects are referred to as **propositional functions** or **open sentences**.

Definition

A **predicate** is a sentence that contains a finite number of variables and becomes a statement when specific values are substituted for the variables. The **domain** of a predicate variable is the set of all values that may be substituted in place of the variable.

Example 3.1.1 Finding Truth Values of a Predicate

Let $P(x)$ be the predicate “ $x^2 > x$ ” with domain the set $\mathbf{R}$ of all real numbers. Write $P(2)$, $P(\frac{1}{2})$, and $P(-\frac{1}{2})$, and indicate which of these statements are true and which are false.

Solution

$$\begin{aligned} P(2): \quad & 2^2 > 2, \text{ or } 4 > 2. \text{ True.} \\ P\left(\frac{1}{2}\right): \quad & \left(\frac{1}{2}\right)^2 > \frac{1}{2}, \text{ or } \frac{1}{4} > \frac{1}{2}. \text{ False.} \\ P\left(-\frac{1}{2}\right): \quad & \left(-\frac{1}{2}\right)^2 > -\frac{1}{2} \text{ or } \frac{1}{4} > -\frac{1}{2}. \text{ True.} \end{aligned}$$

When an element in the domain of the variable of a one-variable predicate is substituted for the variable, the resulting statement is either true or false. The set of all such elements that make the predicate true is called the *truth set* of the predicate.

Definition

If $P(x)$ is a predicate and x has domain D , the **truth set** of $P(x)$ is the set of all elements of D that make $P(x)$ true when they are substituted for x . The truth set of $P(x)$ is denoted

$$\{x \in D \mid P(x)\}.$$

Note Recall that we read these symbols as “the set of all x in D such that $P(x)$.”

Example 3.1.2 Finding the Truth Set of a Predicate

Let $Q(n)$ be the predicate “ n is a factor of 8.” Find the truth set of $Q(n)$ if

- the domain of n is $\mathbf{Z}^+$, the set of all positive integers
- the domain of n is $\mathbf{Z}$, the set of all integers.

Solution

- The truth set is $\{1, 2, 4, 8\}$ because these are exactly the positive integers that divide 8 evenly.
- The truth set is $\{1, 2, 4, 8, -1, -2, -4, -8\}$ because the negative integers $-1, -2, -4$, and -8 also divide into 8 without leaving a remainder.

The Universal Quantifier: $\forall$

One sure way to change predicates into statements is to assign specific values to all their variables. For example, if x represents the number 35, the sentence “ x is (evenly) divisible by 5” is a true statement since $35 = 5 \cdot 7$. Another way to obtain statements from predicates is to add **quantifiers**. Quantifiers are words that refer to quantities such as “some” or “all”



Charles Sanders Peirce
(1839–1914)

Note Think “for every” when you see the symbol $\forall$.

and tell for how many elements a given predicate is true. The formal concept of quantifier was introduced into symbolic logic in the late nineteenth century by the American philosopher, logician, and engineer Charles Sanders Peirce and, independently, by the German logician Gottlob Frege.

The symbol $\forall$ is called the **universal quantifier**. Depending on the context, it is read as “for every,” “for each,” “for any,” “given any,” or “for all.” For example, another way to express the sentence “Every human being is mortal” or “All human beings are mortal” is to write

$$\forall \text{ human beings } x, x \text{ is mortal,}$$

which you would read as “For every human being x , x is mortal.” If you let H be the set of all human beings, then you can symbolize the statement more formally by writing

$$\forall x \in H, x \text{ is mortal.}$$

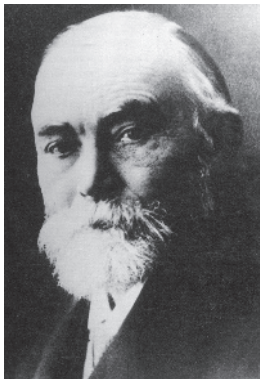
Think of the symbol x as an individual but generic object, with all the properties shared by every human being but with no other properties. Because x is individual, even if you read $\forall$ as “for all,” you should use the singular verb and say, “For all x in H , x is mortal” rather than “For all x in H , x are mortal.”

In a universally quantified sentence the domain of the predicate variable is generally indicated either between the $\forall$ symbol and the variable name (as in $\forall \text{ human being } x$) or immediately following the variable name (as in $\forall x \in H$). In sentences containing a mixture of symbols and words, the $\forall$ symbol can refer to two or more variables. For instance, you could symbolize “For all real numbers x and y , $x + y = y + x$.” as “ $\forall$ real numbers x and y , $x + y = y + x$.”*

Sentences that are quantified universally are defined as statements by giving them the truth values specified in the following definition:

Definition

Let $Q(x)$ be a predicate and D the domain of x . A **universal statement** is a statement of the form “ $\forall x \in D, Q(x)$.” It is defined to be true if, and only if, $Q(x)$ is true for each individual x in D . It is defined to be false if, and only if, $Q(x)$ is false for at least one x in D . A value for x for which $Q(x)$ is false is called a **counterexample** to the universal statement.



Gottlob Frege
(1848–1925)

Example 3.1.3 Truth and Falsity of Universal Statements

- a. Let $D = \{1, 2, 3, 4, 5\}$, and consider the statement

$$\forall x \in D, x^2 \geq x.$$

Write one way to read this statement out loud, and show that it is true.

- b. Consider the statement

$$\forall x \in \mathbf{R}, x^2 \geq x.$$

Find a counterexample to show that this statement is false.

*More formal versions of symbolic logic would require a separate $\forall$ for each variable: “ $\forall x \in \mathbf{R}(\forall y \in \mathbf{R}(x + y = y + x))$.”

Solution

- a. “For every x in the set D , x^2 is greater than or equal to x .” The inequalities below show that “ $x^2 \geq x$ ” is true for each individual x in D .

$$1^2 \geq 1, \quad 2^2 \geq 2, \quad 3^2 \geq 3, \quad 4^2 \geq 4, \quad 5^2 \geq 5.$$

Hence “ $\forall x \in D, x^2 \geq x$ ” is true.

- b. *Counterexample:* The statement claims that $x^2 \geq x$ for every real number x . But when $x = \frac{1}{2}$, for example,

$$\left(\frac{1}{2}\right)^2 = \frac{1}{4} \not\geq \frac{1}{2}.$$

Hence “ $\forall x \in \mathbf{R}, x^2 \geq x$ ” is false. ■

The technique used to show the truth of the universal statement in Example 3.1.3(a) is called the **method of exhaustion**. It consists of showing the truth of the predicate separately for each individual element of the domain. (The idea is to exhaust the possibilities before you exhaust yourself!) This method can, in theory, be used whenever the domain of the predicate variable is finite. In recent years the prevalence of digital computers has greatly increased the convenience of using the method of exhaustion. Computer expert systems, or knowledge-based systems, use this method to arrive at answers to many of the questions posed to them. Because most mathematical sets are infinite, however, the method of exhaustion can rarely be used to derive general mathematical results.

The Existential Quantifier: $\exists$

The symbol $\exists$ denotes “there exists” and is called the **existential quantifier**. For example, the sentence “There is a student in Math 140” can be written as

$\exists$ a person p such that p is a student in Math 140,

or, more formally,

$\exists p \in P$ such that p is a student in Math 140,

where P is the set of all people. The domain of the predicate variable is generally indicated either between the $\exists$ symbol and the variable name or immediately following the variable name, and the words *such that* are inserted just before the predicate. Some other expressions that can be used in place of *there exists* are *there is a*, *we can find a*, *there is at least one*, *for some*, and *for at least one*. In a sentence such as “ $\exists$ integers m and n such that $m + n = m \cdot n$,” the $\exists$ symbol is understood to refer to both m and n .*

Sentences that are quantified existentially are defined as statements by giving them the truth values specified in the following definition.

Definition

Let $Q(x)$ be a predicate and D the domain of x . An **existential statement** is a statement of the form “ $\exists x \in D$ such that $Q(x)$.” It is defined to be true if, and only if, $Q(x)$ is true for at least one x in D . It is false if, and only if, $Q(x)$ is false for all x in D .

*In more formal versions of symbolic logic, the words *such that* are not written out (although they are understood) and a separate $\exists$ symbol is used for each variable: “ $\exists m \in \mathbf{Z} (\exists n \in \mathbf{Z} (m + n = m \cdot n))$.”

Example 3.1.4 Truth and Falsity of Existential Statements

- a. Consider the statement

$$\exists m \in \mathbf{Z}^+ \text{ such that } m^2 = m.$$

Write one way to read this statement out loud, and show that it is true.

- b. Let
- $E = \{5, 6, 7, 8\}$
- and consider the statement

$$\exists m \in E \text{ such that } m^2 = m.$$

Show that this statement is false.

Solution

- a. “There is at least one positive integer m such that $m^2 = m$.” Observe that $1^2 = 1$. Thus “ $m^2 = m$ ” is true for a positive integer m , and so “ $\exists m \in \mathbf{Z}^+ \text{ such that } m^2 = m$ ” is true.
- b. Note that $m^2 = m$ is not true for any integers m from 5 through 8:

$$5^2 = 25 \neq 5, \quad 6^2 = 36 \neq 6, \quad 7^2 = 49 \neq 7, \quad 8^2 = 64 \neq 8.$$

Thus “ $\exists m \in E \text{ such that } m^2 = m$ ” is false. ■

Formal vs. Informal Language

It is important to be able to translate from formal to informal language when trying to make sense of mathematical concepts that are new to you. It is equally important to be able to translate from informal to formal language when thinking out a complicated problem.

Example 3.1.5 Translating from Formal to Informal Language

Rewrite the following formal statements in a variety of equivalent but more informal ways. Do not use the symbol $\forall$ or $\exists$.

- a. $\forall x \in \mathbf{R}, x^2 \geq 0$.
- b. $\forall x \in \mathbf{R}, x^2 \neq -1$.
- c. $\exists m \in \mathbf{Z}^+ \text{ such that } m^2 = m$.

Solution

- a. Every real number has a nonnegative square.
Or: All real numbers have nonnegative squares.
Or: Any real number has a nonnegative square.
Or: The square of each real number is nonnegative.
- b. All real numbers have squares that do not equal -1 .
Or: No real numbers have squares equal to -1 .
 (The words *none are* or *no ... are* are equivalent to the words *all are not*.)
- c. There is a positive integer whose square is equal to itself.
Or: We can find at least one positive integer equal to its own square.
Or: Some positive integer equals its own square.
Or: Some positive integers equal their own squares. ■

Note In ordinary English, the fourth statement in part (c) may be taken to mean that there are at least two positive integers equal to their own squares. In mathematics, we understand the last two statements in part (c) to mean the same thing.

Another way to restate universal and existential statements informally is to place the quantification at the end of the sentence. For instance, instead of saying “For any real number x , x^2 is nonnegative,” you could say “ x^2 is nonnegative for any real number x .” In such a case the quantifier is said to “trail” the rest of the sentence.

Example 3.1.6**Trailing Quantifiers**

Rewrite the following statements so that the quantifier trails the rest of the sentence.

- For any integer n , $2n$ is even.
- There exists at least one real number x such that $x^2 \leq 0$.

Solution

- $2n$ is even for any integer n .
- $x^2 \leq 0$ for some real number x .
Or: $x^2 \leq 0$ for at least one real number x .

Example 3.1.7

Note The following two sentences mean the same thing: “All triangles have three sides” and “Every triangle has three sides.”

Translating from Informal to Formal Language

Rewrite each of the following statements formally. Use quantifiers and variables.

- All triangles have three sides.
- No dogs have wings.
- Some programs are structured.

Solution

- $\forall$ triangle t , t has three sides.
Or: $\forall t \in T$, t has three sides (where T is the set of all triangles).
- $\forall$ dog d , d does not have wings.
Or: $\forall d \in D$, d does not have wings (where D is the set of all dogs).
- $\exists$ a program p such that p is structured.
Or: $\exists p \in P$ such that p is structured (where P is the set of all programs).

Universal Conditional Statements

A reasonable argument can be made that the most important form of statement in mathematics is the **universal conditional statement**:

$$\forall x, \text{ if } P(x) \text{ then } Q(x).$$

Familiarity with statements of this form is essential if you are to learn to speak mathematics.

Example 3.1.8**Writing Universal Conditional Statements Informally**

Rewrite the following statement informally, without quantifiers or variables.

$$\forall x \in \mathbf{R}, \text{ if } x > 2 \text{ then } x^2 > 4.$$

- Solution** If a real number is greater than 2, then its square is greater than 4.
Or: Whenever a real number is greater than 2, its square is greater than 4.
Or: The square of any real number greater than 2 is greater than 4.
Or: The squares of all real numbers greater than 2 are greater than 4.

Example 3.1.9**Writing Universal Conditional Statements Formally**

Rewrite each of the following statements in the form

$$\forall \text{_____, if _____ then _____}.$$

- a. If a real number is an integer, then it is a rational number.
- b. All bytes have eight bits.
- c. No fire trucks are green.

Solution

- a. $\forall$ real number x , if x is an integer, then x is a rational number.
Or: $\forall x \in \mathbf{R}$, if $x \in \mathbf{Z}$ then $x \in \mathbf{Q}$.
- b. $\forall x$, if x is a byte, then x has eight bits.
- c. $\forall x$, if x is a fire truck, then x is not green.

It is common, as in (b) and (c) above, to omit explicit identification of the domain of predicate variables in universal conditional statements. ■

Careful thought about the meaning of universal conditional statements leads to another level of understanding for why the truth table for an if-then statement must be defined as it is. Consider again the statement

$$\forall \text{ real number } x, \text{ if } x > 2 \text{ then } x^2 > 4.$$

Your experience and intuition tell you that this statement is true. But that means that

$$\text{If } x > 2 \text{ then } x^2 > 4$$

must be true for every single real number x . Consequently, it must be true even for values of x that make its hypothesis “ $x > 2$ ” false. In particular, both statements

$$\text{If } 1 > 2 \text{ then } 1^2 > 4 \quad \text{and} \quad \text{If } -3 > 2 \text{ then } (-3)^2 > 4$$

must be true. In both cases the hypothesis is false, but in the first case the conclusion “ $1^2 > 4$ ” is false, and in the second case the conclusion “ $(-3)^2 > 4$ ” is true. Hence, if an if-then statement has a false hypothesis, we have to interpret it as true regardless of whether its conclusion is true or false.

Note also that the definition of valid argument is a universal conditional statement:

For every combination of truth values for the component statements,
if the premises are all true then the conclusion is also true.

Equivalent Forms of Universal and Existential Statements

Observe that the two statements “ $\forall$ real number x , if x is an integer then x is rational” and “ $\forall$ integer x , x is rational” mean the same thing because the set of integers is a subset of the set of real numbers. Both have informal translations “All integers are rational.” In fact, a statement of the form

$$\forall x \in U, \text{ if } P(x) \text{ then } Q(x)$$

can always be rewritten in the form

$$\forall x \in D, Q(x)$$

by narrowing U to be the subset D consisting of all values of the variable x that make $P(x)$ true. Conversely, a statement of the form

$$\forall x \in D, Q(x)$$

can be rewritten as

$$\forall x, \text{ if } x \text{ is in } D \text{ then } Q(x).$$

Example 3.1.10 Equivalent Forms for Universal Statements

Rewrite the following statement in the two forms “ $\forall x$, if _____ then _____” and “ $\forall$ _____ x , _____”: All squares are rectangles.

Solution

$\forall x$, if x is a square then x is a rectangle.

$\forall$ square x , x is a rectangle. ■

Similarly, a statement of the form “ $\exists x$ such that $P(x)$ and $Q(x)$ ” can be rewritten as “ $\exists x \in D$ such that $Q(x)$,” where D is the set of all x for which $P(x)$ is true.

Example 3.1.11 Equivalent Forms for Existential Statements

A **prime number** is an integer greater than 1 whose only positive integer factors are itself and 1. Consider the statement “There is an integer that is both prime and even.” Let $\text{Prime}(n)$ be “ n is prime” and $\text{Even}(n)$ be “ n is even.” Use the notation $\text{Prime}(n)$ and $\text{Even}(n)$ to rewrite this statement in the following two forms:

a. $\exists n$ such that _____ $\wedge$ _____.

b. $\exists$ _____ n such that _____.

Solution

a. $\exists n$ such that $\text{Prime}(n) \wedge \text{Even}(n)$.

b. Two answers: $\exists$ a prime number n such that $\text{Even}(n)$.
 $\exists$ an even number n such that $\text{Prime}(n)$. ■

Bound Variables and Scope

Consider the statement “For every integer x , $x^2 \geq 0$.” First note that you don’t have to call the variable x . You can use any name for it as long as you do so consistently. For instance, all the following statements have the same meaning:

For every integer x , $x^2 \geq 0$. For every integer n , $n^2 \geq 0$. For every integer s , $s^2 \geq 0$.

In each case the variable simply holds a place for any element in the set of all integers. Each way of writing the statement says that whatever integer you might choose, when you square it the result will be nonnegative. It is important to note, however, that once you finish writing the statement, whatever symbol you chose to use in it can be given an entirely different meaning when used in a different context.

For example, consider the following statements:

(1) For every integer x , $x^2 \geq 0$.

(2) There exists a real number x such that $x^3 = 8$.

Statements (1) and (2) both call the variable x , but the x in Statement (1) serves a different function from the x in Statement (2). We say that the variable x is **bound** by the quantifier that controls it and that its **scope** begins when the quantifier introduces it and ends at the end of the quantified statement.

The way variables are used in mathematics is similar to the way they are used in computer programming. A variable in a computer program also serves as a placeholder in the sense that it creates a location in computer memory (either actual or virtual) into which its values can be placed. In addition the way it can be bound in a program is similar to the

way that a mathematical variable can be bound in a statement. For example, consider the following two examples in Python:

Program 1

```
def f():
    X = "Hi"
    print X
def g():
    X = "Bye"
    print X
f()
g()
```

Program 2

```
def f():
    S = "Hi"
    print S
def g():
    S = "Bye"
    print S
f()
g()
```

The output for both programs is

Hi
Bye

In each case the variable—whether X or S —is **local** to the function where it is defined. It is created each time the function is called and destroyed as soon as the call is complete. The local variable is *bound* by the function that defines it, and its *scope* is restricted to that function. Outside of the function definition the variable name can be used for any other purpose. That is why the functions f and g are allowed to use the same name for the variable in their definitions and why f and g define the same functions in both programs.

Implicit Quantification

Consider the statement

If a number is an integer, then it is a rational number.

As shown earlier, this statement is equivalent to a universal statement. However, it does not contain the telltale word *all* or *every* or *any* or *each*. The only clue to indicate its universal quantification comes from the presence of the indefinite article *a*. This is an example of *implicit* universal quantification.

Existential quantification can also be implicit. For instance, the statement “The number 24 can be written as a sum of two even integers” can be expressed formally as “ $\exists$ even integers m and n such that $24 = m + n$.”

Mathematical writing contains many examples of implicitly quantified statements. Some occur, as in the first example above, through the presence of the word *a* or *an*. Others occur in cases where the general context of a sentence supplies part of its meaning. For example, in an algebra course in which the letter x is always used to indicate a real number, the predicate

If $x > 2$ then $x^2 > 4$

is interpreted to mean the same as the statement

For every real number x , if $x > 2$ then $x^2 > 4$.

Mathematicians often use a double arrow to indicate implicit quantification symbolically. For instance, they might express the above statement as

$$x > 2 \Rightarrow x^2 > 4.$$

Notation

Let $P(x)$ and $Q(x)$ be predicates and suppose the common domain of x is D .

- The notation $P(x) \Rightarrow Q(x)$ means that every element in the truth set of $P(x)$ is in the truth set of $Q(x)$, or, equivalently, $\forall x, P(x) \rightarrow Q(x)$.
- The notation $P(x) \Leftrightarrow Q(x)$ means that $P(x)$ and $Q(x)$ have identical truth sets, or, equivalently, $\forall x, P(x) \leftrightarrow Q(x)$.

Example 3.1.12**Using $\Rightarrow$ and $\Leftrightarrow$**

Let

$Q(n)$ be “ n is a factor of 8,”

$R(n)$ be “ n is a factor of 4,”

$S(n)$ be “ $n < 5$ and $n \neq 3$,”

and suppose the domain of n is $\mathbf{Z}^+$, the set of positive integers. Use the $\Rightarrow$ and $\Leftrightarrow$ symbols to indicate true relationships among $Q(n)$, $R(n)$, and $S(n)$.

Solution

1. As noted in Example 3.1.2, the truth set of $Q(n)$ is $\{1, 2, 4, 8\}$ when the domain of n is $\mathbf{Z}^+$. By similar reasoning the truth set of $R(n)$ is $\{1, 2, 4\}$. Thus it is true that every element in the truth set of $R(n)$ is in the truth set of $Q(n)$, or, equivalently, $\forall n$ in $\mathbf{Z}^+$, $R(n) \rightarrow Q(n)$. So $R(n) \Rightarrow Q(n)$, or, equivalently

$$n \text{ is a factor of 4} \Rightarrow n \text{ is a factor of 8.}$$

2. The truth set of $S(n)$ is $\{1, 2, 4\}$, which is identical to the truth set of $R(n)$, or, equivalently, $\forall n$ in $\mathbf{Z}^+$, $R(n) \leftrightarrow S(n)$. So $R(n) \Leftrightarrow S(n)$, or, equivalently,

$$n \text{ is a factor of 4} \Leftrightarrow n < 5 \text{ and } n \neq 3.$$

Moreover, since every element in the truth set of $S(n)$ is in the truth set of $Q(n)$, or, equivalently, $\forall n$ in $\mathbf{Z}^+$, $S(n) \rightarrow Q(n)$, then $S(n) \Rightarrow Q(n)$, or, equivalently,

$$n < 5 \text{ and } n \neq 3 \Rightarrow n \text{ is a factor of 8.} \quad \blacksquare$$

Some questions of quantification can be quite subtle. For instance, a mathematics text might contain the following:

- a. $(x+1)^2 = x^2 + 2x + 1$.
- b. Solve $3x - 4 = 5$.

Although neither (a) nor (b) contains explicit quantification, the reader is supposed to understand that the x in (a) is universally quantified, whereas the x in (b) is existentially quantified. When the quantification is made explicit, (a) and (b) become

- a. $\forall$ real number x , $(x+1)^2 = x^2 + 2x + 1$.
- b. Show (by finding a value) that $\exists$ a real number x such that $3x - 4 = 5$.

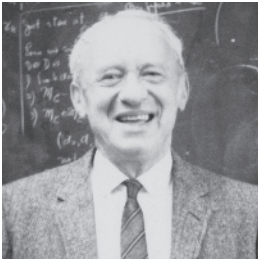
The quantification of a statement—whether universal or existential—crucially determines both how the statement can be applied and what method must be used to establish its truth. Thus it is important to be alert to the presence of hidden quantifiers when you read mathematics so that you will interpret statements in a logically correct way.

Tarski's World

Tarski's World is a computer program developed by information scientists Jon Barwise and John Etchemendy to help teach the principles of logic. It is described in their book

The Language of First-Order Logic, which is accompanied by a CD containing the program Tarski’s World, named after the great logician Alfred Tarski.

Example 3.1.13 Investigating Tarski’s World



Alfred Tarski
(1902–1983)

Briscoe Center for American History

The program for Tarski’s World provides pictures of blocks of various sizes, shapes, and colors, which are located on a grid. Shown in Figure 3.1.1 is a picture of an arrangement of objects in a two-dimensional Tarski world. The configuration can be described using logical operators and—for the two-dimensional version—notation such as $\text{Triangle}(x)$, meaning “ x is a triangle,” $\text{Blue}(y)$, meaning “ y is blue,” and $\text{RightOf}(x, y)$, meaning “ x is to the right of y (but possibly in a different row).” Individual objects can be given names such as a, b , or c .

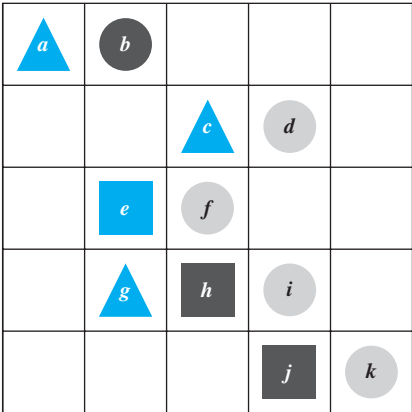


FIGURE 3.1.1

Determine the truth or falsity of each of the following statements. The domain for all variables is the set of objects in the Tarski world shown in Figure 3.1.1.

- a. $\forall t, \text{Triangle}(t) \rightarrow \text{Blue}(t)$.
- b. $\forall x, \text{Blue}(x) \rightarrow \text{Triangle}(x)$.
- c. $\exists y$ such that $\text{Square}(y) \wedge \text{RightOf}(d, y)$.
- d. $\exists z$ such that $\text{Square}(z) \wedge \text{Gray}(z)$.

Solution

- a. This statement is true: Every triangle is blue.
- b. This statement is false. As a counterexample, note that e is blue and it is not a triangle.
- c. This statement is true because e and h are both square and d is to their right.
- d. This statement is false: All the squares are either blue or black.

TEST YOURSELF

Answers to Test Yourself questions are located at the end of each section.

- 1. If $P(x)$ is a predicate with domain D , the truth set of $P(x)$ is denoted _____. We read these symbols out loud as_____.
- 2. Some ways to express the symbol $\forall$ in words are _____.
- 3. Some ways to express the symbol $\exists$ in words are _____.
- 4. A statement of the form $\forall x \in D, Q(x)$ is true if, and only if, $Q(x)$ is _____ for _____.
- 5. A statement of the form $\exists x \in D$ such that $Q(x)$ is true if, and only if, $Q(x)$ is _____ for _____.

EXERCISE SET 3.1*

1. A menagerie consists of seven brown dogs, two black dogs, six gray cats, ten black cats, five blue birds, six yellow birds, and one black bird. Determine which of the following statements are true and which are false.
 - a. There is an animal in the menagerie that is red.
 - b. Every animal in the menagerie is a bird or a mammal.
 - c. Every animal in the menagerie is brown or gray or black.
 - d. There is an animal in the menagerie that is neither a cat nor a dog.
 - e. No animal in the menagerie is blue.
 - f. There are in the menagerie a dog, a cat, and a bird that all have the same color.
 2. Indicate which of the following statements are true and which are false. Justify your answers as best as you can.
 - a. Every integer is a real number.
 - b. 0 is a positive real number.
 - c. For every real number r , $-r$ is a negative real number.
 - d. Every real number is an integer.
 3. Let $R(m, n)$ be the predicate “If m is a factor of n , then m is a factor of n ,” with domain for both m and n being $\mathbf{Z}$ the set of integers.
 - a. Explain why $R(m, n)$ is false if $m = 25$ and $n = 10$.
 - b. Give values different from those in part (a) for which $R(m, n)$ is false.
 - c. Explain why $R(m, n)$ is true if $m = 5$ and $n = 10$.
 - d. Give values different from those in part (c) for which $R(m, n)$ is true.
 4. Let $Q(x, y)$ be the predicate “If $x < y$ then $x^2 < y^2$,” with domain for both x and y being $\mathbf{R}$ the set of real numbers.
 - a. Explain why $Q(x, y)$ is false if $x = -2$ and $y = 1$.
 - b. Give values different from those in part (a) for which $Q(x, y)$ is false.
 - c. Explain why $Q(x, y)$ is true if $x = 3$ and $y = 8$.
 - d. Give values different from those in part (c) for which $Q(x, y)$ is true.
 5. Find the truth set of each predicate.
 - a. Predicate: $6/d$ is an integer, domain: $\mathbf{Z}$
 - b. Predicate: $6/d$ is an integer, domain: $\mathbf{Z}^+$
 - c. Predicate: $1 \leq x^2 \leq 4$, domain: $\mathbf{R}$
 - d. Predicate: $1 \leq x^2 \leq 4$, domain: $\mathbf{Z}$
 6. Let $B(x)$ be “ $-10 < x < 10$.” Find the truth set of $B(x)$ for each of the following domains.
 - a. $\mathbf{Z}$
 - b. $\mathbf{Z}^+$
 - c. The set of all even integers
 7. Let S be the set of all strings of length 3 consisting of a 's, b 's, and c 's. List all the strings in S that satisfy the following conditions:
 1. Every string in S begins with b .
 2. No string in S has more than one c .
 8. Let T be the set of all strings of length 3 consisting of 0's and 1's. List all the strings in T that satisfy the following conditions:
 1. For every string s in T , the second character of s is 1 or the first two characters of s are the same.
 2. No string in T has all three characters the same.
- Find counterexamples to show that the statements in 9–12 are false.
9. $\forall x \in \mathbf{R}, x \geq 1/x$.
 10. $\forall a \in \mathbf{Z}, (a-1)/a$ is not an integer.
 11. $\forall$ positive integers m and n , $m \cdot n \geq m + n$.
 12. $\forall$ real numbers x and y , $\sqrt{x+y} = \sqrt{x} + \sqrt{y}$.
 13. Consider the following statement:

$$\forall \text{ basketball player } x, x \text{ is tall.}$$

Which of the following are equivalent ways of expressing this statement?

 - a. Every basketball player is tall.
 - b. Among all the basketball players, some are tall.
 - c. Some of all the tall people are basketball players.
 - d. Anyone who is tall is a basketball player.
 - e. All people who are basketball players are tall.
 - f. Anyone who is a basketball player is a tall person.

*For exercises with blue numbers or letters, solutions are given in Appendix B. The symbol **H** indicates that only a hint or a partial solution is given. The symbol * signals that an exercise is more challenging than usual.

3.2 Predicates and Quantified Statements II

TOUCHSTONE: *Stand you both forth now: stroke your chins, and swear by your beards that I am a knave.*

CELIA: *By our beards—if we had them—thou art.*

TOUCHSTONE: *By my knavery—if I had it—then I were; but if you swear by that that is not, you are not forsworn.* —William Shakespeare, *As You Like It*

This section continues the discussion of predicates and quantified statements begun in Section 3.1. It contains the rules for negating quantified statements; an exploration of the relation among $\forall$, $\exists$, $\wedge$, and $\vee$; an introduction to the concept of vacuous truth of universal statements; examples of variants of universal conditional statements; and an extension of the meaning of *necessary*, *sufficient*, and *only if* to quantified statements.

Negations of Quantified Statements

Consider the statement “All mathematicians wear glasses.” Many people would say that its negation is “No mathematicians wear glasses,” but if even one mathematician does not wear glasses, then the sweeping statement that *all* mathematicians wear glasses is false. So a correct negation is “There is at least one mathematician who does not wear glasses.”

The general form of the negation of a universal statement follows immediately from the definitions of negation and of the truth values for universal and existential statements.

Theorem 3.2.1 Negation of a Universal Statement

The negation of a statement of the form

$$\forall x \text{ in } D, Q(x)$$

is logically equivalent to a statement of the form

$$\exists x \text{ in } D \text{ such that } \sim Q(x).$$

Symbolically,

$$\sim(\forall x \in D, Q(x)) \equiv \exists x \in D \text{ such that } \sim Q(x).$$

Thus

The negation of a universal statement (“all are”) is logically equivalent to an existential statement (“some are not” or “there is at least one that is not”).

Note that when we speak of **logical equivalence for quantified statements**, we mean that the statements always have identical truth values no matter what predicates are substituted for the predicate symbols and no matter what sets are used for the domains of the predicate variables.

Now consider the statement “Some snowflakes are the same.” What is its negation? For this statement to be false means that not a single snowflake is the same as any other. In other words, “No snowflakes are the same,” or “All snowflakes are different.”

The general form for the negation of an existential statement follows immediately from the definitions of negation and of the truth values for existential and universal statements.

Theorem 3.2.2 Negation of an Existential Statement

The negation of a statement of the form

$$\exists x \text{ in } D \text{ such that } Q(x)$$

is logically equivalent to a statement of the form

$$\forall x \text{ in } D, \sim Q(x).$$

Symbolically,

$$\sim(\exists x \in D \text{ such that } Q(x)) \equiv \forall x \in D, \sim Q(x).$$

Thus

The negation of an existential statement (“some are”) is logically equivalent to a universal statement (“none are” or “all are not”).

Example 3.2.1 Negating Quantified Statements

Write formal negations for the following statements:

- $\forall$ primes p , p is odd.
- $\exists$ a triangle T such that the sum of the angles of T equals 200° .

Solution

- By applying the rule for the negation of a $\forall$ statement, you can see that the answer is

$$\exists \text{ a prime } p \text{ such that } p \text{ is not odd.}$$

- By applying the rule for the negation of a $\exists$ statement, you can see that the answer is

$$\forall \text{ triangles } T, \text{ the sum of the angles of } T \text{ does not equal } 200^\circ. \quad \blacksquare$$

You need to exercise special care to avoid mistakes when writing negations of statements that are given informally. One way to avoid error is to rewrite the statement formally and take the negation using the formal rule.

Example 3.2.2 More Negations

Rewrite the following statements formally. Then write formal and informal negations.

- No politicians are honest.
- The number 1,357 is not divisible by any integer between 1 and 37.

Solution

- Formal version:* $\forall$ politicians x , x is not honest.

Formal negation: $\exists$ a politician x such that x is honest.

Informal negation: Some politicians are honest.

- This statement has a trailing quantifier. Written formally it becomes:

$$\forall \text{ integer } n \text{ between 1 and 37, 1,357 is not divisible by } n.$$

Note Which is true: the statement in part (b) or its negation? Is 1,357 divisible by some integer between 1 and 37? Or is 1,357 not divisible by any integer between 1 and 37?

Its negation is therefore

$\exists$ an integer n between 1 and 37 such that 1,357 is divisible by n .

An informal version of the negation is

The number 1,357 is divisible by some integer between 1 and 37. ■

Another important way to avoid error when taking negations of statements, whether stated formally or informally, is to ask yourself, “What *exactly* would it mean for the given statement to be false? What statement, if true, would be equivalent to saying that the given statement is false?”

Example 3.2.3

Still More Negations

Write informal negations for the following statements:

- All computer programs are finite.
- Some computer hackers are over 40.

Solution

- What exactly would it mean for this statement to be false? The statement asserts that all computer programs satisfy a certain property. So for it to be false, there would have to be at least one computer program that does not satisfy the property. Thus the answer is

There is a computer program that is not finite.

Or: Some computer programs are infinite.

- This statement is equivalent to saying that there is at least one computer hacker with a certain property. So for it to be false, not a single computer hacker can have that property. Thus the negation is

No computer hackers are over 40.

Or: All computer hackers are 40 or under. ■



Caution! Just inserting the word *not* to negate a quantified statement can result in a statement that is ambiguous.

Informal negations of many universal statements can be constructed simply by inserting the word *not* or the words *do not* at an appropriate place. However, the resulting statements may be ambiguous. For example, a possible negation of “All mathematicians wear glasses” is “All mathematicians do not wear glasses.” The problem is that this sentence has two meanings. With the proper verbal stress on the word *not*, it could be interpreted as the logical negation. (What! You say that all mathematicians wear glasses? Nonsense! All mathematicians *do not* wear glasses.) On the other hand, stated in a flat tone of voice (try it!), it would mean that all mathematicians are nonwearers of glasses; that is, not a single mathematician wears glasses. This is a much stronger statement than the logical negation: It implies the negation but is not equivalent to it.

Negations of Universal Conditional Statements

Negations of universal conditional statements are of special importance in mathematics. The form of such negations can be derived from facts that have already been established.

By definition of the negation of a *for all* statement,

$$\sim(\forall x, P(x) \rightarrow Q(x)) \equiv \exists x \text{ such that } \sim(P(x) \rightarrow Q(x)).$$

3.2.1

But the negation of an if-then statement is logically equivalent to an *and* statement. More precisely,

$$\sim(P(x) \rightarrow Q(x)) \equiv P(x) \wedge \sim Q(x). \quad 3.2.2$$

Substituting (3.2.2) into (3.2.1) gives

$$\sim(\forall x, P(x) \rightarrow Q(x)) \equiv \exists x \text{ such that } (P(x) \wedge \sim Q(x)).$$

Written somewhat less symbolically, this becomes

Negation of a Universal Conditional Statement

$$\sim(\forall x, \text{if } P(x) \text{ then } Q(x)) \equiv \exists x \text{ such that } P(x) \text{ and } \sim Q(x).$$

Example 3.2.4

Negating Universal Conditional Statements

Write a formal negation for statement (a) and an informal negation for statement (b).

- $\forall$ person p , if p is blond then p has blue eyes.
- If a computer program has more than 100,000 lines, then it contains a bug.

Solution

- $\exists$ a person p such that p is blond and p does not have blue eyes.
- There is at least one computer program that has more than 100,000 lines and does not contain a bug. ■

The Relation among $\forall$, $\exists$, $\wedge$, and $\vee$

The negation of a *for all* statement is a *there exists* statement, and the negation of a *there exists* statement is a *for all* statement. These facts are analogous to De Morgan's laws, which state that the negation of an *and* statement is an *or* statement and that the negation of an *or* statement is an *and* statement. This similarity is not accidental. In a sense, universal statements are generalizations of *and* statements, and existential statements are generalizations of *or* statements.

If $Q(x)$ is a predicate and the domain D of x is the set $\{x_1, x_2, \dots, x_n\}$, then the statements

$$\forall x \in D, Q(x) \quad \text{and} \quad Q(x_1) \wedge Q(x_2) \wedge \dots \wedge Q(x_n)$$

are logically equivalent. For example, let $Q(x)$ be " $x \cdot x = x$ " and suppose $D = \{0, 1\}$. Then

$$\forall x \in D, Q(x)$$

can be rewritten as

$$\forall \text{ binary digits } x, x \cdot x = x.$$

This is equivalent to

$$0 \cdot 0 = 0 \quad \text{and} \quad 1 \cdot 1 = 1,$$

which can be rewritten in symbols as

$$Q(0) \wedge Q(1).$$

Similarly, if $Q(x)$ is a predicate and $D = \{x_1, x_2, \dots, x_n\}$, then the statements

$$\exists x \in D \text{ such that } Q(x) \quad \text{and} \quad Q(x_1) \vee Q(x_2) \vee \cdots \vee Q(x_n)$$

are logically equivalent. For example, let $Q(x)$ be “ $x + x = x$ ” and suppose $D = \{0, 1\}$. Then

$$\exists x \in D \text{ such that } Q(x)$$

can be rewritten as

$$\exists \text{ a binary digit } x \text{ such that } x + x = x.$$

This is equivalent to

$$0 + 0 = 0 \quad \text{or} \quad 1 + 1 = 1,$$

which can be rewritten in symbols as

$$Q(0) \vee Q(1).$$

Vacuous Truth of Universal Statements

Suppose a bowl sits on a table and next to the bowl is a pile of five blue and five gray balls, any of which may be placed in the bowl. If three blue balls and one gray ball are placed in the bowl, as shown in Figure 3.2.1(a), the statement “All the balls in the bowl are blue” would be false (since one of the balls in the bowl is gray).

Now suppose that no balls at all are placed in the bowl, as shown in Figure 3.2.1(b). Consider the statement

All the balls in the bowl are blue.

Is this statement true or false? The statement is false if, and only if, its negation is true. And its negation is

There exists a ball in the bowl that is not blue.

But the only way this negation can be true is for there actually to be a nonblue ball in the bowl. And there is not! Hence the negation is false, and so the statement is true “by default.”

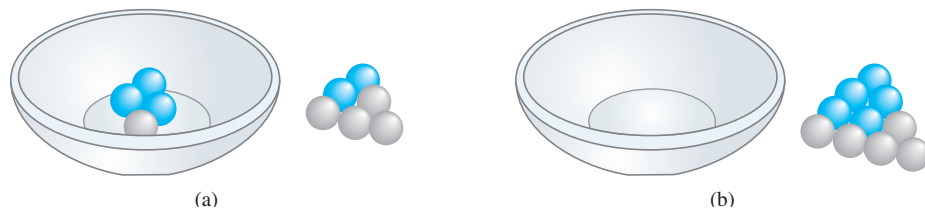


FIGURE 3.2.1

Note In ordinary language the words *in general* mean that something is usually, but not always the case. (In general, I take the bus, but today I walked.) In mathematics the words *in general* mean that something is always true.

In general, a statement of the form

$$\forall x \text{ in } D, \text{ if } P(x) \text{ then } Q(x)$$

is called **vacuously true** or **true by default** if, and only if, $P(x)$ is false for every x in D .

In mathematics, the words *in general* signal that what is to follow is a generalization of some aspect of the example that always holds true.

Variants of Universal Conditional Statements

Recall from Section 2.2 that a conditional statement has a contrapositive, a converse, and an inverse. The definitions of these terms can be extended to universal conditional statements.

Definition

Consider a statement of the form $\forall x \in D$, if $P(x)$ then $Q(x)$.

1. Its **contrapositive** is the statement $\forall x \in D$, if $\sim Q(x)$ then $\sim P(x)$.
2. Its **converse** is the statement $\forall x \in D$, if $Q(x)$ then $P(x)$.
3. Its **inverse** is the statement $\forall x \in D$, if $\sim P(x)$ then $\sim Q(x)$.

Example 3.2.5**Contrapositive, Converse, and Inverse of a Universal Conditional Statement**

Write a formal and an informal contrapositive, converse, and inverse for the following statement:

If a real number is greater than 2, then its square is greater than 4.

Solution The formal version of this statement is $\forall x \in \mathbf{R}$, if $x > 2$ then $x^2 > 4$.

Contrapositive: $\forall x \in \mathbf{R}$, if $x^2 \leq 4$ then $x \leq 2$.

Or: If the square of a real number is less than or equal to 4, then the number is less than or equal to 2.

Converse: $\forall x \in \mathbf{R}$, if $x^2 > 4$ then $x > 2$.

Or: If the square of a real number is greater than 4, then the number is greater than 2.

Inverse: $\forall x \in \mathbf{R}$, if $x \leq 2$ then $x^2 \leq 4$.

Or: If a real number is less than or equal to 2, then the square of the number is less than or equal to 4.

Note that in solving this example, we have used the equivalence of “ $x \not> a$ ” and “ $x \leq a$ ” for all real numbers x and a . (See page 47.) ■

In Section 2.2 we showed that a conditional statement is logically equivalent to its contrapositive and that it is not logically equivalent to either its converse or its inverse. The following discussion shows that these facts generalize to the case of universal conditional statements and their contrapositives, converses, and inverses.

Let $P(x)$ and $Q(x)$ be any predicates, let D be the domain of x , and consider the statement

$$\forall x \in D, \text{ if } P(x) \text{ then } Q(x)$$

and its contrapositive

$$\forall x \in D, \text{ if } \sim Q(x) \text{ then } \sim P(x).$$

Any particular x in D that makes “if $P(x)$ then $Q(x)$ ” true also makes “if $\sim Q(x)$ then $\sim P(x)$ ” true (by the logical equivalence between $p \rightarrow q$ and $\sim q \rightarrow \sim p$). It follows that the sentence “If $P(x)$ then $Q(x)$ ” is true for all x in D if, and only if, the sentence “If $\sim Q(x)$ then $\sim P(x)$ ” is true for all x in D .

Thus we write the following and say that a universal conditional statement is logically equivalent to its contrapositive:

$$\forall x \in D, \text{ if } P(x) \text{ then } Q(x) \equiv \forall x \in D, \text{ if } \sim Q(x) \text{ then } \sim P(x)$$

In Example 3.2.5 we noted that the statement

$$\forall x \in \mathbf{R}, \text{ if } x > 2 \text{ then } x^2 > 4$$

has the converse

$$\forall x \in \mathbf{R}, \text{ if } x^2 > 4 \text{ then } x > 2.$$

Observe that the statement is true whereas its converse is false (since, for instance, $(-3)^2 = 9 > 4$ but $-3 \not> 2$). This shows that a universal conditional statement may have a different truth value from its converse. Hence a universal conditional statement is not logically equivalent to its converse. This is written in symbols as follows:

$$\forall x \in D, \text{ if } P(x) \text{ then } Q(x) \not\equiv \forall x \in D, \text{ if } Q(x) \text{ then } P(x).$$

In exercise 35 at the end of this section, you are asked to provide an example to show that a universal conditional statement is not logically equivalent to its inverse.

$$\forall x \in D, \text{ if } P(x) \text{ then } Q(x) \not\equiv \forall x \in D, \text{ if } \sim P(x) \text{ then } \sim Q(x).$$

Necessary and Sufficient Conditions, Only If

The definitions of *necessary*, *sufficient*, and *only if* can also be extended to apply to universal conditional statements.

Definition

- “ $\forall x, r(x)$ is a **sufficient condition** for $s(x)$ ” means “ $\forall x, \text{ if } r(x) \text{ then } s(x)$.”
- “ $\forall x, r(x)$ is a **necessary condition** for $s(x)$ ” means “ $\forall x, \text{ if } \sim r(x) \text{ then } \sim s(x)$ ” or, equivalently, “ $\forall x, \text{ if } s(x) \text{ then } r(x)$.”
- “ $\forall x, r(x)$ **only if** $s(x)$ ” means “ $\forall x, \text{ if } \sim s(x) \text{ then } \sim r(x)$ ” or, equivalently, “ $\forall x, \text{ if } r(x) \text{ then } s(x)$.”

Example 3.2.6 Necessary and Sufficient Conditions

Rewrite each of the following as a universal conditional statement, quantified either explicitly or implicitly. Do not use the word *necessary* or *sufficient*.

- Squareness is a sufficient condition for rectangularity.
- Being at least 35 years old is a necessary condition for being president of the United States.

Solution

- A formal version of the statement is

$$\forall x, \text{ if } x \text{ is a square, then } x \text{ is a rectangle.}$$

Or, with implicit universal quantification:

If a figure is a square, then it is a rectangle.

- Using formal language, you could write the answer as

$$\forall \text{ person } x, \text{ if } x \text{ is younger than 35, then } x \text{ cannot be president of the United States.}$$

Or, by the equivalence between a statement and its contrapositive:

$$\forall \text{ person } x, \text{ if } x \text{ is president of the United States, then } x \text{ is at least 35 years old.}$$

Example 3.2.7 Only If

Rewrite the following as a universal conditional statement:

A product of two numbers is 0 only if one of the numbers is 0.

Solution Using informal language, you could write the answer as

If it is not the case that one of two numbers is 0,
then the product of the numbers is not 0.

In other words,

If neither of two numbers is 0, then the product of the numbers is not 0.

Or, by the equivalence between a statement and its contrapositive:

If a product of two numbers is 0, then one of the numbers is 0. ■

TEST YOURSELF

1. A negation for “All R have property S ” is “There is _____ R that _____.”
2. A negation for “Some R have property S ” is “_____.”
3. A negation for “For every x , if x has property P then x has property Q ” is “_____.”
4. The converse of “For every x , if x has property P then x has property Q ” is “_____.”
5. The contrapositive of “For every x , if x has property P then x has property Q ” is “_____.”
6. The inverse of “For every x , if x has property P then x has property Q ” is “_____.”

EXERCISE SET 3.2

1. Which of the following is a negation for “All discrete mathematics students are athletic”? More than one answer may be correct.
 - a. There is a discrete mathematics student who is nonathletic.
 - b. All discrete mathematics students are nonathletic.
 - c. There is an athletic person who is not a discrete mathematics student.
 - d. No discrete mathematics students are athletic.
 - e. Some discrete mathematics students are nonathletic.
 - f. No athletic people are discrete mathematics students.
2. Which of the following is a negation for “All dogs are loyal”? More than one answer may be correct.
 - a. All dogs are disloyal.
 - b. No dogs are loyal.
 - c. Some dogs are disloyal.
 - d. Some dogs are loyal.
 - e. There is a disloyal animal that is not a dog.
 - f. There is a dog that is disloyal.
 - g. No animals that are not dogs are loyal.
 - h. Some animals that are not dogs are loyal.
3. Write a formal negation for each of the following statements.
 - a. $\forall$ string s , s has at least one character.
 - b. $\forall$ computer c , c has a CPU.
 - c. $\exists$ a movie m such that m is over 6 hours long.
 - d. $\exists$ a band b such that b has won at least 10 Grammy awards.
4. Write an informal negation for each of the following statements. Be careful to avoid negations that are ambiguous.
 - a. All dogs are friendly.
 - b. All graphs are connected.
 - c. Some suspicions were substantiated.
 - d. Some estimates are accurate.
5. Write a negation for each of the following statements.
 - a. Every valid argument has a true conclusion.
 - b. All real numbers are positive, negative, or zero.

35. Give an example to show that a universal conditional statement is not logically equivalent to its inverse.
- *36. If $P(x)$ is a predicate and the domain of x is the set of all real numbers, let R be " $\forall x \in \mathbf{Z}, P(x)$," let S be " $\forall x \in \mathbf{Q}, P(x)$," and let T be " $\forall x \in \mathbf{R}, P(x)$."
- Find a definition for $P(x)$ (but do not use " $x \in \mathbf{Z}$ ") so that R is true and both S and T are false.
 - Find a definition for $P(x)$ (but do not use " $x \in \mathbf{Q}$ ") so that both R and S are true and T is false.
37. Consider the following sequence of digits: 0204. A person claims that all the 1's in the sequence are to the left of all the 0's in the sequence. Is this true? Justify your answer. (*Hint: Write the claim formally and write a formal negation for it. Is the negation true or false?*)
38. True or false? All occurrences of the letter u in *Discrete Mathematics* are lowercase. Justify your answer.
- Rewrite each statement of 39–44 in if-then form.**
39. Earning a grade of C– in this course is a sufficient condition for it to count toward graduation.
40. Being divisible by 8 is a sufficient condition for being divisible by 4.
41. Being on time each day is a necessary condition for keeping this job.
42. Passing a comprehensive exam is a necessary condition for obtaining a master's degree.
43. A number is prime only if it is greater than 1.
44. A polygon is square only if it has four sides.
Use the facts that the negation of a $\forall$ statement is a $\exists$ statement and that the negation of an if-then statement is an *and* statement to rewrite each of the statements 45–48 without using the word *necessary* or *sufficient*.
45. Being divisible by 8 is not a necessary condition for being divisible by 4.
46. Having a large income is not a necessary condition for a person to be happy.
47. Having a large income is not a sufficient condition for a person to be happy.
48. Being a polynomial is not a sufficient condition for a function to have a real root.
49. The computer scientists Richard Conway and David Gries once wrote:
 The absence of error messages during translation of a computer program is only a necessary and not a sufficient condition for reasonable [program] correctness.
 Rewrite this statement without using the words *necessary* or *sufficient*.
50. A frequent-flyer club brochure states, "You may select among carriers only if they offer the same lowest fare." Assuming that "only if" has its formal, logical meaning, does this statement guarantee that if two carriers offer the same lowest fare, the customer will be free to choose between them? Explain.

ANSWERS FOR TEST YOURSELF

1. some (*Alternative answers:* at least one; an); does not have property S . 2. No R have property S . 3. There is an x such that x has property P and x does not have property Q . 4. For every x , if x has property Q then x has property P . 5. For every x , if x does not have property Q then x does not have property P . 6. For every x , if x does not have property P then x does not have property Q .

3.3 Statements with Multiple Quantifiers

It is not enough to have a good mind. The main thing is to use it well. —René Descartes

Imagine you are visiting a factory that manufactures computer microchips. The factory guide tells you,

"There is a person supervising every detail of the production process."

Note that this statement contains informal versions of both the existential quantifier *there is* and the universal quantifier *every*. Which of the following best describes its meaning?

- There is one single person who supervises all the details of the production process.
- For any particular production detail, there is a person who supervises that detail, but there might be different supervisors for different details.

As it happens, either interpretation could be what the guide meant. (Reread the sentence to be sure you agree!) Taken by itself, his statement is genuinely ambiguous, although other things he may have said (the context for his statement) might have clarified it. In our ordinary lives, we deal with this kind of ambiguity all the time. Usually context helps resolve it, but sometimes we simply misunderstand each other.

In mathematics, formal logic, and computer science, by contrast, it is essential that we all interpret statements in exactly the same way. For instance, the initial stage of software development typically involves careful discussion between a programmer analyst and a client to turn vague descriptions of what the client wants into unambiguous program specifications that client and programmer can mutually agree on.

Because many important technical statements contain both $\exists$ and $\forall$, a convention has developed for interpreting them uniformly. **When a statement contains more than one kind of quantifier, we imagine the actions suggested by the quantifiers as being performed in the order in which the quantifiers occur.** For instance, consider a statement of the form

$$\forall x \text{ in set } D, \exists y \text{ in set } E \text{ such that } x \text{ and } y \text{ satisfy property } P(x, y).$$

To show that such a statement is true, you must be able to meet the following challenge:

- Imagine that someone is allowed to choose any element whatsoever from the set D , and imagine that the person gives you that element. Call it x .
- The challenge for you is to find an element y in E so that the person's x and your y , taken together, satisfy property $P(x, y)$.

Because you do not have to specify the y until after the other person has specified the x , you are allowed to find a different value of y for each different x you are given.

Note The scope of $\forall x$ extends throughout the statement, whereas the scope of $\exists y$ starts in the middle. That is why the value of y depends on the value of x .

Example 3.3.1 Truth of a $\forall \exists$ Statement in a Tarski World

Consider the Tarski world shown in Figure 3.3.1.

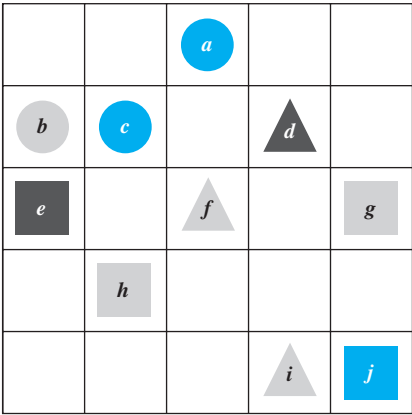


FIGURE 3.3.1

Show that the following statement is true in this world:

For every triangle x , there is a square y such that x and y have the same color.

Solution The statement says that no matter which triangle someone gives you, you will be able to find a square of the same color. There are only three triangles, d , f , and i . The following table shows that for each of these triangles a square of the same color can be found.

Given $x =$	choose $y =$	and check that y is the same color as x .
d	e	yes ✓
f or i	h or g	yes ✓

Now consider a statement containing both $\forall$ and $\exists$, where the $\exists$ comes before the $\forall$:

$\exists x$ in set D such that $\forall y$ in set E , x and y satisfy property $P(x, y)$.

To show that a statement of this form is true:

You must find one single element (call it x) in D with the following property:

- After you have found your x , someone is allowed to choose any element whatsoever from E . The person challenges you by giving you that element. Call it y .
- Your job is to show that your x together with the person's y satisfy property $P(x, y)$.

Your x has to work for *any* y the person might give you; ***you are not allowed to change your x once you have specified it initially.***

Note The value of x cannot be changed once it is specified because the scope of $\exists x$ extends throughout the entire statement.

Example 3.3.2

Truth of a $\exists\forall$ Statement in a Tarski World

Consider again the Tarski world in Figure 3.3.1. Show that the following statement is true: There is a triangle x such that for every circle y , x is to the right of y .

Solution The statement says that you can find a triangle that is to the right of all the circles. Actually, either d or i would work for all of the three circles, a , b , and c , as you can see in the following table.

Choose $x =$	Then: given $y =$	check that x is to the right of y .
d or i	a	yes ✓
	b	yes ✓
	c	yes ✓

Here is a summary of the convention for interpreting statements with two different quantifiers:

Interpreting Statements with Two Different Quantifiers

If you want to establish the truth of a statement of the form

$\forall x$ in D , $\exists y$ in E such that $P(x, y)$

your challenge is to allow someone else to pick whatever element x in D they wish and then you must find an element y in E that “works” for that particular x .

If you want to establish the truth of a statement of the form

$\exists x$ in D such that $\forall y$ in E , $P(x, y)$

your job is to find one particular x in D that will “work” no matter what y in E anyone might choose to challenge you with.

Example 3.3.3 Interpreting Statements with More Than One Quantifier

A college cafeteria line has four stations: salads, main courses, desserts, and beverages. The salad station offers a choice of green salad or fruit salad; the main course station offers spaghetti or fish; the dessert station offers pie or cake; and the beverage station offers milk, soda, or coffee. Three students, Uta, Tim, and Yuen, go through the line and make the following choices:

Uta: green salad, spaghetti, pie, milk

Tim: fruit salad, fish, pie, cake, milk, coffee

Yuen: spaghetti, fish, pie, soda

These choices are illustrated in Figure 3.3.2.

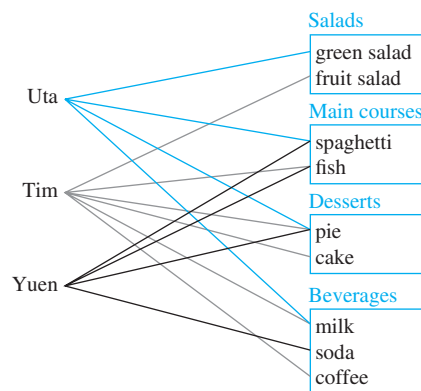


FIGURE 3.3.2

Write each of following statements informally and find its truth value.

- $\exists$ an item I such that $\forall$ student S , S chose I .
- $\exists$ a student S such that $\forall$ item I , S chose I .
- $\exists$ a student S such that $\forall$ station Z , $\exists$ an item I in Z such that S chose I .
- $\forall$ student S and $\forall$ station Z , $\exists$ an item I in Z such that S chose I .

Solution

- There is an item that was chosen by every student. This is true; every student chose pie.
- There is a student who chose every available item. This is false; no student chose all nine items.
- There is a student who chose at least one item from every station. This is true; both Uta and Tim chose at least one item from every station.
- Every student chose at least one item from every station. This is false; Yuen did not choose a salad.

Translating from Informal to Formal Language

Most problems are stated in informal language, but solving them often requires translating them into more formal terms.

Example 3.3.4 Translating Statements with Multiple Quantifiers from Informal to Formal Language

The **reciprocal** of a real number a is a real number b such that $ab = 1$. The following two statements are true. Rewrite them formally using quantifiers and variables.

- Every nonzero real number has a reciprocal.
- There is a real number with no reciprocal.

Note The number 0 has no reciprocal.

Solution

- $\forall$ nonzero real number u , $\exists$ a real number v such that $uv = 1$.
- $\exists$ a real number c such that $\forall$ real number d , $cd \neq 1$.

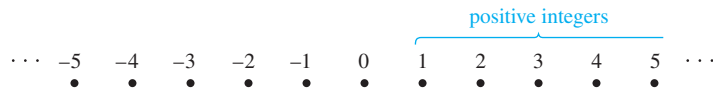
Example 3.3.5 There Is a Smallest Positive Integer

Recall that every integer is a real number and that real numbers are of three types: positive, negative, and zero (zero being neither positive nor negative). Consider the statement “There is a smallest positive integer.” Write this statement formally using both symbols $\exists$ and $\forall$.

Solution To say that there is a smallest positive integer means that there is a positive integer m with the property that no matter what positive integer n a person might pick, m will be less than or equal to n :

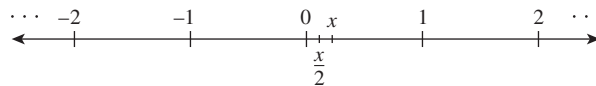
$\exists$ a positive integer m such that $\forall$ positive integer n , $m \leq n$.

Note that this statement is true because 1 is a positive integer that is less than or equal to every positive integer.



Example 3.3.6 There Is No Smallest Positive Real Number

Imagine the positive real numbers on the real number line. These numbers correspond to all the points to the right of 0. Observe that no matter how small a real number x is, the number $x/2$ will be both positive and less than x .*



Thus the following statement is true: “There is no smallest positive real number.” Write this statement formally using both symbols $\forall$ and $\exists$.

Solution $\forall$ positive real number x , $\exists$ a positive real number y such that $y < x$.

Example 3.3.7 The Definition of Limit of a Sequence

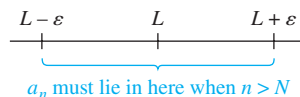
The definition of limit of a sequence, studied in calculus, uses both quantifiers $\forall$ and $\exists$ and also if-then. We say that the limit of the sequence a_n as n goes to infinity equals L and write

$$\lim_{n \rightarrow \infty} a_n = L$$

if, and only if, the values of a_n become *arbitrarily* close to L as n gets larger and larger without bound. More precisely, this means that given any positive number ε , we can find

*This can be deduced from the properties of the real numbers given in Appendix A. Because x is positive, $0 < x$. Add x to both sides to obtain $x < 2x$. Then $0 < x < 2x$. Now multiply all parts of the inequality by the positive number $1/2$. This does not change the direction of the inequality, so $0 < x/2 < x$.

an integer N such that whenever n is larger than N , the number a_n sits between $L - \varepsilon$ and $L + \varepsilon$ on the number line.



Symbolically:

$\forall \varepsilon > 0, \exists$ an integer N such that $\forall$ integer n ,
if $n > N$ then $L - \varepsilon < a_n < L + \varepsilon$.

Considering the logical complexity of this definition, it is no wonder that many students find it hard to understand. ■

Ambiguous Language

The drawing in Figure 3.3.3 is a famous example of visual ambiguity. When you look at it for a while, you will probably see either a silhouette of a young woman wearing a large hat or an elderly woman with a large nose. Whichever image first pops into your mind, try to see how the drawing can be interpreted in the other way. (*Hint:* The mouth of the elderly woman is the necklace on the young woman.)



Chronicle/Alamy Stock Photo

FIGURE 3.3.3

Once most people see one of the images, it is difficult for them to perceive the other. So it is with ambiguous language. Once you interpreted the sentence at the beginning of this section in one way, it may have been hard for you to see that it could be understood in the other way. Perhaps you had difficulty even though the two possible meanings were explained, just as many people have difficulty seeing the second interpretation for the drawing even when they are told what to look for.

Although statements written informally may be open to multiple interpretations, we cannot determine their truth or falsity without interpreting them one way or another. Therefore, we have to use context to try to ascertain their meaning as best we can.

Negations of Statements with More Than One Quantifier

You can use the same rules to negate statements with several quantifiers that you used to negate simpler quantified statements. Recall that

$$\sim(\forall x \text{ in } D, P(x)) \equiv \exists x \text{ in } D \text{ such that } \sim P(x).$$

and

$$\sim(\exists x \text{ in } D \text{ such that } P(x)) \equiv \forall x \text{ in } D, \sim P(x).$$

Thus

$$\begin{aligned} \sim(\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } P(x, y)) &\equiv \exists x \text{ in } D \text{ such that } \sim(\exists y \text{ in } E \text{ such that } P(x, y)) \\ &\equiv \exists x \text{ in } D \text{ such that } \forall y \text{ in } E, \sim P(x, y) \end{aligned}$$

Similarly,

$$\begin{aligned} \sim(\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, P(x, y)) &\equiv \forall x \text{ in } D, \sim(\forall y \text{ in } E, P(x, y)) \\ &\equiv \forall x \text{ in } D, \exists y \text{ in } E \text{ such that } \sim P(x, y) \end{aligned}$$

These facts are summarized as follows:

Negations of Statements with Two Different Quantifiers

$$\begin{aligned} \sim(\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } P(x, y)) &\equiv \exists x \text{ in } D \text{ such that } \forall y \text{ in } E, \sim P(x, y) \\ \sim(\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, P(x, y)) &\equiv \forall x \text{ in } D, \exists y \text{ in } E \text{ such that } \sim P(x, y) \end{aligned}$$

Example 3.3.8

Negating Statements in a Tarski World

Refer to the Tarski world of Figure 3.3.1, which is reprinted here for reference.

Write a negation for each of the following statements, and determine which is true, the given statement or its negation.

- a. For every square x , there is a circle y such that x and y have the same color.
- b. There is a triangle x such that for every square y , x is to the right of y .

Solution

- a. *First version of negation:* $\exists$ a square x such that $\sim(\exists$ a circle y such that x and y have the same color).
Final version of negation: $\exists$ a square x such that $\forall$ circle y , x and y do not have the same color.

The negation is true. Square e is black and no circle in this Tarski world is black, so there is a square that does not have the same color as any circle.

- b. *First version of negation:* $\forall$ triangle x , $\sim(\forall$ square y , x is to the right of y).
Final version of negation: $\forall$ triangle x , $\exists$ a square y such that x is not to the right of y .

The negation is true because no matter what triangle is chosen, it is not to the right of square g or square j , which are the only squares in this Tarski world. ■

Order of Quantifiers

Consider the following two statements:

$$\forall \text{ person } x, \exists \text{ a person } y \text{ such that } x \text{ loves } y.$$

$$\exists \text{ a person } y \text{ such that } \forall \text{ person } x, x \text{ loves } y.$$

Note that except for the order of the quantifiers, these statements are identical. However, the first means that given any person, it is possible to find someone whom that person loves, whereas the second means that there is one amazing individual who is loved by all people. (Reread the statements carefully to verify these interpretations!) The two sentences illustrate an extremely important property about statements with two different quantifiers.

In a statement containing both $\forall$ and $\exists$, changing the order of the quantifiers can significantly change the meaning of the statement.

Interestingly, however, if one quantifier immediately follows another quantifier *of the same type*, then the order of the quantifiers does not affect the meaning. Consider the commutative property of addition of real numbers, for example:

$$\forall \text{ real number } x \text{ and } \forall \text{ real number } y, x + y = y + x.$$

This means the same as

$$\forall \text{ real number } y \text{ and } \forall \text{ real number } x, x + y = y + x.$$

Thus the property can be expressed a little less formally as

$$\forall \text{ real numbers } x \text{ and } y, x + y = y + x.$$

Example 3.3.9 Quantifier Order in a Tarski World

Look again at the Tarski world of Figure 3.3.1. Do the following two statements have the same truth value?

- For every square x there is a triangle y such that x and y have different colors.
- There exists a triangle y such that for every square x , x and y have different colors.



Caution! If a statement contains two different quantifiers, reversing their order may change the truth value of the statement to its opposite.

Solution Statement (a) says that if someone gives you one of the squares from the Tarski world, you can find a triangle that has a different color. This is true. If someone gives you square g or h (which are gray), you can use triangle d (which is black); if someone gives you square e (which is black), you can use either triangle f or i (which are gray); and if someone gives you square j (which is blue), you can use triangle d (which is black) or triangle f or i (which are gray).

Statement (b) says that there is one particular triangle in the Tarski world that has a different color from every one of the squares in the world. This is false. Two of the triangles are gray, but they cannot be used to show the truth of the statement because the Tarski world contains gray squares. The only other triangle is black, but it cannot be used either because there is a black square in the Tarski world.

Thus one of the statements is true and the other is false, and so they have opposite truth values. ■

Formal Logical Notation

In some areas of computer science, logical statements are expressed in purely symbolic notation. The notation involves using predicates to describe all properties of variables and omitting the words *such that* in existential statements. (When you try to figure out the meaning of a formal statement, however, it is helpful to think the words *such that* to yourself each time they are appropriate.) The formalism also depends on the following facts:

“ $\forall x \text{ in } D, P(x)$ ” can be written as “ $\forall x (x \text{ in } D \rightarrow P(x))$,” and

“ $\exists x \text{ in } D \text{ such that } P(x)$ ” can be written as “ $\exists x (x \text{ in } D \wedge P(x))$.”

We illustrate the use of these facts in Example 3.3.10.

Example 3.3.10 Formalizing Statements in a Tarski World

Consider once more the Tarski world of Figure 3.3.1:

Let $\text{Triangle}(x)$, $\text{Circle}(x)$, and $\text{Square}(x)$ mean “ x is a triangle,” “ x is a circle,” and “ x is a square”; let $\text{Blue}(x)$, $\text{Gray}(x)$, and $\text{Black}(x)$ mean “ x is blue,” “ x is gray,” and “ x is black”; let $\text{RightOf}(x, y)$, $\text{Above}(x, y)$, and $\text{SameColorAs}(x, y)$ mean “ x is to the right of y ,” “ x is above y ,” and “ x has the same color as y ”; and use the notation $x = y$ to denote the predicate “ x is equal to y .” Let the common domain D of all variables be the set of all the objects in the Tarski world. Use formal logical notation to write each of the following statements, and write a formal negation for each statement.

- For every circle x , x is above f .
- There is a square x such that x is black.
- For every circle x , there is a square y such that x and y have the same color.
- There is a square x such that for every triangle y , x is to the right of y .

Solution

- Statement:* $\forall x(\text{Circle}(x) \rightarrow \text{Above}(x, f))$
Negation: $\sim(\forall x(\text{Circle}(x) \rightarrow \text{Above}(x, f)))$
 $\equiv \exists x \sim(\text{Circle}(x) \rightarrow \text{Above}(x, f))$ by the law for negating a $\forall$ statement
 $\equiv \exists x(\text{Circle}(x) \wedge \sim \text{Above}(x, f))$ by the law of negating an if-then statement
- Statement:* $\exists x(\text{Square}(x) \wedge \text{Black}(x))$
Negation: $\sim(\exists x(\text{Square}(x) \wedge \text{Black}(x)))$
 $\equiv \forall x \sim(\text{Square}(x) \wedge \text{Black}(x))$ by the law for negating a $\exists$ statement
 $\equiv \forall x(\sim \text{Square}(x) \vee \sim \text{Black}(x))$ by De Morgan's law
- Statement:* $\forall x(\text{Circle}(x) \rightarrow \exists y(\text{Square}(y) \wedge \text{SameColor}(x, y)))$
Negation: $\sim(\forall x(\text{Circle}(x) \rightarrow \exists y(\text{Square}(y) \wedge \text{SameColor}(x, y))))$
 $\equiv \exists x \sim(\text{Circle}(x) \rightarrow \exists y(\text{Square}(y) \wedge \text{SameColor}(x, y)))$ by the law for negating a $\forall$ statement
 $\equiv \exists x(\text{Circle}(x) \wedge \sim(\exists y(\text{Square}(y) \wedge \text{SameColor}(x, y))))$ by the law for negating an if-then statement
 $\equiv \exists x(\text{Circle}(x) \wedge \forall y(\sim(\text{Square}(y) \wedge \text{SameColor}(x, y))))$ by the law for negating a $\exists$ statement
 $\equiv \exists x(\text{Circle}(x) \wedge \forall y(\sim \text{Square}(y) \vee \sim \text{SameColor}(x, y)))$ by De Morgan's law
- Statement:* $\exists x(\text{Square}(x) \wedge \forall y(\text{Triangle}(y) \rightarrow \text{RightOf}(x, y)))$
Negation: $\sim(\exists x(\text{Square}(x) \wedge \forall y(\text{Triangle}(y) \rightarrow \text{RightOf}(x, y))))$
 $\equiv \forall x \sim(\text{Square}(x) \wedge \forall y(\text{Triangle}(y) \rightarrow \text{RightOf}(x, y)))$ by the law for negating a $\exists$ statement
 $\equiv \forall x(\sim \text{Square}(x) \vee \sim(\forall y(\text{Triangle}(y) \rightarrow \text{RightOf}(x, y))))$ by De Morgan's law
 $\equiv \forall x(\sim \text{Square}(x) \vee \exists y(\sim(\text{Triangle}(y) \rightarrow \text{RightOf}(x, y))))$ by the law for negating a $\forall$ statement
 $\equiv \forall x(\sim \text{Square}(x) \vee \exists y(\text{Triangle}(y) \wedge \sim \text{RightOf}(x, y)))$ by the law for negating an if-then statement

The disadvantage of the fully formal notation is that because it is complex and somewhat remote from intuitive understanding, when we use it, we may make errors that go unrecognized. The advantage, however, is that operations, such as taking negations, can be made completely mechanical and programmed on a computer. Also, when we become comfortable with formal manipulations, we can use them to check our intuition, and then we can use our intuition to check our formal manipulations. Formal logical notation is used in branches of computer science such as artificial intelligence, program verification, and automata theory and formal languages.

Taken together, the symbols for quantifiers, variables, predicates, and logical connectives make up what is known as the **language of first-order logic**. Even though this language is simpler in many respects than the language we use every day, learning it requires the same kind of practice needed to acquire any foreign language.

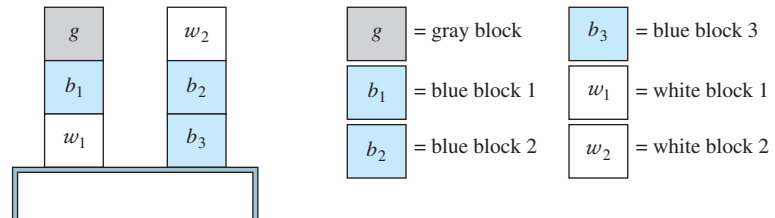
Prolog

The programming language Prolog (short for *programming in logic*) was developed in France in the 1970s by A. Colmerauer and P. Roussel to help programmers working in the field of artificial intelligence. A simple Prolog program consists of a set of statements describing some situation together with questions about the situation. Built into the language are search and inference techniques needed to answer the questions by deriving the answers from the given statements. This frees the programmer from the necessity of having to write separate programs to answer each type of question. Example 3.3.11 gives a very simple example of a Prolog program.

Example 3.3.11

A Prolog Program

Consider the following picture, which shows colored blocks stacked on a table.



Note Different Prolog implementations follow different conventions as to how to represent constant, variable, and predicate names and forms of questions and answers. The conventions used here are similar to those of Edinburgh Prolog.

The following are statements in Prolog that describe this picture and ask two questions about it.

isabove(g, b ₁)	color(g, gray)	color(b ₃ , blue)
isabove(b ₁ , w ₁)	color(b ₁ , blue)	color(w ₁ , white)
isabove(w ₂ , b ₂)	color(b ₂ , blue)	color(w ₂ , white)
isabove(b ₂ , b ₃)	isabove(X, Z) if isabove(X, Y) and isabove(Y, Z)	
1. ?color(b ₁ , blue)	2. ?isabove(X, w ₁)	

The statements “isabove(g, b₁)” and “color(g, gray)” are to be interpreted as “g is above b₁” and “g is colored gray.” The statement “isabove(X, Z) if isabove(X, Y) and isabove(Y, Z)” is to be interpreted as “For all X, Y, and Z, if X is above Y and Y is above Z, then X is above Z.”

Statement 1

?color(b₁, blue)

asks whether block b_1 is colored blue. Prolog answers this by writing

Yes.

Statement 2

?isabove(X, w_1)

asks for which blocks X the predicate “ X is above w_1 ” is true. Prolog answers by giving a list of all such blocks. In this case, the answer is

$X = b_1, X = g.$

Note that Prolog can find the solution $X = b_1$ by merely searching the original set of given facts. However, Prolog must *infer* the solution $X = g$ from the following statements:

isabove(g, b_1),
isabove(b_1, w_1),
isabove(X, Z) if isabove(X, Y) and isabove(Y, Z).

Write the answers Prolog would give if the following questions were added to the program above.

- a. ?isabove(b_2, w_1) b. ?color(w_1, X) c. ?color(X, blue)

Solution

- The question means “Is b_2 above w_1 ?”; so the answer is “No.”
- The question means “For what colors X is the predicate ‘ w_1 is colored X ’ true?”; so the answer is “ $X = \text{white}.$ ”
- The question means “For what blocks is the predicate ‘ X is colored blue’ true?”; so the answer is “ $X = b_1,$ ” “ $X = b_2,$ ” and “ $X = b_3.$ ” ■

TEST YOURSELF

- To establish the truth of a statement of the form “ $\forall x$ in $D, \exists y$ in E such that $P(x, y)$,” you imagine that someone has given you an element x from D but that you have no control over what that element is. Then you need to find _____ with the property that the x the person gave you together with the _____ you subsequently found satisfy _____.
- To establish the truth of a statement of the form “ $\exists x$ in D such that $\forall y$ in $E, P(x, y)$,” you need to find _____ so that no matter what _____ a person might subsequently give you, _____ will be true.
- Consider the statement “ $\forall x, \exists y$ such that $P(x, y)$, a property involving x and y , is true.” A negation for this statement is “_____.”
- Consider the statement “ $\exists x$ such that $\forall y, P(x, y)$, a property involving x and y , is true.” A negation for this statement is “_____.”
- Suppose $P(x, y)$ is some property involving x and y , and suppose the statement “ $\forall x$ in $D, \exists y$ in E such that $P(x, y)$ ” is true. Then the statement “ $\exists x$ in D such that $\forall y$ in $E, P(x, y)$ ”
 - is true.
 - is false.
 - may be true or may be false.

EXERCISE SET 3.3

- Let C be the set of cities in the world, let N be the set of nations in the world, and let $P(c, n)$ be “ c is the capital city of n .” Determine the truth values of the following statements.
 - $P(\text{Tokyo, Japan})$
 - $P(\text{Athens, Egypt})$
 - $P(\text{Paris, France})$
 - $P(\text{Miami, Brazil})$
 - Let $G(x, y)$ be “ $x^2 > y$.” Indicate which of the following statements are true and which are false.
 - $G(2, 3)$
 - $G(1, 1)$
 - $G(\frac{1}{2}, \frac{1}{2})$
 - $G(-2, 2)$
 - The following statement is true: “ $\forall$ nonzero number x , $\exists$ a real number y such that $xy = 1$.” For each x given below, find a y to make the predicate “ $xy = 1$ ” true.
 - $x = 2$
 - $x = -1$
 - $x = 3/4$
 - The following statement is true: “ $\forall$ real number x , $\exists$ an integer n such that $n > x$.”* For each x given below, find an n to make the predicate “ $n > x$ ” true.
 - $x = 15.83$
 - $x = 10^8$
 - $x = 10^{10^{10}}$
- The statements in exercises 5–8 refer to the Tarski world given in Figure 3.3.1. Explain why each is true.
- For every circle x there is a square y such that x and y have the same color.
 - For every square x there is a circle y such that x and y have different colors and y is above x .
 - There is a triangle x such that for every square y , x is above y .
 - There is a triangle x such that for every circle y , y is above x .
 - Let $D = E = \{-2, -1, 0, 1, 2\}$. Explain why the following statements are true.
 - $\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } x + y = 0.$
 - $\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, x + y = y.$
 - This exercise refers to Example 3.3.3. Determine whether each of the following statements is true or false.
 - $\forall$ student S , $\exists$ a dessert D such that S chose D .
 - $\forall$ student S , $\exists$ a salad T such that S chose T .
 - $\exists$ a dessert D such that $\forall$ student S , S chose D .
 - $\exists$ a beverage B such that $\forall$ student D , D chose B .
 - $\exists$ an item I such that $\forall$ student S , S did not choose I .
 - $\exists$ a station Z such that $\forall$ student S , $\exists$ an item I such that S chose I from Z .
 - Let S be the set of students at your school, let M be the set of movies that have ever been released, and let $V(s, m)$ be “student s has seen movie m .” Rewrite each of the following statements without using the symbol $\forall$, the symbol $\exists$, or variables.
 - $\exists s \in S$ such that $V(s, \text{Casablanca})$.
 - $\forall s \in S, V(s, \text{Star Wars})$.
 - $\forall s \in S, \exists m \in M$ such that $V(s, m)$.
 - $\exists m \in M$ such that $\forall s \in S, V(s, m)$.
 - $\exists s \in S, \exists t \in S$, and $\exists m \in M$ such that $s \neq t$ and $V(s, m) \wedge V(t, m)$.
 - $\exists s \in S$ and $\exists t \in S$ such that $s \neq t$ and $\forall m \in M, V(s, m) \rightarrow V(t, m)$.
 - Let $D = E = \{-2, -1, 0, 1, 2\}$. Write negations for each of the following statements and determine which is true, the given statement or its negation.
 - $\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } x + y = 1.$
 - $\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, x + y = -y.$
 - $\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } xy \geq y.$
 - $\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, x \leq y.$
- In each of 13–19, (a) rewrite the statement in English without using the symbol $\forall$ or $\exists$ or variables and expressing your answer as simply as possible, and (b) write a negation for the statement.
- $\forall$ color C , $\exists$ an animal A such that A is colored C .
 - $\exists$ a book b such that $\forall$ person p , p has read b .
 - $\forall$ odd integer n , $\exists$ an integer k such that $n = 2k + 1$.
 - $\exists$ a real number u such that $\forall$ real number v , $uv = v$.
 - $\forall r \in \mathbf{Q}$, $\exists$ integers a and b such that $r = a/b$.
 - $\forall x \in \mathbf{R}$, $\exists$ a real number y such that $x + y = 0$.
 - $\exists x \in \mathbf{R}$ such that for every real number y , $x + y = 0$.
 - Recall that reversing the order of the quantifiers in a statement with two different quantifiers may

*This is called the Archimedean principle because it was first formulated (in geometric terms) by the great Greek mathematician Archimedes of Syracuse, who lived from about 287 to 212 B.C.E.

56. $\exists x \in D, (P(x) \wedge Q(x))$, and
 $(\exists x \in D, P(x)) \wedge (\exists x \in D, Q(x))$
57. $\forall x \in D, (P(x) \vee Q(x))$, and
 $(\forall x \in D, P(x)) \vee (\forall x \in D, Q(x))$
58. $\exists x \in D, (P(x) \vee Q(x))$, and
 $(\exists x \in D, P(x)) \vee (\exists x \in D, Q(x))$

In 59–61, find the answers Prolog would give if the following questions were added to the program given in Example 3.3.11.

59. a. $?isabove(b_1, w_1)$ 60. a. $?isabove(w_1, g)$
 b. $?color(X, white)$ b. $?color(w_2, blue)$
 c. $?isabove(X, b_3)$ c. $?isabove(X, b_1)$
61. a. $?isabove(w_2, b_3)$
 b. $?color(X, gray)$
 c. $?isabove(g, X)$

ANSWERS FOR TEST YOURSELF

1. an element y in E ; y ; $P(x, y)$ 2. an element x in D ; y in E ; $P(x, y)$ 3. $\exists x$ such that $\forall y$, the property $P(x, y)$ is false. 4. $\forall x, \exists y$ such that the property $P(x, y)$ is false.
5. The answer is (c): the truth or falsity of a statement in which the quantifiers are reversed depends on the nature of the property involving x and y .

3.4 Arguments with Quantified Statements

The only complete safeguard against reasoning ill, is the habit of reasoning well; familiarity with the principles of correct reasoning; and practice in applying those principles. —John Stuart Mill

The rule of *universal instantiation* (in-stan-she-AY-shun) says the following:

Universal Instantiation

If a property is true of *everything* in a set, then it is true of *any particular* thing in the set.

Use of the words *universal instantiation* indicates that the truth of a property in a particular case follows as a special instance of its more general or universal truth. The validity of this argument form follows immediately from the definition of truth values for a universal statement. One of the most famous examples of universal instantiation is the following:

All men are mortal.
 Socrates is a man.
 $\therefore$ Socrates is mortal.

Universal instantiation is *the* fundamental tool of deductive reasoning. Mathematical formulas, definitions, and theorems are like general templates that are used over and over in a wide variety of particular situations. A given theorem says that such and such is true for all things of a certain type. If, in a given situation, you have a particular object of that type, then by universal instantiation, you conclude that such and such is true for that particular object. You may repeat this process 10, 20, or more times in a single proof or problem solution.

As an example of universal instantiation, suppose you are doing a problem that requires you to simplify

$$r^{k+1} \cdot r,$$

where r is a particular real number and k is a particular integer. You know from your study of algebra that the following universal statements are true:

1. For every real number x and for all integers m and n , $x^m \cdot x^n = x^{m+n}$.
2. For every real number x , $x^1 = x$.

So you proceed as follows:

$$\begin{aligned} r^{k+1} \cdot r &= r^{k+1} \cdot r^1 && \text{Step 1} \\ &= r^{(k+1)+1} && \text{Step 2} \\ &= r^{k+2} && \text{by basic algebra.} \end{aligned}$$

Here is the reasoning behind steps 1 and 2.

Step 1: For every real number x , $x^1 = x$. universal truth
 r is a particular real number. particular instance
 $\therefore r^1 = r$. conclusion

Step 2: For every real number x and for all integers m and n , $x^m \cdot x^n = x^{m+n}$. universal truth
 r is a particular real number and $k+1$ and 1 are particular integers. particular instance
 $\therefore r^{k+1} \cdot r^1 = r^{(k+1)+1}$. conclusion

Both arguments are examples of universal instantiation.

Universal Modus Ponens

The rule of universal instantiation can be combined with modus ponens to obtain the valid form of argument called *universal modus ponens*.

Universal Modus Ponens

Formal Version	Informal Version
$\forall x$, if $P(x)$ then $Q(x)$.	If x makes $P(x)$ true, then x makes $Q(x)$ true.
$P(a)$ for a particular a .	a makes $P(x)$ true.
$\therefore Q(a)$.	$\therefore a$ makes $Q(x)$ true.

Note that the first, or major, premise of universal modus ponens could be written “All things that make $P(x)$ true make $Q(x)$ true,” in which case the conclusion would follow by universal instantiation alone. However, the if-then form is more natural to use in the majority of mathematical situations.

Example 3.4.1

Recognizing Universal Modus Ponens

Rewrite the following argument using quantifiers, variables, and predicate symbols. Is this argument valid? Why?

If an integer is even, then its square is even.
 k is a particular integer that is even.
 $\therefore k^2$ is even.

Solution The major premise of this argument can be rewritten as

$\forall x$, if x is an even integer then x^2 is even.

Let $E(x)$ be “ x is an even integer,” let $S(x)$ be “ x^2 is even,” and let k stand for a particular integer that is even. Then the argument has the following form:

$$\begin{aligned} &\forall x, \text{ if } E(x) \text{ then } S(x). \\ &E(k), \text{ for a particular } k. \\ &\therefore S(k). \end{aligned}$$

This argument has the form of universal modus ponens and is therefore valid. ■

Example 3.4.2 Drawing Conclusions Using Universal Modus Ponens

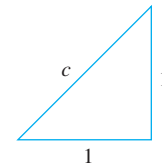
Write the conclusion that can be inferred using universal modus ponens.

If T is any right triangle with hypotenuse c and legs a and b , then $c^2 = a^2 + b^2$.

The triangle shown at the right is a right triangle with both legs equal to 1 and hypotenuse c .

$\therefore$ _____

Pythagorean Theorem



Solution $c^2 = 1^2 + 1^2 = 2$

Note that if you take the nonnegative square root of both sides of this equation, you obtain $c = \sqrt{2}$. This shows that there is a line segment whose length is $\sqrt{2}$. Section 4.7 contains a proof that $\sqrt{2}$ is not a rational number. ■

Use of Universal Modus Ponens in a Proof

In Chapter 4 we discuss methods of proving quantified statements. Here is a proof that the sum of any two even integers is even. It makes use of the definition of even integer, namely, that an integer is *even* if, and only if, it equals twice some integer. (Or, more formally: $\forall$ integers x , x is even if, and only if, $\exists$ an integer—say, k —such that $x = 2k$.)

Suppose m and n are particular but arbitrarily chosen even integers. Then $m = 2r$ for some integer r ,⁽¹⁾ and $n = 2s$ for some integer s .⁽²⁾ Hence

$$\begin{aligned} m + n &= 2r + 2s && \text{by substitution} \\ &= 2(r + s) && \text{by factoring out the 2.} \end{aligned}$$

Now $r + s$ is an integer,⁽⁴⁾ and so $2(r + s)$ is even.⁽⁵⁾ Thus $m + n$ is even.

The following expansion of the proof shows how each of the numbered steps is justified by arguments that are valid by universal modus ponens.

Note The logical principle of **existential instantiation** says that if we know or have deduced that something exists, we may give it a name. This is the principle that allows us to call the integers r and s .

- (1) If an integer is even, then it equals twice some integer.
 m is a particular even integer.
 $\therefore m$ equals twice some integer, say, r .
- (2) If an integer is even, then it equals twice some integer.
 n is a particular even integer.
 $\therefore n$ equals twice some integer, say, s .
- (3) If a quantity is an integer, then it is a real number.
 r and s are particular integers.
 $\therefore r$ and s are real numbers.

For all a , b , and c , if a , b , and c are real numbers, then $ab + ac = a(b + c)$.

2 , r , and s are particular real numbers.

$\therefore 2r + 2s = 2(r + s)$.

- (4) For all u and v , if u and v are integers, then $u + v$ is an integer.
 r and s are two particular integers.
 $\therefore r + s$ is an integer.
- (5) If a number equals twice some integer, then that number is even.
 $2(r + s)$ equals twice the integer $r + s$.
 $\therefore 2(r + s)$ is even.

Of course, the actual proof that the sum of even integers is even does not explicitly contain the sequence of arguments given above. In fact, people who are good at analytical thinking are normally not even conscious that they are reasoning in this way because they have absorbed the method so completely that it has become almost as automatic as breathing.

Universal Modus Tollens

Another crucially important rule of inference is *universal modus tollens*. Its validity results from combining universal instantiation with modus tollens. Universal modus tollens is the heart of proof of contradiction, which is one of the most important methods of mathematical argument.

Universal Modus Tollens

Formal Version

$\forall x$, if $P(x)$ then $Q(x)$.
 $\sim Q(a)$, for a particular a .
 $\therefore \sim P(a)$.

Informal Version

If x makes $P(x)$ true, then x makes $Q(x)$ true.
 a does not make $Q(x)$ true.
 $\therefore a$ does not make $P(x)$ true.

Example 3.4.3

Recognizing the Form of Universal Modus Tollens

Rewrite the following argument using quantifiers, variables, and predicate symbols. Write the major premise in conditional form. Is this argument valid? Why?

All human beings are mortal.
 Zeus is not mortal.
 $\therefore$ Zeus is not human.

Solution The major premise can be rewritten as

$\forall x$, if x is human then x is mortal.

Let $H(x)$ be “ x is human,” let $M(x)$ be “ x is mortal,” and let Z stand for Zeus. The argument becomes

$\forall x$, if $H(x)$ then $M(x)$
 $\sim M(Z)$
 $\therefore \sim H(Z)$.

This argument has the form of universal modus tollens and is therefore valid. ■

Example 3.4.4 Drawing Conclusions Using Universal Modus Tollens

Write the conclusion that can be inferred using universal modus tollens.

All professors are absent-minded.

Tom Hutchins is not absent-minded.

$\therefore$ _____.

Solution Tom Hutchins is not a professor. ■

Proving Validity of Arguments with Quantified Statements

The intuitive definition of validity for arguments with quantified statements is the same as for arguments with compound statements. An argument is valid if, and only if, the truth of its conclusion follows *necessarily* from the truth of its premises. The formal definition is as follows:

Definition

To say that an *argument form* is **valid** means the following: No matter what particular predicates are substituted for the predicate symbols in its premises, if the resulting premise statements are all true, then the conclusion is also true. An *argument* is called **valid** if, and only if, its form is valid. It is called **sound** if, and only if, its form is valid and its premises are true.

As already noted, the validity of universal instantiation follows immediately from the definition of the truth value of a universal statement. General formal proofs of validity of arguments in the predicate calculus are beyond the scope of this book. We give the proof of the validity of universal modus ponens as an example to show that such proofs are possible and to give an idea of how they look.

Universal modus ponens asserts that

$\forall x$, if $P(x)$ then $Q(x)$.

$P(a)$ for a particular a .

$\therefore Q(a)$.

To prove that this form of argument is valid, suppose the major and minor premises are both true. [*We must show that the conclusion “ $Q(a)$ ” is also true.*] By the minor premise, $P(a)$ is true for a particular value of a . By the major premise and universal instantiation, the statement “If $P(a)$ then $Q(a)$ ” is true for that particular a . But by modus ponens, since the statements “If $P(a)$ then $Q(a)$ ” and “ $P(a)$ ” are both true, it follows that $Q(a)$ is true also. [*This is what was to be shown.*]

The proof of validity given above is abstract and somewhat subtle. We include the proof not because we expect that you will be able to make up such proofs yourself at this stage of your study. Rather, it is intended as a glimpse of a more advanced treatment of the subject, which you can try your hand at in exercises 35 and 36 at the end of this section if you wish.

One of the paradoxes of the formal study of logic is that the laws of logic are used to prove that the laws of logic are valid!

In the next part of this section we show how you can use diagrams to analyze the validity or invalidity of arguments that contain quantified statements. Diagrams do not provide totally rigorous proofs of validity and invalidity, and in some complex settings they may even be confusing, but in many situations they are helpful and convincing.

Using Diagrams to Test for Validity

Consider the statement

All integers are rational numbers.

Or, formally,

$\forall$ integer n , n is a rational number.

Picture the set of all integers and the set of all rational numbers as disks. The truth of the given statement is represented by placing the integers disk entirely inside the rationals disk, as shown in Figure 3.4.1.

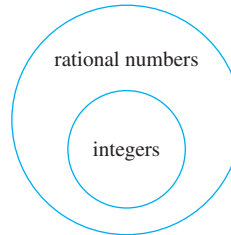


FIGURE 3.4.1



G. W. Leibniz
(1646–1716)

Bettmann/Getty Images

Because the two statements “ $\forall x \in D, Q(x)$ ” and “ $\forall x$, if x is in D then $Q(x)$ ” are logically equivalent, both can be represented by diagrams like the foregoing.

Perhaps the first person to use diagrams like these to analyze arguments was the German mathematician and philosopher Gottfried Wilhelm Leibniz. Leibniz (LIPE-nits) was far ahead of his time in anticipating modern symbolic logic. He also developed the main ideas of the differential and integral calculus at approximately the same time as (and independently of) Isaac Newton (1642–1727).

To test the validity of an argument diagrammatically, represent the truth of both premises with diagrams. Then analyze the diagrams to see whether they necessarily represent the truth of the conclusion as well.

Example 3.4.5 Using a Diagram to Show Validity

Use diagrams to show the validity of the following syllogism:

All human beings are mortal.

Zeus is not mortal.

$\therefore$ Zeus is not a human being.

Solution The major premise is pictured on the left in Figure 3.4.2 by placing a disk labeled “human beings” inside a disk labeled “mortals.” The minor premise is pictured on the right in Figure 3.4.2 by placing a dot labeled “Zeus” outside the disk labeled “mortals.”

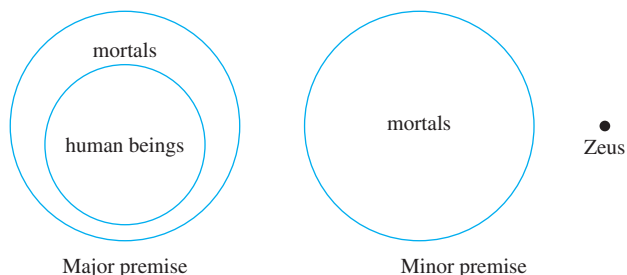


FIGURE 3.4.2

The two diagrams fit together in only one way, as shown in Figure 3.4.3.

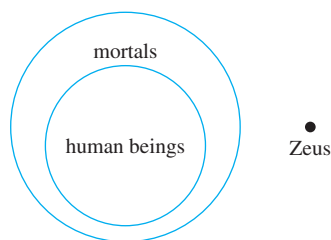


FIGURE 3.4.3

Since the Zeus dot is outside the mortals disk, it is necessarily outside the human beings disk. Thus the truth of the conclusion follows necessarily from the truth of the premises. It is impossible for the premises of this argument to be true and the conclusion false; hence the argument is valid. ■

Example 3.4.6 Using Diagrams to Show Invalidity

Use a diagram to show the invalidity of the following argument:

All human beings are mortal.

Felix is mortal.

∴ Felix is a human being.

Solution The major and minor premises are represented diagrammatically in Figure 3.4.4.

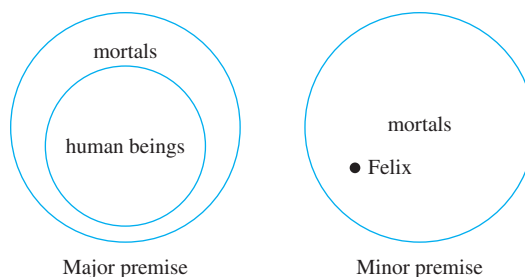


FIGURE 3.4.4

All that is known is that the Felix dot is located *somewhere* inside the mortals disk. Where it is located with respect to the human beings disk cannot be determined. Either one of the situations shown in Figure 3.4.5 might be the case.

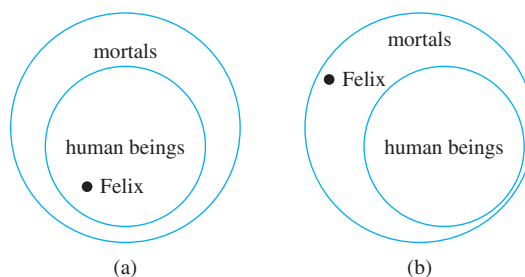


FIGURE 3.4.5



Caution! Be careful when using diagrams to test for validity! For instance, in this example if you put the diagrams for the premises together to obtain only Figure 3.4.5(a) and not Figure 3.4.5(b), you would conclude erroneously that the argument was valid.

The conclusion “Felix is a human being” is true in the first case but not in the second (Felix might, for example, be a cat). Because the conclusion does not necessarily follow from the premises, the argument is invalid.

The argument of Example 3.4.6 would be valid if the major premise were replaced by its converse. But since a universal conditional statement is not logically equivalent to its converse, such a replacement cannot, in general, be made. We say that this argument exhibits the converse error.

Converse Error (Quantified Form)

Formal Version

$\forall x$, if $P(x)$ then $Q(x)$.
 $Q(a)$ for a particular a .
 $\therefore P(a)$. $\leftarrow$ invalid conclusion

Informal Version

If x makes $P(x)$ true, then x makes $Q(x)$ true.
 a makes $Q(x)$ true.
 $\therefore a$ makes $P(x)$ true. $\leftarrow$ invalid conclusion

The following form of argument would be valid if a conditional statement were logically equivalent to its inverse. But it is not, and the argument form is invalid. We say that it exhibits the inverse error. You are asked to show the invalidity of this argument form in the exercises at the end of this section.

Inverse Error (Quantified Form)

Formal Version

$\forall x$, if $P(x)$ then $Q(x)$.
 $\sim P(a)$, for a particular a .
 $\therefore \sim Q(a)$. $\leftarrow$ invalid conclusion

Informal Version

If x makes $P(x)$ true, then x makes $Q(x)$ true.
 a does not make $P(x)$ true.
 $\therefore a$ does not make $Q(x)$ true. $\leftarrow$ invalid conclusion

Example 3.4.7

An Argument with “No”

Use diagrams to test the following argument for validity:

No polynomial functions have horizontal asymptotes.

This function has a horizontal asymptote.

$\therefore$ This function is not a polynomial function.

Solution A good way to represent the major premise diagrammatically is shown in Figure 3.4.6, two disks—a disk for polynomial functions and a disk for functions with horizontal asymptotes—that do not overlap at all. The minor premise is represented by placing a dot labeled “this function” inside the disk for functions with horizontal asymptotes.

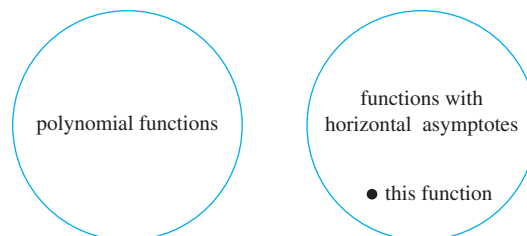


FIGURE 3.4.6

The diagram shows that “this function” must lie outside the polynomial functions disk, and so the truth of the conclusion necessarily follows from the truth of the premises. Hence the argument is valid. ■

An alternative way to solve Example 3.4.7 is to transform “No polynomial functions have horizontal asymptotes” into the equivalent statement “ $\forall x$, if x is a polynomial function, then x does not have a horizontal asymptote.” If this is done, the argument can be seen to have the form

$$\begin{aligned} &\forall x, \text{ if } P(x) \text{ then } Q(x). \\ &\sim Q(a), \text{ for a particular } a. \\ \therefore &\sim P(a). \end{aligned}$$

where $P(x)$ is “ x is a polynomial function” and $Q(x)$ is “ x does not have a horizontal asymptote.” This is valid by universal modus tollens.

Creating Additional Forms of Argument

Universal modus ponens and modus tollens were obtained by combining universal instantiation with modus ponens and modus tollens. In the same way, additional forms of arguments involving universally quantified statements can be obtained by combining universal instantiation with other of the valid argument forms given in Section 2.3. For instance, in Section 2.3 the argument form called transitivity was introduced:

$$\begin{aligned} &p \rightarrow q \\ &q \rightarrow r \\ \therefore &p \rightarrow r \end{aligned}$$

This argument form can be combined with universal instantiation to obtain the following valid argument form.

Universal Transitivity

Formal Version

$$\begin{aligned} &\forall x P(x) \rightarrow Q(x). \\ &\forall x Q(x) \rightarrow R(x). \\ \therefore &\forall x P(x) \rightarrow R(x). \end{aligned}$$

Informal Version

$$\begin{aligned} &\text{Any } x \text{ that makes } P(x) \text{ true makes } Q(x) \text{ true.} \\ &\text{Any } x \text{ that makes } Q(x) \text{ true makes } R(x) \text{ true.} \\ \therefore &\text{Any } x \text{ that makes } P(x) \text{ true makes } R(x) \text{ true.} \end{aligned}$$

Example 3.4.8

Evaluating an Argument for Tarski's World

The following argument refers to the kind of arrangement of objects of various types and colors described in Examples 3.1.13 and 3.3.1. Reorder and rewrite the premises to show that the conclusion follows as a valid consequence from the premises.

1. All the triangles are blue.
 2. If an object is to the right of all the squares, then it is above all the circles.
 3. If an object is not to the right of all the squares, then it is not blue.
- $\therefore$ All the triangles are above all the circles.

Solution It is helpful to begin by rewriting the premises and the conclusion in if-then form:

1. $\forall x$, if x is a triangle, then x is blue.
2. $\forall x$, if x is to the right of all the squares, then x is above all the circles.

3. $\forall x$, if x is not to the right of all the squares, then x is not blue.
 $\therefore \forall x$, if x is a triangle, then x is above all the circles.

The goal is to reorder the premises so that the conclusion of each is the same as the hypothesis of the next. Also, the hypothesis of the argument's conclusion should be the same as the hypothesis of the first premise, and the conclusion of the argument's conclusion should be the same as the conclusion of the last premise. To achieve this goal, it may be necessary to rewrite some of the statements in contrapositive form.

In this example you can see that the first premise should remain where it is, but the second and third premises should be interchanged. Then the hypothesis of the argument is the same as the hypothesis of the first premise, and the conclusion of the argument's conclusion is the same as the conclusion of the third premise. But the hypotheses and conclusions of the premises do not quite line up. This is remedied by rewriting the third premise in contrapositive form.

1. $\forall x$, if x is a triangle, then x is blue.
 3. $\forall x$, if x is blue, then x is to the right of all the squares.
 2. $\forall x$, if x is to the right of all the squares, then x is above all the circles.

Putting 1 and 3 together and using universal transitivity gives that

4. $\forall x$, if x is a triangle, then x is to the right of all the squares.

And putting 4 together with 2 and using universal transitivity gives that

$$\therefore \forall x, \text{ if } x \text{ is a triangle, then } x \text{ is above all the circles,}$$

which is the conclusion of the argument. ■

Remark on the Converse and Inverse Errors

One reason why so many people make converse and inverse errors is that the forms of the resulting arguments would be valid if the major premise were a biconditional rather than a simple conditional. And, as we noted in Section 2.2, many people tend to conflate biconditionals and conditionals.

Consider, for example, the following argument:

All the town criminals frequent the Den of Iniquity bar.
 John frequents the Den of Iniquity bar.
 $\therefore$ John is one of the town criminals.

The conclusion of this argument is invalid—it results from making the converse error. Therefore, it may be false even when the premises of the argument are true. This type of argument attempts unfairly to establish guilt by association.

The closer, however, the major premise comes to being a biconditional, the more likely the conclusion is to be true. If hardly anyone but criminals frequent the bar and John also frequents the bar, then it is likely (though not certain) that John is a criminal. On the basis of the given premises, it might be sensible to be suspicious of John, but it would be wrong to convict him.

A variation of the converse error is a very useful reasoning tool, provided that it is used with caution. It is the type of reasoning that is used by doctors to make medical diagnoses and by auto mechanics to repair cars. It is the type of reasoning used to generate explanations for phenomena. It goes like this:

If a statement of the form

For every x , if $P(x)$ then $Q(x)$

is true, and if

$Q(a)$ is true, for a particular a ,

then check out the statement $P(a)$; it just might be true. For instance, suppose a doctor knows that

For every x , if x has pneumonia, then x has a fever and chills,
coughs deeply, and feels exceptionally tired and miserable.

And suppose the doctor also knows that

John has a fever and chills, coughs deeply,
and feels exceptionally tired and miserable.

On the basis of these data, the doctor concludes that a diagnosis of pneumonia is a strong possibility, though not a certainty. The doctor will probably attempt to gain further support for this diagnosis through laboratory testing that is specifically designed to detect pneumonia. Note that the closer a set of symptoms comes to being a necessary and sufficient condition for an illness, the more nearly certain the doctor can be of his or her diagnosis.

This form of reasoning has been named **abduction** by researchers working in artificial intelligence. It is used in certain computer programs, called expert systems, that attempt to duplicate the functioning of an expert in some field of knowledge.

TEST YOURSELF

1. The rule of universal instantiation says that if some property is true for _____ in a domain, then it is true for _____.
2. If the first two premises of universal modus ponens are written as “If x makes $P(x)$ true, then x makes $Q(x)$ true” and “For a particular value of a _____,” then the conclusion can be written as “_____.”
3. If the first two premises of universal modus tollens are written as “If x makes $P(x)$ true, then x makes $Q(x)$ true” and “For a particular value of a _____,” then the conclusion can be written as “_____.”
4. If the first two premises of universal transitivity are written as “Any x that makes $P(x)$ true makes $Q(x)$ true” and “Any x that makes $Q(x)$ true makes $R(x)$ true,” then the conclusion can be written as “_____.”
5. Diagrams can be helpful in testing an argument for validity. However, if some possible configurations of the premises are not drawn, a person could conclude that an argument was _____ when it was actually _____.

EXERCISE SET 3.4

1. Let the following law of algebra be the first statement of an argument: For all real numbers a and b ,

$$(a + b)^2 = a^2 + 2ab + b^2.$$

Suppose each of the following statements is, in turn, the second statement of the argument. Use universal instantiation or universal modus ponens to write the conclusion that follows in each case.

 - a. $a = x$ and $b = y$ are particular real numbers.
 - b. $a = f_i$ and $b = f_j$ are particular real numbers.
 - c. $a = 3u$ and $b = 5v$ are particular real numbers.
 - d. $a = g(r)$ and $b = g(s)$ are particular real numbers.
 - e. $a = \log(t_1)$ and $b = \log(t_2)$ are particular real numbers.

Use universal instantiation or universal modus ponens to fill in valid conclusions for the arguments in 2–4.
2. If an integer n equals $2 \cdot k$ and k is an integer, then n is even.
 0 equals $2 \cdot 0$ and 0 is an integer.
 $\therefore$ _____.

ELEMENTARY NUMBER THEORY AND METHODS OF PROOF

The underlying content of this chapter consists of properties of integers (whole numbers), rational numbers (integer fractions), and real numbers. The underlying theme of the chapter is how to determine the truth or falsity of a mathematical statement.

Here is an example involving a concept used frequently in computer science. Given any real number x , the floor of x , or greatest integer in x , denoted $\lfloor x \rfloor$, is the largest integer that is less than or equal to x . On the number line, $\lfloor x \rfloor$ is the integer immediately to the left of x (or equal to x if x is, itself, an integer). Thus $\lfloor 2.3 \rfloor = 2$, $\lfloor 12.99999 \rfloor = 12$, and $\lfloor -1.5 \rfloor = -2$. Consider the following two questions:

1. For any real number x , is $\lfloor x - 1 \rfloor = \lfloor x \rfloor - 1$?
2. For any real numbers x and y , is $\lfloor x - y \rfloor = \lfloor x \rfloor - \lfloor y \rfloor$?

Take a few minutes to try to answer these questions for yourself.

It turns out that the answer to (1) is yes, whereas the answer to (2) is no. Are these the answers you got? If not, don't worry. In Section 4.6 you will learn the techniques you need to answer these questions and more. If you did get the correct answers, congratulations! You have excellent mathematical intuition. Now ask yourself, "How sure am I of my answers? Were they plausible guesses or absolute certainties? Was there any difference in certainty between my answers to (1) and (2)? Would I have been willing to bet a large sum of money on the correctness of my answers?"

One of the best ways to think of a mathematical proof is as a carefully reasoned argument to convince a skeptical listener (often yourself) that a given statement is true. Imagine the listener challenging your reasoning every step of the way, constantly asking, "Why is that so?" If you can counter every possible challenge, then your proof as a whole will be correct.

As an example, imagine proving to someone not very familiar with mathematical notation that if x is a number with $5x + 3 = 33$, then $x = 6$. You could argue as follows:

If $5x + 3 = 33$, then $5x + 3$ minus 3 will equal $33 - 3$ because subtracting the same number from two equal quantities gives equal results. But $5x + 3$ minus 3 equals $5x$ because adding 3 to $5x$ and then subtracting 3 just leaves $5x$. Also, $33 - 3 = 30$. Hence $5x = 30$. This means that x is a number which when multiplied by 5 equals 30. But the only number with this property is 6. Therefore, if $5x + 3 = 33$ then $x = 6$.

Of course there are other ways to phrase this proof, depending on the level of mathematical sophistication of the intended reader. In practice, mathematicians often omit reasons for certain steps of an argument when they are confident that the reader can easily supply them. When you are first learning to write proofs, however, it is better to err on the side of supplying too many reasons rather than too few. All too frequently, when even the best mathematicians carefully examine some "details" in their arguments, they discover that those details are actually false. One of the most important reasons for requiring proof in

mathematics is that writing a proof forces us to become aware of weaknesses in our arguments and in the unconscious assumptions we have made.

Sometimes correctness of a mathematical argument can be a matter of life or death. Suppose, for example, that a mathematician is part of a team charged with designing a new type of airplane engine, and suppose that the mathematician is given the job of determining whether the thrust delivered by various engine types is adequate. If you knew that the mathematician was only fairly sure, but not positive, of the correctness of his analysis, would you want to ride in the resulting aircraft?

At a certain point in Lewis Carroll's *Alice in Wonderland* (see exercise 28 in Section 2.2), the March Hare tells Alice to "say what you mean." In other words, if she means a thing, then that is exactly what she should say. In this chapter, perhaps more than in any other mathematics course you have ever taken, you will need to say what you mean. Precision of thought and language is essential to achieve the mathematical certainty that is necessary for you to have complete confidence in your solutions to mathematical problems.

4.1 Direct Proof and Counterexample I: Introduction

Mathematics, as a science, commenced when first someone, probably a Greek, proved propositions about "any" things or about "some" things without specification of definite particular things. —Alfred North Whitehead, 1861–1947

Both discovery and proof are integral parts of problem solving. When you think you have discovered that a certain statement is true, try to figure out why it is true. If you succeed, you will know that your discovery is genuine. Even if you fail, the process of trying will give you insight into the nature of the problem and may lead you to discover that the statement is false. For complex problems, the interplay between discovery and proof is not reserved to the end of the problem-solving process but, rather, is an important part of each step.

Assumptions

- In this text we assume a familiarity with the laws of basic algebra, which are listed in Appendix A.
- We also use the three properties of equality: For all objects A , B , and C , (1) $A = A$, (2) if $A = B$, then $B = A$, and (3) if $A = B$ and $B = C$, then $A = C$.
- And we use the principle of substitution: For all objects A and B , if $A = B$, then we may substitute B wherever we have A .
- In addition, we assume that there is no integer between 0 and 1 and that the set of all integers is closed under addition, subtraction, and multiplication. This means that sums, differences, and products of integers are integers.

Note Most quotients of integers are not integers. For example, $3 \div 2$, which equals $3/2$, is not an integer, and $3 \div 0$ is not even a number.

The mathematical content of this section primarily concerns even and odd integers and prime and composite numbers.

Even, Odd, Prime, and Composite Integers

In order to evaluate the truth or falsity of a statement, you must understand what the statement is about. In other words, you must know the meanings of all terms that occur in

the statement. Mathematicians define terms very carefully and precisely and consider it important to learn definitions virtually word for word.

Definitions

An integer n is **even** if, and only if, n equals twice some integer. An integer n is **odd** if, and only if, n equals twice some integer plus 1.

Symbolically, for any integer, n

$$n \text{ is even} \Leftrightarrow n = 2k \text{ for some integer } k$$

$$n \text{ is odd} \Leftrightarrow n = 2k + 1 \text{ for some integer } k$$

It follows from the definition that if you are doing a problem in which you know that a particular integer is even, you can deduce that it has the form $2 \cdot (\text{some integer})$. Conversely, if you know that an integer equals $2 \cdot (\text{some integer})$, then you can deduce that the integer is even.

Know a particular integer n is even. $\xrightarrow{\text{deduce}}$ n has the form $2 \cdot (\text{some integer})$.

Know n has the form $2 \cdot (\text{some integer})$. $\xrightarrow{\text{deduce}}$ n is even.

This illustrates why both the *if* and the *only-if* parts of definitions are important in mathematical reasoning. In stating definitions, however, mathematics books often replace the words *if-and-only-if* by the single word *if*, perhaps to seem less formal. For instance, the definition of even might be given as “An integer is even if it equals twice some integer.” But when the definition is actually used in a proof, both the *if* and the *only-if* parts are usually needed. So, even when the *only-if* part of a definition is not stated explicitly, you are supposed to understand intuitively that it should be included.

Also observe that the definitions of even and odd integers are quantified statements. In Section 3.1 we pointed out that variables used in quantified statements are local, which means that they are bound by the quantifier to which they are attached and that their scopes extend only to the end of the quantified statements that contain them. As a result, the particular names used for the variables have no meaning themselves and are freely replaceable by other names. For example, you can substitute any symbols you like in place of n and k in the definitions of even and odd without changing the meaning of the definitions.

For every integer n , n is *even* if, and only if, $n = 2r$ for some integer r .

For every integer m , m is *even* if, and only if, $m = 2a$ for some integer a .

For every integer a , a is *odd* if, and only if, $a = 2s + 1$ for some integer s .

For every integer k , k is *odd* if, and only if, $k = 2n + 1$ for some integer n .

Example 4.1.1 Even and Odd Integers

Use the definitions of *even* and *odd* to justify your answers to the following questions.

- Is 0 even?
- Is -301 odd?
- If a and b are integers, is $6a^2b$ even?

- d. If a and b are integers, is $10a + 8b + 1$ odd?
- e. Is every integer either even or odd?

Solution

- a. Yes, 0 is even because $0 = 2 \cdot 0$.
- b. Yes, -301 is odd because $-301 = 2(-151) + 1$ and -151 is an integer.
- c. Yes, $6a^2b$ is even because $6a^2b = 2(3a^2b)$ and $3a^2b$ is an integer since it is a product of integers.
- d. Yes, $10a + 8b + 1$ is odd because $10a + 8b + 1 = 2(5a + 4b) + 1$ and $5a + 4b$ is an integer since it is a sum of products of integers.
- e. Yes, every integer is either even or odd. However, the reason for this fact is not immediately apparent. It can be deduced using the method of proof by contradiction, which is introduced in Section 4.7. It is also a consequence of the quotient-remainder theorem, which is stated in Section 4.5. ■

The integer 6, which equals $2 \cdot 3$, is a product of two smaller positive integers. On the other hand, 7 cannot be written as a product of two smaller positive integers; its only positive factors are 1 and 7. A positive integer, such as 7, that cannot be written as a product of two smaller positive integers is called *prime*.

Definition

An integer n is **prime** if, and only if, $n > 1$ and for all positive integers r and s , if $n = rs$, then either r or s equals n . An integer n is **composite** if, and only if, $n > 1$ and $n = rs$ for some integers r and s with $1 < r < n$ and $1 < s < n$.

In symbols: For each integer n with $n > 1$,

$$\begin{aligned} n \text{ is prime} &\Leftrightarrow \forall \text{ positive integers } r \text{ and } s, \text{ if } n = rs \\ &\quad \text{then either } r = 1 \text{ and } s = n \text{ or } r = n \text{ and } s = 1. \\ n \text{ is composite} &\Leftrightarrow \exists \text{ positive integers } r \text{ and } s \text{ such that } n = rs \\ &\quad \text{and } 1 < r < n \text{ and } 1 < s < n. \end{aligned}$$

Example 4.1.2**Prime and Composite Numbers**

- a. Is 1 prime?
- b. Is every integer greater than 1 either prime or composite?
- c. Write the first six prime numbers.
- d. Write the first six composite numbers.

Solution

- a. No. A prime number is required to be greater than 1.
- b. Yes. Let n be any integer that is greater than 1. Consider all pairs of positive integers r and s such that $n = rs$. There exist at least two such pairs, namely, $r = n$ and $s = 1$ and $r = 1$ and $s = n$. Moreover, since $n = rs$, all such pairs satisfy the inequalities $1 \leq r \leq n$ and $1 \leq s \leq n$. If n is prime, then these two pairs are the only ways to write n as rs . Otherwise, there exists a pair of positive integers r and s such that $n = rs$ and neither r nor s equals either 1 or n . Therefore, in this case $1 < r < n$ and $1 < s < n$, and hence n is composite.

Note The reason for not allowing 1 to be prime is discussed in Section 4.4.

- c. 2, 3, 5, 7, 11, 13
- d. 4, 6, 8, 9, 10, 12



Proving Existential Statements

According to the definition given in Section 3.1, a statement in the form

$$\exists x \in D \text{ such that } Q(x)$$

is true if, and only if,

$Q(x)$ is true for at least one x in D .

One way to prove this is to find an x in D that makes $Q(x)$ true. Another way is to give a set of directions for finding such an x . Both of these methods are called **constructive proofs of existence**. The logical principle underlying such a proof is called **existential generalization**. It says that if you know a certain property is true for a particular object, then you may conclude that “there exists an object for which the property is true.”

Example 4.1.3 Constructive Proofs of Existence

- a. Prove: $\exists$ an even integer n that can be written in two ways as a sum of two prime numbers.
- b. Suppose that r and s are integers. Prove: $\exists$ an integer k such that $22r + 18s = 2k$.

Solution

- a. Let $n = 10$. Then $10 = 5 + 5 = 3 + 7$ and 3, 5, and 7 are all prime numbers. Thus $\exists$ an even integer—namely, 10—that can be written in two ways as a sum of two prime numbers.
- b. Let $k = 11r + 9s$. Then k is an integer because it is a sum of products of integers, and by substitution, and the distributive law of algebra,

$$2k = 2(11r + 9s) = 22r + 18s.$$

Thus $\exists$ an integer, namely k , such that $22r + 18s = 2k$.



A **nonconstructive proof of existence** involves showing either (a) that the existence of a value of x that makes $Q(x)$ true is guaranteed by an axiom or a previously proved theorem or (b) that the assumption that there is no such x leads to a contradiction. The disadvantage of a nonconstructive proof is that it may give virtually no clue about where or how x may be found. The widespread use of digital computers in recent years has led to some dissatisfaction with this aspect of nonconstructive proofs and to increased efforts to produce constructive proofs containing directions for computer calculation of the quantity in question.

Disproving Universal Statements by Counterexample

To disprove a statement means to show that it is false. Consider the question of disproving a statement of the form

$$\forall x \text{ in } D, \text{ if } P(x) \text{ then } Q(x).$$

Showing that this statement is false is equivalent to showing that its negation is true. The negation of the statement is existential:

$$\exists x \text{ in } D \text{ such that } P(x) \text{ and not } Q(x).$$

But to show that an existential statement is true, we generally give an example, and because the example is used to show that the original statement is false, we call it a *counterexample*. Thus the method of disproof by *counterexample* can be written as follows:

Disproof by Counterexample

To disprove a statement of the form “ $\forall x \in D$, if $P(x)$ then $Q(x)$,” find a value of x in D for which the hypothesis $P(x)$ is true and the conclusion $Q(x)$ is false. Such an x is called a **counterexample**.

Example 4.1.4 Disproof by Counterexample

Disprove the following statement by finding a counterexample:

$$\forall \text{ real numbers } a \text{ and } b, \text{ if } a^2 = b^2 \text{ then } a = b.$$

Solution To disprove this statement, you need to find real numbers a and b such that the hypothesis $a^2 = b^2$ is true and the conclusion $a = b$ is false. The fact that both positive and negative integers have positive squares helps in the search. If you flip through some possibilities in your mind, you will quickly see that 1 and -1 will work (or 2 and -2 , or 0.5 and -0.5 , and so forth). You only need one such pair to give a counterexample.

Note Example 4.1.4 shows that it is not always true that if the squares of two numbers are equal, then the numbers are equal. However, it is true that if the squares of two *positive* numbers are equal, then the numbers are equal.

Statement: $\forall$ real numbers a and b , if $a^2 = b^2$, then $a = b$.

Counterexample: Let $a = 1$ and $b = -1$. Then $a^2 = 1^2 = 1$ and $b^2 = (-1)^2 = 1$, and so $a^2 = b^2$. But $a \neq b$ since $1 \neq -1$.

After observing that a property holds in a large number of cases, you may guess that it holds in all cases. You may, however, run into difficulty when you try to prove your guess. Perhaps you just have not figured out the key to the proof, or perhaps your guess is false. Consequently, when you are having serious difficulty proving a general statement, you should interrupt your efforts to look for a counterexample. Analyzing the kinds of problems you are encountering in your proof efforts may help in the search. It may even happen that if you find a counterexample and therefore prove the statement false, your understanding may be sufficiently clarified so that you can formulate a more limited but true version of the statement by changing the hypothesis.

Proving Universal Statements

The vast majority of mathematical statements to be proved are universal. In discussing how to prove such statements, it is helpful to imagine them in a standard form:

$$\forall x \in D, \text{ if } P(x) \text{ then } Q(x).$$

Sections 1.1 and 3.1 give examples showing how to write any universal statement in this form. When D is finite or when only a finite number of elements satisfy $P(x)$, such a statement can be proved by the method of exhaustion.

Example 4.1.5 The Method of Exhaustion

Use the method of exhaustion to prove the following statement:

$\forall n \in \mathbf{Z}$, if n is even and $4 \leq n \leq 26$ then n can be written as a sum of two prime numbers.

Solution

$4 = 2 + 2$	$6 = 3 + 3$	$8 = 3 + 5$	$10 = 5 + 5$
$12 = 5 + 7$	$14 = 11 + 3$	$16 = 5 + 11$	$18 = 7 + 11$
$20 = 7 + 13$	$22 = 5 + 17$	$24 = 5 + 19$	$26 = 7 + 19$

In most mathematical situations, however, the method of exhaustion cannot be used. For instance, to prove by exhaustion that *every* even integer greater than 2 can be written as a sum of two prime numbers you would have to check every even integer. But this is impossible because there are infinitely many such numbers.

Even when the domain is finite, it may be infeasible to use the method of exhaustion. Imagine, for example, trying to check by exhaustion that the multiplication circuitry of a particular computer gives the correct result for every pair of numbers in the computer's range. Since a typical computer would require thousands of years just to compute all possible products of all numbers in its range (not to mention the time it would take to check the accuracy of the answers), checking correctness by the method of exhaustion is obviously impractical.

The most powerful technique for proving a universal statement is one that works regardless of the size of the domain over which the statement is quantified. It is based on a logical principle sometimes called *universal generalization*. A more descriptive name is *generalizing from the generic particular*.

Generalizing from the Generic Particular

To show that *every* element of a set satisfies a certain property, suppose x is a *particular* but *arbitrarily chosen* element of the set, and show that x satisfies the property.

The principle of generalizing from the generic particular is not a typical part of everyday reasoning. Its main use is to determine that a general mathematical statement is correct. The example below introduces the idea.

Example 4.1.6 Generalizing from the Generic Particular

At some time you may have been shown a “mathematical trick” like the following. You ask a person to pick any number, add 5, multiply by 4, subtract 6, divide by 2, and subtract twice the original number. Then you astound the person by announcing that their final result was 7. How does this “trick” work? Imagine that an empty box $\square$ contains whatever number the person picked. The table shows that by the end of the calculations, whatever was in the empty box was subtracted out of the answer.

Step	Visual Result	Algebraic Result
Pick a number.	□	x
Add 5.	□	$x + 5$
Multiply by 4.	□ □ □ □	$(x + 5) \cdot 4 = 4x + 20$
Subtract 6.	□ □ □ □	$(4x + 20) - 6 = 4x + 14$
Divide by 2.	□ □	$\frac{4x + 14}{2} = 2x + 7$
Subtract twice the original number.	 	$(2x + 7) - 2x = 7$

The x in the table above is another way of holding a place for the number the person picked. It is *particular* (because it is a single object), but it is also *arbitrarily chosen* or *generic* (because any number whatsoever can be put in its place). So you can generalize from the generic particular to conclude that if you follow the steps you will always get 7, regardless of the initial value you put in place of x or inside the empty box. ■

The point of having x be arbitrarily chosen (or generic) is to make a proof that can be generalized to all elements of the domain. By choosing x arbitrarily, you are making no special assumptions about x that are not also true of all other elements of the domain. The word *generic* means “sharing all the common characteristics of a group or class.” Thus everything you deduce about a generic element x of the domain is equally true of any other element of the domain.

When the method of generalizing from the generic particular is applied to a property of the form “If $P(x)$ then $Q(x)$,” the result is the method of *direct proof*. Recall that the only way an if-then statement can be false is for the hypothesis to be true and the conclusion to be false. Thus, given the statement “If $P(x)$ then $Q(x)$,” if you can show that the truth of $P(x)$ compels the truth of $Q(x)$, then you will have proved the statement. It follows by the method of generalizing from the generic particular that to show that “ $\forall x$, if $P(x)$ then $Q(x)$,” is true for *every* element x in a set D , you suppose x is a particular but arbitrarily chosen element of D that makes $P(x)$ true, and then you show that x makes $Q(x)$ true.

Method of Direct Proof

- Express the statement to be proved in the form “For every $x \in D$, if $P(x)$ then $Q(x)$.” (This step is often done mentally.)
- Start the proof by supposing x is a particular but arbitrarily chosen element of D for which the hypothesis $P(x)$ is true. (This step is often abbreviated “Suppose $x \in D$ and $P(x)$.”)
- Show that the conclusion $Q(x)$ is true by using definitions, previously established results, and the rules for logical inference.

Example 4.1.7**A Direct Proof of a Theorem**

Caution! The word *two* in this statement does not necessarily refer to two distinct integers. If a choice of integers is made arbitrarily, the integers are very likely to be distinct, but they might be the same.

Prove that the sum of any two even integers is even.

Solution Whenever you are presented with a statement to be proved, it is a good idea to ask yourself whether you believe it to be true. In this case you might imagine some pairs of even integers—say $2 + 4$, $6 + 10$, $12 + 12$, $28 + 54$ —and mentally check that their sums are even. However, since you cannot possibly check all pairs of even numbers, you cannot know for sure that the statement is true in general by checking its truth in these particular instances. Many properties hold for a large number of examples and yet fail to be true in general.

To prove this statement in general, you need to show that no matter what even integers are given, their sum is even. But given any two even integers, it is possible to represent them as $2r$ and $2s$ for some integers r and s . And by the distributive law of algebra, $2r + 2s = 2(r + s)$, which is even because $r + s$ is an integer. Thus the statement is true in general.

Suppose the statement to be proved is much more complicated than this. What method can you use to derive a proof? You can begin by expressing the statement formally.

Formal Restatement: $\forall$ integers m and n , if m and n are even then $m + n$ is even.

This statement is universally quantified over an infinite domain. Thus to prove it in general, you need to show that no matter what two integers you might be given, if both of them are even then their sum will also be even.

Next ask yourself, “How should I start the proof?” or “What am I supposing?” The answer to such a question gives you the starting point, or first sentence, of the proof.

Starting Point: Suppose m and n are any particular but arbitrarily chosen integers that are even.

Or, in abbreviated form:

Suppose m and n are any even integers.

Then ask yourself, “What conclusion do I need to show in order to complete the proof?”

To Show: $m + n$ is even.

At this point you need to ask yourself, “How do I get from the starting point to the conclusion?” Since both involve the term *even integer*, you must use the definition of this term—and thus you must know what it means for an integer to be even. It follows from the definition that since m and n are even, each equals twice some integer. One of the basic laws of logic, called *existential instantiation*, says, in effect, that if you know something exists, you can give it a name. However, you cannot use the same name to refer to two different things, both of which are currently under discussion.

Existential Instantiation

If the existence of a certain kind of object is assumed or has been deduced, then it can be given a name, as long as that name is not currently being used to refer to something else in the same discussion.

Thus since m equals twice some integer, you can give that integer a name, and since n equals twice some integer, you can also give that integer a name:

$$m = 2r, \text{ for some integer } r \quad \text{and} \quad n = 2s, \text{ for some integer } s.$$



Caution! Because m and n are arbitrarily chosen they can be any pair of even integers whatsoever. But if you write $m = 2r$ and $n = 2r$, then m would equal n , which is not usually the case.

Now what you want to show is that $m + n$ is even. In other words, you want to show that $m + n$ equals $2 \cdot (\text{some integer})$. Having just found alternative representations for m (as $2r$) and n (as $2s$), it seems reasonable to substitute these representations in place of m and n :

$$m + n = 2r + 2s.$$

Your goal is to show that $m + n$ is even. By definition of even, this means that $m + n$ can be written in the form

$$2 \cdot (\text{some integer}).$$

This analysis narrows the gap between the starting point and what is to be shown to showing that

$$2r + 2s = 2 \cdot (\text{some integer}).$$

Why is this true? First, because of the distributive law from algebra, which says that

$$2r + 2s = 2(r + s),$$

and, second, because the sum of any two integers is an integer, which implies that $r + s$ is an integer.

This discussion is summarized by rewriting the statement as a theorem and giving a formal proof of it. (In mathematics, the word *theorem* refers to a statement that is known to be true because it has been proved.) The formal proof, as well as many others in this text, includes explanatory notes to make its logical flow apparent. Such comments are purely a convenience for the reader and could be omitted entirely. For this reason they are italicized and enclosed in italic square brackets: [].

Donald Knuth, one of the pioneers of the science of computing, has compared constructing a computer program from a set of specifications to writing a mathematical proof based on a set of axioms.* In keeping with this analogy, the bracketed comments can be thought of as similar to the explanatory documentation provided by a good programmer. Documentation is not necessary for a program to run, but it helps a human reader understand what is going on.

Theorem 4.1.1

The sum of any two even integers is even.

Proof: Suppose m and n are any [particular but arbitrarily chosen] even integers. [We must show that $m + n$ is even.] By definition of even, $m = 2r$ and $n = 2s$ for some integers r and s . Then

$$\begin{aligned} m + n &= 2r + 2s && \text{by substitution} \\ &= 2(r + s) && \text{by factoring out a 2.} \end{aligned}$$

Let $t = r + s$. Note that t is an integer because it is a sum of integers. Hence

$$m + n = 2t \quad \text{where } t \text{ is an integer.}$$

It follows by definition of even that $m + n$ is even. [This is what we needed to show.][†]

Note Introducing t to equal $r + s$ is another use of existential instantiation.

*Donald E. Knuth, *The Art of Computer Programming*, 2nd ed., Vol. I (Reading, MA: Addison-Wesley, 1973), p. ix.

[†]See page 148 for a discussion of the role of universal modus ponens in this proof.

Most theorems, like Theorem 4.1.1, can be analyzed to a point where you realize that as soon as a certain thing is shown, the theorem will be proved. When that thing has been shown, it is natural to end the proof with the words “this is what we needed to show” or “as was to be shown.” The Latin words for this are *quod erat demonstrandum*, or Q.E.D. for short. Proofs in older mathematics books end with these initials.

Note that both the *if* and the *only if* parts of the definition of even were used in the proof of Theorem 4.1.1. Since m and n were known to be even, the *only if* ($\Rightarrow$) part of the definition was used to deduce that m and n had a certain general form. Then, after some algebraic substitution and manipulation, the *if* ($\Leftarrow$) part of the definition was used to deduce that $m + n$ was even.

Getting Proofs Started

Believe it or not, once you understand the idea of generalizing from the generic particular and the method of direct proof, you can write the beginnings of proofs even for theorems you do not understand. The reason is that the starting point and what is to be shown in a proof depend only on the linguistic form of the statement to be proved, not on the content of the statement.

Example 4.1.8

Identifying the “Starting Point” and the “Conclusion to Be Shown”

Write the first sentence of a proof (the “starting point”) and the last sentence of a proof (the “conclusion to be shown”) for the following statement:

Every complete bipartite graph is connected.

Note You are not expected to know anything about complete bipartite graphs.

Solution It is helpful to rewrite the statement formally using a quantifier and a variable:

Formal Restatement: For every $\overbrace{\text{graph } G}^{\text{domain}}$, if $\overbrace{G \text{ is complete bipartite}}^{\text{hypothesis}}$, then $\overbrace{G \text{ is connected}}^{\text{conclusion}}$.

The first sentence, or starting point, of a proof supposes the existence of an object (in this case G) in the domain (in this case the set of all graphs) that satisfies the hypothesis of the if-then part of the statement (in this case that G is complete bipartite). The conclusion to be shown is just the conclusion of the if-then part of the statement (in this case that G is connected).

Starting Point: Suppose G is a [particular but arbitrarily chosen] graph such that G is complete bipartite.

Conclusion to Be Shown: G is connected.

Thus the proof has the following shape:

Proof:

Suppose G is a [particular but arbitrarily chosen] graph such that G is complete bipartite.

$\vdots$

Therefore, G is connected. ■

Example 4.1.9

Fill in the Blanks for a Proof

Fill in the blanks in the proof of the following theorem.

Theorem: For all integers r and s , if r is even and s is odd then $3r + 2s$ is even.

Proof:

Suppose r and s are any [particular but arbitrarily chosen] integers such that r is even and s is odd.

[We must show that $3r + 2s$ is even.]

By (a), $r = 2m$ and $s = 2n + 1$ for some integers m and n .

Then

$$\begin{aligned} 3r + 2s &= 3(2m) + 2(2n + 1) && \text{by (b)} \\ &= 6m + 4n + 2 && \text{by multiplying out} \\ &= 2(3m + 2n + 1) && \text{by factoring out 2} \end{aligned}$$

Let $t = 3m + 2n + 1$.

Then t is an integer because m , n , 3, 2, and 1 are integers and because (c).

Hence $3r + 2s = 2t$, where t is an integer, and so by (d), $3r + 2s$ is even [as was to be shown].

Solution

(a) definition of even and odd, (b) substitution, (c) products and sums of integers are integers, (d) definition of even.

TEST YOURSELF

Answers to Test Yourself questions are located at the end of each section.

- An integer is even if, and only if, _____.
- An integer is odd if, and only if, _____.
- An integer n is prime if, and only if, _____.
- The most common way to disprove a universal statement is to find _____.
- According to the method of generalizing from the generic particular, to show that every element of a set satisfies a certain property, suppose x is a _____, and show that _____.
- To use the method of direct proof to prove a statement of the form, "For every x in a set D , if $P(x)$ then $Q(x)$," one supposes that _____ and one shows that _____.

EXERCISE SET 4.1*

In 1–4 justify your answers by using the definitions of even, odd, prime, and composite numbers.

- Assume that k is a particular integer.
 - Is -17 an odd integer?
 - Is 0 neither even nor odd?
 - Is $2k - 1$ odd?
- Assume that c is a particular integer.
 - Is $-6c$ an even integer?
 - Is $8c + 5$ an odd integer?
 - Is $(c^2 + 1) - (c^2 - 1) - 2$ an even integer?
- Assume that m and n are particular integers.
 - Is $6m + 8n$ even?
 - Is $10mn + 7$ odd?
 - If $m > n > 0$, is $m^2 - n^2$ composite?

- Assume that r and s are particular integers.
 - Is $4rs$ even?
 - Is $6r + 4s^2 + 3$ odd?
 - If r and s are both positive, is $r^2 + 2rs + s^2$ composite?

Prove the statements in 5–11.

- There are integers m and n such that $m > 1$ and $n > 1$ and $\frac{1}{m} + \frac{1}{n}$ is an integer.
- There are distinct integers m and n such that $\frac{1}{m} + \frac{1}{n}$ is an integer.
- There are real numbers a and b such that

$$\sqrt{a+b} = \sqrt{a} + \sqrt{b}.$$

*For exercises with blue numbers, solutions are given in Appendix B. The symbol **H** indicates that only a hint or partial solution is given. The symbol * signals that an exercise is more challenging than usual.

Hence $m + n = 2u$, where u is an integer, and so, by (d), $m + n$ is even [as was to be shown].

- 29. Theorem:** The negative of any even integer is even.

Proof: Suppose n is any [particular but arbitrarily chosen] even integer.

[We must show that $-n$ is even.]

By (a), $n = 2k$ for some integer k .

Then

$$\begin{aligned} -n &= -(2k) && \text{by (b)} \\ &= 2(-k) && \text{by algebra.} \end{aligned}$$

Let $r = -k$. Then r is an integer because (-1) and k are integers and (c).

Hence $-n = 2r$, where r is an integer, and so $-n$ is even by (d) [as was to be shown].

- 30. Theorem 4.1.2:** The sum of any even integer and any odd integer is odd.

Proof: Suppose m is any even integer and n is (a). By definition of even, $m = 2r$ for some (b), and by definition of odd, $n = 2s + 1$ for

some integer s . By substitution and algebra,

$$m + n = \text{(c)} = 2(r + s) + 1.$$

Since r and s are both integers, so is their sum $r + s$. Hence $m + n$ has the form twice some integer plus one, and so (d) by definition of odd.

- 31. Theorem:** Whenever n is an odd integer, $5n^2 + 7$ is even.

Proof: Suppose n is any [particular but arbitrarily chosen] odd integer.

[We must show that $5n^2 + 7$ is even.]

By definition of odd, $n = \text{(a)}$ for some integer k .

Then

$$\begin{aligned} 5n^2 + 7 &= \text{(b)} && \text{by substitution} \\ &= 5(4k^2 + 4k + 1) + 7 \\ &= 20k^2 + 20k + 12 \\ &= 2(10k^2 + 10k + 6) && \text{by algebra.} \end{aligned}$$

Let $t = \text{(c)}$. Then t is an integer because products and sums of integers are integers.

Hence $5n^2 + 7 = 2t$, where t is an integer, and thus (d) by definition of even [as was to be shown].

ANSWERS FOR TEST YOURSELF

1. it equals twice some integer 2. it equals twice some integer plus 1 3. n is greater than 1 and if n equals the product of any two positive integers, then one of the integers equals 1 and the other equals n . 4. a counterexample

5. particular but arbitrarily chosen element of the set; x satisfies the given property 6. x is a particular but arbitrarily chosen element of the set D that makes the hypothesis $P(x)$ true; x makes the conclusion $Q(x)$ true.

4.2 Direct Proof and Counterexample II: Writing Advice

“... it is demanded for proof that every doubt becomes impossible.” —Carl Friedrich Gauss (1777–1855)

Think of a proof as a way to communicate a convincing argument for the truth of a mathematical statement. When you write a proof, try to be clear and complete. Keep in mind that a classmate reading your proof will see only what you actually write down, not any unexpressed thoughts behind it. Ideally, your proof will lead your reader to understand *why* the given statement is true.

Directions for Writing Proofs of Universal Statements

Over the years, the following rules of style have become fairly standard for writing the final versions of proofs:

1. **Copy the statement of the theorem to be proved on your paper.**

This makes the theorem statement available for reference to anyone reading the proof.

2. **Clearly mark the beginning of your proof with the word Proof.**

This word separates general discussion about the theorem from its actual proof.

3. **Make your proof self-contained.**

This means that you should explain the meaning of each variable used in your proof in the body of the proof. Thus you will begin proofs by introducing the initial variables and stating what kind of objects they are. The first sentence of your proof would be something like “Suppose m and n are any even integers” or “Let x be a real number such that x is greater than 2.” This is similar to declaring variables and their data types at the beginning of a computer program.

At a later point in your proof, you may introduce a new variable to represent a quantity that is known at that point to exist. For example, if you have assumed that a particular integer n is even, then you know that n equals 2 times some integer, and you can give this integer a name so that you can work with it concretely later in the proof. Thus if you decide to call the integer, say, s , you would write, “Since n is even, $n = 2s$ for some integer s ,” or “since n is even, there exists an integer, say s , such that $n = 2s$.”

4. **Write your proof in complete, grammatically correct sentences.**

This does not mean that you should avoid using symbols and shorthand abbreviations, just that you should incorporate them into sentences. For example, the proof of Theorem 4.1.1 contains the sentence

$$\begin{aligned} \text{Then } m + n &= 2r + 2s && \text{by substitution} \\ &= 2(r + s) && \text{by factoring out 2.} \end{aligned}$$

To read such text as a sentence, read the first equals sign as “equals” and each subsequent equals sign as “which equals.”

5. **Keep your reader informed about the status of each statement in your proof.**

Your reader should never be in doubt about whether something in your proof has been assumed or established or is still to be deduced. If something is assumed, preface it with a word like *Suppose* or *Assume*. If it is still to be shown, preface it with words like, *We must show that* or *In other words, we must show that*. This is especially important if you introduce a variable in rephrasing what you need to show. (See Common Mistakes on the next page.)

6. **Give a reason for each assertion in your proof.**

Each assertion in a proof should come directly from the hypothesis of the theorem, or follow from the definition of one of the terms in the theorem, or be a result obtained earlier in the proof, or be a mathematical result that has previously been established or is agreed to be assumed. Indicate the reason for each step of your proof using phrases such as *by hypothesis*, *by definition of* . . . *by theorem* . . . and so forth.

It is best to refer to definitions and theorems by name or number. If you need to state one in the body of your proof, avoid using a variable when you write it because otherwise your proof could end up with a variable that has two conflicting meanings.*

Proofs in more advanced mathematical contexts often omit reasons for some steps because it is assumed that students either understand them or can easily figure them out for themselves. However, in a course that introduces mathematical proof, you should make sure to provide the details of your arguments because you cannot guarantee that your readers have the necessary mathematical knowledge and sophistication to supply them on their own.

*When a variable is used to state a definition, the scope of the variable extends only to the end of the definition. After that, the symbol for the variable no longer has the same meaning. Confusion can result from thinking that the meaning of the symbol continues into other parts of the proof.

7. Include the “little words and phrases” that make the logic of your arguments clear.

When writing a mathematical argument, especially a proof, indicate how each sentence is related to the previous one. Does it follow from the previous sentence or from a combination of the previous sentence and earlier ones? If so, start the sentence with the word *Because* or *Since* and state the reason why it follows, or write *Then*, or *Thus*, or *So*, or *Hence*, or *Therefore*, or *Consequently*, or *It follows that*, and include the reason at the end of the sentence. For instance, in the proof of Theorem 4.1.1, once you know that m is even, you can write: “By definition of even, $m = 2r$ for some integer r ,” or you can write, “Then $m = 2r$ for some integer r by definition of even.” And when you write “Then $m + n = 2r + 2s$,” add the words *by substitution* to explain why you are allowed to write $2r$ in place of m and $2s$ in place of n .

If a sentence expresses a new thought or fact that does not follow as an immediate consequence of the preceding statement but is needed for a later part of a proof, introduce it by writing *Observe that*, or *Note that*, or *Recall that*, or *But*, or *Now*.

Sometimes in a proof it is desirable to define a new variable in terms of previous variables. In such a case, introduce the new variable with the word *Let*. For instance, in the proof of Theorem 4.1.1, once it is known that $m + n = 2(r + s)$, where r and s are integers, a new variable t is introduced to represent $r + s$. The convention in mathematics and computer science is to put a new variable to the left of the equal sign and the expression that defines it to the right of the sign. Thus the proof goes on to say, “Let $t = r + s$. Then t is an integer because it is a sum of two integers.”

8. Display equations and inequalities.

The convention is to display equations and inequalities on separate lines to increase readability, both for other people and for ourselves so that we can more easily check our work for accuracy. We follow the convention in the text of this book, but in order to save space, we violate it in a few of the exercises and in many of the solutions contained in Appendix B. So you may need to copy out some parts of solutions on scratch paper to understand them fully. Please follow the convention in your own work. Leave plenty of empty space, and don’t be stingy with paper!

Variations among Proofs

It is rare that two proofs of a given statement, written by two different people, are identical. Even when the basic mathematical steps are the same, the two people may use different notation or may give differing amounts of explanation for their steps, or may choose different words to link the steps together into paragraph form. An important question is how detailed to make the explanations for the steps of a proof. This must ultimately be worked out between the writer of a proof and the intended reader, whether they be student and teacher, teacher and student, student and fellow student, or mathematician and colleague. Your teacher may provide explicit guidelines for you to use in your course. Or you may follow the example of the proofs in this book (which are generally explained rather fully in order to be understood by students at various stages of mathematical development). Remember that the phrases written inside brackets [] are intended to elucidate the logical flow or underlying assumptions of the proof and need not be written down at all. It is your decision whether to include such phrases in your own proofs.

Common Mistakes

The following are some of the most common mistakes people make when writing mathematical proofs.

1. Arguing from examples.

Looking at examples is one of the most helpful practices a problem solver can engage in and is encouraged by all good mathematics teachers. However, it is a mistake to think that a general statement can be proved by showing it to be true for some individual cases. A property referred to in a universal statement may be true in many instances without being true in general.

Consider the following “proof” that the sum of any two even integers is even (Theorem 4.1.1).

This is true because if $m = 14$ and $n = 6$, which are both even,
then $m + n = 20$, which is also even.

Some people find this kind of argument convincing because it does, after all, consist of evidence in support of a true conclusion. But remember that when we discussed valid arguments, we pointed out that an argument may be invalid and yet have a true conclusion. In the same way, an argument from examples may be mistakenly used to “prove” a true statement. In the previous example, it is not sufficient to show that the conclusion “ $m + n$ is even” is true for $m = 14$ and $n = 6$. You must give an argument to show that the conclusion is true for any arbitrarily chosen even integers m and n .

2. Using the same letter to mean two different things.

Some beginning theorem provers give a new variable quantity the same letter name as a previously introduced variable. Consider the following “proof” fragment:

Suppose m and n are any odd integers. Then by definition of odd,
 $m = 2k + 1$ and $n = 2k + 1$ where k is an integer.

You might think of a variable in a mathematical proof as similar to a global variable in a computer program: once introduced, it has the same meaning throughout the program. In other words, its scope extends to the end of the program. In this example, using the symbol k in the expressions for both m and n makes k a global variable. As a result, both m and n equal $2k + 1$, and thus are equal to each other. The proof then only shows that a sum of two identical odd integers is even, not that the sum of two arbitrarily chosen odd integers is even.

3. Jumping to a conclusion.

To jump to a conclusion means to allege the truth of something without giving an adequate reason. Consider the following “proof” that the sum of any two even integers is even.

Suppose m and n are any even integers. By definition of even,
 $m = 2r$ and $n = 2s$ for some integers r and s . Then $m + n = 2r + 2s$.
So $m + n$ is even.

The problem with this “proof” is that to show an integer is even one needs to show that it equals twice some integer. This proof jumps to the conclusion that $m + n$ is even without having found an integer that, when doubled, equals $m + n$.

4. Assuming what is to be proved.

To assume what is to be proved is a variation of jumping to a conclusion. As an example, consider the following “proof” of the fact that the product of any two odd integers is odd:

Suppose m and n are any odd integers. When any odd integers are
multiplied, their product is odd. Hence mn is odd.

5. Confusion between what is known and what is still to be shown.

A more subtle way to jump to a conclusion occurs when the conclusion is restated using a variable. Here is an example in a “proof” that the product of any two odd integers is odd:

Suppose m and n are any odd integers. We must show that mn is odd. This means that there exists an integer s such that

$$mn = 2s + 1.$$

Also by definition of odd, there exist integers a and b such that

$$m = 2a + 1 \text{ and } n = 2b + 1.$$

Then

$$mn = (2a + 1)(2b + 1) = 2s + 1.$$

Note At this point in the proof, we do not know whether there is an integer s making this equality true.

So, since s is an integer, mn is odd by definition of odd.

In this example, when the author restated the conclusion to be shown (that mn is odd), the author wrote “there exists an integer s such that $mn = 2s + 1$.” But we only know that the integer s exists if we know that mn is odd, which is what the author is trying to show. Thus, in the sentence starting with the word “Then,” the author jumped to an unjustified conclusion. This mistake might have been avoided if the author had written

“This means we must show that there exists an integer s such that $mn = 2s + 1$.”

An even better way to avoid this kind of error is not to introduce a variable into a proof unless it is either part of the hypothesis or deducible from it.

6. Use of any when the correct word is some.

There are a few situations in which the words *any* and *some* can be used interchangeably. For instance, in starting a proof that the square of any odd integer is odd, one could correctly write, “Suppose m is any odd integer” or “Suppose m is some odd integer.” In most situations, however, the words *any* and *some* are not interchangeable. Here is the start of a “proof” that the square of any odd integer is odd, which uses *any* when the correct word is *some*:

Suppose m is a particular but arbitrarily chosen odd integer.

By definition of odd, $m = 2a + 1$ for any integer a .

In the second sentence it is incorrect to say that “ $m = 2a + 1$ for any integer a ” because a cannot be just “any” integer; in fact, solving $m = 2a + 1$ for a shows that the only possible value for a is $(m - 1)/2$. The correct way to finish the second sentence is, “ $m = 2a + 1$ for some integer a ” or “there exists an integer a such that $m = 2a + 1$.”

7. Misuse of the word if.

Another common error is not serious in itself, but it reflects imprecise thinking that sometimes leads to problems later in a proof. This error involves using the word *if* when the word *because* is really meant. Consider the following proof fragment:

Suppose p is a prime number. If p is prime, then p cannot be written as a product of two smaller positive integers.

The use of the word *if* in the second sentence is inappropriate. It suggests that the primeness of p is in doubt. But p is known to be prime by the first sentence. It cannot



Caution! A reader of your proof should never be in doubt about what you have already shown and what still needs to be shown.

be written as a product of two smaller positive integers *because* it is prime. Here is a correct version of the fragment:

Suppose p is a prime number. Because p is prime, p cannot be written as a product of two smaller positive integers.

Example 4.2.1 An Odd Integer Minus an Even Integer

Prove that the difference of any odd integer and any even integer is odd. Use only the definitions of odd and even and the Assumptions listed on page 161, not any other properties of odd and even integers. Follow the directions given in this section for writing proofs of universal statements.

Solution

You may already have a sense that the statement to be proved is true, but to make sure your intuition is correct and to develop a careful proof, rewrite the statement using names such as a and b for the odd and even integers so that you will have a convenient way to refer to them:

For all integers a and b , if a is odd and b is even, then $a - b$ is odd.

Or: For every odd integer a and every even integer b , the difference $a - b$ is odd.

Or: If a is any odd integer and b is any even integer, then $a - b$ is odd.

Thus the starting point for your proof would be something like, “Suppose a is any odd integer and b is any even integer,” and the conclusion to be shown would be “We must show that $a - b$ is odd.” If, in addition, you know how to use the definitions of odd and even, you will have reduced the creative part of developing the proof to a small, but crucial, section in the middle.

Theorem 4.2.1

The difference of any odd integer and any even integer is odd.

Proof:

1. Suppose a is any odd integer and b is any even integer. [*We must show that $a - b$ is odd.*]
2. By definition of odd, $a = 2r + 1$ for some integer r , and $b = 2s$ for some integer s .
3. Then $a - b = (2r + 1) - 2s$ by substitution
4. $\quad \quad \quad = 2r - 2s + 1$ by combining like terms
5. $\quad \quad \quad = 2(r - s) + 1$ by factoring out 2.
6. Let $t = r - s$.
7. Then t is an integer because it is a difference of integers.
8. So, by substitution, $a - b = 2t + 1$, where t is an integer.
9. Hence $a - b$ is odd [*as was to be shown*].

Note that lines 1–3 follow immediately from the general structure of the proof, the definitions of odd and even, and substitution. In order to figure out your next steps, it can be helpful to refer to what must be shown—namely, that $a - b$ is odd. According to the definition of odd, you can conclude that $a - b$ is odd if you can show that it equals $2 \cdot (\text{some integer}) + 1$. So showing that $a - b$ is odd involves transforming $(2r + 1) - 2s$ into $2 \cdot (\text{some integer}) + 1$. Lines 4–8 show the steps for doing this, and line 9 concludes that what was to be shown has been achieved. ■

Some of the exercises at the end of the section are based on actual student work and ask you to identify mistakes in “proofs” that have been proposed. Example 4.2.2 illustrates the kind of care that must be taken in evaluating a proof.

Example 4.2.2 Identifying a Mistake in a Proposed Proof

Find the mistake in the following “proof.”

Theorem: If n is any even integer, then $(-1)^n = 1$.

Proof:

1. Suppose n is any even integer. [*We must show that $(-1)^n$ is even.*]
2. By definition of even, $n = 2a$ for some integer a .
3. Then $(-1)^n = (-1)^{2a}$ by substitution
4. $= ((-1)^a)^2$ by a law of exponents
5. $= 1$ because any nonzero real number squared is positive.

Solution

This “proof” incorrectly jumps to a conclusion in line 5. Although it is true that the square of any nonzero real number is positive, it does not follow that the square of $(-1)^a$ is 1. Exercise 10 at the end of this section asks you to give a correct proof of this theorem. ■

Showing That an Existential Statement Is False

Recall that the negation of an existential statement is universal. It follows that to prove an existential statement is false, you must prove a universal statement (its negation) is true.

Example 4.2.3 Disproving an Existential Statement

Show that the following statement is false:

There is a positive integer n such that $n^2 + 3n + 2$ is prime.

Solution Proving that the given statement is false is equivalent to proving its negation is true. The negation is

For all positive integers n , $n^2 + 3n + 2$ is not prime.

Because the negation is universal, it is proved by generalizing from the generic particular.

Claim: The statement “There is a positive integer n such that $n^2 + 3n + 2$ is prime” is false.

Proof:

Suppose n is any [particular but arbitrarily chosen] positive integer. [*We will show that $n^2 + 3n + 2$ is not prime.*] Factoring shows that

$$n^2 + 3n + 2 = (n + 1)(n + 2).$$

In addition, $n + 1$ and $n + 2$ are integers (because they are sums of integers), and both $n + 1 > 1$ and $n + 2 > 1$ (because $n \geq 1$). Thus $n^2 + 3n + 2$ is a product of two integers each greater than 1, and so $n^2 + 3n + 2$ is not prime. ■

Conjecture, Proof, and Disproof

More than 350 years ago, the French mathematician Pierre de Fermat claimed that it is impossible to find positive integers x , y , and z with $x^n + y^n = z^n$ if n is an integer that is at least 3. (For $n = 2$, the equation has many integer solutions, such as $3^2 + 4^2 = 5^2$ and



Bettmann/Getty Images

Pierre de Fermat
(1601–1665)

$5^2 + 12^2 = 13^2$.) Fermat wrote his claim in the margin of a book, along with the comment “I have discovered a truly remarkable PROOF of this theorem which this margin is too small to contain.” No proof, however, was found among his papers, and over the years some of the greatest mathematical minds tried and failed to discover a proof or a counterexample for what came to be known as Fermat’s last theorem.

In 1986 Kenneth Ribet of the University of California at Berkeley showed that if a certain other statement, the Taniyama–Shimura conjecture, could be proved, then Fermat’s theorem would follow. Andrew Wiles, an English mathematician and faculty member at Princeton University, had become intrigued by Fermat’s claim while still a child and, as an adult, had come to work in the branch of mathematics to which the Taniyama–Shimura conjecture belonged. As soon as he heard of Ribet’s result, Wiles immediately set to work to prove the conjecture. In June of 1993, after 7 years of concentrated effort, he presented a proof to worldwide acclaim.

During the summer of 1993, however, while the proof was being carefully checked to prepare for publication, Wiles found a step he had difficulty justifying and which he ultimately realized was an error. Having worked alone for so long, he decided to call on a former student, Richard Taylor, then at Cambridge University in England, who agreed to join him in Princeton, and, together, they worked ceaselessly for months to resolve the problem. After almost a year without a breakthrough, Taylor encouraged Wiles to revisit an approach that had been abandoned years earlier, and, as Wiles examined the details, he suddenly saw that the reason it had failed was the exact reason another approach he had previously abandoned would succeed. By the end of 1994, the revised proof had been thoroughly checked and pronounced correct by experts in the field. It was published in the *Annals of Mathematics* in 1995. Several books and an excellent documentary have been produced that convey the drama and excitement of the discovery.*



AP Images/Charles Rex Arbogast

Andrew Wiles
(born 1953)

One of the oldest problems in mathematics that remains unsolved is the Goldbach conjecture. In Example 4.1.5 it was shown that every even integer from 4 to 26 can be represented as a sum of two prime numbers. More than 250 years ago, Christian Goldbach (1690–1764) conjectured that every even integer greater than 2 can be so represented. Explicit computer-aided calculations have shown the conjecture to be true up to at least 10^{18} . But there is a huge chasm between 10^{18} and infinity. As pointed out by James Gleick of the *New York Times*, many other plausible conjectures in number theory have proved false. Leonhard Euler (1707–1783), for example, proposed in the eighteenth century that $a^4 + b^4 + c^4 = d^4$ had no nontrivial whole number solutions. In other words, no three perfect fourth powers add up to another perfect fourth power. For many numbers, Euler’s conjecture looked good. But in 1987 a Harvard mathematician, Noam Elkies, proved it wrong. One counterexample, found by Roger Frye of Thinking Machines Corporation in a long computer search, is $95,800^4 + 217,519^4 + 414,560^4 = 422,481^4$.†

In May 2000, “to celebrate mathematics in the new millennium,” the Clay Mathematics Institute of Cambridge, Massachusetts, announced that it would award prizes of \$1 million each for the solutions to seven longstanding, classical mathematical questions. One of them, “P vs. NP,” asks whether problems belonging to a certain class can be solved on a computer using more efficient methods than the very inefficient methods that are presently known to work for them. This question is discussed briefly at the end of Chapter 11.

*“The Proof,” produced in 1997, for the series *Nova* on the Public Broadcasting System; *Fermat’s Enigma: The Epic Quest to Solve the World’s Greatest Mathematical Problem*, by Simon Singh and John Lynch (New York: Bantam Books, 1998); *Fermat’s Last Theorem: Unlocking the Secret of an Ancient Mathematical Problem* by Amir D. Aczel (New York: Delacorte Press, 1997).

†James Gleick, “Fermat’s Last Theorem Still Has Zero Solutions,” *New York Times*, April 17, 1988.

TEST YOURSELF

1. The meaning of every variable used in a proof should be explained within _____.
2. Proofs should be written in sentences that are _____ and _____.
3. Every assertion in a proof should be supported by a _____.
4. The following are some useful “little words and phrases” that clarify the reasoning in a proof: _____, _____, _____, _____, and _____.
5. A new thought or fact that does not follow as an immediate consequence of the preceding statement can be introduced by writing _____, _____, _____, or _____.
6. To introduce a new variable that is defined in terms of previous variables, use the word _____.
7. Displaying equations and inequalities increases the _____ of a proof.
8. Some proof-writing mistakes are _____, _____, _____, _____, and _____.

EXERCISE SET 4.2

Prove the statements in 1–11. In each case use only the definitions of the terms and the Assumptions listed on page 161, not any previously established properties of odd and even integers. Follow the directions given in this section for writing proofs of universal statements.

1. For every integer n , if n is odd then $3n + 5$ is even.
2. For every integer m , if m is even then $3m + 5$ is odd.
- H** 3. For every integer n , $2n - 1$ is odd.
4. **Theorem 4.2.2:** The difference of any even integer minus any odd integer is odd.
5. If a and b are any odd integers, then $a^2 + b^2$ is even.
6. If k is any odd integer and m is any even integer, then $k^2 + m^2$ is odd.
- H** 7. The difference between the squares of any two consecutive integers is odd.
8. For any integers m and n , if m is even and n is odd then $5m + 3n$ is odd.
9. If an integer greater than 4 is a perfect square, then the immediately preceding integer is not prime.
10. If n is any even integer, then $(-1)^n = 1$.
11. If n is any odd integer, then $(-1)^n = -1$.

Prove that the statements in 12–14 are false.

12. There exists an integer $m \geq 3$ such that $m^2 - 1$ is prime.
13. There exists an integer n such that $6n^2 + 27$ is prime.

14. There exists an integer $k \geq 4$ such that $2k^2 - 5k + 2$ is prime.

Find the mistakes in the “proofs” shown in 15–19.

15. **Theorem:** For every integer k , if $k > 0$ then $k^2 + 2k + 1$ is composite.
“Proof: For $k = 2$, $k > 0$ and $k^2 + 2k + 1 = 2^2 + 2 \cdot 2 + 1 = 9$. And since $9 = 3 \cdot 3$, then 9 is composite. Hence the theorem is true.”
16. **Theorem:** The difference between any odd integer and any even integer is odd.
“Proof: Suppose n is any odd integer, and m is any even integer. By definition of odd, $n = 2k + 1$ where k is an integer, and by definition of even, $m = 2k$ where k is an integer. Then

$$n - m = (2k + 1) - 2k = 1,$$
and 1 is odd. Therefore, the difference between any odd integer and any even integer is odd.”
17. **Theorem:** For every integer k , if $k > 0$ then $k^2 + 2k + 1$ is composite.

“Proof: Suppose k is any integer such that $k > 0$. If $k^2 + 2k + 1$ is composite, then $k^2 + 2k + 1 = rs$ for some integers r and s such that

$$1 < r < k^2 + 2k + 1$$

$$\text{and } 1 < s < k^2 + 2k + 1.$$

$$\text{Since } k^2 + 2k + 1 = rs$$

and both r and s are strictly between 1 and $k^2 + 2k + 1$, then $k^2 + 2k + 1$ is not prime. Hence $k^2 + 2k + 1$ is composite as was to be shown.”

4.3 Direct Proof and Counterexample III: Rational Numbers

Such, then, is the whole art of convincing. It is contained in two principles: to define all notations used, and to prove everything by replacing mentally the defined terms by their definitions. —Blaise Pascal, 1623–1662

Sums, differences, and products of integers are integers. But most quotients of integers are not integers. Quotients of integers are, however, important; they are known as *rational numbers*.

Definition

A real number r is **rational** if, and only if, it can be expressed as a quotient of two integers with a nonzero denominator. A real number that is not rational is **irrational**. More formally, if r is a real number, then

$$r \text{ is rational} \Leftrightarrow \exists \text{ integers } a \text{ and } b \text{ such that } r = \frac{a}{b} \text{ and } b \neq 0.$$

The word *rational* contains the word *ratio*, which is another word for quotient. A rational number can be written as a ratio of integers.

Example 4.3.1

Determining Whether Numbers Are Rational or Irrational

- Is $10/3$ a rational number?
- Is $-\frac{5}{39}$ a rational number?
- Is 0.281 a rational number?
- Is 7 a rational number?
- Is 0 a rational number?
- Is $2/0$ a rational number?
- Is $2/0$ an irrational number?
- Is $0.12121212 \dots$ a rational number (where the digits 12 are assumed to repeat forever)?
- If m and n are integers and neither m nor n is zero, is $(m+n)/mn$ a rational number?

Solution

- Yes, $10/3$ is a quotient of the integers 10 and 3 and hence is rational.
- Yes, $-\frac{5}{39} = \frac{-5}{39}$, which is a quotient of the integers -5 and 39 and hence is rational.
- Yes, $0.281 = 281/1000$. Note that the numbers shown on a typical calculator display are all finite decimals. An explanation similar to the one in this example shows that any such number is rational. It follows that a calculator with such a display can accurately represent only rational numbers.
- Yes, $7 = 7/1$.
- Yes, $0 = 0/1$.
- No, $2/0$ is not a real number (division by 0 is not allowed).

g. No, because every irrational number is a real number, and $2/0$ is not a real number. We discuss additional techniques for determining whether numbers are irrational in Sections 4.7, 4.8, and 7.4.

h. Yes. Let $x = 0.12121212 \dots$. Then $100x = 12.12121212 \dots$. Thus

$$100x - x = 12.12121212 \dots - 0.12121212 \dots = 12.$$

But also $100x - x = 99x$ by basic algebra.

Hence $99x = 12$,

and so $x = \frac{12}{99}$.

Therefore, $0.12121212 \dots = 12/99$, which is a ratio of two nonzero integers and thus is a rational number.

Note that you can use an argument similar to this one to show that any repeating decimal is a rational number. In Section 9.4 we show that any rational number can be written as a repeating or terminating decimal.

i. Yes, since m and n are integers, so are $m + n$ and mn (because sums and products of integers are integers). Also $mn \neq 0$ by the *zero product property*. One version of this property says the following:

Zero Product Property

If neither of two real numbers is zero, then their product is also not zero.

(See Theorem T11 in Appendix A and exercise 8 at the end of this section.) It follows that $(m + n)/mn$ is a quotient of two integers with a nonzero denominator and hence is a rational number. ■

More on Generalizing from the Generic Particular

If you claim a property holds for all elements in a domain, then someone can challenge your claim by picking any element in the domain and asking you to prove that that element satisfies the property. To prove your claim, you must be able to meet all such challenges. In other words, you must have a way to convince the challenger that the property is true for an *arbitrarily chosen* element in the domain.

For example, suppose “A” claims that every integer is a rational number. “B” challenges this claim by asking “A” to prove it for $n = 7$. “A” observes that

$$7 = \frac{7}{1} \quad \text{which is a quotient of integers and hence rational.}$$

“B” accepts this explanation but challenges again with $n = -12$. “A” responds that

$$-12 = \frac{-12}{1} \quad \text{which is a quotient of integers and hence rational.}$$

Next “B” tries to trip up “A” by challenging with $n = 0$, but “A” answers that

$$0 = \frac{0}{1} \quad \text{which is a quotient of integers and hence rational.}$$

As you can see, “A” is able to respond effectively to all “B”’s challenges because “A” has a general procedure for putting integers into the form of rational numbers: “A” just divides whatever integer “B” gives by 1. That is, no matter what integer n “B” gives “A”, “A” writes

$$n = \frac{n}{1} \quad \text{which is a quotient of integers and hence rational.}$$

This discussion is an informal proof for the following theorem.

Theorem 4.3.1

Every integer is a rational number.

In exercise 11 at the end of this section you are asked to condense the above discussion into a formal proof.

Proving Properties of Rational Numbers

The next example shows how to use the method of generalizing from the generic particular to prove a property of rational numbers.

Example 4.3.2

Any Sum of Rational Numbers Is Rational

Prove that the sum of any two rational numbers is rational.

Solution Begin by mentally or explicitly rewriting the statement to be proved in the form “ $\forall$ _____, if _____ then _____.”

Formal Restatement: $\forall$ real numbers r and s , if r and s are rational then $r + s$ is rational.

Next ask yourself, “Where am I starting from?” or “What am I supposing?” The answer gives you the starting point, or first sentence, of the proof.

Starting Point: Suppose r and s are any particular but arbitrarily chosen real numbers such that r and s are rational; *or, more simply,*

Suppose r and s are any rational numbers.

Then ask yourself, “What must I show to complete the proof?”

To Show: $r + s$ is rational.

Finally ask, “How do I get from the starting point to the conclusion?” or “Why must $r + s$ be rational if both r and s are rational?” The answer depends in an essential way on the definition of rational.

Rational numbers are quotients of integers, so to say that r and s are rational means that

$$r = \frac{a}{b} \quad \text{and} \quad s = \frac{c}{d} \quad \text{for some integers } a, b, c, \text{ and } d \\ \text{where } b \neq 0 \text{ and } d \neq 0.$$

It follows by substitution that

$$r + s = \frac{a}{b} + \frac{c}{d}. \quad (4.3.1)$$

You need to show that $r + s$ is rational, which means that $r + s$ can be written as a single fraction or ratio of two integers with a nonzero denominator. But the right-hand side of equation (4.3.1) is

$$\begin{aligned}\frac{a}{b} + \frac{c}{d} &= \frac{ad}{bd} + \frac{bc}{bd} && \text{by rewriting the fraction with a common denominator} \\ &= \frac{ad + bc}{bd} && \text{by adding fractions with a common denominator.}\end{aligned}$$

Is this fraction a ratio of integers? Yes. Because products and sums of integers are integers, $ad + bc$ and bd are both integers. Is the denominator $bd \neq 0$? Yes, by the zero product property (since $b \neq 0$ and $d \neq 0$). Thus $r + s$ is a rational number.

This discussion is summarized as follows:

Theorem 4.3.2

The sum of any two rational numbers is rational.

Proof: Suppose r and s are any rational numbers. [We must show that $r + s$ is rational.] Then, by definition of rational, $r = a/b$ and $s = c/d$ for some integers a, b, c , and d with $b \neq 0$ and $d \neq 0$. Thus

$$\begin{aligned}r + s &= \frac{a}{b} + \frac{c}{d} && \text{by substitution} \\ &= \frac{ad + bc}{bd} && \text{by basic algebra.}\end{aligned}$$

Let $p = ad + bc$ and $q = bd$. Then p and q are integers because products and sums of integers are integers and because a, b, c , and d are all integers. Also $q \neq 0$ by the zero product property. Thus

$$r + s = \frac{p}{q} \text{ where } p \text{ and } q \text{ are integers and } q \neq 0.$$

Therefore, $r + s$ is rational by definition of a rational number [as was to be shown].

Deriving New Mathematics from Old

Section 4.1 focused on establishing truth and falsity of mathematical theorems using only the basic algebra normally taught in secondary school; the fact that the integers are closed under addition, subtraction, and multiplication; and the definitions of the terms in the theorems themselves. In the future, when we ask you to **prove something directly from the definitions**, we will mean that you should restrict yourself to this approach. However, once a collection of statements has been proved directly from the definitions, another method of proof becomes possible. The statements in the collection can be used to derive additional results.

Example 4.3.3

Deriving Additional Results about Even and Odd Integers

Suppose that you have already proved the following properties of even and odd integers:

1. The sum, product, and difference of any two even integers are even.
2. The sum and difference of any two odd integers are even.

3. The product of any two odd integers is odd.
4. The product of any even integer and any odd integer is even.
5. The sum of any odd integer and any even integer is odd.
6. The difference of any odd integer minus any even integer is odd.
7. The difference of any even integer minus any odd integer is odd.

Use the properties listed above to prove that if a is any even integer and b is any odd integer, then $\frac{a^2 + b^2 + 1}{2}$ is an integer.

Solution Suppose a is any even integer and b is any odd integer. By property 3, b^2 is odd, and by property 1, a^2 is even. Then by property 5, $a^2 + b^2$ is odd, and because 1 is also odd, the sum $(a^2 + b^2) + 1 = a^2 + b^2 + 1$ is even by property 2. Hence, by definition of even, there exists an integer k such that $a^2 + b^2 + 1 = 2k$. Dividing both sides by 2 gives $\frac{a^2 + b^2 + 1}{2} = k$, which is an integer. Thus $\frac{a^2 + b^2 + 1}{2}$ is an integer [as was to be shown]. ■

A **corollary** is a statement whose truth can be immediately deduced from a theorem that has already been proved.

Example 4.3.4

The Double of a Rational Number

Derive the following as a corollary of Theorem 4.3.2.

Corollary 4.2.3

The double of a rational number is rational.

Solution The double of a number is just its sum with itself. But since the sum of any two rational numbers is rational (Theorem 4.3.2), the sum of a rational number with itself is rational. Hence the double of a rational number is rational. Here is a formal version of this argument:

Proof: Suppose r is any rational number. Then $2r = r + r$ is a sum of two rational numbers. So, by Theorem 4.3.2, $2r$ is rational. ■

TEST YOURSELF

1. To show that a real number is rational, we must show that we can write it as _____.
2. An irrational number is a _____ that is _____.
3. Zero is a rational number because _____.

EXERCISE SET 4.3

The numbers in 1–7 are all rational. Write each number as a ratio of two integers.

1. $-\frac{35}{6}$
2. 4.6037
3. $\frac{4}{5} + \frac{2}{9}$
4. 0.37373737...
5. 0.56565656...

6. 320.5492492492...
7. 52.4672167216721...
8. The zero product property, says that if a product of two real numbers is 0, then one of the numbers must be 0.

ANSWERS FOR TEST YOURSELF

1. a ratio of integers with a nonzero denominator 2. real number; not rational 3. $0 = \frac{0}{1}$

4.4 Direct Proof and Counterexample IV: Divisibility

The essential quality of a proof is to compel belief. —Pierre de Fermat

When you were first introduced to the concept of division in elementary school, you were probably taught that 12 divided by 3 is 4 because if you separate 12 objects into groups of 3, you get 4 groups with nothing left over.

xxx

xxx

xxx

xxx

You may also have been taught to describe this fact by saying that “12 is evenly divisible by 3” or “3 divides 12 evenly.”

The notion of divisibility is the central concept of one of the most beautiful subjects in advanced mathematics: **number theory**, the study of properties of integers.

Note According to the definition of divisibility if you know that n and d are any integers such that d divides n , then you may assume that d is not equal to zero.

Definition

If n and d are integers then

n is **divisible by** d if, and only if, n equals d times some integer and $d \neq 0$.

Instead of “ n is divisible by d ,” we can say that

n is a **multiple of** d , or

d is a **factor of** n , or

d is a **divisor of** n , or

d **divides** n .

The notation $d \mid n$ is read “ d divides n .” Symbolically, if n and d are integers:

$$d \mid n \iff \exists \text{ an integer, say } k, \text{ such that } n = dk \text{ and } d \neq 0.$$

The notation $d \nmid n$ is read “ d does not divide n .”

Example 4.4.1

Divisibility

- a. Is 21 divisible by 3? b. Does 5 divide 40? c. Does $7 \mid 42$?
d. Is 32 a multiple of -16 ? e. Is 6 a factor of 54? f. Is 7 a factor of -7 ?

Solution

- a. Yes, $21 = 3 \cdot 7$. b. Yes, $40 = 5 \cdot 8$. c. Yes, $42 = 7 \cdot 6$.
d. Yes, $32 = (-16) \cdot (-2)$. e. Yes, $54 = 6 \cdot 9$. f. Yes, $-7 = 7 \cdot (-1)$. ■

Example 4.4.2

Divisors of Zero

If k is any nonzero integer, does k divide 0?

Solution Yes, because $0 = k \cdot 0$. ■

Two useful properties of divisibility are (1) that if one positive integer divides a second positive integer, then the first is less than or equal to the second, and (2) that the only divisors of 1 are 1 and -1 .

Theorem 4.4.1 A Positive Divisor of a Positive Integer

For all integers a and b , if a and b are positive and a divides b then $a \leq b$.

Proof: Suppose a and b are any positive integers such that a divides b . [We must show that $a \leq b$.] By definition of divisibility, there exists an integer k so that $b = ak$. By property T25 of Appendix A, k must be positive because both a and b are positive. It follows that

$$1 \leq k$$

because every positive integer is greater than or equal to 1. Multiplying both sides by a gives

$$a \leq ka = b$$

because multiplying both sides of an inequality by a positive number preserves the inequality by property T20 of Appendix A. Thus $a \leq b$ [as was to be shown].

Theorem 4.4.2 Divisors of 1

The only divisors of 1 are 1 and -1 .

Proof: Since $1 \cdot 1 = 1$ and $(-1)(-1) = 1$, both 1 and -1 are divisors of 1. Now suppose m is any integer that divides 1. Then there exists an integer n such that $1 = mn$. By Theorem T25 in Appendix A, either both m and n are positive or both m and n are negative. If both m and n are positive, then m is a positive integer divisor of 1. By Theorem 4.4.1, $m \leq 1$, and, since the only positive integer that is less than or equal to 1 is 1 itself, it follows that $m = 1$. On the other hand, if both m and n are negative, then, by Theorem T12 in Appendix A, $(-m)(-n) = mn = 1$. In this case $-m$ is a positive integer divisor of 1, and so, by the same reasoning, $-m = 1$ and thus $m = -1$. Therefore there are only two possibilities: either $m = 1$ or $m = -1$. So the only divisors of 1 are 1 and -1 .

Example 4.4.3 Divisibility and Algebraic Expressions

- If a and b are integers, is $3a + 3b$ divisible by 3?
- If k and m are integers, is $10km$ divisible by 5?

Solution

- Yes. By the distributive law of algebra, $3a + 3b = 3(a + b)$ and $a + b$ is an integer because it is a sum of two integers.
- Yes. By the associative law of algebra, $10km = 5 \cdot (2km)$ and $2km$ is an integer because it is a product of three integers. ■

When the definition of divides is rewritten formally using the existential quantifier, the result is

$$d|n \Leftrightarrow \exists \text{ an integer } k \text{ such that } n = dk \text{ and } d \neq 0.$$

Note If $d = 0$, then $\frac{n}{d}$ is not defined, and so it is certainly not an integer.

Since the negation of an existential statement is universal, it follows that d does not divide n (denoted $d \nmid n$) if, and only if, $\forall$ integer k , $n \neq dk$ or $d = 0$; in other words, the quotient n/d is not an integer.

$$\text{For all integers } n \text{ and } d, \quad d \nmid n \Leftrightarrow \frac{n}{d} \text{ is not an integer.}$$

Example 4.4.4 Checking Nondivisibility

Does $4|15$?

Solution No, $\frac{15}{4} = 3.75$, which is not an integer. ■



Caution! $a|b$ denotes the sentence “ a divides b ,” whereas a/b denotes the number a divided by b .

Be careful to distinguish between the notation $a|b$ and the notation a/b . The notation $a|b$ stands for the sentence “ a divides b ,” which means that there is an integer k such that $b = ak$. Dividing both sides by a gives $b/a = k$, an integer. Thus, when $a \neq 0$, $a|b$ if, and only if, b/a is an integer. On the other hand, the notation a/b stands for the number a/b which is the result of dividing a by b and which may or may not be an integer. In particular, since the symbol $|$ stands for the word “divides,” be sure to avoid writing something like

$$\cancel{4|(3+5)} \neq 4|8.$$

If read out loud, this becomes “4 divides the quantity 3 plus 5 equals 4 divides 8,” which is nonsense.

Example 4.4.5 Prime Numbers and Divisibility

An alternative way to define a prime number is to say that an integer $n > 1$ is prime if, and only if, its only positive integer divisors are 1 and itself. ■

Proving Properties of Divisibility

One of the most useful properties of divisibility is that it is transitive. If one number divides a second and the second number divides a third, then the first number divides the third.

Example 4.4.6 Transitivity of Divisibility

Prove that for all integers a , b , and c , if $a|b$ and $b|c$, then $a|c$.

Solution Since the statement to be proved is already written formally, you can immediately pick out the starting point, or first sentence of the proof, and the conclusion that must be shown.

Starting Point: Suppose a , b , and c are particular but arbitrarily chosen integers such that $a|b$ and $b|c$.

To Show: $a|c$.

You need to show that $a|c$, or, in other words, that

$$c = a \cdot (\text{some integer}).$$

But since $a|b$,

$$b = ar \quad \text{for some integer } r. \quad 4.4.1$$

And since $b|c$,

$$c = bs \quad \text{for some integer } s. \quad 4.4.2$$

Equation 4.4.2 expresses c in terms of b , and equation 4.4.1 expresses b in terms of a . Thus if you substitute 4.4.1 into 4.4.2, you will have an equation that expresses c in terms of a .

$$\begin{aligned} c &= bs && \text{by equation 4.4.2} \\ &= (ar)s && \text{by equation 4.4.1.} \end{aligned}$$

But $(ar)s = a(rs)$ by the associative law for multiplication. Hence

$$c = a(rs).$$

Now you are almost finished. You have expressed c as $a \cdot (\text{something})$. It remains only to verify that that something is an integer. But of course it is, because it is a product of two integers.

This discussion is summarized as follows:

Theorem 4.4.3 Transitivity of Divisibility

For all integers a , b , and c , if a divides b and b divides c , then a divides c .

Proof: Suppose a , b , and c are any [particular but arbitrarily chosen] integers such that a divides b and b divides c . [We must show that a divides c .] By definition of divisibility,

$$b = ar \quad \text{and} \quad c = bs \quad \text{for some integers } r \text{ and } s.$$

By substitution

$$\begin{aligned} c &= bs \\ &= (ar)s \\ &= a(rs) && \text{by basic algebra.} \end{aligned}$$

Let $k = rs$. Then k is an integer since it is a product of integers, and therefore

$$c = ak \quad \text{where } k \text{ is an integer.}$$

Thus a divides c by definition of divisibility. [This is what was to be shown.]

It would appear from the definition of prime that to show that an integer is prime you would need to show that it is not divisible by any integer greater than 1 and less than itself. In fact, you need only check whether it is divisible by a prime number less than or equal to itself. This follows from Theorems 4.4.1, 4.4.3, and the following theorem, which says that any integer greater than 1 is divisible by a prime number. The idea of the proof is quite simple. You start with a positive integer. If it is prime, you are done; if not, it is a product of two smaller positive factors. If one of these is prime, you are done; if not, you can pick one of the factors and write it as a product of still smaller positive factors. You can continue in this way, factoring the factors of the number you started with, until one of them turns out to

be prime. This must happen eventually because all the factors can be chosen to be positive and each is smaller than the preceding one.

Theorem 4.4.4 Divisibility by a Prime

Any integer $n > 1$ is divisible by a prime number.

Proof: Suppose n is a [particular but arbitrarily chosen] integer that is greater than 1. [We must show that there is a prime number that divides n .] If n is prime, then n is divisible by a prime number (namely itself), and we are done. If n is not prime, then, as discussed in Example 4.1.2b,

$$n = r_0 s_0 \quad \text{where } r_0 \text{ and } s_0 \text{ are integers and} \\ 1 < r_0 < n \text{ and } 1 < s_0 < n.$$

It follows by definition of divisibility that $r_0 | n$.

If r_0 is prime, then r_0 is a prime number that divides n , and we are done. If r_0 is not prime, then

$$r_0 = r_1 s_1 \quad \text{where } r_1 \text{ and } s_1 \text{ are integers and} \\ 1 < r_1 < r_0 \text{ and } 1 < s_1 < r_0.$$

It follows by the definition of divisibility that $r_1 | r_0$. But we already know that $r_0 | n$. Consequently, by transitivity of divisibility, $r_1 | n$.

If r_1 is prime, then r_1 is a prime number that divides n , and we are done. If r_1 is not prime, then

$$r_1 = r_2 s_2 \quad \text{where } r_2 \text{ and } s_2 \text{ are integers and} \\ 1 < r_2 < r_1 \text{ and } 1 < s_2 < r_1.$$

It follows by definition of divisibility that $r_2 | r_1$. But we already know that $r_1 | n$. Consequently, by transitivity of divisibility, $r_2 | n$.

If r_2 is prime, then r_2 is a prime number that divides n , and we are done. If r_2 is not prime, then we may repeat the previous process by factoring r_2 as $r_3 s_3$.

We may continue in this way, factoring successive factors of n until we find a prime factor. We must succeed in a finite number of steps because each new factor is both less than the previous one (which is less than n) and greater than 1, and there are fewer than n integers strictly between 1 and n .* Thus we obtain a sequence

$$r_0, r_1, r_2, \dots, r_k,$$

where $k \geq 0$, $1 < r_k < r_{k-1} < \dots < r_2 < r_1 < r_0 < n$, and $r_i | n$ for each $i = 0, 1, 2, \dots, k$. The condition for termination is that r_k should be prime. Hence r_k is a prime number that divides n . [This is what we were to show.]

Counterexamples and Divisibility

To show that a proposed divisibility property is not universally true, you need only find one pair of integers for which it is false.

*Strictly speaking, this statement is justified by an axiom for the integers called the well-ordering principle, which is discussed in Section 5.4. Theorem 4.4.4 can also be proved using strong mathematical induction, as shown in Example 5.4.1.

Example 4.4.7 Checking a Proposed Divisibility Property

Is the following statement true or false? For all integers a and b , if $a|b$ and $b|a$ then $a = b$.

Solution This statement is false. Can you think of a counterexample just by concentrating for a minute or so?

The following discussion describes a mental process that may take just a few seconds. It is helpful to be able to use it consciously, however, to solve more difficult problems.

To discover the truth or falsity of a statement such as the one given above, start off much as you would if you were trying to prove it.

Starting Point: Suppose a and b are integers such that $a|b$ and $b|a$.

Ask yourself, “*Must* it follow that $a = b$, or *could* it happen that $a \neq b$ for some a and b ?” Focus on the supposition. What does it mean? By definition of divisibility, the conditions $a|b$ and $b|a$ mean that

$$b = ha \quad \text{and} \quad a = kb \quad \text{for some integers } h \text{ and } k.$$

Must it follow that $a = b$, or can you find integers a and b that satisfy these equations for which $a \neq b$? The equations imply that

$$b = ha = h(kb) = (hk)b.$$

Since $b|a$, $b \neq 0$, and so you can cancel b from the extreme left and right sides to obtain

$$1 = hk.$$

In other words, h and k are divisors of 1. But, by Theorem 4.4.2, the only divisors of 1 are 1 and -1 . Thus h and k are both 1 or are both -1 . If $h = k = 1$, then $b = a$. But if $h = k = -1$, then $b = -a$ and so $a \neq b$. This analysis suggests that you can find a counterexample by taking $b = -a$. Here is a formal answer:

Proposed Divisibility Property: For all integers a and b , if $a|b$ and $b|a$ then $a = b$.

Counterexample: Let $a = 2$ and $b = -2$. Then $-2 = (-1) \cdot 2$ and $2 = (-1) \cdot (-2)$, and thus

$$a|b \text{ and } b|a, \text{ but } a \neq b \text{ because } 2 \neq -2.$$

Therefore, the statement is false.

The search for a proof will frequently help you discover a counterexample (provided the statement you are trying to prove is, in fact, false). Conversely, in trying to find a counterexample for a statement, you may come to realize the reason why it is true (if it is, in fact, true). The important thing is to keep an open mind until you are convinced by the evidence of your own careful reasoning.

The Unique Factorization of Integers Theorem

The most comprehensive statement about divisibility of integers is contained in the *unique factorization of integers theorem*. Because of its importance, this theorem is also called the

fundamental theorem of arithmetic. Although Euclid, who lived about 300 B.C.E., seems to have been acquainted with the theorem, it was first stated precisely by the great German mathematician Carl Friedrich Gauss (rhymes with *house*) in 1801.

The unique factorization of integers theorem says that any integer greater than 1 either is prime or can be written as a product of prime numbers in a way that is unique except, perhaps, for the order in which the primes are written. For example,

$$72 = 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 = 2 \cdot 3 \cdot 3 \cdot 2 \cdot 2 = 3 \cdot 2 \cdot 2 \cdot 3 \cdot 2,$$

and so forth. The three 2's and two 3's may be written in any order, but any factorization of 72 as a product of primes must contain exactly three 2's and two 3's—no other collection of prime numbers besides three 2's and two 3's multiplies out to 72.

Note This theorem is the reason the number 1 is not allowed to be prime. If 1 were prime, then factorizations would not be unique. For example, $6 = 2 \cdot 3 = 1 \cdot 2 \cdot 3$, and so forth.

Theorem 4.4.5 Unique Factorization of Integers Theorem (Fundamental Theorem of Arithmetic)

Given any integer $n > 1$, there exist a positive integer k , distinct prime numbers $p_1, p_2, \dots, p_k$, and positive integers $e_1, e_2, \dots, e_k$ such that

$$n = p_1^{e_1} p_2^{e_2} p_3^{e_3} \cdots p_k^{e_k},$$

and any other expression for n as a product of prime numbers is identical to this except, perhaps, for the order in which the factors are written.

The proof of the unique factorization of integers theorem is outlined in the exercises for Sections 5.4 and 8.4.

Because of the unique factorization theorem, any integer $n > 1$ can be put into a *standard factored form* in which the prime factors are written in ascending order from left to right.

Definition

Given any integer $n > 1$, the **standard factored form** of n is an expression of the form

$$n = p_1^{e_1} p_2^{e_2} p_3^{e_3} \cdots p_k^{e_k},$$

where k is a positive integer, $p_1, p_2, \dots, p_k$ are prime numbers, $e_1, e_2, \dots, e_k$ are positive integers, and $p_1 < p_2 < \cdots < p_k$.

Example 4.4.8 Writing Integers in Standard Factored Form

Write 3,300 in standard factored form.

Solution First find all the factors of 3,300. Then write them in ascending order:

$$\begin{aligned} 3,300 &= 100 \cdot 33 = 4 \cdot 25 \cdot 3 \cdot 11 \\ &= 2 \cdot 2 \cdot 5 \cdot 5 \cdot 3 \cdot 11 = 2^2 \cdot 3^1 \cdot 5^2 \cdot 11^1. \end{aligned}$$

Example 4.4.9**Using Unique Factorization to Solve a Problem**

Suppose m is an integer such that

$$8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot m = 17 \cdot 16 \cdot 15 \cdot 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10.$$

Does $17 \mid m$?

Solution Since 17 is one of the prime factors of the right-hand side of the equation, it is also a prime factor of the left-hand side (by the unique factorization of integers theorem). But 17 does not equal any prime factor of 8, 7, 6, 5, 4, 3, or 2 (because it is too large). Hence 17 must occur as one of the prime factors of m , and so $17 \mid m$. ■

TEST YOURSELF

- To show that a nonzero integer d divides an integer n , we must show that _____.
- To say that d divides n means the same as saying that _____ is divisible by _____.
- If a and b are positive integers and $a \mid b$, then _____ is less than or equal to _____.
- For all integers n and d , $d \nmid n$ if, and only if, _____.
- If a and b are integers, the notation $a \mid b$ denotes _____ and the notation a/b denotes _____.
- The transitivity of divisibility theorem says that for all integers a , b , and c , if _____ then _____.
- The divisibility by a prime theorem says that every integer greater than 1 is _____.
- The unique factorization of integers theorem says that any integer greater than 1 is either _____ or can be written as _____ in a way that is unique except possibly for the _____ in which the numbers are written.

EXERCISE SET 4.4

Give a reason for your answer in each of 1–13. Assume that all variables represent integers.

- Is 52 divisible by 13?
- Does $7 \mid 56$?
- Does $5 \mid 0$?
- Does 3 divide $(3k+1)(3k+2)(3k+3)$?
- Is $6m(2m+10)$ divisible by 4?
- Is 29 a multiple of 3?
- Is -3 a factor of 66?
- Is $6a(a+b)$ a multiple of $3a$?
- Is 4 a factor of $2a \cdot 34b$?
- Does $7 \mid 34$?
- Does $13 \mid 73$?
- If $n = 4k + 1$, does 8 divide $n^2 - 1$?
- If $n = 4k + 3$, does 8 divide $n^2 - 1$?

- Fill in the blanks in the following proof that for all integers a and b , if $a \mid b$ then $a \mid (-b)$.

Proof: Suppose a and b are any integers such that (a). By definition of divisibility, there exists an integer r such that (b). By substitution,

$$-b = -(ar) = a(-r).$$

Let $t = \underline{(c)}$. Then t is an integer because $t = (-1) \cdot r$, and both -1 and r are integers. Thus, by substitution, $-b = at$, where t is an integer, and so by definition of divisibility, (d), as was to be shown.

Prove statements 15 and 16 directly from the definition of divisibility.

- For all integers a , b , and c , if $a \mid b$ and $a \mid c$ then $a \mid (b+c)$.
- H** For all integers a , b , and c , if $a \mid b$ then $a \mid c$ then $a \mid (b-c)$.
- For all integers a , b , c , and d , if $a \mid c$ and $b \mid d$ then $ab \mid cd$.

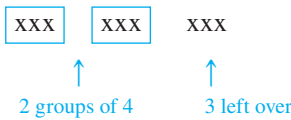
4.5 Direct Proof and Counterexample V: Division into Cases and the Quotient-Remainder Theorem

Be especially critical of any statement following the word “obviously.”
—Anna Pell Wheeler, 1883–1966

When you divide 11 by 4, you get a quotient of 2 and a remainder of 3.

$$\begin{array}{r} 2 \leftarrow \text{quotient} \\ 4 \overline{)11} \\ \underline{8} \\ 3 \leftarrow \text{remainder} \end{array}$$

Another way to say this is that 11 equals 2 groups of 4 with 3 left over:



Or,

$$\begin{array}{c} 11 = 2 \cdot 4 + 3. \\ \uparrow \quad \uparrow \\ \text{2 groups of 4} \quad \text{3 left over} \end{array}$$

The number left over (3) is less than the size of the groups (4) because if 4 or more were left over, another group of 4 could be formed.

The quotient-remainder theorem says that when any integer n is divided by any positive integer d , the result is a quotient q and a nonnegative integer remainder r that is smaller than d .

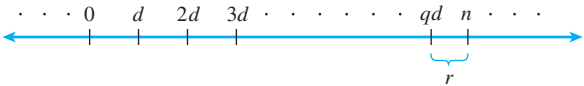
Theorem 4.5.1 The Quotient-Remainder Theorem

Given any integer n and positive integer d , there exist unique integers q and r such that

$$n = dq + r \quad \text{and} \quad 0 \leq r < d.$$

The proof that there exist integers q and r with the given properties is in Section 5.4; the proof that q and r are unique is outlined in exercise 21 in Section 4.8.

If n is positive, the quotient-remainder theorem can be illustrated on the number line as follows:



If n is negative, the picture changes. Since $n = dq + r$, where r is nonnegative, d must be multiplied by a negative integer q to bring dq either exactly to n (in which case $r = 0$) or to a point below n (in which case the positive integer r is added to bring $dq + r$ back up to n). This is illustrated as follows:



Example 4.5.1 **The Quotient-Remainder Theorem**

For each of the following values of n and d , find integers q and r such that $n = dq + r$ and $0 \leq r < d$.

- a. $n = 54, d = 4$ b. $n = -54, d = 4$ c. $n = 54, d = 70$

Solution

- a. $54 = 4 \cdot 13 + 2$; hence $q = 13$ and $r = 2$.
 b. $-54 = 4 \cdot (-14) + 2$; hence $q = -14$ and $r = 2$.
 c. $54 = 70 \cdot 0 + 54$; hence $q = 0$ and $r = 54$. ■

div* and *mod

A number of computer languages have built-in functions that enable you to compute values of the quotients and remainders for the quotient-remainder theorem. In Python $n \text{ div } d$ is written $n // d$ and $n \text{ mod } d$ is written $n \% d$, and for all integer inputs both operators give the values that satisfy the quotient-remainder theorem. In C, C#, and Java, $n \text{ div } d$ is written n / d and $n \text{ mod } d$ is written $n \% d$. For all nonnegative integer inputs for n and positive integer inputs for d , both operators give the values that satisfy the quotient-remainder theorem, but for negative integer inputs for n or d the resulting values differ from their mathematical counterparts.

Definition

Given an integer n and a positive integer d ,

$n \text{ div } d$ = the integer quotient obtained
when n is divided by d , and

$n \text{ mod } d$ = the nonnegative integer remainder obtained
when n is divided by d .

Symbolically, if n and d are integers and $d > 0$, then

$$n \text{ div } d = q \quad \text{and} \quad n \text{ mod } d = r \quad \Leftrightarrow \quad n = dq + r,$$

where q and r are integers and $0 \leq r < d$.

Note In 1801 Carl Friedrich Gauss introduced the word *modulus* and its abbreviation *mod* with the meaning shown here.

It follows from the quotient-remainder theorem that $n \text{ mod } d$ equals one of the integers from 0 through $d - 1$ (since the remainder of the division of n by d must be one of these integers). Also a necessary and sufficient condition for an integer n to be divisible by an integer d is that $n \text{ mod } d = 0$. You are asked to prove this in the exercises at the end of this section.

Example 4.5.2 **Computing *div* and *mod* by Hand or with a Four-Function Calculator**

Compute $32 \text{ div } 9$ and $32 \text{ mod } 9$ by hand or with a four-function calculator.

Solution Performing the division by hand gives the following results:

$$\begin{array}{r} 3 \leftarrow 32 \text{ div } 9 \\ 9 \overline{)32} \\ \underline{27} \\ 5 \leftarrow 32 \text{ mod } 9 \end{array}$$

To use a four-function calculator to compute $n \text{ div } d$ for a nonnegative integer n and a positive integer d , just divide n by d and ignore the part of the answer to the right of the decimal point. To compute $n \text{ mod } d$, substitute $n \text{ div } d$ in place of q and $n \text{ mod } d$ in place of r in the equation $n = dq + r$. The result is

$$n = d \cdot (n \text{ div } d) + n \text{ mod } d.$$

Solving for $n \text{ mod } d$ gives

$$n \text{ mod } d = n - d \cdot (n \text{ div } d).$$

Thus when you use a four-function calculator to divide 32 by 9, you obtain an expression like 3.55555556. Discarding the fractional part gives $32 \text{ div } 9 = 3$, and so

$$32 \text{ mod } 9 = 32 - 9 \cdot (32 \text{ div } 9) = 32 - 27 = 5. \quad \blacksquare$$

Example 4.5.3 Computing the Day of the Week

Suppose today is Tuesday, and neither this year nor next year is a leap year. What day of the week will it be 1 year from today?

Solution There are 365 days in a year that is not a leap year, and each week has 7 days. Now

$$365 \text{ div } 7 = 52 \quad \text{and} \quad 365 \text{ mod } 7 = 1$$

because $365 = 52 \cdot 7 + 1$. Thus 52 weeks, or 364 days, from today will be a Tuesday, and so 365 days from today will be 1 day later, namely, Wednesday.

More generally, if $\text{Day}T$ is the day of the week today and $\text{Day}N$ is the day of the week in N days, then

$$\text{Day}N = (\text{Day}T + N) \text{ mod } 7, \quad 4.5.1$$

where Sunday = 0, Monday = 1, ..., Saturday = 6. \blacksquare

Example 4.5.4 Solving Problems about mod

- Prove that if n is a positive integer, then $n \text{ mod } 10$ is the digit in the ones place in the decimal representation for n . (See Section 2.5 or the preamble to exercises 44–49 in Section 4.4 for discussion about the decimal representation of integers.)
- Suppose m is an integer. If $m \text{ mod } 11 = 6$, what is $4m \text{ mod } 11$?

Solution

- Proof:** Suppose n is any positive integer. The decimal representation for n is $d_k d_{k-1} \dots d_2 d_1 d_0$, where $d_0, d_1, d_2, \dots, d_k$ are integers from 0 to 9 inclusive, $d_k \neq 0$ unless $n = 0$ and $k = 0$,

$$n = d_k \cdot 10^k + d_{k-1} \cdot 10^{k-1} + \dots + d_2 \cdot 10^2 + d_1 \cdot 10 + d_0,$$

and d_0 is the digit in the ones place. Factoring out 10 from all but the final term gives

$$n = 10 \cdot (d_k \cdot 10^{k-1} + d_{k-1} \cdot 10^{k-2} + \dots + d_2 \cdot 10^1 + d_1) + d_0.$$

Thus $n = 10 \cdot (\text{an integer}) + d_0$, and so $n \text{ mod } 10 = d_0$, which is the digit in the ones place in the decimal representation for n .

- Because $m \text{ mod } 11 = 6$, the remainder obtained when m is divided by 11 is 6. This means that there is some integer q so that

$$m = 11q + 6.$$

Thus $4m = 44q + 24 = 44q + 22 + 2 = 11(4q + 2) + 2$.

Since $4q + 2$ is an integer (because products and sums of integers are integers) and since $2 < 11$, the remainder obtained when $4m$ is divided by 11 is 2. Therefore,

$$4m \bmod 11 = 2.$$

Representations of Integers

In Section 4.1 we defined an even integer to have the form twice some integer. At that time we could have defined an odd integer to be one that was not even. Instead, because it was more useful for proving theorems, we specified that an odd integer has the form twice some integer plus 1. The quotient-remainder theorem brings these two ways of describing odd integers together by guaranteeing that any integer is either even or odd. To see why, let n be any integer, and consider what happens when n is divided by 2. By the quotient-remainder theorem (with $d = 2$), there exist unique integers q and r such that

$$n = 2q + r \quad \text{and} \quad 0 \leq r < 2.$$

But the only integers that satisfy $0 \leq r < 2$ are $r = 0$ and $r = 1$. It follows that given any integer n , there exists an integer q with

$$n = 2q + 0 \quad \text{or} \quad n = 2q + 1.$$

In the case that $n = 2q + 0 = 2q$, n is even. In the case that $n = 2q + 1$, n is odd. Hence n is either even or odd, and, because of the uniqueness of q and r , n cannot be both even and odd.

The **parity** of an integer refers to whether the integer is even or odd. For instance, 5 has odd parity and 28 has even parity.

Note The fact that any integer is either even or odd is called the **parity property**.

Example 4.5.5

Consecutive Integers Have Opposite Parity

Prove that given any two consecutive integers, one is even and the other is odd.

Solution Two integers are called *consecutive* if, and only if, one is one more than the other. So if one integer is m , the next consecutive integer is $m + 1$.

To prove the given statement, you can divide the analysis into two cases: case 1, where the smaller of the two integers is even, and case 2, where the smaller of the two integers is odd.

Theorem 4.5.2 The Parity Property

Any two consecutive integers have opposite parity.

Proof:

Suppose that two [particular but arbitrarily chosen] consecutive integers are given; call them m and $m + 1$. [We must show that one of m and $m + 1$ is even and that the other is odd.] By the parity property, either m is even or m is odd. [We break the proof into two cases depending on whether m is even or odd.]

Case 1 (m is even): In this case, $m = 2k$ for some integer k , and so $m + 1 = 2k + 1$, which is odd [by definition of odd]. Hence in this case, one of m and $m + 1$ is even and the other is odd.

(continued on page 204)

Case 2 (m is odd): In this case, $m = 2k + 1$ for some integer k , and so $m + 1 = (2k + 1) + 1 = 2k + 2 = 2(k + 1)$. But $k + 1$ is an integer because it is a sum of two integers. Therefore, $m + 1$ equals twice some integer, and thus $m + 1$ is even. Hence in this case also, one of m and $m + 1$ is even and the other is odd.

It follows that regardless of which case actually occurs for the particular m and $m + 1$ that are chosen, one of m and $m + 1$ is even and the other is odd. [*This is what was to be shown.*]

The division into cases in a proof is like the transfer of control for an **if-then-else** statement in a computer program. If m is even, control transfers to case 1; if not, control transfers to case 2. For any given integer, only one of the cases will apply. You must consider both cases, however, to obtain a proof that is valid for an arbitrarily given integer whether even or not.

There are times when division into more than two cases is called for. Suppose that at some stage of developing a proof, you know that a statement of the form

$$A_1 \text{ or } A_2 \text{ or } A_3 \text{ or } \dots \text{ or } A_n$$

is true, and suppose you want to deduce a conclusion C . By definition of *or*, you know that at least one of the statements A_i is true (although you may not know which). In this situation, you should use the method of division into cases. First assume A_1 is true and deduce C ; next assume A_2 is true and deduce C ; and so forth, until you have assumed A_n is true and deduced C . At that point, you can conclude that regardless of which statement A_i happens to be true, the truth of C follows.

Method of Proof by Division into Cases

To prove a statement of the form “If A_1 or A_2 or $\dots$ or A_n , then C ,” prove all of the following:

If A_1 , then C ,
 If A_2 , then C ,
 $\vdots$
 If A_n , then C .

This process shows that C is true regardless of which of $A_1, A_2, \dots, A_n$ happens to be the case.

Proof by division into cases is a generalization of the argument form shown in Example 2.3.7, whose validity you were asked to establish in exercise 21 of Section 2.3. This method of proof was combined with the quotient-remainder theorem for $d = 2$ to prove Theorem 4.5.2. Allowing d to take on additional values makes it possible to obtain a variety of other results. We begin by showing what happens when $a = 4$.

Example 4.5.6

Representing Integers mod 4

Show that any integer can be written in one of the four forms

$$n = 4q \quad \text{or} \quad n = 4q + 1 \quad \text{or} \quad n = 4q + 2 \quad \text{or} \quad n = 4q + 3$$

for some integer q .

Solution Given any integer n , apply the quotient-remainder theorem to n with the divisor equal to 4. This implies that there exist an integer quotient q and a remainder r such that

$$n = 4q + r \quad \text{and} \quad 0 \leq r < 4.$$

But the only nonnegative remainders r that are less than 4 are 0, 1, 2, and 3. Hence

$$n = 4q \quad \text{or} \quad n = 4q + 1 \quad \text{or} \quad n = 4q + 2 \quad \text{or} \quad n = 4q + 3$$

for some integer q . In other words, $n \bmod 4$ equals 0, 1, 2, or 3. ■

The next example illustrates how the alternative representations for integers $\bmod 4$ can help establish a result in number theory. The solution is broken into two parts: a discussion and a formal proof. These correspond to the stages of actual proof development. Very few people, when asked to prove an unfamiliar theorem, immediately write down the kind of formal proof you find in a mathematics text. They may first check some examples to explore whether the theorem is believable. If it passes that test, they often need to experiment with several possible approaches before finding one that works. A formal proof is much like the ending of a mystery story—the part in which the action of the story is systematically reviewed and all the loose ends are carefully tied together.

Example 4.5.7

The Square of an Odd Integer

Note Another way to state this fact is that if you square an odd integer and divide by 8, you will always get a remainder of 1. Try a few examples!

Prove: The square of any odd integer has the form $8m + 1$ for some integer m .

Solution If checking some examples convinces you that the statement may be true, begin to develop a proof by asking, “Where am I starting from?” and “What do I need to show?” To help answer these questions, introduce variables to rewrite the statement more formally.

Formal Restatement: $\forall$ odd integer n , $\exists$ an integer m such that $n^2 = 8m + 1$.

From this, you can immediately identify the starting point and what is to be shown.

Starting Point: Suppose n is a particular but arbitrarily chosen odd integer.

To Show: $\exists$ an integer m such that $n^2 = 8m + 1$.

This looks tough. Why should there be an integer m with the property that $n^2 = 8m + 1$? That would say that $(n^2 - 1)/8$ is an integer, or that 8 divides $n^2 - 1$. Perhaps you could make use of the fact that $n^2 - 1 = (n - 1)(n + 1)$. Does 8 divide $(n - 1)(n + 1)$? Since n is odd, both $(n - 1)$ and $(n + 1)$ are even. That means that their product is divisible by 4. But that’s not enough. You need to show that the product is divisible by 8.

You could try another approach by arguing that since n is odd, you can represent it as $2q + 1$ for some integer q . Then $n^2 = (2q + 1)^2 = 4q^2 + 4q + 1 = 4(q^2 + q) + 1$. It is clear from this analysis that n^2 can be written in the form $4m + 1$, but it may not be clear that it can be written as $8m + 1$.*

Yet another possibility is to use the result of Example 4.5.6. That example showed that any integer can be written in one of the four forms $4q$, $4q + 1$, $4q + 2$, or $4q + 3$. Two of these, $4q + 1$ and $4q + 3$, are odd. Thus any odd integer can be written in the form $4q + 1$ or $4q + 3$ for some integer q . You could try breaking into cases based on these two different forms.

Note Desperation can spur creativity. When you have tried all the obvious approaches without success and you really care about solving a problem, you reach into the odd corners of your memory for anything that may help.

*See exercise 18 for a different perspective about this approach.

It turns out that this last possibility works! In each of the two cases, the conclusion follows readily by direct calculation. The details are shown in the following formal proof:

Theorem 4.5.3

The square of any odd integer has the form $8m + 1$ for some integer m .

Proof: Suppose n is a [particular but arbitrarily chosen] odd integer. By the quotient-remainder theorem with the divisor equal to 4, n can be written in one of the forms

$$4q \quad \text{or} \quad 4q + 1 \quad \text{or} \quad 4q + 2 \quad \text{or} \quad 4q + 3$$

for some integer q . In fact, since n is odd and $4q$ and $4q + 2$ are even, n must have one of the forms

$$4q + 1 \quad \text{or} \quad 4q + 3.$$

Case 1 ($n = 4q + 1$ for some integer q): [We must find an integer m such that $n^2 = 8m + 1$.]
Since $n = 4q + 1$,

$$\begin{aligned} n^2 &= (4q + 1)^2 && \text{by substitution} \\ &= (4q + 1)(4q + 1) && \text{by definition of square} \\ &= 16q^2 + 8q + 1 \\ &= 8(2q^2 + q) + 1 && \text{by the laws of algebra.} \end{aligned}$$

Let $m = 2q^2 + q$. Then m is an integer since 2 and q are integers and sums and products of integers are integers. Thus, substituting,

$$n^2 = 8m + 1 \quad \text{where } m \text{ is an integer.}$$

Case 2 ($n = 4q + 3$ for some integer q): [We must find an integer m such that $n^2 = 8m + 1$.]
Since $n = 4q + 3$,

$$\begin{aligned} n^2 &= (4q + 3)^2 && \text{by substitution} \\ &= (4q + 3)(4q + 3) && \text{by definition of square} \\ &= 16q^2 + 24q + 9 \\ &= 16q^2 + 24q + (8 + 1) \\ &= 8(2q^2 + 3q + 1) + 1 && \text{by the laws of algebra.} \end{aligned}$$

[The motivation for the choice of algebra steps was the desire to write the expression in the form $8 \cdot (\text{some integer}) + 1$.]

Let $m = 2q^2 + 3q + 1$. Then m is an integer since 1, 2, 3, and q are integers and sums and products of integers are integers. Thus, substituting,

$$n^2 = 8m + 1 \quad \text{where } m \text{ is an integer.}$$

Cases 1 and 2 show that given any odd integer, whether of the form $4q + 1$ or $4q + 3$, $n^2 = 8m + 1$ for some integer m . [This is what we needed to show.]

Note that the result of Theorem 4.5.3 can also be written, “For any odd integer n , $n^2 \bmod 8 = 1$.”

In general, according to the quotient-remainder theorem, if an integer n is divided by an integer d , the possible remainders are $0, 1, 2, \dots, (d-1)$. This implies that n can be written in one of the forms

$$dq, dq+1, dq+2, \dots, dq+(d-1) \quad \text{for some integer } q.$$

Many properties of integers can be obtained by giving d a variety of different values and analyzing the cases that result.

Absolute Value and the Triangle Inequality

The triangle inequality is one of the most important results involving absolute value. It has applications in many areas of mathematics.

Definition

For any real number x , the **absolute value of x** , denoted $|x|$, is defined as follows:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0. \end{cases}$$

The triangle inequality says that the absolute value of the sum of two numbers is less than or equal to the sum of their absolute values. We give a proof based on the following two facts, both of which are derived using division into cases. We state both as lemmas. A **lemma** is a statement that does not have much intrinsic interest but is helpful in deriving other results.

Lemma 4.5.4

For every real number r , $-|r| \leq r \leq |r|$.

Proof: Suppose r is any real number. We divide into cases according to whether $r = 0$, $r > 0$, or $r < 0$.

Case 1 ($r = 0$): In this case, by definition of absolute value, $|r| = r = 0$. since $0 = -0$, we have that $-0 = -|r| = 0 = r = |r|$, and so it is true that

$$-|r| \leq r \leq |r|.$$

Case 2 ($r > 0$): In this case, by definition of absolute value, $|r| = r$. Also, since r is positive and $-|r|$ is negative, $-|r| < r$. Thus it is true that

$$-|r| \leq r \leq |r|.$$

Case 3 ($r < 0$): In this case, by definition of absolute value, $|r| = -r$. Multiplying both sides by -1 gives that $-|r| = r$. Also, since r is negative and $|r|$ is positive, $r < |r|$. Thus it is also true in this case that

$$-|r| \leq r \leq |r|.$$

Hence, in every case,

$$-|r| \leq r \leq |r|$$

[as was to be shown].

Lemma 4.5.5

For every real number r , $|-r| = |r|$.

Proof: Suppose r is any real number. By Theorem T23 in Appendix A, if $r > 0$, then $-r < 0$, and if $r < 0$, then $-r > 0$. Thus

$$\begin{aligned}
 |-r| &= \begin{cases} -r & \text{if } -r > 0 \\ 0 & \text{if } -r = 0 \\ -(-r) & \text{if } -r < 0 \end{cases} && \text{by definition of absolute value} \\
 &= \begin{cases} -r & \text{if } -r > 0 \\ 0 & \text{if } r = 0 \\ r & \text{if } -r < 0 \end{cases} && \begin{array}{l} \text{because } -(-r) = r \text{ by Theorem T4 in} \\ \text{Appendix A, and when } -r = 0, \text{ then } r = 0 \end{array} \\
 &= \begin{cases} -r & \text{if } r < 0 \\ 0 & \text{if } r = 0 \\ r & \text{if } r > 0 \end{cases} && \begin{array}{l} \text{because, by Theorem T24 in Appendix A,} \\ \text{when } -r > 0, \text{ then } r < 0, \text{ when } -r < 0, \\ \text{then } r > 0 \end{array} \\
 &= \begin{cases} r & \text{if } r \geq 0 \\ -r & \text{if } r < 0 \end{cases} && \text{by reformatting the previous result} \\
 &= |r| && \text{by definition of absolute value.}
 \end{aligned}$$

Lemmas 4.5.4 and 4.5.5 now provide a basis for proving the triangle inequality.

Theorem 4.5.6 The Triangle Inequality

For all real numbers x and y , $|x + y| \leq |x| + |y|$.

Proof: Suppose x and y are any real numbers.

Case 1 ($x + y \geq 0$): In this case, $|x + y| = x + y$, and so, by Lemma 4.5.4,

$$x \leq |x| \quad \text{and} \quad y \leq |y|.$$

Hence, by Theorem T26 of Appendix A,

$$|x + y| = x + y \leq |x| + |y|.$$

Case 2 ($x + y < 0$): In this case, $|x + y| = -(x + y) = (-x) + (-y)$, and so, by Lemmas 4.5.4 and 4.5.5,

$$-x \leq |-x| = |x| \quad \text{and} \quad -y \leq |-y| = |y|.$$

It follows, by Theorem T26 of Appendix A, that

$$|x + y| = (-x) + (-y) \leq |x| + |y|.$$

Hence in both cases $|x + y| \leq |x| + |y|$ [as was to be shown].

TEST YOURSELF

- The quotient-remainder theorem says that for all integers n and d with $d \geq 0$, there exist _____ q and r such that _____ and _____.
- If n and d are integers with $d > 0$, $n \text{ div } d$ is _____ and $n \text{ mod } d$ is _____.
- The parity of an integer indicates whether the integer is _____.
- According to the quotient-remainder theorem, if an integer n is divided by a positive integer d , the possible remainders are _____. This implies that n can be written in one of the forms _____ for some integer q .
- To prove a statement of the form “If A_1 or A_2 or A_3 , then C ,” prove _____ and _____ and _____.
- The triangle inequality says that for all real numbers x and y , _____.

EXERCISE SET 4.5

For each of the values of n and d given in 1–6, find integers q and r such that $n = dq + r$ and $0 \leq r < d$.

- | | |
|----------------------|---------------------|
| 1. $n = 70, d = 9$ | 2. $n = 62, d = 7$ |
| 3. $n = 36, d = 40$ | 4. $n = 3, d = 11$ |
| 5. $n = -45, d = 11$ | 6. $n = -27, d = 8$ |

Evaluate the expressions in 7–10.

- | | |
|----------------------------|------------------------|
| 7. a. $43 \text{ div } 9$ | b. $43 \text{ mod } 9$ |
| 8. a. $50 \text{ div } 7$ | b. $50 \text{ mod } 7$ |
| 9. a. $28 \text{ div } 5$ | b. $28 \text{ mod } 5$ |
| 10. a. $30 \text{ div } 2$ | b. $30 \text{ mod } 2$ |
- Check the correctness of formula (4.5.1) given in Example 4.5.3 for the following values of $\text{Day}T$ and N .
 - $\text{Day}T = 6$ (Saturday) and $N = 15$
 - $\text{Day}T = 0$ (Sunday) and $N = 7$
 - $\text{Day}T = 4$ (Thursday) and $N = 12$
 - * Justify formula (4.5.1) for general values of $\text{Day}T$ and N .
 - On a Monday a friend says he will meet you again in 30 days. What day of the week will that be?
 - H 14. If today is Tuesday, what day of the week will it be 1,000 days from today?
 - January 1, 2000, was a Saturday, and 2000 was a leap year. What day of the week will January 1, 2050, be?
 - Suppose d is a positive and n is any integer. If $d \mid n$, what is the remainder obtained when the quotient-remainder theorem is applied to n with divisor d ?

- Prove directly from the definitions that for every integer n , $n^2 - n + 3$ is odd. Use division into two cases: n is even and n is odd.
- a. Prove that the product of any two consecutive integers is even.
b. The result of part (a) suggests that the second approach in the discussion of Example 4.5.7 might be possible after all. Write a new proof of Theorem 4.5.3 based on this observation.
- Prove directly from the definitions that for all integers m and n , if m and n have the same parity, then $5m + 7n$ is even. Divide into two cases: m and n are both even and m and n are both odd.
- Suppose a is any integer. If $a \text{ mod } 7 = 4$, what is $5a \text{ mod } 7$? In other words, if division of a by 7 gives a remainder of 4, what is the remainder when $5a$ is divided by 7? Your solution should show that you obtain the same answer no matter what integer you start with.
- Suppose b is any integer. If $b \text{ mod } 12 = 5$, what is $8b \text{ mod } 12$? In other words, if division of b by 12 gives a remainder of 5, what is the remainder when $8b$ is divided by 12? Your solution should show that you obtain the same answer no matter what integer you start with.
- Suppose c is any integer. If $c \text{ mod } 15 = 3$, what is $10c \text{ mod } 15$? In other words, if division of c by 15 gives a remainder of 3, what is the remainder when $10c$ is divided by 15? Your solution should show that you obtain the same answer no matter what integer you start with.

H 30. For every integer n , $\left\lfloor \frac{n}{2} \right\rfloor + \left\lceil \frac{n}{2} \right\rceil = n$.

H 31. For every integer n , $\left\lfloor \frac{\left\lfloor \frac{n}{2} \right\rfloor}{3} \right\rfloor = \left\lfloor \frac{n}{6} \right\rfloor$.

32. For every integer n , $\left\lceil \frac{\left\lceil \frac{n}{2} \right\rceil}{3} \right\rceil = \left\lceil \frac{n}{6} \right\rceil$.

33. A necessary and sufficient condition for an integer n to be divisible by a nonzero integer d is that $n = \lfloor n/d \rfloor \cdot d$. In other words, for every integer n and nonzero integer d ,

a. if $d \mid n$, then $n = \lfloor n/d \rfloor \cdot d$.

b. if $n = \lfloor n/d \rfloor \cdot d$ then $d \mid n$.

ANSWERS FOR TEST YOURSELF

1. $n \leq x < n + 1$ 2. $n - 1 < x \leq n$

4.7 Indirect Argument: Contradiction and Contraposition

Reductio ad absurdum . . . is one of a mathematician's finest weapons. It is a far finer gambit than any chess gambit: a chess player may offer the sacrifice of a pawn or even a piece, but the mathematician offers the game. —G. H. Hardy, 1877–1947

In a direct proof you start with the hypothesis of a statement and make one deduction after another until you reach the conclusion. Indirect proofs are more roundabout. One kind of indirect proof, *argument by contradiction*, is based on the fact that either a statement is true or it is false but not both. So if you can show that the assumption that a given statement is not true leads logically to a contradiction, impossibility, or absurdity, then that assumption must be false; and, hence, the given statement must be true. This method of proof is also known as *reductio ad impossibile* or *reductio ad absurdum* because it relies on reducing a given assumption to an impossibility or absurdity.

Argument by contradiction occurs in many different settings. For example, if a man accused of holding up a bank can prove that he was someplace else at the time the crime was committed, he will certainly be acquitted. The logic of his defense is as follows:

Suppose I did commit the crime. Then at the time of the crime, I would have had to be at the scene of the crime. In fact, at the time of the crime I was in a meeting with 20 people far from the crime scene, as they will testify. This contradicts the supposition that I committed the crime since it is impossible to be in two places at one time. Hence that supposition is false.

Another example occurs in debate. One technique of debate is to say, “Suppose for a moment that what my opponent says is correct.” Starting from this supposition, the debater then deduces one statement after another until finally arriving at a statement that is completely ridiculous and unacceptable to the audience. By this means the debater shows the opponent’s statement to be false.

The point of departure for a proof by contradiction is the supposition that the statement to be proved is false. The goal is to reason to a contradiction. Thus proof by contradiction has the following outline:

Method of Proof by Contradiction

1. Suppose the statement to be proved is false. That is, suppose that the negation of the statement is true.
2. Show that this supposition leads logically to a contradiction.
3. Conclude that the statement to be proved is true.



Caution! People often make mistakes when they write a negation for a statement they want to prove by contradiction.

There are no clear-cut rules for when to try a direct proof and when to try a proof by contradiction, but there are some general guidelines. Proof by contradiction is indicated if you want to show that there is no object with a certain property, or if you want to show that a certain object does not have a certain property. The next three examples illustrate these situations.

Example 4.7.1**There Is No Greatest Integer**

Use proof by contradiction to show that there is no greatest integer.

Solution For this proof, the “certain property” is the property of being the greatest integer. So prove that there is no object with this property, begin by supposing the negation: that there is an object with the property.

Starting Point: Suppose not. Suppose there is a greatest integer; call it N . This means that $N \geq n$ for every integer n .

To Show: This supposition leads logically to a contradiction.

Theorem 4.7.1

There is no greatest integer.

Proof: [We take the negation of the theorem and suppose it to be true.] Suppose not. That is, suppose there is a greatest integer N . [We must deduce a contradiction.] Then $N \geq n$ for every integer n . Let $M = N + 1$. Now M is an integer since it is a sum of integers. Also $M > N$ since $M = N + 1$. Thus M is an integer that is greater than N . So N is the greatest integer and N is not the greatest integer, which is a contradiction. [This contradiction shows that the supposition is false and, hence, that the theorem is true.]

After a contradiction has been reached, the logic of the argument is always the same: “This is a contradiction. Hence the supposition is false and the theorem is true.” Because of this, most mathematics texts end proofs by contradiction at the point at which the contradiction has been obtained.

The contradiction in the next example is based on the fact that $1/2$ is not an integer.

Example 4.7.2**No Integer Can Be Both Even and Odd**

The fact that no integer is both even and odd can also be deduced from the uniqueness part of the quotient-remainder theorem. A full proof of this part of the theorem is outlined in exercise 21 of Section 4.8.

Theorem 4.7.2

There is no integer that is both even and odd.

Proof: [We take the negation of the theorem and suppose it to be true.] Suppose not. That is, suppose there is at least one integer n that is both even and odd.

(continued on page 220)

[We must deduce a contradiction.] By definition of even, $n = 2a$ for some integer a , and by definition of odd, $n = 2b + 1$ for some integer b . Consequently,

$$2a = 2b + 1 \quad \text{by equating the two expressions for } n,$$

and so

$$2a - 2b = 1$$

$$2(a - b) = 1$$

$$a - b = 1/2 \quad \text{by algebra.}$$

Now since a and b are integers, the difference $a - b$ must also be an integer. But $a - b = 1/2$, and $1/2$ is not an integer. Thus $a - b$ is an integer and $a - b$ is not an integer, which is a contradiction. [This contradiction shows that the supposition is false and, hence, that the theorem is true.]

If you want to prove that a certain object does not have a certain property, you may need to assume that it does have the property and deduce a contradiction. The next example illustrates this strategy. It asks you to prove that a certain object (the sum of a rational and an irrational number) does not have the property being rational.

Example 4.7.3

The Sum of a Rational Number and an Irrational Number

Use proof by contradiction to show that the sum of any rational number and any irrational number is irrational.



Caution! The negation of “The sum of any irrational number and any rational number is irrational” is NOT “The sum of any irrational number and any rational number is rational.”

Solution Begin by supposing the negation of what you are to prove. Be very careful when writing down what this means. If you take the negation incorrectly, the entire rest of the proof will be flawed. In this example, the statement to be proved can be written formally as

$\forall$ real numbers r and s , if r is rational and s is irrational, then $r + s$ is irrational.

From this you can see that the negation is

$\exists$ a rational number r and an irrational number s such that $r + s$ is rational.

It follows that the starting point and what is to be shown are as follows:

Starting Point: Suppose not. That is, suppose there is a rational number r and an irrational number s such that $r + s$ is rational.

To Show: This supposition leads to a contradiction.

To derive a contradiction, you need to understand what you are supposing: that there are numbers r and s such that r is rational, s is irrational, and $r + s$ is rational. By definition of rational and irrational, this means there are convenient expressions that can be substituted for r and $r + s$, but all you can say about s is that it cannot be written as a quotient of integers

$$r = \frac{a}{b} \quad \text{for some integers } a \text{ and } b \text{ with } b \neq 0, \text{ and} \quad 4.7.1$$

$$r + s = \frac{c}{d} \quad \text{for some integers } c \text{ and } d \text{ with } d \neq 0. \quad 4.7.2$$

Note There is no convenient expression that can be substituted for an irrational number.

If you substitute (4.7.1) into (4.7.2), you obtain

$$\frac{a}{b} + s = \frac{c}{d}.$$

Subtracting a/b from both sides gives

$$\begin{aligned} s &= \frac{c}{d} - \frac{a}{b} \\ &= \frac{bc}{bd} - \frac{ad}{bd} && \text{by rewriting } c/d \text{ and } a/b \text{ as equivalent fractions} \\ &= \frac{bc - ad}{bd} && \begin{array}{l} \text{by the rule for subtracting fractions} \\ \text{with the same denominator.} \end{array} \end{aligned}$$

Now both $bc - ad$ and bd are integers because products and differences of integers are integers, and $bd \neq 0$ by the zero product property. Hence s can be expressed as a quotient of two integers with a nonzero denominator, and so s is rational, which contradicts the supposition that it is irrational.

This discussion is summarized in a formal proof.

Theorem 4.7.3

The sum of any rational number and any irrational number is irrational.

Proof:

[We take the negation of the theorem and suppose it to be true.] Suppose not. That is, suppose there is a rational number r and an irrational number s such that $r + s$ is rational. [We must deduce a contradiction.] By definition of rational, $r = a/b$ and $r + s = c/d$ for some integers a, b, c , and d with $b \neq 0$ and $d \neq 0$. By substitution,

$$\frac{a}{b} + s = \frac{c}{d},$$

and so

$$\begin{aligned} s &= \frac{c}{d} - \frac{a}{b} && \text{by subtracting } a/b \text{ from both sides} \\ &= \frac{bc - ad}{bd} && \text{by the laws of algebra.} \end{aligned}$$

Now $bc - ad$ and bd are both integers [since a, b, c , and d are integers and since products and differences of integers are integers], and $bd \neq 0$ [by the zero product property]. Hence s is a quotient of the two integers $bc - ad$ and bd with $bd \neq 0$. Thus, by definition of rational, s is rational, which contradicts the supposition that s is irrational. [Hence the supposition is false and the theorem is true.]

Argument by Contraposition

A second form of indirect argument, *argument by contraposition*, is based on the logical equivalence between a statement and its contrapositive. To prove a statement by contraposition, you take the contrapositive of the statement, prove the contrapositive by a direct proof, and conclude that the original statement is true. The underlying reasoning is that

since a conditional statement is logically equivalent to its contrapositive, if the contrapositive is true then the statement must also be true.

Method of Proof by Contraposition

1. Express the statement to be proved in the form

$$\forall x \text{ in } D, \text{ if } P(x) \text{ then } Q(x).$$

(This step may be done mentally.)

2. Rewrite this statement in the contrapositive form

$$\forall x \text{ in } D, \text{ if } Q(x) \text{ is false then } P(x) \text{ is false.}$$

(This step may also be done mentally.)

3. Prove the contrapositive by a direct proof.

- a. Suppose x is a (particular but arbitrarily chosen) element of D such that $Q(x)$ is false.

- b. Show that $P(x)$ is false.

Example 4.7.4

If the Square of an Integer Is Even, Then the Integer Is Even

Prove that for every integer n , if n^2 is even then n is even.

Solution First form the contrapositive of the statement to be proved.

Contrapositive: For every integer n , if n is not even then n^2 is not even.

By the quotient-remainder theorem with divisor equal to 2, any integer is even or odd, and, by Theorem 4.7.2, no integer is both even and odd. So if an integer is not even, then it is odd. Thus the contrapositive can be restated as follows:

Contrapositive: For every integer n , if n is odd then n^2 is odd.

A straightforward computation is the heart of a direct proof for this statement, as shown below.

Proposition 4.7.4

For every integer n , if n^2 is even then n is even.

Proof (by contraposition): Suppose n is any odd integer. [We must show that n^2 is odd.] By definition of odd, $n = 2k + 1$ for some integer k . By substitution and algebra,

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Now $2k^2 + 2k$ is an integer because products and sums of integers are integers. So $n^2 = 2 \cdot (\text{an integer}) + 1$, and thus, by definition of odd, n^2 is odd [as was to be shown].

We used the word *proposition* here rather than *theorem* because although the word *theorem* can refer to any statement that has been proved, mathematicians often restrict it to especially important statements that have many and varied consequences. Then they use

the word **proposition** to refer to a statement that is somewhat less consequential but nonetheless worth writing down. We will use Proposition 4.7.4 in Section 4.8 to prove that $\sqrt{2}$ is irrational. ■

Relation between Proof by Contradiction and Proof by Contraposition

Observe that any proof by contraposition can be recast in the language of proof by contradiction. In a proof by contraposition, the statement

$$\forall x \text{ in } D, \text{ if } P(x) \text{ then } Q(x)$$

is proved by giving a direct proof of the equivalent statement

$$\forall x \text{ in } D, \text{ if } \sim Q(x) \text{ then } \sim P(x).$$

To do this, you suppose you are given an arbitrary element x of D such that $\sim Q(x)$. You then show that $\sim P(x)$. This is illustrated in Figure 4.7.1.

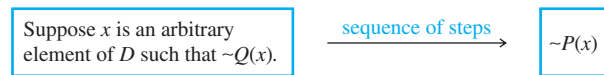


FIGURE 4.7.1 Proof by Contraposition

Exactly the same sequence of steps can be used as the heart of a proof by contradiction for the given statement. The only thing that changes is the context in which the steps are written down.

To rewrite the proof as a proof by contradiction, you suppose there is an x in D such that $P(x)$ and $\sim Q(x)$. You then follow the steps of the proof by contraposition to deduce the statement $\sim P(x)$. But $\sim P(x)$ is a contradiction to the supposition that $P(x)$ and $\sim Q(x)$. (Because to contradict a conjunction of two statements, it is only necessary to contradict one of them.) This process is illustrated in Figure 4.7.2.



FIGURE 4.7.2 Proof by Contradiction

As an example, here is a proof by contradiction of Proposition 4.7.4.

Proposition 4.7.4

For every integer n , if n^2 is even then n is even.

Proof (by contradiction): [We take the negation of the theorem and suppose it to be true.] Suppose not. That is, suppose there is an integer n such that n^2 is even and n is not even. [We must deduce a contradiction.] By the quotient-remainder theorem with divisor equal to 2, any integer is even or odd. Hence, since n is not even it is odd, and thus, by definition of odd, $n = 2k + 1$ for some integer k . By substitution and algebra,

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

(continued on page 224)

Now $2k^2 + 2k$ is an integer because products and sums of integers are integers. So $n^2 = 2 \cdot (\text{an integer}) + 1$, and thus, by definition of odd, n^2 is odd. Therefore, n^2 is both even and odd. This contradicts Theorem 4.7.2, which states that no integer can be both even and odd. *[This contradiction shows that the supposition is false and, hence, that the proposition is true.]*

Note that when you use proof by contraposition, you know exactly what conclusion you need to show, namely, the negation of the hypothesis; whereas in proof by contradiction, it may be difficult to know what contradiction to head for. On the other hand, when you use proof by contradiction, once you have deduced any contradiction whatsoever, you are done. The main advantage of contraposition over contradiction is that you avoid having to take (possibly incorrectly) the negation of a complicated statement. The disadvantage of contraposition as compared with contradiction is that you can use contraposition only for a specific class of statements—those that are universal and conditional. The previous discussion shows that any statement that can be proved by contraposition can be proved by contradiction. But the converse is not true. Statements such as “ $\sqrt{2}$ is irrational” (discussed in the next section) can be proved by contradiction but not by contraposition.

Proof as a Problem-Solving Tool

Direct proof, disproof by counterexample, proof by contradiction, and proof by contraposition are all tools that may be used to help determine whether statements are true or false. Working with examples might have given you a sense that a statement of the form

For all elements in a domain, if (hypothesis) then (conclusion),

might be true. To explore further, imagine elements in the domain that satisfy the hypothesis. Ask yourself: Must they satisfy the conclusion? If you can see that the answer is “yes” in all cases, then the statement is true and your insight will form the basis for a direct proof. If after some thought it is not clear that the answer is “yes,” ask yourself whether there are elements of the domain that satisfy the hypothesis and *not* the conclusion. If you are successful in finding some, then the statement is false and you have a counterexample. On the other hand, if you are not successful in finding such elements, perhaps none exist. Perhaps you can show that assuming the existence of elements in the domain that satisfy the hypothesis and not the conclusion leads logically to a contradiction. If so, then the given statement is true and you have the basis for a proof by contradiction. Alternatively, you could imagine elements of the domain for which the conclusion is false and ask whether such elements also fail to satisfy the hypothesis. If the answer in all cases is “yes,” then you have a basis for a proof by contraposition.

Solving problems, especially difficult problems, is rarely a straightforward process. At any stage of following the guidelines above, you might want to try the method of a previous stage again. If, for example, you fail to find a counterexample for a certain statement, your experience in trying to find it might help you decide to reattempt a direct argument rather than trying an indirect one. Psychologists who have studied problem solving have found that the most successful problem solvers are those who are flexible and willing to use a variety of approaches without getting stuck in any one of them for very long. Mathematicians sometimes work for months (or longer) on difficult problems. Don’t be discouraged if some problems in this book take you quite a while to solve.

Learning the skills of proof and disproof is much like learning other skills, such as those used in a sport or playing a musical instrument. When you first start out, you may

feel bewildered by all the rules, and you may not feel confident as you attempt new things. But with practice the rules become internalized and you can use them in conjunction with all your other powers—of balance, coordination, judgment, aesthetic sense—to concentrate on winning a competition or performing in public.

Now that you have worked through several sections of this chapter, return to the idea that, above all, a proof or disproof should be a convincing argument. You need to know how direct and indirect proofs and counterexamples are structured. But to use this knowledge effectively, you must use it in conjunction with your imaginative powers, your intuition, and especially your common sense.

TEST YOURSELF

1. To prove a statement by contradiction, you suppose that _____ and you show that _____.
2. A proof by contraposition of a statement of the form “ $\forall x \in D$, if $P(x)$ then $Q(x)$ ” is a direct proof of _____.
3. To prove a statement of the form “ $\forall x \in D$, if $P(x)$ then $Q(x)$ ” by contraposition, you suppose that _____ and you show that _____.

EXERCISE SET 4.7

1. Fill in the blanks in the following proof by contradiction that there is no least positive real number.

Proof: Suppose not. That is, suppose that there is a least positive real number x . [We must deduce (a)] Consider the number $x/2$. Since x is a positive real number, $x/2$ is also (b). In addition, we can deduce that $x/2 < x$ by multiplying both sides of the inequality $1 < 2$ by (c) and dividing (d). Hence $x/2$ is a positive real number that is less than the least positive real number. This is a (e). [Thus the supposition is false, and so there is no least positive real number.]

2. Is $\frac{1}{0}$ an irrational number? Explain.
3. Use proof by contradiction to show that for every integer n , $3n + 2$ is not divisible by 3.
4. Use proof by contradiction to show that for every integer m , $7m + 4$ is not divisible by 7.

Carefully formulate the negations of each of the statements in 5–7. Then prove each statement by contradiction.

5. There is no greatest even integer.
6. There is no greatest negative real number.
7. There is no least positive rational number.
8. Fill in the blanks for the following proof that the difference of any rational number and any irrational number is irrational.

Proof (by contradiction): Suppose not. That is, suppose that there exist (a) x and (b) y such that $x - y$ is rational. By definition of rational, there exist integers a, b, c , and d with $b \neq 0$ and $d \neq 0$ so that $x = \frac{a}{b}$ and $x - y = \frac{c}{d}$. By substitution,

$$\frac{a}{b} - y = \frac{c}{d}$$

Adding y and subtracting $\frac{c}{d}$ on both sides gives

$$\begin{aligned} y &= \frac{(e)}{\quad} && \text{by substitution} \\ &= \frac{ad}{bd} - \frac{bc}{bd} \\ &= \frac{ad - bc}{bd} && \text{by algebra.} \end{aligned}$$

Now both $ad - bc$ and bd are integers because products and differences of (f) are (g). And $bd \neq 0$ by the (h). Hence y is a ratio of integers with a nonzero denominator, and thus y is (i) by definition of rational. We therefore have both that y is irrational and that y is rational, which is a contradiction. [Thus the supposition is false and the statement to be proved is true.]

9. a. When asked to prove that the difference of any irrational number and any rational number is irrational, a student began, “Suppose not. That is, suppose the difference of any irrational number and any rational number is rational.”

ANSWERS FOR TEST YOURSELF

1. the statement is false; this supposition leads to a contradiction
2. the contrapositive of the statement, namely, $\forall x \in D$, if $\sim Q(x)$ then $\sim P(x)$
3. x is any [particular but arbitrarily chosen] element of D for which $Q(x)$ is false; $P(x)$ is false

4.8 Indirect Argument: Two Famous Theorems

He is unworthy of the name of man who does not know that the diagonal of a square is incommensurable with its side. —Plato (ca. 428–347 B.C.E.)

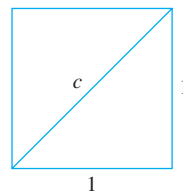
This section contains proofs of two of the most famous theorems in mathematics: that $\sqrt{2}$ is irrational and that there are infinitely many prime numbers. Both proofs are examples of indirect arguments and were well known more than 2,000 years ago, but they remain exemplary models of mathematical argument to this day.

The Irrationality of $\sqrt{2}$

When mathematics flourished at the time of the ancient Greeks, mathematicians believed that given any two line segments, say A : _____ and B : _____, a certain unit of length could be found so that segment A was exactly m units long and segment B was exactly n units long, for some integers m and n . (The segments were said to be *commensurable* with respect to this special unit of length.) Then the ratio of the lengths of A and B would be in the same proportion as the ratio of the integers m and n . Symbolically:

$$\frac{\text{length } A}{\text{length } B} = \frac{m}{n}.$$

Now it is easy to find a line segment of length $\sqrt{2}$; just take the diagonal of a unit square:



By the Pythagorean theorem, $c^2 = 1^2 + 1^2 = 2$, and so $c = \sqrt{2}$. If the belief of the ancient Greeks were correct, there would be integers m and n such that

$$\frac{\text{length (diagonal)}}{\text{length (side)}} = \frac{m}{n}.$$

And this would imply that

$$\frac{c}{1} = \frac{\sqrt{2}}{1} = \sqrt{2} = \frac{m}{n}.$$

But then $\sqrt{2}$ would be a ratio of two integers, or, in other words, $\sqrt{2}$ would be rational.

In the fourth or fifth century B.C.E., the followers of the Greek mathematician and philosopher Pythagoras discovered that $\sqrt{2}$ was not rational. This discovery was very upsetting to them, for it undermined their deep, quasi-religious belief in the power of whole numbers to describe phenomena.



Euclid
(fl. 300 B.C.E.)

Note Strictly speaking, being able to assume that m and n have no common factors is a consequence of the “well-ordering principle for the integers,” which is discussed in Section 5.4.

The following proof of the irrationality of $\sqrt{2}$ was known to Aristotle and is similar to that in the tenth book of Euclid’s *Elements of Geometry*. The Greek mathematician Euclid is best known as a geometer, and knowledge of the geometry in the first six books of his *Elements* was considered an essential part of a liberal education for more than 2,000 years. However, books 7–10 of his *Elements* contain much that we would now call number theory.

The proof begins by supposing the negation: $\sqrt{2}$ is rational. This means that there exist integers m and n such that $\sqrt{2} = m/n$. Now if m and n have any common factors, these may be factored out to obtain a new fraction, equal to m/n , in which the numerator and denominator have no common factors. (For example, $18/12 = (6 \cdot 3)/(6 \cdot 2) = 3/2$, which is a fraction whose numerator and denominator have no common factors.) Thus, without loss of generality, we may assume that m and n had no common factors in the first place. We will then derive the contradiction that m and n *do* have a common factor of 2. The argument makes use of Proposition 4.7.4: If the square of an integer is even, then that integer is even.

Theorem 4.8.1 Irrationality of $\sqrt{2}$

$\sqrt{2}$ is irrational.

Proof (by contradiction): [We take the negation and suppose it to be true.] Suppose not. That is, suppose $\sqrt{2}$ is rational. Then there are integers m and n with no common factors such that

$$\sqrt{2} = \frac{m}{n} \quad 4.8.1$$

[by dividing m and n by any common factors if necessary]. [We must derive a contradiction.] Squaring both sides of equation (4.8.1) gives

$$2 = \frac{m^2}{n^2}.$$

Or, equivalently,

$$m^2 = 2n^2. \quad 4.8.2$$

Note that equation (4.8.2) implies that m^2 is even (by definition of even). It follows that m is even (by Proposition 4.7.4). We file this fact away for future reference and also deduce (by definition of even) that

$$m = 2k \quad \text{for some integer } k. \quad 4.8.3$$

Substituting equation (4.8.3) into equation (4.8.2), we see that

$$m^2 = (2k)^2 = 4k^2 = 2n^2.$$

Dividing both sides of the right-most equation by 2 gives

$$n^2 = 2k^2.$$

Consequently, n^2 is even, and so n is even (by Proposition 4.7.4). But we also know that m is even. [This is the fact we filed away.] Hence both m and n have a common factor of 2. But this contradicts the supposition that m and n have no common factors. [Hence the supposition is false and so the theorem is true.]

Now that you have seen the proof that $\sqrt{2}$ is irrational, you can use the irrationality of $\sqrt{2}$ to derive the irrationality of certain other real numbers.

Example 4.8.1 Irrationality of $1 + 3\sqrt{2}$

Prove by contradiction that $1 + 3\sqrt{2}$ is irrational.

Solution The essence of the argument is the observation that if $1 + 3\sqrt{2}$ could be written as a ratio of integers, then so could $\sqrt{2}$. But by Theorem 4.8.1, we know that to be impossible.

Proposition 4.8.2

$1 + 3\sqrt{2}$ is irrational.

Proof: Suppose not. Suppose $1 + 3\sqrt{2}$ is rational. [We must derive a contradiction.] Then by definition of rational,

$$1 + 3\sqrt{2} = \frac{a}{b} \quad \text{for some integers } a \text{ and } b \text{ with } b \neq 0.$$

It follows that

$$\begin{aligned} 3\sqrt{2} &= \frac{a}{b} - 1 && \text{by subtracting 1 from both sides} \\ &= \frac{a}{b} - \frac{b}{b} && \text{by substitution} \\ &= \frac{a-b}{b} && \text{by the rule for subtracting fractions} \\ &&& \text{with a common denominator.} \end{aligned}$$

Hence

$$\sqrt{2} = \frac{a-b}{3b} \quad \text{by dividing both sides by 3.}$$

But $a - b$ and $3b$ are integers (since a and b are integers and differences and products of integers are integers), and $3b \neq 0$ by the zero product property. Hence $\sqrt{2}$ is a quotient of the two integers $a - b$ and $3b$ with $3b \neq 0$, and so $\sqrt{2}$ is rational (by definition of rational). This contradicts the fact that $\sqrt{2}$ is irrational. [The contradiction shows that the supposition is false.] Hence $1 + 3\sqrt{2}$ is irrational.

Are There Infinitely Many Prime Numbers?

You know that a prime number is a positive integer that cannot be factored as a product of two smaller positive integers. Is the set of all such numbers infinite, or is there a largest prime number? The answer was known to Euclid, and a proof that the set of all prime numbers is infinite appears in Book 9 of his *Elements of Geometry*.

Euclid's proof requires one additional fact we have not yet established: If a prime number divides an integer, then it does not divide the next successive integer.

Proposition 4.8.3

For any integer a and any prime number p , if $p|a$ then $p \nmid (a+1)$.

Proof (by contradiction): Suppose not. That is, suppose there exist an integer a and a prime number p such that $p|a$ and $p|(a+1)$. Then, by definition of divisibility, there exist integers r and s such that $a = pr$ and $a+1 = ps$. It follows that

$$1 = (a+1) - a = ps - pr = p(s-r),$$

and so (since $s-r$ is an integer) $p|1$. But, by Theorem 4.4.2, the only integer divisors of 1 are 1 and -1 , and $p > 1$ because p is prime. Thus $p \leq 1$ and $p > 1$, which is a contradiction. [Hence the supposition is false, and the proposition is true.]

The idea of Euclid's proof is this: Suppose the set of prime numbers were finite. Then you could take the product of all the prime numbers and add 1. By Theorem 4.4.4 this number must be divisible by some prime number. But by Proposition 4.8.3, this number is not divisible by any of the prime numbers in the set. Hence there must be a prime number that is not in the set of all prime numbers, which is impossible.

The following formal proof fills in the details of this outline.

Theorem 4.8.4 Infinitude of the Primes

The set of prime numbers is infinite.

Proof (by contradiction): Suppose not. That is, suppose the set of prime numbers is finite. [We must deduce a contradiction.] Then some prime number p is the largest of all the prime numbers, and hence we can list the prime numbers in ascending order:

$$2, 3, 5, 7, 11, \dots, p.$$

Let N be the product of all the prime numbers plus 1:

$$N = (2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdots p) + 1$$

Then $N > 1$, and so, by Theorem 4.4.4, N is divisible by some prime number q . Because q is prime, q must equal one of the prime numbers $2, 3, 5, 7, 11, \dots, p$. Thus, by definition of divisibility, q divides $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdots p$, and so, by Proposition 4.8.3, q does not divide $(2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdots p) + 1$, which equals N . Hence N is divisible by q and N is not divisible by q , and we have reached a contradiction. [Therefore, the supposition is false and the theorem is true.]

The proof of Theorem 4.8.4 shows that if you form the product of all the prime numbers up to a certain point and add one, the result, N , is divisible by a prime not on the list. The proof does not show that N is itself prime. In exercise 30 at the end of the section you will find a value for N that is not prime, although it is divisible by a prime.

When to Use Indirect Proof

The examples in this section and Section 4.7 have not provided a definitive answer to the question of when to prove a statement directly and when to prove it indirectly. Many



Ben Joseph Green
(born 1977)

The Royal Society



Terence Chi-Shen Tao
(born 1975)

Philippe Desmazes/AP/Getty Images



Yitang Zhang
(born 1955)

Peter Bohler/Redux



Marie-Sophie Germain
(1776–1831)

Science & Society Picture Library/SSPL/Getty Images

theorems can be proved either way. Usually, however, when both types of proof are possible, indirect proof is clumsier than direct proof. In the absence of obvious clues suggesting indirect argument, try first to prove a statement directly. Then, if that does not succeed, look for a counterexample. If the search for a counterexample is unsuccessful, look for a proof by contradiction or contraposition.

Open Questions in Number Theory

In this section we proved that there are infinitely many prime numbers. There is no known formula for obtaining primes, but a few formulas have been found to be more successful at producing them than other formulas. One such is due to Marin Mersenne, a French monk who lived from 1588 to 1648. *Mersenne primes* have the form $2^p - 1$, where p is prime. Not all numbers of this form are prime, but because Mersenne primes are easier to test for primality than are other numbers, most of the largest known prime numbers are Mersenne primes.

An interesting question is whether there are infinitely many Mersenne primes. As of the date of publication of this book, the answer is not known, but new mathematical discoveries are being made every day and by the time you read this someone may have discovered the answer. Another formula that seems to produce a relatively large number of prime numbers is due to Fermat. *Fermat primes* are prime numbers of the form $2^{2^n} + 1$, where n is a positive integer. Are there infinitely many Fermat primes? Again, as of now, no one knows. Similarly unknown are whether there are infinitely many primes of the form $n^2 + 1$, where n is a positive integer, and whether there is always a prime number between integers n^2 and $(n + 1)^2$.

Another famous open question involving primes is the *twin primes conjecture*, which states that there are infinitely many pairs of prime numbers of the form p and $p + 2$. As with other well-known problems in number theory, this conjecture has withstood computer testing up to extremely large numbers, and some progress has been made toward a proof. In 2004, Ben Green and Terence Tao showed that for any integer $m > 1$, there is a sequence of m equally spaced integers all of which are prime. In 2013 Yitang Zhang proved that there are infinitely many pairs of prime numbers that differ by no more than 70,000,000. This is a lot more than 2, but Zhang's was the first discovery of any fixed upper bound between infinitely many pairs of prime numbers. In 2014 a group of mathematicians working collaboratively showed that the bound could be reduced from 70,000,000 to 246; still more than 2 but giving hope that a proof of the twin primes conjecture may be attainable.

Related to the twin primes conjecture is a conjecture made by Sophie Germain, a French mathematician born in 1776, who made significant progress toward a proof of Fermat's Last Theorem. Germain conjectured that there are infinitely many prime number pairs of the form p and $2p + 1$. Initial values of p with this property are 2, 3, 5, 11, 23, 29, 41, and 53, and computer testing has verified the conjecture for many additional values. In fact, as of the writing of this book, the largest prime p for which $2p + 1$ is also known to be prime is $2618163402417 \times 2^{1290000} - 1$. This is a number with 388,341 decimal digits! But compared with infinity, any number, no matter how large, is less than a drop in the bucket.

In 1844, the Belgian mathematician Eugène Catalan conjectured that the only solutions to the equation $x^n - y^m = 1$, where x, y, n , and m are all integers greater than 1, is $3^2 - 2^3 = 1$. This conjecture was finally proved by Preda Mihăilescu in 2002.

In 1993, while trying to prove Fermat's Last Theorem, an amateur number theorist, Andrew Beal, became intrigued by the equation $x^m + y^n = z^k$, where no two of x, y , or z have any common factor other than ± 1 . When diligent effort, first by hand and then by computer, failed to reveal any solutions, Beal conjectured that no solutions exist. His conjecture has become known as the *Beal conjecture*, and he has offered a prize of \$100,000 to anyone who can either prove or disprove it.

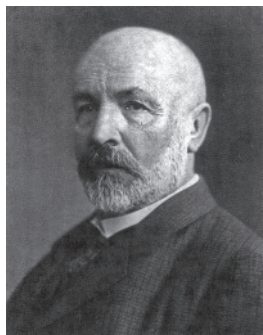
These are just a few of a large number of open questions in number theory. Many people believe that mathematics is a fixed subject that changes very little from one century to the next. In fact, more mathematical questions are being raised and more results are being discovered now than ever before in history.

TEST YOURSELF

1. The ancient Greeks discovered that in a right triangle where both legs have length 1, the ratio of the length of the hypotenuse to the length of one of the legs is not equal to a ratio of _____.
2. One way to prove that $\sqrt{2}$ is an irrational number is to assume that $\sqrt{2} = m/n$ for some integers m and n that have no common factor greater than 1, use the lemma that says that if the square of an integer is even then _____, and eventually show that m and n _____.
3. One way to prove that there are infinitely many prime numbers is to assume that there is a largest prime number p , construct the number _____, and then show that this number has to be divisible by a prime number that is greater than _____.

EXERCISE SET 4.8

1. A calculator display shows that $\sqrt{2} = 1.414213562$. Because $1.414213562 = \frac{1414213562}{1000000000}$, this suggests that $\sqrt{2}$ is a rational number, which contradicts Theorem 4.8.1. Explain the discrepancy.
 2. Example 4.3.1(h) illustrates a technique for showing that any repeating decimal number is rational. A calculator display shows the result of a certain calculation as 40.7272727272. Can you be sure that the result of the calculation is a rational number? Explain.
 3. Could there be a rational number whose first trillion digits are the same as the first trillion digits of $\sqrt{2}$? Explain.
 4. A calculator display shows that the result of a certain calculation is 0.2. Can you be sure that the result of the calculation is a rational number?
 5. Let S be the statement: The cube root of every irrational number is irrational. This statement is true, but the following “proof” is incorrect. Explain the mistake.
“Proof (by contradiction): Suppose not. Suppose the cube root of every irrational number is rational. But $2\sqrt{2}$ is irrational because it is a product of a rational and an irrational number, and the cube root of $2\sqrt{2}$ is $\sqrt{2}$, which is irrational. This is a contradiction, and hence it is not true that the cube root of every irrational number is rational. Thus the statement to be proved is true.”
- Determine which statements in 6–16 are true and which are false. Prove those that are true and disprove those that are false.
6. $6 - 7\sqrt{2}$ is irrational.
 7. $3\sqrt{2} - 7$ is irrational.
 8. $\sqrt{4}$ is irrational.
 9. $\sqrt{2}/6$ is irrational.
 10. The sum of any two irrational numbers is irrational.
 11. The difference of any two irrational numbers is irrational.
 12. The positive square root of a positive irrational number is irrational.
 13. If r is any rational number and s is any irrational number, then r/s is irrational.
 14. The sum of any two positive irrational numbers is irrational.
 15. The product of two irrational numbers is irrational.
 - H 16. If an integer greater than 1 is a perfect square, then its cube root is irrational.
 17. Consider the following sentence: If x is rational then $\sqrt{x}$ is irrational. Is this sentence always true, sometimes true and sometimes false, or always false? Justify your answer.
 18. a. Prove that for every integer a , if a^3 is even then a is even.
 b. Prove that $\sqrt[3]{2}$ is irrational.



Georg Cantor
(1845–1918)

Pictorial Press Ltd./Alamy Stock Photo

In the late nineteenth century, Georg Cantor was the first to realize the potential usefulness of investigating properties of sets in general as distinct from properties of the elements that comprise them. Many mathematicians of his time resisted accepting the validity of Cantor's work. Now, however, abstract set theory is regarded as the foundation of mathematical thought. All mathematical objects (even numbers!) can be defined in terms of sets, and the language of set theory is used in every mathematical subject.

In this chapter we add to the basic definitions and notation of set theory introduced in Chapter 1 and show how to establish properties of sets through the use of proofs and counterexamples. We also introduce the notion of a Boolean algebra, explain how to derive its properties, and discuss their relationships to logical equivalencies and set identities. The chapter ends with a discussion of a famous “paradox” of set theory and its relation to computer science.

Source: David Eugene Smith Collection, Columbia University.

6.1 Set Theory: Definitions and the Element Method of Proof

The introduction of suitable abstractions is our only mental aid to organize and master complexity. —E. W. Dijkstra, 1930–2002

The words *set* and *element* are undefined terms of set theory just as *sentence*, *true*, and *false* are undefined terms of logic. The founder of set theory, Georg Cantor, suggested imagining a set as a “collection into a whole M of definite and separate objects of our intuition or our thought. These objects are called the elements of M .” Cantor used the letter M because it is the first letter of the German word for set: *Menge*.

Following the spirit of Cantor's notation (though not the letter), let S denote a set and a an element of S . Then, as indicated in Section 1.2, $a \in S$ means that a is an element of S , $a \notin S$ means that a is not an element of S , $\{1, 2, 3\}$ refers to the set whose elements are 1, 2, and 3, and $\{1, 2, 3, \dots\}$ refers to the set of all positive integers. If S is a set and $P(x)$ is a property that elements of S may or may not satisfy, then a set A may be defined by writing

$$A = \{x \in S \mid P(x)\},$$

↑ the set of all ↑ such that

which is read “ A is the set of all x in S such that P of x .”



Caution! Don't forget to include the words “the set of all.”

Subsets: Proof and Disproof

In Section 1.2 we defined what it means for a set A to be a subset of the set B . Here we rewrite the definition as a formal universal conditional statement:

$$A \subseteq B \iff \forall x, \text{ if } x \in A \text{ then } x \in B.$$

The negation is, therefore, existential:

$$A \not\subseteq B \iff \exists x \text{ such that } x \in A \text{ and } x \notin B.$$

Recall that a *proper subset* of a set is a subset that is not equal to its containing set. That is:

$$\begin{aligned} A \text{ is a proper subset of } B &\iff \\ (1) \quad &A \subseteq B, \text{ and} \\ (2) \quad &\text{there is at least one element in } B \text{ that is not in } A. \end{aligned}$$

Example 6.1.1 Testing Whether One Set Is a Subset of Another

Let $A = \{1\}$ and $B = \{1, \{1\}\}$.

- Is $A \subseteq B$?
- If so, is A a proper subset of B ?

Solution

- Because $A = \{1\}$, A has only one element—namely, the symbol 1. This element is also one of the elements in set B . Hence every element in A is in B , and so $A \subseteq B$.
- B has two distinct elements, the symbol 1 and the set $\{1\}$ whose only element is 1. Since $1 \neq \{1\}$, the set $\{1\}$ is not an element of A , and so there is an element of B that is not an element of A . Hence A is a proper subset of B . ■

Note A set like $\{1\}$, with just one element, is called a **singleton set**.

Because we define what it means for one set to be a subset of another by means of a universal conditional statement, we can use the method of direct proof to establish a subset relationship. Such a proof is called an *element argument* and is the fundamental proof technique of set theory.

Element Argument: The Basic Method for Proving That One Set Is a Subset of Another

Let sets X and Y be given. To prove that $X \subseteq Y$,

- suppose** that x is a particular but arbitrarily chosen element of X ,
- show** that x is an element of Y .

Example 6.1.2 Proving and Disproving Subset Relations

Define sets A and B as follows:

$$A = \{m \in \mathbf{Z} \mid m = 6r + 12 \text{ for some } r \in \mathbf{Z}\}$$

$$B = \{n \in \mathbf{Z} \mid n = 3s \text{ for some } s \in \mathbf{Z}\}.$$

- Outline a proof that $A \subseteq B$.
- Prove that $A \subseteq B$.
- Disprove that $B \subseteq A$.

Solutiona. **Proof Outline:**

Starting Point: Suppose x is a particular but arbitrarily chosen element of A .

To Show: Therefore, x is an element of B .

b. **Proof:**

Suppose x is a particular but arbitrarily chosen element of A .

[We must show that $x \in B$. By definition of B , this means we must show that $x = 3 \cdot (\text{some integer})$.]

By definition of A , there is an integer, say r , such that $x = 6r + 12$.

[Given that $x = 6r + 12$, can we express x as $3 \cdot (\text{some integer})$? That is, does $6r + 12 = 3 \cdot (\text{some integer})$? Yes, $6r + 12 = 3 \cdot (2r + 4)$.]

Let $s = 2r + 4$.

[We must check that s is an integer.]

Then s is an integer because products and sums of integers are integers, and so $3s \in B$ by definition of B .

[Now we must check that $x = 3s$.]

Also $3s = 3(2r + 4) = 6r + 12 = x$,

Thus, by definition of B , x is an element of B ,

[as was to be shown].

- c. To disprove a statement means to show that it is false. And to show that $B \subseteq A$ is false, you must find an element of B that is not an element of A . By the definitions of A and B , this means that you must find an integer x of the form $3 \cdot (\text{some integer})$ that cannot be written in the form $6 \cdot (\text{some integer}) + 12$. A little experimentation reveals that various numbers work. For instance, you could let $x = 3$. Then $x \in B$ because $3 = 3 \cdot 1$, but $x \notin A$ because there is no integer r such that $3 = 6r + 12$. For if there were such an integer, then

Note Recall that the notation $P(x) \Rightarrow Q(x)$ means that every element that makes $P(x)$ true also makes $Q(x)$ true.

$$\begin{aligned} 6r + 12 &= 3 && \text{by assumption} \\ \Rightarrow 2r + 4 &= 1 && \text{by dividing both sides by 3} \\ \Rightarrow 2r &= -3 && \text{by subtracting 4 from both sides} \\ \Rightarrow r &= -3/2 && \text{by dividing both sides by 2.} \end{aligned}$$

But $-3/2$ is not an integer. Thus $3 \in B$ whereas $3 \notin A$, and so $B \not\subseteq A$. ■

Set Equality

Recall that by the axiom of extension, sets A and B are equal if, and only if, they have exactly the same elements. We restate this as a definition that uses the language of subsets.

Definition

Given sets A and B , A **equals** B , written $A = B$, if, and only if, every element of A is in B and every element of B is in A .

Symbolically:

$$A = B \iff A \subseteq B \text{ and } B \subseteq A.$$

This version of the definition of equality implies the following:

To know that a set A equals a set B , you must know that $A \subseteq B$ and you must also know that $B \subseteq A$.

Example 6.1.3

Set Equality

Define sets A and B as follows:

$$A = \{m \in \mathbf{Z} \mid m = 2a \text{ for some integer } a\}$$

$$B = \{n \in \mathbf{Z} \mid n = 2b - 2 \text{ for some integer } b\}.$$

Is $A = B$?

Solution Yes. To prove this, both subset relations $A \subseteq B$ and $B \subseteq A$ must be proved.

Part 1, Proof That $A \subseteq B$:

Suppose x is a particular but arbitrarily chosen element of A .

[We must show that $x \in B$. By definition of B , this means we must show that $x = 2 \cdot (\text{some integer}) - 2$.]

By definition of A , there is an integer, say a , such that $x = 2a$.

[Given that $x = 2a$, can x also be expressed as $2 \cdot (\text{some integer}) - 2$? In other words, is there an integer—say, b —such that $2a = 2b - 2$? Solve for b to obtain $b = (2a + 2)/2 = a + 1$. Check to see if this works.]

Let $b = a + 1$.

[First check that b is an integer.]

Then b is an integer because it is a sum of integers.

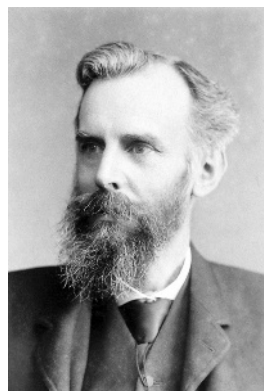
[Then check that $x = 2b - 2$.]

Also, $2b - 2 = 2(a + 1) - 2 = 2a + 2 - 2 = 2a = x$.

Thus, by definition of B , x is an element of B

[as was to be shown].

Part 2, Proof That $B \subseteq A$: This part of the proof is left as exercise 2 at the end of this section. ■



Pictures From History/Newscom

John Venn
(1834–1923)

Venn Diagrams

If sets A and B are represented as regions in the plane, relationships between A and B can be represented by pictures called **Venn diagrams**, which were introduced by the British mathematician John Venn in 1881. For instance, the relationship $A \subseteq B$ can be pictured in one of two ways, as shown in Figure 6.1.1.

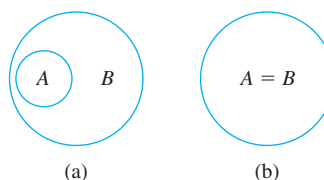


FIGURE 6.1.1 $A \subseteq B$

The relationship $A \not\subseteq B$ can be represented in three different ways, as shown in Figure 6.1.2.

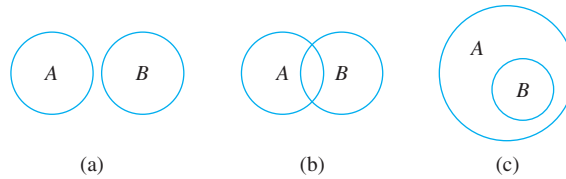


FIGURE 6.1.2 $A \not\subseteq B$

If we allow the possibility that some subregions of Venn diagrams do not contain any points, then in Figure 6.1.1 diagram (b) can be viewed as a special case of diagram (a) by imagining that the part of B outside A does not contain any points. Similarly, diagrams (a) and (c) of Figure 6.1.2 can be viewed as special cases of diagram (b). To obtain (a) from (b), imagine that the region of overlap between A and B does not contain any points. To obtain (c), imagine that the part of B that lies outside A does not contain any points. However, in all three diagrams it would be necessary to specify that there is a point in A that is not in B .

Example 6.1.4 Relations among Sets of Numbers



FIGURE 6.1.3

Since $\mathbf{Z}$, $\mathbf{Q}$, and $\mathbf{R}$ denote the sets of integers, rational numbers, and real numbers, respectively, then $\mathbf{Z}$ is a subset of $\mathbf{Q}$ because every integer is rational (any integer n can be written in the form $\frac{n}{1}$), and $\mathbf{Q}$ is a subset of $\mathbf{R}$ because every rational number is real (any rational number can be represented as a length on the number line). $\mathbf{Z}$ is a proper subset of $\mathbf{Q}$ because there are rational numbers that are not integers (for example, $\frac{1}{2}$), and $\mathbf{Q}$ is a proper subset of $\mathbf{R}$ because there are real numbers that are not rational (for example, $\sqrt{2}$). These relationships are shown diagrammatically in Figure 6.1.3. ■

Operations on Sets

Most mathematical discussions are carried on within some context. For example, in a certain situation all sets being considered might be sets of real numbers. In such a situation, the set of real numbers would be called a **universal set** or a **universe of discourse** for the discussion.

Definition

Let A and B be subsets of a universal set U .

1. The **union** of A and B , denoted $A \cup B$, is the set of all elements that are in at least one of A or B .
2. The **intersection** of A and B , denoted $A \cap B$, is the set of all elements that are common to both A and B .
3. The **difference** of B minus A (or **relative complement** of A in B), denoted $B - A$, is the set of all elements that are in B and not A .
4. The **complement** of A , denoted A^c , is the set of all elements in U that are not in A .

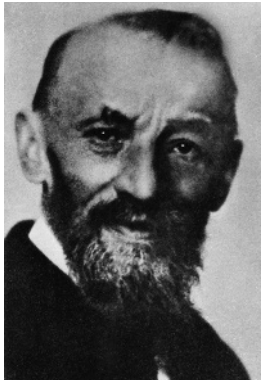
Symbolically:

$$A \cup B = \{x \in U \mid x \in A \text{ or } x \in B\}$$

$$A \cap B = \{x \in U \mid x \in A \text{ and } x \in B\}$$

$$B - A = \{x \in U \mid x \in B \text{ and } x \notin A\}$$

$$A^c = \{x \in U \mid x \notin A\}.$$



Giuseppe Peano
(1858–1932)

The symbols $\in$, $\cup$, and $\cap$ were introduced in 1889 by the Italian mathematician Giuseppe Peano.

Venn diagram representations for union, intersection, difference, and complement are shown in Figure 6.1.4.

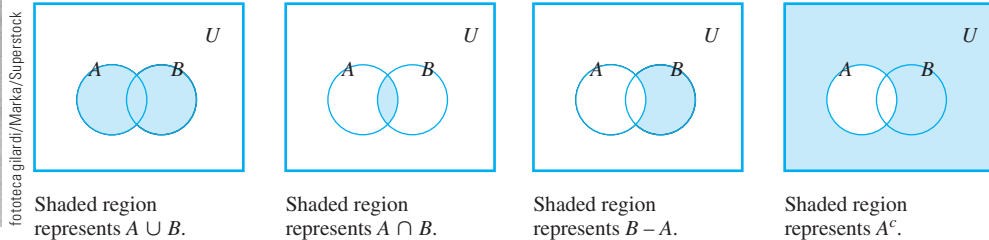


FIGURE 6.1.4

Example 6.1.5 Unions, Intersections, Differences, and Complements

Let the universal set be the set $U = \{a, b, c, d, e, f, g\}$, and let $A = \{a, c, e, g\}$ and $B = \{d, e, f, g\}$. Find $A \cup B$, $A \cap B$, $B - A$, and A^c .

Solution

$$\begin{aligned} A \cup B &= \{a, c, d, e, f, g\} & A \cap B &= \{e, g\} \\ B - A &= \{d, f\} & A^c &= \{b, d, f\} \end{aligned}$$

There is a convenient notation for subsets of real numbers that are intervals.

Interval Notation

Given real numbers a and b with $a \leq b$:

$$\begin{aligned} (a, b) &= \{x \in \mathbf{R} \mid a < x < b\} & [a, b] &= \{x \in \mathbf{R} \mid a \leq x \leq b\} \\ (a, b] &= \{x \in \mathbf{R} \mid a < x \leq b\} & [a, b) &= \{x \in \mathbf{R} \mid a \leq x < b\}. \end{aligned}$$

The symbols ∞ and $-\infty$ are used to indicate intervals that are unbounded either on the right or on the left:

$$\begin{aligned} (a, \infty) &= \{x \in \mathbf{R} \mid x > a\} & [a, \infty) &= \{x \in \mathbf{R} \mid x \geq a\} \\ (-\infty, b) &= \{x \in \mathbf{R} \mid x < b\} & (-\infty, b] &= \{x \in \mathbf{R} \mid x \leq b\}. \end{aligned}$$

Note The symbol ∞ does not represent a number. It just indicates the unboundedness of the interval.

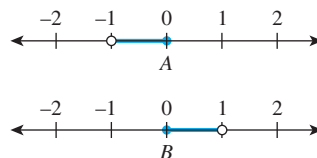
Although the notation for the interval (a, b) is identical to the notation for the ordered pair (a, b) , context makes it unlikely that the two will be confused.

Example 6.1.6 An Example with Intervals

Let the universal set be $\mathbf{R}$, the set of all real numbers, and let

$$A = (-1, 0] = \{x \in \mathbf{R} \mid -1 < x \leq 0\} \quad \text{and} \quad B = [0, 1) = \{x \in \mathbf{R} \mid 0 \leq x < 1\}.$$

These sets are shown on the number lines below.



Find $A \cup B$, $A \cap B$, $B - A$, and A^c .

Solution

$$A \cup B = \{x \in \mathbf{R} \mid x \in (-1, 0] \text{ or } x \in [0, 1)\} = \{x \in \mathbf{R} \mid x \in (-1, 1)\} = (-1, 1).$$

$$A \cap B = \{x \in \mathbf{R} \mid x \in (-1, 0] \text{ and } x \in [0, 1)\} = \{0\}.$$

$$B - A = \{x \in \mathbf{R} \mid x \in [0, 1) \text{ and } x \notin (-1, 0]\} = \{x \in \mathbf{R} \mid 0 < x < 1\} = (0, 1)$$

$$A^c = \{x \in \mathbf{R} \mid \text{it is not the case that } x \in (-1, 0]\}$$

$$= \{x \in \mathbf{R} \mid \text{it is not the case that } (-1 < x \text{ and } x \leq 0)\} \quad \text{by definition of the double inequality}$$

$$= \{x \in \mathbf{R} \mid x \leq -1 \text{ or } x > 0\} = (-\infty, -1] \cup (0, \infty) \quad \text{by De Morgan's law}$$

The definitions of unions and intersections for more than two sets are very similar to the definitions for two sets.

Note $\bigcup_{i=0}^n A_i$ is read “the union of the A -sub- i from i equals zero to n .”

Definition

Unions and Intersections of an Indexed Collection of Sets

Given sets $A_0, A_1, A_2, \dots$ that are subsets of a universal set U and given a nonnegative integer n ,

$$\bigcup_{i=0}^n A_i = \{x \in U \mid x \in A_i \text{ for at least one } i = 0, 1, 2, \dots, n\}$$

$$\bigcup_{i=0}^{\infty} A_i = \{x \in U \mid x \in A_i \text{ for at least one nonnegative integer } i\}$$

$$\bigcap_{i=0}^n A_i = \{x \in U \mid x \in A_i \text{ for every } i = 0, 1, 2, \dots, n\}$$

$$\bigcap_{i=0}^{\infty} A_i = \{x \in U \mid x \in A_i \text{ for every nonnegative integer } i\}.$$

An alternative notation for $\bigcup_{i=0}^n A_i$ is $A_0 \cup A_1 \cup \dots \cup A_n$, and an alternative notation for $\bigcap_{i=0}^n A_i$ is $A_0 \cap A_1 \cap \dots \cap A_n$.

Example 6.1.7

Finding Unions and Intersections of More than Two Sets

For each positive integer i , let $A_i = \{x \in \mathbf{R} \mid -\frac{1}{i} < x < \frac{1}{i}\} = (-\frac{1}{i}, \frac{1}{i})$.

- a. Find $A_1 \cup A_2 \cup A_3$ and $A_1 \cap A_2 \cap A_3$. b. Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$.

Solution

- a. $A_1 \cup A_2 \cup A_3 = \{x \in \mathbf{R} \mid x \text{ is in at least one of the intervals } (-1, 1),$
 $\text{or } (-\frac{1}{2}, \frac{1}{2}), \text{ or } (-\frac{1}{3}, \frac{1}{3})\}$
 $= \{x \in \mathbf{R} \mid -1 < x < 1\} \quad \text{because all the elements in } (-\frac{1}{2}, \frac{1}{2})$
 $= (-1, 1) \quad \text{and } (-\frac{1}{3}, \frac{1}{3}) \text{ are in } (-1, 1)$

$$\begin{aligned}
A_1 \cap A_2 \cap A_3 &= \{x \in \mathbf{R} \mid x \text{ is in all of the intervals } (-1, 1), \text{ and } (-\tfrac{1}{2}, \tfrac{1}{2}), \text{ and } (-\tfrac{1}{3}, \tfrac{1}{3})\} \\
&= \{x \in \mathbf{R} \mid -\tfrac{1}{3} < x < \tfrac{1}{3}\} && \text{because } (-\tfrac{1}{3}, \tfrac{1}{3}) \subseteq (-\tfrac{1}{2}, \tfrac{1}{2}) \subseteq (-1, 1) \\
&= (-\tfrac{1}{3}, \tfrac{1}{3})
\end{aligned}$$

$$\begin{aligned}
\text{b. } \bigcup_{i=1}^{\infty} A_i &= \{x \in \mathbf{R} \mid x \text{ is in at least one of the intervals } (-\tfrac{1}{i}, \tfrac{1}{i}), \\
&\hspace{15em} \text{where } i \text{ is a positive integer}\} \\
&= \{x \in \mathbf{R} \mid -1 < x < 1\} && \text{because all the elements in every interval} \\
&= (-1, 1) && (-\tfrac{1}{i}, \tfrac{1}{i}) \text{ are in } (-1, 1)
\end{aligned}$$

$$\begin{aligned}
\bigcap_{i=1}^{\infty} A_i &= \{x \in \mathbf{R} \mid x \text{ is in all of the intervals } (-\tfrac{1}{i}, \tfrac{1}{i}), \text{ where } i \text{ is a positive integer}\} \\
&= \{0\} && \text{because the only element in every interval is } 0
\end{aligned}$$

The Empty Set

We have stated that a set is defined by the elements that compose it. This being so, can there be a set that has no elements? It turns out that it is convenient to allow such a set. Otherwise, every time we wanted to take the intersection of two sets or to define a set by specifying a property, we would have to check that the result had elements and hence could be defined as a set. For example, if $A = \{1, 3\}$ and $B = \{2, 4\}$, then $A \cap B$ has no elements. Neither does $\{x \in \mathbf{R} \mid x^2 = -1\}$ because no real numbers have negative squares.

It may seem strange to talk about a set with no elements, but it often happens in mathematics that the definitions formulated to fit one set of circumstances are satisfied by some extreme cases not originally anticipated. Yet changing the definitions to exclude those cases would seriously undermine the simplicity and elegance of the theory taken as a whole.

In Section 6.2 we will show that there is only one set with no elements. Because it is unique, we can give it a special name. We call it the **empty set** (or **null set**) and denote it by the symbol $\emptyset$. Thus $\{1, 3\} \cap \{2, 4\} = \emptyset$ and $\{x \in \mathbf{R} \mid x^2 = -1\} = \emptyset$.

Example 6.1.8 A Set with No Elements

Describe the following sets.

$$\text{a. } D = \{x \in \mathbf{R} \mid 3 < x < 2\}. \qquad \text{b. } E = \{x \in \mathbf{Z} \mid 2 < x < 3\}.$$

Solution:

- Recall that $a < x < b$ means that $a < x$ and $x < b$. So D consists of all real numbers that are both greater than 3 and less than 2. Since there are no such numbers, D has no elements and thus $D = \emptyset$.
- E is the set of all integers that are both greater than 2 and less than 3. Since no integers satisfy this condition, E has no elements, and so $E = \emptyset$. ■

Partitions of Sets

In many applications of set theory, sets are divided into nonoverlapping (or *disjoint*) pieces. Such a division is called a *partition*.

Definition

Two sets are called **disjoint** if, and only if, they have no elements in common. Symbolically:

$$A \text{ and } B \text{ are disjoint} \Leftrightarrow A \cap B = \emptyset.$$

Example 6.1.9**Disjoint Sets**

Let $A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$. Are A and B disjoint?

Solution Yes. By inspection A and B have no elements in common, or, in other words, $\{1, 3, 5\} \cap \{2, 4, 6\} = \emptyset$. ■

Definition

Sets $A_1, A_2, A_3, \dots$ are **mutually disjoint** (or **pairwise disjoint** or **nonoverlapping**) if, and only if, no two sets A_i and A_j with distinct subscripts have any elements in common. More precisely, for all integers i and $j = 1, 2, 3, \dots$

$$A_i \cap A_j = \emptyset \text{ whenever } i \neq j.$$

Example 6.1.10**Mutually Disjoint Sets**

- Let $A_1 = \{3, 5\}$, $A_2 = \{1, 4, 6\}$, and $A_3 = \{2\}$. Are A_1, A_2 , and A_3 mutually disjoint?
- Let $B_1 = \{2, 4, 6\}$, $B_2 = \{3, 7\}$, and $B_3 = \{4, 5\}$. Are B_1, B_2 , and B_3 mutually disjoint?

Solution

- Yes. A_1 and A_2 have no elements in common, A_1 and A_3 have no elements in common, and A_2 and A_3 have no elements in common.
- No. B_1 and B_3 both contain 4. ■

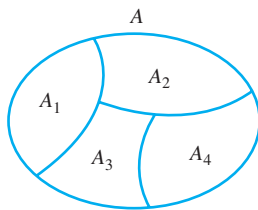


FIGURE 6.1.5 A Partition of a Set

Suppose A, A_1, A_2, A_3 , and A_4 are the sets of points represented by the regions shown in Figure 6.1.5. Then A_1, A_2, A_3 , and A_4 are subsets of A , and $A = A_1 \cup A_2 \cup A_3 \cup A_4$. Suppose further that boundaries are assigned to the regions representing A_1, A_2, A_3 , and A_4 in such a way that these sets are mutually disjoint. Then A is called a *union of mutually disjoint subsets*, and the collection of sets $\{A_1, A_2, A_3, A_4\}$ is said to be a *partition* of A .

Definition

A finite or infinite collection of nonempty sets $\{A_1, A_2, A_3, \dots\}$ is a **partition** of a set A if, and only if,

1. A is the union of all the A_i ;
2. the sets $A_1, A_2, A_3, \dots$ are mutually disjoint.

Example 6.1.11**Partitions of Sets**

- Let $A = \{1, 2, 3, 4, 5, 6\}$, $A_1 = \{1, 2\}$, $A_2 = \{3, 4\}$, and $A_3 = \{5, 6\}$. Is $\{A_1, A_2, A_3\}$ a partition of A ?

b. Let $\mathbf{Z}$ be the set of all integers and let

$$\begin{aligned} T_0 &= \{n \in \mathbf{Z} \mid n = 3k, \text{ for some integer } k\}, \\ T_1 &= \{n \in \mathbf{Z} \mid n = 3k + 1, \text{ for some integer } k\}, \text{ and} \\ T_2 &= \{n \in \mathbf{Z} \mid n = 3k + 2, \text{ for some integer } k\}. \end{aligned}$$

Is $\{T_0, T_1, T_2\}$ a partition of $\mathbf{Z}$?

Solution

- a. Yes. By inspection, $A = A_1 \cup A_2 \cup A_3$ and the sets A_1 , A_2 , and A_3 are mutually disjoint.
 b. Yes. By the quotient-remainder theorem, every integer n can be represented in exactly one of the three forms

$$n = 3k \quad \text{or} \quad n = 3k + 1 \quad \text{or} \quad n = 3k + 2,$$

for some integer k . This implies that no integer can be in any two of the sets T_0 , T_1 , or T_2 . So T_0 , T_1 , and T_2 are mutually disjoint. The theorem also implies that every integer is in one of the sets T_0 , T_1 , or T_2 . So $\mathbf{Z} = T_0 \cup T_1 \cup T_2$. ■

Power Sets

There are various situations in which it is useful to consider the set of all subsets of a particular set. The **power set axiom** guarantees that this is a set.

Definition

Given a set A , the **power set** of A , denoted $\mathcal{P}(A)$, is the set of all subsets of A .

Example 6.1.12 Power Set of a Set

Find the power set of the set $\{x, y\}$. That is, find $\mathcal{P}(\{x, y\})$.

Solution $\mathcal{P}(\{x, y\})$ is the set of all subsets of $\{x, y\}$. In Section 6.2 we will show that $\emptyset$ is a subset of every set, and so $\emptyset \in \mathcal{P}(\{x, y\})$. Also any set is a subset of itself, so $\{x, y\} \in \mathcal{P}(\{x, y\})$. The only other subsets of $\{x, y\}$ are $\{x\}$ and $\{y\}$, so

$$\mathcal{P}(\{x, y\}) = \{\emptyset, \{x\}, \{y\}, \{x, y\}\} \quad \blacksquare$$

An Algorithm to Check Whether One Set Is a Subset of Another (Optional)

You may get some additional insight into the concept of subset by considering an algorithm for checking whether one finite set is a subset of another. Order the elements of both sets and successively compare each element of the first set with each element of the second set. If some element of the first set is not found to equal any element of the second, then the first set is not a subset of the second. But if each element of the first set is found to equal an element of the second set, then the first set is a subset of the second. The following algorithm formalizes this reasoning.

Algorithm 6.1.1 Testing Whether $A \subseteq B$

[The input sets A and B are represented as one-dimensional arrays $a[1], a[2], \dots, a[m]$ and $b[1], b[2], \dots, b[n]$, respectively. Starting with $a[1]$ and for each successive $a[i]$ in A , a check is made to see whether $a[i]$ is in B . To do this, $a[i]$ is compared to successive elements of B . If $a[i]$ is not equal to any element of B , then the output string, called answer, is given the value " $A \not\subseteq B$." If $a[i]$ equals some element of B , the next successive element in A is checked to see whether it is in B . If every successive element of A is found to be in B , then the answer never changes from its initial value " $A \subseteq B$."]

Input: m [a positive integer], $a[1], a[2], \dots, a[m]$ [a one-dimensional array representing the set A], n [a positive integer], $b[1], b[2], \dots, b[n]$ [a one-dimensional array representing the set B]

Algorithm Body:

```

 $i := 1$ , answer := " $A \subseteq B$ "
while ( $i \leq m$  and answer = " $A \subseteq B$ ")
     $j := 1$ , found := "no"
    while ( $j \leq n$  and found = "no")
        if  $a[i] = b[j]$  then found := "yes"
         $j := j + 1$ 
    end while
    [If found has not been given the value "yes" when execution reaches this point,
    then  $a[i] \notin B$ .]
    if found = "no" then answer := " $A \not\subseteq B$ "
     $i := i + 1$ 
end while
    
```

Output: answer [a string]

Example 6.1.13

Tracing Algorithm 6.1.1

Trace the action of Algorithm 6.1.1 on the variables i , j , found, and answer for $m = 3$, $n = 4$, and sets A and B represented as the arrays $a[1] = u$, $a[2] = v$, $a[3] = w$, $b[1] = w$, $b[2] = x$, $b[3] = y$, and $b[4] = u$.

Solution

i	1					2					3
j	1	2	3	4	5	1	2	3	4	5	
found	no			yes		no					
answer	$A \subseteq B$									$A \not\subseteq B$	

In the exercises at the end of this section, you are asked to write an algorithm to check whether a given element is in a given set. To do this, you can represent the set as a one-dimensional array and compare the given element with successive elements of the array to determine whether the two elements are equal. If they are, then the element is in the set; if the given element does not equal any element of the array, then the element is not in the set.

TEST YOURSELF

Answers to Test Yourself questions are located at the end of each section.

- The notation $A \subseteq B$ is read “_____” and means that _____.
- To use an element argument for proving that a set X is a subset of a set Y , you suppose that _____ and show that _____.
- To disprove that a set X is a subset of a set Y , you show that there is _____.
- An element x is in $A \cup B$ if, and only if, _____.
- An element x is in $A \cap B$ if, and only if, _____.
- An element x is in $B - A$ if, and only if, _____.
- An element x is in A^c if, and only if, _____.
- The empty set is a set with _____.
- The power set of a set A is _____.
- Sets A and B are disjoint if, and only if, _____.
- A collection of nonempty sets $A_1, A_2, A_3, \dots$ is a partition of a set A if, and only if, _____.

EXERCISE SET 6.1*

- In each of (a)–(f), answer the following questions: Is $A \subseteq B$? Is $B \subseteq A$? Is either A or B a proper subset of the other?
 - $A = \{2, \{2\}, (\sqrt{2})^2\}$, $B = \{2, \{2\}, \{\{2\}\}\}$
 - $A = \{3, \sqrt{5^2 - 4^2}, 24 \bmod 7\}$, $B = \{8 \bmod 5\}$
 - $A = \{\{1, 2\}, \{2, 3\}\}$, $B = \{1, 2, 3\}$
 - $A = \{a, b, c\}$, $B = \{\{a\}, \{b\}, \{c\}\}$
 - $A = \{\sqrt{16}, \{4\}\}$, $B = \{4\}$
 - $A = \{x \in \mathbf{R} \mid \cos x \in \mathbf{Z}\}$,
 $B = \{x \in \mathbf{R} \mid \sin x \in \mathbf{Z}\}$
- Complete the proof from Example 6.1.3: Prove that $B \subseteq A$ where

$$A = \{m \in \mathbf{Z} \mid m = 2a \text{ for some integer } a\}$$
 and

$$B = \{n \in \mathbf{Z} \mid n = 2b - 2 \text{ for some integer } b\}$$
- Let sets R , S , and T be defined as follows:

$$R = \{x \in \mathbf{Z} \mid x \text{ is divisible by } 2\}$$

$$S = \{y \in \mathbf{Z} \mid y \text{ is divisible by } 3\}$$

$$T = \{z \in \mathbf{Z} \mid z \text{ is divisible by } 6\}.$$
 Prove or disprove each of the following statements.
 - $R \subseteq T$
 - $T \subseteq R$
 - $T \subseteq S$
- Let $A = \{n \in \mathbf{Z} \mid n = 5r \text{ for some integer } r\}$ and $B = \{m \in \mathbf{Z} \mid m = 20s \text{ for some integer } s\}$. Prove or disprove each of the following statements.
 - $A \subseteq B$
 - $B \subseteq A$
- Let $C = \{n \in \mathbf{Z} \mid n = 6r - 5 \text{ for some integer } r\}$ and $D = \{m \in \mathbf{Z} \mid m = 3s + 1 \text{ for some integer } s\}$. Prove or disprove each of the following statements.
 - $C \subseteq D$
 - $D \subseteq C$
- Let $A = \{x \in \mathbf{Z} \mid x = 5a + 2 \text{ for some integer } a\}$, $B = \{y \in \mathbf{Z} \mid y = 10b - 3 \text{ for some integer } b\}$, and $C = \{z \in \mathbf{Z} \mid z = 10c + 7 \text{ for some integer } c\}$. Prove or disprove each of the following statements.
 - $A \subseteq B$
 - $B \subseteq A$
 - $B = C$
- Write in words how to read each of the following out loud. Then write each set using the symbols for union, intersection, set difference, or set complement.
 - $\{x \in U \mid x \in A \text{ and } x \in B\}$
 - $\{x \in U \mid x \in A \text{ or } x \in B\}$
 - $\{x \in U \mid x \in A \text{ and } x \notin B\}$
 - $\{x \in U \mid x \notin A\}$
- Complete the following sentences without using the symbols $\cup$, $\cap$, or $-$.
 - $x \notin A \cup B$ if, and only if, _____.
 - $x \notin A \cap B$ if, and only if, _____.
 - $x \notin A - B$ if, and only if, _____.

*For exercises with blue numbers or letters, solutions are given in Appendix B. The symbol **H** indicates that only a hint or a partial solution is given. The symbol * signals that an exercise is more challenging than usual.

ANSWERS FOR TEST YOURSELF

1. the set A is a subset of the set B ; for every x , if $x \in A$ then $x \in B$ (Or: every element of A is also an element of B)
2. x is any [particular but arbitrarily chosen] element of X ; x is an element of Y
3. an element in X that is not in Y
4. x is in A or x is in B (Or: x is in at least one of the sets A and B)
5. x is in A and x is in B (Or: x is in both A and B)
6. x is in B and x is not in A
7. x is in the universal set and is not in A
8. no elements
9. the set of all subsets of A
10. $A \cap B = \emptyset$ (Or: A and B have no elements in common)
11. A is the union of all the sets $A_1, A_2, A_3, \dots$ and $A_i \cap A_j = \emptyset$ whenever $i \neq j$

6.2 Properties of Sets

... only the last line is a genuine theorem here—everything else is in the fantasy.

—Douglas Hofstadter, *Gödel, Escher, Bach*, 1979

It is possible to list many relations involving unions, intersections, complements, and differences of sets. Some of these are true for all sets, whereas others fail to hold in some cases. In this section we show how to establish basic set properties using *element arguments*, and we discuss a variation used to prove that a set is empty. In the next section we will show how to disprove a proposed set property by constructing a counterexample and how to use an algebraic technique to derive new set properties from set properties already known to be true.

We begin by listing some set properties that involve subset relations. As you read them, keep in mind that the operations of union, intersection, and difference take precedence over set inclusion. Thus, for example, $A \cap B \subseteq C$ means $(A \cap B) \subseteq C$.

Theorem 6.2.1 Some Subset Relations

1. *Inclusion of Intersection:* For all sets A and B ,

$$(a) A \cap B \subseteq A \quad \text{and} \quad (b) A \cap B \subseteq B.$$

2. *Inclusion in Union:* For all sets A and B ,

$$(a) A \subseteq A \cup B \quad \text{and} \quad (b) B \subseteq A \cup B.$$

3. *Transitive Property of Subsets:* For all sets A , B , and C ,

$$\text{if } A \subseteq B \text{ and } B \subseteq C, \text{ then } A \subseteq C.$$

The conclusion of each part of Theorem 6.2.1 states that one set X is a subset of another set Y and so to prove them, you suppose that x is any [particular but arbitrarily chosen] element of X , and you show that x is an element of Y .

In most proofs of set properties, the secret of getting from the assumption that x is in X to the conclusion that x is in Y is to think of the definitions of basic set operations in terms of how they act on elements, that is, in procedural terms. For example, the union of sets X and Y , $X \cup Y$, is defined as

$$X \cup Y = \{x \mid x \in X \quad \text{or} \quad x \in Y\}.$$

This means that any time you know an element x is in $X \cup Y$, you can conclude that x must be in X or x must be in Y . Conversely, any time you know that a particular x is in some set X or is in some set Y , you can conclude that x is in $X \cup Y$. Thus, for any sets X and Y and any element x ,

$$x \in X \cup Y \quad \text{if, and only if,} \quad x \in X \quad \text{or} \quad x \in Y.$$

Procedural versions of the definitions of the other set operations are derived similarly and are summarized below.

Procedural Versions of Set Definitions

Let X and Y be subsets of a universal set U and suppose x and y are elements of U .

1.

$x \in X \cup Y \iff x \in X \text{ or } x \in Y$

2.

$x \in X \cap Y \iff x \in X \text{ and } x \in Y$

3.

$x \in X - Y \iff x \in X \text{ and } x \notin Y$

4.

$x \in X^c \iff x \notin X$

5.

$(x, y) \in X \times Y \iff x \in X \text{ and } y \in Y$

Proving a Subset Relation

Consider trying to prove Theorem 6.2.1(a): For all sets A and B , $A \cap B \subseteq A$. First notice that the statement is universal. It makes a claim about *all* sets A and B . So the proof has the following outline:

Starting Point: Suppose A and B are any [particular but arbitrarily chosen] sets.

To Show: $A \cap B \subseteq A$

Now to prove $A \cap B \subseteq A$ you must use the definition of subset. In other words, you must show that

$\forall x, \text{ if } x \in A \cap B, \text{ then } x \in A.$

This statement is also universal, and to prove it you use an element argument:

suppose x is any element in $A \cap B$
and
show that x is in A .

You can fill in the gap between “suppose” and the “show” by using the procedural version of the definition of intersection along with your knowledge of logic and the definition of subset. Examples 6.2.1(a) and 6.2.1(b) show proofs for Theorem 6.2.1(1)(a) and Theorem 6.2.1(2)(a). Each contains blanks to fill in with explanations or parts of proof steps.

Example 6.2.1 Fill in the Blanks for Proofs of Subset Relations

Fill in the blanks in the proofs shown below.

a. **Theorem 6.2.1(1)(a):** For all sets A and B , $A \cap B \subseteq A$.

Proof:

Statement	Explanation
Suppose A and B are any sets.	starting point
We must show that $A \cap B \subseteq A$.	conclusion to be shown
Let x be any element in $A \cap B$.	start of an element argument
Then x is in A and x is in B .	<u>(i)</u>
In particular, x is in A .	<u>(ii)</u>
Thus every element in $A \cap B$ is in A .	because x could be any element of $A \cap B$
Therefore, $A \cap B \subseteq A$.	<u>(iii)</u>

b. **Theorem 6.2.1(2)(a):** For all sets A and B , $A \subseteq A \cup B$.

Proof:

Statement	Explanation
Suppose <u>(i)</u> .	starting point
We must show that $A \subseteq A \cup B$.	conclusion to be shown
<u>(ii)</u>	start of an element argument
Then the following statement is true: “ x is in A or x is in B .”	For an <i>or</i> statement to be true only <u>(iii)</u> component needs to be true.
Thus x is in $A \cup B$.	by definition of union
Hence every element in A is in <u>(iv)</u> .	because x could be any element of A
Therefore, <u>(v)</u> .	by definition of subset

Solution

- a. (i) by definition of intersection (ii) If an *and* statement is true, then each individual component is true. (iii) by definition of subset
- b. (i) A and B are any sets. (ii) Let x be any element in A . (Or: Suppose x is any element in A .) (iii) one (iv) $A \cup B$ (v) $A \subseteq A \cup B$ ■

In his book *Gödel, Escher, Bach*,* Douglas Hofstadter introduces the fantasy rule for mathematical proof. Hofstadter points out that when you start a mathematical argument with *if*, *let*, or *suppose*, you are stepping into a fantasy world where not only are all the facts of the real world true but whatever you are supposing is also true. Once you are in that world, you can suppose something else. That sends you into a subfantasy world where not only is everything in the fantasy world true but also the new thing you are supposing. Of course, you can continue stepping into new subfantasy worlds in this way indefinitely. You return one level closer to the real world each time you derive a conclusion that makes a whole if-then or universal statement true. Your aim in a proof is to continue deriving such conclusions until you return to the world from which you made your first supposition.

Occasionally, mathematical problems are stated in the following form:

Suppose (*statement 1*). Prove that (*statement 2*).

When this phrasing is used, the author intends the reader to add statement 1 to his or her general mathematical knowledge and not to make explicit reference to it in the proof. In Hofstadter's terms, the author invites the reader to enter a fantasy world where statement 1 is known to be true and to prove statement 2 in this fantasy world. Thus the solver of such a problem would begin a proof with the starting point for a proof of statement 2. Consider, for instance, the following restatement from Example 6.2.1(a):

Suppose A and B are arbitrarily chosen sets.

Prove that $A \cap B \subseteq A$.

The proof would begin “Suppose $x \in A \cap B$,” it being *understood* that sets A and B have already been chosen arbitrarily.

The proofs in Example 6.2.1 are called element arguments because they show one set to be a subset of another by demonstrating that every element in the one set is also an element in the other. In higher mathematics, element arguments are the standard method for establishing relations among sets. High-school students are often allowed to justify set properties

**Gödel, Escher, Bach: An Eternal Golden Braid* (New York: Basic Books, 1979).

by using Venn diagrams. This method is appealing, but for it to be mathematically rigorous may be more complicated than you might expect. Appropriate Venn diagrams can be drawn for two or three sets, but the verbal explanations needed to justify conclusions inferred from them are normally as long or longer than a straightforward element proof.

In general, Venn diagrams are not very helpful when the number of sets is four or more. For instance, if the requirement is made that a Venn diagram must show every possible intersection of the sets, it is impossible to draw a symmetric Venn diagram for four sets, or, in fact, for any nonprime number of sets. In 2002, computer scientists/mathematicians Carla Savage and Jerrold Griggs and undergraduate student Charles Killian solved a long-standing open problem by proving that it *is* possible to draw such a symmetric Venn diagram for any prime number of sets. For $n > 5$, however, the resulting pictures are extremely complicated! However, the fact that such symmetric diagrams exist has applications in the area of computer science called coding theory.

Set Identities

An **identity** is an equation that is universally true for all elements in some set. For example, the equation $a + b = b + a$ is an identity for real numbers because it is true for all real numbers a and b . The collection of set properties in the next theorem consists entirely of set identities. That is, they are equations that are true for all sets in some universal set.

Theorem 6.2.2 Set Identities

Let all sets referred to below be subsets of a universal set U .

1. *Commutative Laws:* For all sets A and B ,

$$(a) A \cup B = B \cup A \quad \text{and} \quad (b) A \cap B = B \cap A.$$

2. *Associative Laws:* For all sets A , B , and C ,

$$(a) (A \cup B) \cup C = A \cup (B \cup C) \quad \text{and}$$

$$(b) (A \cap B) \cap C = A \cap (B \cap C).$$

3. *Distributive Laws:* For all sets A , B , and C ,

$$(a) A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \quad \text{and}$$

$$(b) A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

4. *Identity Laws:* For every set A ,

$$(a) A \cup \emptyset = A \quad \text{and} \quad (b) A \cap U = A.$$

5. *Complement Laws:* For every set A ,

$$(a) A \cup A^c = U \quad \text{and} \quad (b) A \cap A^c = \emptyset.$$

6. *Double Complement Law:* For every set A ,

$$(A^c)^c = A.$$

7. *Idempotent Laws:* For every set A ,

$$(a) A \cup A = A \quad \text{and} \quad (b) A \cap A = A.$$

8. *Universal Bound Laws:* For every set A ,

$$(a) A \cup U = U \quad \text{and} \quad (b) A \cap \emptyset = \emptyset.$$

9. *De Morgan's Laws:* For all sets A and B ,

$$(a) (A \cup B)^c = A^c \cap B^c \quad \text{and} \quad (b) (A \cap B)^c = A^c \cup B^c.$$

10. *Absorption Laws:* For all sets A and B ,

$$(a) A \cup (A \cap B) = A \quad \text{and} \quad (b) A \cap (A \cup B) = A.$$

11. *Complements of U and $\emptyset$:*

$$(a) U^c = \emptyset \quad \text{and} \quad (b) \emptyset^c = U.$$

12. *Set Difference Law:* For all sets A and B ,

$$A - B = A \cap B^c.$$

The conclusion of each part of Theorem 6.2.2 is that one set equals another set. As we noted in Section 6.1,

Two sets are equal $\Leftrightarrow$ each is a subset of the other.

The method derived from this fact is the most basic way to prove equality of sets.

Basic Method for Proving That Sets Are Equal

Let sets X and Y be given. To prove that $X = Y$:

1. Prove that $X \subseteq Y$.
2. Prove that $Y \subseteq X$.

Example 6.2.2

Proof of a Distributive Law

Consider trying to prove that for all sets A , B , and C ,

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Solution The proof of this fact is somewhat more complicated than the proofs in Example 6.2.1, so we first derive its logical structure, then find the core arguments, and end with a formal proof as a summary. As in the subset relation examples, the statement to be proved is universal. Thus, by the method of generalizing from the generic particular, the proof has the following outline:

Starting Point: Suppose A , B , and C are arbitrarily chosen sets.

To Show: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Now two sets are equal if, and only if, each is a subset of the other. Hence, the following two statements must be proved:

$$A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$$

and

$$(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C).$$

Showing the first subset relation requires showing that

$$\forall x, \text{ if } x \in A \cup (B \cap C) \text{ then } x \in (A \cup B) \cap (A \cup C).$$

Showing the second containment requires showing that

$$\forall x, \text{ if } x \in (A \cup B) \cap (A \cup C) \text{ then } x \in A \cup (B \cap C).$$

Note that both of these statements are universal. So to prove the first containment, you

suppose you have any element x in $A \cup (B \cap C)$,

and then you

show that $x \in (A \cup B) \cap (A \cup C)$.

And to prove the second containment, you

suppose you have any element x in $(A \cup B) \cap (A \cup C)$,

and then you

show that $x \in A \cup (B \cap C)$.

In Figure 6.2.1, the structure of the proof is illustrated by the kind of diagram that is often used in connection with structured programs. The analysis in the diagram reduces the proof to two concrete tasks: filling in the steps indicated by dots in the two center boxes of Figure 6.2.1.

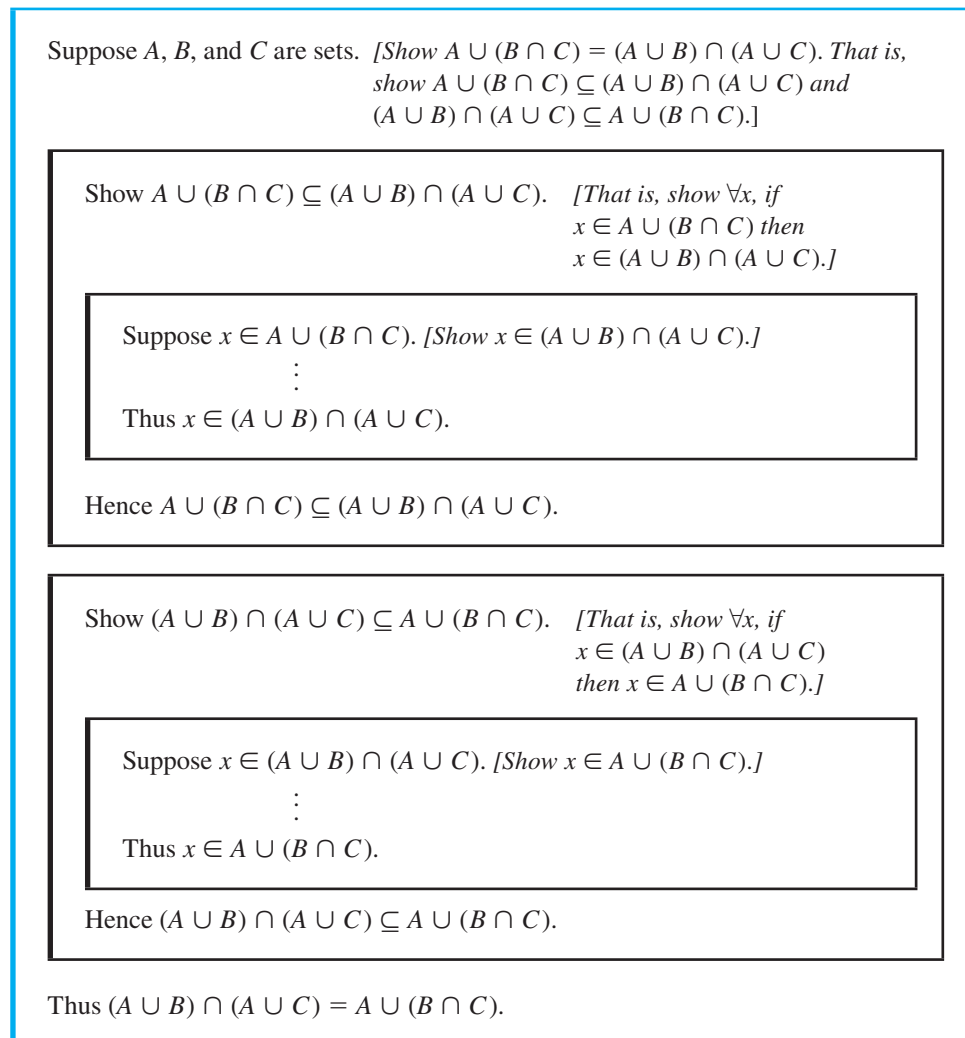


FIGURE 6.2.1

The following proof shows the steps for filling in the two innermost boxes of Figure 6.2.1. As you read it, notice how the procedural version of the definition of union is used. For example:

If you know that $x \in B$, then you can conclude that $x \in A \cup B$ because the statement “ $x \in A$ or $x \in B$ ” is true.

Similarly:

If you know that $x \in A$, then you can conclude that $x \in A \cup (B \cap C)$ because the statement “ $x \in A$ or $x \in B \cap C$ ” is true.

Also suppose you know that an element—say x —is in a union of two sets but you don’t know which set x is in. If you want to deduce a conclusion about x , you need to show that the conclusion follows regardless of which set x is in. So you need to break your argument into two cases: x is in the first set and x is in the second set.

The proof has a few blanks for you to fill in as practice for writing set theory proofs on your own. To make the proof more concise, the symbols $\cap$ and $\cup$ are used in place of the words “intersection” and “union,” respectively.

Theorem 6.2.2(3)(a) A Distributive Law for Sets

For all sets A , B , and C ,

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Proof: Suppose A , B , and C are any sets.

(1) Proof that $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$:

Let $x \in A \cup (B \cap C)$. [We must show that $x \in$ (a).]

By definition of $\cup$, $x \in$ (b) or $x \in B \cap C$.

Case 1 ($x \in A$): Since $x \in A$, then both statements $x \in A \cup B$ and $x \in A \cup C$ are true by definition of $\cup$. Hence $x \in (A \cup B) \cap (A \cup C)$ by definition of $\cap$.

Case 2 ($x \in B \cap C$): Since $x \in B \cap C$, then $x \in B$ and $x \in C$ by definition of $\cap$. Since $x \in B$, then $x \in A \cup B$ by definition of $\cup$. Similarly, since $x \in C$, then $x \in A \cup C$ by definition of $\cup$. Hence $x \in (A \cup B) \cap (A \cup C)$ by definition of $\cap$.

Therefore, in both cases 1 and 2, $x \in (A \cup B) \cap (A \cup C)$.

Because x could be any element in $A \cup (B \cap C)$, this argument shows that every element of $A \cup (B \cap C)$ is in $(A \cup B) \cap (A \cup C)$. Hence,

$$A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$$

by definition of (d).

(2) Proof that $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$:

Let $x \in (A \cup B) \cap (A \cup C)$. [We must show that $x \in A \cup (B \cap C)$.]

We consider the two cases: $x \in A$ and $x \notin A$.*

Case 1 ($x \in A$): In this case, because x is in A , we can conclude immediately that $x \in A \cup (B \cap C)$ by definition of $\cup$.

(continued on page 398)

*The reason to consider the two cases $x \in A$ and $x \notin A$ is that when $x \in (A \cup B) \cap (A \cup C)$, then, by definition of $\cap$, $x \in (A \cup B)$ and $x \in (A \cup C)$. Now one way for this statement to be true is for x to be in A , but, since that may not be the case, the proof must also consider the possibility that x is not in A .

Case 2 ($x \notin A$): In this case, we know that $x \in (A \cup B) \cap (A \cup C)$. Thus, by definition of (a), $x \in A \cup B$ and $x \in A \cup C$.

Because x is in $A \cup B$, then x is in at least one of A or B , and since x is not in A , then x is in B . Similarly, because x is in $A \cup C$, then x is in at least one of A or C , and since x is not in A , then x is in C .

It follows that $x \in B$ (b) $x \in C$, and, thus, $x \in B \cap C$ by definition of $\cap$.

Since $x \in B \cap C$, then by definition of (c), $x \in A \cup (B \cap C)$.

Therefore, in both cases 1 and 2, $x \in A \cup (B \cap C)$.

Because x could be any element in $(A \cup B) \cap (A \cup C)$, this argument shows that every element of $(A \cup B) \cap (A \cup C)$ is in $A \cup (B \cap C)$. Hence, $(A \cup B) \cap (A \cup C)$ (d) $A \cup (B \cap C)$. Thus, $(A \cup B) \cap (A \cup C)$ (d) $A \cup (B \cap C)$ by definition of subset.

(3) Conclusion: Since both subset relations have been proved, it follows, by definition of set equality, that (a).

Solution

- (1) a. $(A \cup B) \cap (A \cup C)$ b. A c. $\cap$ d. subset
 (2) a. $\cap$ b. and c. $\cup$ d. $\subseteq$
 (3) a. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

In the study of artificial intelligence, the types of reasoning used previously to derive the proof of the distributive law are called *forward chaining* and *backward chaining*. First what is to be shown is viewed as a goal to be reached starting from a certain initial position: the starting point. Analysis of this goal leads to the realization that if a certain job is accomplished, then the goal will be reached. Call this job subgoal 1: SG_1 . (For instance, if the goal is to show that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, then SG_1 would be to show that each set is a subset of the other.) Analysis of SG_1 shows that when yet another job is completed, SG_1 will be reached. Call this job subgoal 2: SG_2 . Continuing in this way, a chain of argument leading backward from the goal is constructed.

$$\boxed{\text{starting point}} \rightarrow SG_3 \rightarrow SG_2 \rightarrow SG_1 \rightarrow \boxed{\text{goal}}$$

At a certain point, backward chaining becomes difficult, but analysis of the current subgoal suggests it may be reachable by a direct line of argument, or, in other words, by forward chaining, beginning at the starting point. Using the information contained in the starting point, another piece of information, I_1 , is deduced; from that another piece of information, I_2 , is deduced; and so forth until finally one of the subgoals is reached. This completes the chain and proves the theorem. A completed chain is illustrated below.

$$\boxed{\text{starting point}} \rightarrow I_1 \rightarrow I_2 \rightarrow I_3 \rightarrow I_4 \rightarrow SG_3 \rightarrow SG_2 \rightarrow SG_1 \rightarrow \boxed{\text{goal}}$$

Since set complement is defined in terms of *not*, and since unions and intersections are defined in terms of *or* and *and*, it is not surprising that there are analogues of De Morgan's laws of logic for sets.

Example 6.2.3 Proof of a De Morgan's Law for Sets

Prove that for all sets A and B , $(A \cup B)^c = A^c \cap B^c$.

Solution As in previous examples, the statement to be proved is universal, and so the starting point of the proof and the conclusion to be shown are as follows:

Starting Point: Suppose A and B are arbitrarily chosen sets.

To Show: $(A \cup B)^c = A^c \cap B^c$

To do this, you must prove both that $(A \cup B)^c \subseteq A^c \cap B^c$ and that $A^c \cap B^c \subseteq (A \cup B)^c$. To prove the first subset relation means to show that

$$\forall x, \text{ if } x \in (A \cup B)^c \text{ then } x \in A^c \cap B^c.$$

And to prove the second subset relation means to show that

$$\forall x, \text{ if } x \in A^c \cap B^c \text{ then } x \in (A \cup B)^c.$$

Since each of these statements is universal and conditional, for the first subset relation, you

suppose $x \in (A \cup B)^c$,

and then you

show that $x \in A^c \cap B^c$.

And for the second subset relation, you

suppose $x \in A^c \cap B^c$,

and then you

show that $x \in (A \cup B)^c$.

To fill in the steps of these arguments, you use the procedural versions of the definitions of complement, union, and intersection, and at crucial points you use De Morgan's laws of logic.

Theorem 6.2.2(9)(a) A De Morgan's Law for Sets

For all sets A and B , $(A \cup B)^c = A^c \cap B^c$.

Proof: Suppose A and B are sets.

Proof that $(A \cup B)^c \subseteq A^c \cap B^c$:

[We must show that $\forall x$, if $x \in (A \cup B)^c$ then $x \in A^c \cap B^c$.]

Suppose $x \in (A \cup B)^c$. [We must show that $x \in A^c \cap B^c$.] By definition of complement,

$$x \notin A \cup B.$$

Now to say that $x \notin A \cup B$ means that

it is false that $(x \text{ is in } A \text{ or } x \text{ is in } B)$.

By De Morgan's laws of logic, this implies that

x is not in A and x is not in B ,

which can be written

$$x \notin A \quad \text{and} \quad x \notin B.$$

(continued on page 400)

Hence $x \in A^c$ and $x \in B^c$ by definition of complement. It follows, by definition of intersection, that $x \in A^c \cap B^c$ [as was to be shown]. So $(A \cup B)^c \subseteq A^c \cap B^c$ by definition of subset.

Proof that $A^c \cap B^c \subseteq (A \cup B)^c$:

[We must show that $\forall x$, if $x \in A^c \cap B^c$ then $x \in (A \cup B)^c$.]

Suppose $x \in A^c \cap B^c$. [We must show that $x \in (A \cup B)^c$.] By definition of intersection, $x \in A^c$ and $x \in B^c$, and by definition of complement,

$$x \notin A \quad \text{and} \quad x \notin B.$$

In other words,

$$x \text{ is not in } A \text{ and } x \text{ is not in } B.$$

By De Morgan's laws of logic this implies that

$$\text{it is false that } (x \text{ is in } A \text{ or } x \text{ is in } B),$$

which can be written

$$x \notin A \cup B$$

by definition of union. Hence, by definition of complement, $x \in (A \cup B)^c$ [as was to be shown]. It follows that $A^c \cap B^c \subseteq (A \cup B)^c$ by definition of subset.

Conclusion: Since both set containments have been proved, $(A \cup B)^c = A^c \cap B^c$ by definition of set equality.

The set property given in the next theorem says that if one set is a subset of another, then their intersection is the smaller of the two sets and their union is the larger of the two sets.

Theorem 6.2.3 Intersection and Union with a Subset

For any sets A and B , if $A \subseteq B$, then

$$(a) A \cap B = A \quad \text{and} \quad (b) A \cup B = B.$$

Proof:

Part (a): Suppose A and B are sets with $A \subseteq B$. To show part (a) we must show both that $A \cap B \subseteq A$ and that $A \subseteq A \cap B$. We already know that $A \cap B \subseteq A$ by the inclusion of intersection property. To show that $A \subseteq A \cap B$, let x be any element in A . [We must show that x is in $A \cap B$.] But, because of the hypothesis that $A \subseteq B$, we can conclude that x is also in B by definition of subset. Hence

$$x \in A \quad \text{and} \quad x \in B,$$

and thus

$$x \in A \cap B$$

by definition of intersection [as was to be shown].

Proof:

Part (b): The proof of part (b) is left as an exercise.

The Empty Set

In Section 6.1 we introduced the concept of a set with no elements and promised that in this section we would show that there is only one such set. To do so, we start with the most basic property of a set with no elements: It is a subset of every set. To see why this is true, just ask yourself, “Could it possibly be false? Could there be a set without elements that is *not* a subset of some given set?” The crucial fact is that the negation of a universal statement is existential: If a set B is not a subset of a set A , then there exists an element in B that is not in A . But if B has no elements, then no such element can exist.

Theorem 6.2.4 A Set with No Elements Is a Subset of Every Set

If E is a set with no elements and A is any set, then $E \subseteq A$.

Proof (by contradiction): Suppose not. *[We take the negation of the theorem and suppose it to be true.]* Suppose there exists a set E with no elements and a set A such that $E \not\subseteq A$. *[We must deduce a contradiction.]* Then there would be an element of E that is not an element of A *[by definition of subset]*. But there can be no such element since E has no elements. This is a contradiction. *[Hence the supposition that there are sets E and A , where E has no elements and $E \not\subseteq A$, is false, and so the theorem is true.]*

The truth of Theorem 6.2.4 can also be understood by appeal to the notion of vacuous truth. If E is a set with no elements and A is any set, then to say that $E \subseteq A$ is the same as saying that

$$\forall x, \text{ if } x \in E, \text{ then } x \in A.$$

But since E has no elements, this conditional statement is vacuously true.

How many sets with no elements are there? Only one.

Corollary 6.2.5 Uniqueness of the Empty Set

There is only one set with no elements.

Proof: Suppose E_1 and E_2 are both sets with no elements. By Theorem 6.2.4, $E_1 \subseteq E_2$ since E_1 has no elements. Also $E_2 \subseteq E_1$ since E_2 has no elements. Thus $E_1 = E_2$ by definition of set equality.

It follows from Corollary 6.2.5 that the set of all the pink elephants on earth is equal to the set of all the real numbers whose square is -1 because neither set has any elements! Since there is only one set with no elements, we are justified in calling it by a special name, the empty set (or null set) and in denoting it by the special symbol $\emptyset$.

Note that whereas $\emptyset$ is the set with no elements, the set $\{\emptyset\}$ has one element, the empty set. This is similar to the convention in the computer programming languages Lisp and Scheme, in which $()$ denotes the empty list and $(())$ denotes the list whose one element is the empty list.

Suppose you need to show that a certain set equals the empty set. By Corollary 6.2.5 it suffices to show that the set has no elements. For since there is only one set with no elements (namely $\emptyset$), if the given set has no elements, then it must equal $\emptyset$.

Element Method for Proving a Set Equals the Empty Set

To prove that a set X is equal to the empty set $\emptyset$, prove that X has no elements. To do this, suppose X has an element and derive a contradiction.

Example 6.2.4 Proving That a Set Is Empty

Prove Theorem 6.2.2(8)(b). That is, prove that for any set A , $A \cap \emptyset = \emptyset$.

Solution Let A be a [particular, but arbitrarily chosen] set. To show that $A \cap \emptyset = \emptyset$, it suffices to show that $A \cap \emptyset$ has no elements [by the element method for proving a set equals the empty set]. Suppose not. That is, suppose there is at least one element—say x —such that $x \in A \cap \emptyset$. Then, by definition of intersection, $x \in A$ and $x \in \emptyset$. In particular, $x \in \emptyset$. But this is impossible since $\emptyset$ has no elements. [This contradiction shows that the supposition that there is an element x in $A \cap \emptyset$ is false. So $A \cap \emptyset$ has no elements, as was to be shown.] Thus $A \cap \emptyset = \emptyset$. ■

Example 6.2.5 A Proof for a Conditional Statement

Prove that for all sets A , B , and C , if $A \subseteq B$ and $B \subseteq C^c$, then $A \cap C = \emptyset$.

Solution Because the statement to be proved is both universal and conditional, you start with the method of direct proof:

Suppose A , B , and C are arbitrarily chosen sets
that satisfy the condition: $A \subseteq B$ and $B \subseteq C^c$.

Show that $A \cap C = \emptyset$.

Since the conclusion to be shown is that a certain set is empty, you can use the principle for proving that a set equals the empty set. A complete proof is shown below.

Proposition 6.2.6

For all sets A , B , and C , if $A \subseteq B$ and $B \subseteq C^c$, then $A \cap C = \emptyset$.

Proof: Suppose A , B , and C are any sets such that $A \subseteq B$ and $B \subseteq C^c$. We must show that $A \cap C = \emptyset$. Suppose not. That is, suppose there is an element x in $A \cap C$. By definition of intersection, $x \in A$ and $x \in C$. Then, since $A \subseteq B$, $x \in B$ by definition of subset. Also, since $B \subseteq C^c$, then $x \in C^c$ by definition of subset again. It follows by definition of complement that $x \notin C$. Thus $x \in C$ and $x \notin C$, which is a contradiction. So the supposition that there is an element x in $A \cap C$ is false, and thus $A \cap C = \emptyset$ [as was to be shown]. ■

Example 6.2.6 A Generalized Distributive Law

Prove that for all sets A and $B_1, B_2, B_3, \dots, B_n$, where n is a positive integer,

$$A \cup \left(\bigcap_{i=1}^n B_i \right) = \bigcap_{i=1}^n (A \cup B_i).$$

Solution Compare this proof to the one given in Example 6.2.2. Although the notation is more complex, the basic ideas are the same.

Proof: Suppose A and $B_1, B_2, B_3, \dots, B_n$ are any sets and n is a positive integer.

Part 1, Proof that $A \cup \left(\bigcap_{i=1}^n B_i \right) \subseteq \bigcap_{i=1}^n (A \cup B_i)$:

Suppose x is any element in $A \cup \left(\bigcap_{i=1}^n B_i \right)$. [We must show that x is in $\bigcap_{i=1}^n (A \cup B_i)$.]

By definition of union, $x \in A$ or $x \in \bigcap_{i=1}^n B_i$.

Case 1, $x \in A$: In this case, it is true by definition of union that for every integer $i = 1, 2, \dots, n$, $x \in A \cup B_i$. Hence $x \in \bigcap_{i=1}^n (A \cup B_i)$.

Case 2, $x \in \bigcap_{i=1}^n B_i$: In this case, by definition of the general intersection, we have that for every integer $i = 1, 2, \dots, n$, $x \in B_i$. Hence, by definition of union, for every integer $i = 1, 2, \dots, n$, $x \in A \cup B_i$, and so, by definition of general intersection, $x \in \bigcap_{i=1}^n (A \cup B_i)$.

Thus, in either case, $x \in \bigcap_{i=1}^n (A \cup B_i)$ [as was to be shown].

Part 2, Proof that $\bigcap_{i=1}^n (A \cup B_i) \subseteq A \cup \left(\bigcap_{i=1}^n B_i \right)$:

Suppose x is any element in $\bigcap_{i=1}^n (A \cup B_i)$ [We must show that x is in $A \cup \bigcap_{i=1}^n B_i$.]

By definition of intersection, $x \in A \cup B_i$ for every integer $i = 1, 2, \dots, n$. Either $x \in A$ or $x \notin A$.

Case 1, $x \in A$: In this case, $x \in A \cup \left(\bigcap_{i=1}^n B_i \right)$ by definition of union.

Case 2, $x \notin A$: By definition of intersection, $x \in A \cup B_i$ for every integer $i = 1, 2, \dots, n$. Since $x \notin A$, x must be in each B_i for every integer $i = 1, 2, \dots, n$. Hence, by definition of

intersection, $x \in \bigcap_{i=1}^n B_i$, and so, by definition of union, $x \in A \cup \left(\bigcap_{i=1}^n B_i \right)$.

Conclusion: Since both set containments have been proved, it follows by definition of set equality that $A \cup \left(\bigcap_{i=1}^n B_i \right) = \bigcap_{i=1}^n (A \cup B_i)$. ■

TEST YOURSELF

1. To prove that a set X is a subset of a set $A \cap B$, you suppose that x is any element of X and you show that $x \in A$ _____ $x \in B$.
2. To prove that a set X is a subset of a set $A \cup B$, you suppose that x is any element of X and you show that $x \in A$ _____ $x \in B$.
3. To prove that a set $A \cup B$ is a subset of a set X , you start with any element x in $A \cup B$ and consider the two cases _____ and _____. You then show that in either case _____.
4. To prove that a set $A \cap B$ is a subset of a set X , you suppose that _____ and you show that _____.

5. To prove that a set X equals a set Y , you prove that _____ and that _____.
6. To prove that a set X does not equal a set Y , you need to find an element that is in _____ and not _____ or that is in _____ and not _____.

EXERCISE SET 6.2

1.

a. To say that an element is in $A \cap (B \cup C)$ means that it is in (1) _____ and in (2) _____.

b. To say that an element is in $(A \cap B) \cup C$ means that it is in (1) _____ or in (2) _____.

c. To say that an element is in $A - (B \cap C)$ means that it is in (1) _____ and not in (2) _____.

d. To prove that $(A \cup B) \cap C \subseteq A \cup (B \cap C)$, we suppose that x is any element in (1) _____. Then we must show that (2) _____.

e. If A , B , and C are any sets such that $B \subseteq C$, to prove that $A \cap B \subseteq A \cap C$, we suppose that x is any element in (1) _____. Then we must show that (2) _____.
2. The following are two proofs that for all sets A and B , $A - B \subseteq A$. The first is less formal, and the second is more formal. Fill in the blanks.

a. Proof: Suppose A and B are any sets. To show that $A - B \subseteq A$, we must show that every element in (1) _____ is in (2) _____. But any element in $A - B$ is in (3) _____ and not in (4) _____ (by definition of $A - B$). In particular, such an element is in A .

b. Proof: Suppose A and B are any sets and $x \in A - B$. [We must show that (1) _____.] By definition of set difference, $x \in$ (2) _____ and $x \notin$ (3) _____. In particular, $x \in$ (4) _____ [which is what was to be shown].

In 3 and 4, supply explanations of the steps in the given proofs.

3. Theorem: For all sets A , B , and C , if $A \subseteq B$, $B \subseteq C$, then $A \subseteq C$.
- Proof:

Statement	Explanation
Suppose A , B , and C are any sets such that $A \subseteq B$ and $B \subseteq C$.	starting point
We must show that $A \subseteq C$.	conclusion to be shown
Let x be any element in A .	start of an element proof

Then x is in B .	(a) _____
It follows that x is in C .	(b) _____
Thus every element in A is in C .	since x could be any element of A
Therefore, $A \subseteq C$ [as was to be shown].	(c) _____

4. Theorem: For all sets A and B , if $A \subseteq B$, then $A \cup B \subseteq B$.
- Proof:

Statement	Explanation
Suppose A , B , and C are any sets such that $A \subseteq B$.	starting point
We must show that $A \cup B \subseteq B$.	conclusion to be shown
Let x be any element in $A \cup B$.	start of an element proof
Then x is in A or x is in B .	(a) _____
In case x is in A , then x is in B .	(b) _____
In case x is in B , then x is in B .	tautology ($p \rightarrow p$)
So in either case x is in B .	proof by division into cases
Thus every element in $A \cup B$ is in B .	since x could be any element of $A \cup B$
Therefore, $A \cup B \subseteq B$ [as was to be shown].	(c) _____

5. Prove that for all sets A and B , $(B - A) = B \cap A^c$.
- H 6. Let $\cap$ and $\cup$ stand for the words “intersection” and “union,” respectively. Fill in the blanks in the following proof that for all sets A , B , and C , $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

43. For every positive integer n , if A and $B_1, B_2, B_3, \dots$ are any sets, then

$$\bigcup_{i=1}^n (A \times B_i) = A \times \left(\bigcup_{i=1}^n B_i \right).$$

44. For every positive integer n , if A and $B_1, B_2, B_3, \dots$ are any sets, then

$$\bigcap_{i=1}^n (A \times B_i) = A \times \left(\bigcap_{i=1}^n B_i \right).$$

ANSWERS FOR TEST YOURSELF

1. and 2. or 3. $x \in A; x \in B; x \in X$
 4. $x \in A \cap B$ (Or: x is an element of both

- A and B); $x \in X$ 5. $X \subseteq Y; Y \subseteq X$
 6. X ; in Y ; Y ; in X

6.3 Disproofs and Algebraic Proofs

If a fact goes against common sense, and we are nevertheless compelled to accept and deal with this fact, we learn to alter our notion of common sense.

—Phillip J. Davis and Reuben Hersh, *The Mathematical Experience*, 1981

In Section 6.2 we gave examples only of set properties that were true. Occasionally, however, a proposed set property is false. We begin this section by discussing how to disprove such a proposed property. Then we prove an important theorem about the power set of a set and go on to discuss an “algebraic” method for deriving new set properties from set properties already known to be true. We finish the section with an introduction to Boolean algebras.

Disproving an Alleged Set Property

Recall that to show a universal statement is false, it suffices to find one example (called a counterexample) for which it is false.

Example 6.3.1 Finding a Counterexample for a Set Identity

Is the following set property true?

For all sets A, B , and C , $(A - B) \cup (B - C) = A - C$.

Solution Observe that the property is true if, and only if, the given equality holds for *all* sets A, B , and C .

So it is false if, and only if,

there are sets A, B , and C for which the equality does *not* hold.

One way to solve this problem is to picture sets A, B , and C by drawing a Venn diagram such as that shown in Figure 6.3.1 on the next page. If you assume that any of the eight regions of the diagram may be empty of points, then the diagram is quite general.

Find and shade the region corresponding to $(A - B) \cup (B - C)$. Then shade the region corresponding to $A - C$. These are shown in Figure 6.3.2 on the next page.

Comparing the shaded regions seems to indicate that the property is false. For instance, if there is an element in B that is not in either A or C then this element would be in $(A - B) \cup (B - C)$ (because of being in B and not C), but it would not be in $A - C$ since $A - C$ contains nothing outside A . Similarly, an element that is in both A and C but not B

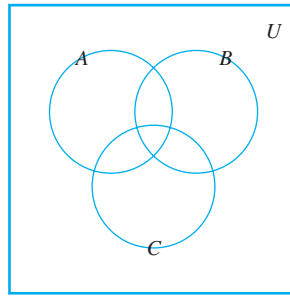


FIGURE 6.3.1

would be in $(A - B) \cup (B - C)$ (because of being in A and not B), but it would not be in $A - C$ (because of being in both A and C).

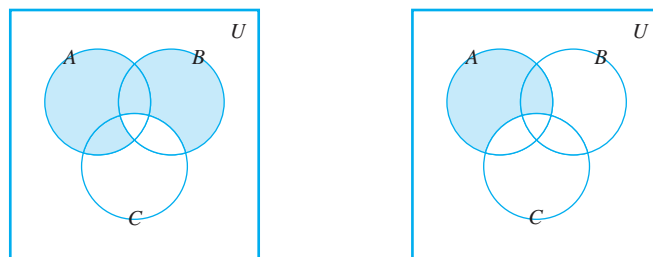


FIGURE 6.3.2

Construct a concrete counterexample in order to confirm your answer and make sure that you did not make a mistake either in drawing or analyzing your diagrams. One way is to put one of the integers from 1 through 7 into each of the seven subregions enclosed by the circles representing A , B , and C . If the proposed set property had involved set complements, it would also be helpful to label the region outside the circles, and so we place the number 8 there. (See Figure 6.3.3.) Then define discrete sets A , B , and C to consist of all the numbers in their respective subregions.

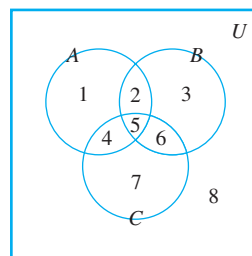


FIGURE 6.3.3

Counterexample 1: Let $A = \{1, 2, 4, 5\}$, $B = \{2, 3, 5, 6\}$, and $C = \{4, 5, 6, 7\}$. Then

$$A - B = \{1, 4\}, \quad B - C = \{2, 3\}, \quad \text{and} \quad A - C = \{1, 2\}.$$

Hence

$$(A - B) \cup (B - C) = \{1, 4\} \cup \{2, 3\} = \{1, 2, 3, 4\}, \quad \text{whereas} \quad A - C = \{1, 2\}.$$

Since $\{1, 2, 3, 4\} \neq \{1, 2\}$, we have that $(A - B) \cup (B - C) \neq A - C$.

A more economical counterexample can be obtained by observing that as long as the set B contains an element, such as 3, that is not in A , then regardless of whether B contains any other elements and regardless of whether A and C contain any elements at all, $(A - B) \cup (B - C) \neq A - C$.

Counterexample 2: Let $A = \emptyset$, $B = \{3\}$, and $C = \emptyset$. Then

$$A - B = \emptyset, \quad B - C = \{3\}, \quad \text{and} \quad A - C = \emptyset.$$

Hence $(A - B) \cup (B - C) = \emptyset \cup \{3\} = \{3\}$, whereas $A - C = \emptyset$.

Since $\{3\} \neq \emptyset$, we have that $(A - B) \cup (B - C) \neq A - C$.

Note Check that when $A = C = \{4\}$ and $B = \emptyset$, $(A - B) \cup (B - C) \neq A - C$.

Another economical counterexample requires only that $A = C$ = a singleton set, such as $\{4\}$, while B is the empty set. ■

Problem-Solving Strategy

How can you discover whether a given universal statement about sets is true or false? There are two basic approaches. Either you plunge in and start trying to prove the statement, asking yourself, “What do I need to show?” and “How do I show it?” or you try to find a set of conditions that must be fulfilled to construct a counterexample. With either approach you may have immediate success or you may run into difficulty. The trick is to be ready to switch to the other approach if the one you are working on does not look promising. For more difficult questions, you may alternate several times between the two approaches before arriving at the correct answer.

The Number of Subsets of a Set

The following theorem states the important fact that if a set has n elements then its power set has 2^n elements. The proof uses mathematical induction and is based on the following observations. Suppose X is a set and z is an element of X .

1. The subsets of X can be split into two groups: those that do not contain z and those that do contain z .
2. The subsets of X that do not contain z are the same as the subsets of $X - \{z\}$.
3. The subsets of X that do not contain z can be matched up one for one with the subsets of X that do contain z by matching each subset A that does not contain z to the subset $A \cup \{z\}$ that contains z . Thus there are as many subsets of X that contain z as there are subsets of X that do not contain z . For instance, if $X = \{x, y, z\}$, the following table shows the correspondence between subsets of X that do not contain z and subsets of X that contain z .

Subsets of X That Do Not Contain z		Subsets of X That Contain z
$\emptyset$	$\leftrightarrow$	$\emptyset \cup \{z\} = \{z\}$
$\{x\}$	$\leftrightarrow$	$\{x\} \cup \{z\} = \{x, z\}$
$\{y\}$	$\leftrightarrow$	$\{y\} \cup \{z\} = \{y, z\}$
$\{x, y\}$	$\leftrightarrow$	$\{x, y\} \cup \{z\} = \{x, y, z\}$

Theorem 6.3.1

For every integer $n \geq 0$, if a set X has n elements, then $\mathcal{P}(X)$ has 2^n elements.

Proof (by mathematical induction): Let the property $P(n)$ be the sentence

Any set with n elements has 2^n subsets. $\leftarrow P(n)$

Show that $P(0)$ is true:

To establish $P(0)$, we must show that

Any set with 0 elements has 2^0 subsets. $\leftarrow P(0)$

Now the only set with zero elements is the empty set, and the only subset of the empty set is itself. Thus a set with zero elements has one subset. Since $1 = 2^0$, we have that $P(0)$ is true.

Show that for every integer $k \geq 0$, if $P(k)$ is true then $P(k+1)$ is also true:

[Suppose that $P(k)$ is true for a particular but arbitrarily chosen integer $k \geq 0$. That is:]

Suppose that k is any integer with $k \geq 0$ such that

Any set with k elements has 2^k subsets. $\leftarrow P(k)$ inductive hypothesis

[We must show that $P(k+1)$ is true. That is:] We must show that

Any set with $k+1$ elements has 2^{k+1} subsets. $\leftarrow P(k+1)$

Let X be a set with $k+1$ elements. Since $k+1 \geq 1$, we may pick an element z in X . Observe that any subset of X either contains z or does not. Furthermore, any subset of X that does not contain z is a subset of $X - \{z\}$. And any subset A of $X - \{z\}$ can be matched up with a subset B , equal to $A \cup \{z\}$, of X that contains z . Consequently, there are as many subsets of X that contain z as do not, and thus there are twice as many subsets of X as there are subsets of $X - \{z\}$. It follows that since $X - \{z\}$ has k elements, then, by inductive hypothesis,

$$\text{the number of subsets of } X - \{z\} = 2^k$$

Therefore,

$$\begin{aligned} \text{the number of subsets of } X &= 2 \cdot (\text{the number of subsets of } X - \{z\}) \\ &= 2 \cdot (2^k) && \text{by substitution} \\ &= 2^{k+1} && \text{by basic algebra.} \end{aligned}$$

[This is what was to be shown.]

[Since we have proved both the basis step and the inductive step, we conclude that the theorem is true.]

“Algebraic” Proofs of Set Identities

Let U be a universal set and consider the power set of U , $\mathcal{P}(U)$. The set identities given in Theorem 6.2.2 hold for all elements of $\mathcal{P}(U)$. Once a certain number of identities and other properties have been established, new properties can be derived from them algebraically without having to use element method arguments. It turns out that only identities (1–5) of Theorem 6.2.2 are needed to prove any other identity involving only unions, intersections, and complements. With the addition of identity (12), the set difference law,

any set identity involving unions, intersections, complements, and set differences can be established.

To use known properties to derive new ones, you need to use the fact that such properties are universal statements. Like the laws of algebra for real numbers, they apply to a wide variety of different situations. Assume that all sets are subsets of $\mathcal{P}(U)$, then, for instance, one of the distributive laws states that

$$\text{for all sets } A, B, \text{ and } C, \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

This law can be viewed as a general template into which *any* three particular sets can be placed. Thus, for example, if A_1 , A_2 , and A_3 represent particular sets, then

$$\begin{array}{c} A_1 \cap (A_2 \cup A_3) = (A_1 \cap A_2) \cup (A_1 \cap A_3), \\ \underbrace{\quad}_A \cap \underbrace{(B \cup C)}_{(B \cup C)} = \underbrace{(A \cap B)}_{(A \cap B)} \cup \underbrace{(A \cap C)}_{(A \cap C)} \end{array}$$

where A_1 takes the place of A , A_2 takes the place of B , and A_3 takes the place of C . Similarly, if W , X , Y , and Z are any particular sets, then, by the distributive law,

$$\begin{array}{c} (W \cap X) \cap (Y \cup Z) = ((W \cap X) \cap Y) \cup ((W \cap X) \cap Z), \\ \underbrace{\quad}_A \cap \underbrace{(B \cup C)}_{(B \cup C)} = \underbrace{(A \cap B)}_{(A \cap B)} \cup \underbrace{(A \cap C)}_{(A \cap C)} \end{array}$$

where $W \cap X$ takes the place of A , Y takes the place of B , and Z takes the place of C .

Example 6.3.2 Deriving a Set Difference Property

Construct an algebraic proof that for all sets A , B , and C ,

$$(A \cup B) - C = (A - C) \cup (B - C).$$

Cite a property from Theorem 6.2.2 for each step of the proof.

Solution Let A , B , and C be any sets. Then

$$\begin{aligned} (A \cup B) - C &= (A \cup B) \cap C^c && \text{by the set difference law} \\ &= C^c \cap (A \cup B) && \text{by the commutative law for } \cap \\ &= (C^c \cap A) \cup (C^c \cap B) && \text{by the distributive law} \\ &= (A \cap C^c) \cup (B \cap C^c) && \text{by the commutative law for } \cap \\ &= (A - C) \cup (B - C) && \text{by the set difference law.} \end{aligned}$$

Example 6.3.3 Deriving a Set Identity Using Properties of $\emptyset$

Construct an algebraic proof that for all sets A and B ,

$$A - (A \cap B) = A - B.$$

Cite a property from Theorem 6.2.2 for every step of the proof.

Solution Suppose A and B are any sets. Then

$$\begin{aligned} A - (A \cap B) &= A \cap (A \cap B)^c && \text{by the set difference law} \\ &= A \cap (A^c \cup B^c) && \text{by De Morgan's law} \\ &= (A \cap A^c) \cup (A \cap B^c) && \text{by the distributive law} \\ &= \emptyset \cup (A \cap B^c) && \text{by the complement law} \\ &= (A \cap B^c) \cup \emptyset && \text{by the commutative law for } \cup \\ &= A \cap B^c && \text{by the identity law for } \cup \\ &= A - B && \text{by the set difference law.} \end{aligned}$$

To many people an algebraic proof seems simpler than an element proof, but often an element proof is actually easier to understand. For instance, in Example 6.3.3 above, you could see immediately that $A - (A \cap B) = A - B$ because for an element to be in $A - (A \cap B)$ means that it is in A and not in both A and B , and this is equivalent to saying that it is in A and not in B .

Example 6.3.4 Deriving a Generalized Associative Law

Prove that for any sets A_1, A_2, A_3 , and A_4 ,

$$((A_1 \cup A_2) \cup A_3) \cup A_4 = A_1 \cup ((A_2 \cup A_3) \cup A_4).$$

Cite a property from Theorem 6.2.2 for every step of the proof.

Solution Let A_1, A_2, A_3 , and A_4 be any sets. Then

$$\begin{aligned} ((A_1 \cup A_2) \cup A_3) \cup A_4 &= (A_1 \cup (A_2 \cup A_3)) \cup A_4 && \text{by the associative law for } \cup \text{ with } A_1 \\ &&& \text{taking the place of } A, A_2 \text{ taking the place of } B, \text{ and } A_3 \text{ taking the place of } C \\ &= A_1 \cup ((A_2 \cup A_3) \cup A_4) && \text{by the associative law for } \cup \text{ with } A_1 \\ &&& \text{taking the place of } A, A_2 \cup A_3 \text{ taking the place of } B, \text{ and } A_4 \text{ taking the place of } C. \end{aligned}$$



Caution! When doing problems similar to Examples 6.3.2–6.3.4, be sure to use the set properties exactly as they are stated in Theorem 6.2.2.

TEST YOURSELF

- Given a proposed set identity involving set variables A, B , and C , the most common way to show that the equation does not hold in general is to find concrete sets A, B , and C that, when substituted for the set variables in the equation, _____.
- When using the algebraic method for proving a set identity, it is important to _____ for every step.
- When applying a property from Theorem 6.2.2, it must be used _____ as it is stated.

EXERCISESET 6.3

For each of 1–4 find a counterexample to show that the statement is false. Assume all sets are subsets of a universal set U .

- For all sets A, B , and C ,
 $(A \cup B) \cap C = A \cup (B \cap C)$.
- For all sets A and B , $(A \cup B)^c = A^c \cup B^c$.
- For all sets A, B , and C , if $A \not\subseteq B$ and $B \not\subseteq C$ then $A \not\subseteq C$.
- For all sets A, B , and C , if $B \cup C \subseteq A$ then
 $(A - B) \cap (A - C) = \emptyset$.

For each of 5–21 prove each statement that is true and find a counterexample for each statement that is false. Assume all sets are subsets of a universal set U .

- For all sets A, B , and C ,
 $A - (B - C) = (A - B) - C$.

- For all sets A and B , $A \cap (A \cup B) = A$.
- For all sets A, B , and C ,
 $(A - B) \cap (C - B) = A - (B \cup C)$.
- For all sets A and B , if $A^c \subseteq B$ then $A \cup B = U$.
- For all sets A, B , and C , if $A \subseteq C$ and $B \subseteq C$ then $A \cup B \subseteq C$.
- For all sets A and B , if $A \subseteq B$ then $A \cap B^c = \emptyset$.
- For all sets A, B , and C , if $A \subseteq B$ then
 $A \cap (B \cap C)^c = \emptyset$.

H 12. For all sets A, B , and C ,

$$A \cap (B - C) = (A \cap B) - (A \cap C).$$

- For all sets A, B , and C ,

$$A \cup (B - C) = (A \cup B) - (A \cup C).$$

36. For all sets A and B , $((A^c \cup B^c) - A)^c = A$.
37. For all sets A and B , $(B^c \cup (B^c - A))^c = B$.
38. For all sets A and B , $(A \cap B)^c \cap A = A - B$.
39. For all sets A and B ,
- $$(A - B) \cup (B - A) = (A \cup B) - (A \cap B).$$

40. For all sets A , B , and C ,

$$(A - B) - (B - C) = A - B.$$

In 41–43 simplify the given expression. Cite a property from Theorem 6.2.2 for every step.

- H 41. $A \cap ((B \cup A^c) \cap B^c)$
42. $(A - (A \cap B)) \cap (B - (A \cap B))$
43. $((A \cap (B \cup C)) \cap (A - B)) \cap (B \cup C^c)$
44. Consider the following set property: For all sets A and B , $A - B$ and B are disjoint.
- Use an element argument to derive the property.
 - Use an algebraic argument to derive the property (by applying properties from Theorem 6.2.2).
 - Comment on which method you found easier.
45. Consider the following set property: For all sets A , B , and C , $(A - B) \cup (B - C) = (A \cup B) - (B \cap C)$.
- Use an element argument to derive the property.
 - Use an algebraic argument to derive the property (by applying properties from Theorem 6.2.2).
 - Comment on which method you found easier.

Definition: Given sets A and B , the **symmetric difference of A and B** , denoted $A \Delta B$, is

$$A \Delta B = (A - B) \cup (B - A).$$

46. Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{5, 6, 7, 8\}$. Find each of the following sets:
- $A \Delta B$
 - $B \Delta C$
 - $A \Delta C$
 - $(A \Delta B) \Delta C$

Refer to the definition of symmetric difference given above. Prove each of 47–52, assuming that A , B , and C are all subsets of a universal set U .

47. $A \Delta B = B \Delta A$
48. $A \Delta \emptyset = A$
49. $A \Delta A^c = U$
50. $A \Delta A = \emptyset$

H* 51. If $A \Delta C = B \Delta C$, then $A = B$.

H* 52. $(A \Delta B) \Delta C = A \Delta (B \Delta C)$

H 53. Derive the set identity $A \cup (A \cap B) = A$ from the properties listed in Theorem 6.2.2(1)–(9). Start by showing that for every subset B of a universal set U , $U \cup B = U$. Then intersect both sides with A and deduce the identity.

54. Derive the set identity $A \cap (A \cup B) = A$ from the properties listed in Theorem 6.2.2(1)–(9). Start by showing that for every subset B of a universal set U , $\emptyset = \emptyset \cap B$. Then take the union of both sides with A and deduce the identity.

ANSWERS FOR TEST YOURSELF

1. make the left-hand side unequal to the right-hand side
(Or: result in different values on the two sides of the

equation) 2. cite one of the properties from Theorem 6.2.2
(Or: give a precise reason) 3. exactly

6.4 Boolean Algebras, Russell's Paradox, and the Halting Problem

From the paradise created for us by Cantor, no one will drive us out.
—David Hilbert (1862–1943)

Table 6.4.1 summarizes the main features of the logical equivalences from Theorem 2.1.1 and the set properties from Theorem 6.2.2. Notice how similar the entries in the two columns are.

TABLE 6.4.1

Logical Equivalences	Set Properties
For all statement variables p, q , and r :	For all sets A, B , and C :
a. $p \vee q \equiv q \vee p$ b. $p \wedge q \equiv q \wedge p$	a. $A \cup B = B \cup A$ b. $A \cap B = B \cap A$
a. $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$ b. $p \vee (q \vee r) \equiv (p \vee q) \vee r$	a. $A \cap (B \cap C) = (A \cap B) \cap C$ b. $A \cup (B \cup C) = (A \cup B) \cup C$
a. $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ b. $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	a. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ b. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
a. $p \vee \mathbf{c} \equiv p$ b. $p \wedge \mathbf{t} \equiv p$	a. $A \cup \emptyset = A$ b. $A \cap U = A$
a. $p \vee \sim p \equiv \mathbf{t}$ b. $p \wedge \sim p \equiv \mathbf{c}$	a. $A \cup A^c = U$ b. $A \cap A^c = \emptyset$
$\sim(\sim p) \equiv p$	$(A^c)^c = A$
a. $p \vee p \equiv p$ b. $p \wedge p \equiv p$	a. $A \cup A = A$ b. $A \cap A = A$
a. $p \vee \mathbf{t} \equiv \mathbf{t}$ b. $p \wedge \mathbf{c} \equiv \mathbf{c}$	a. $A \cup U = U$ b. $A \cap \emptyset = \emptyset$
a. $\sim(p \vee q) \equiv \sim p \wedge \sim q$ b. $\sim(p \wedge q) \equiv \sim p \vee \sim q$	a. $(A \cup B)^c = A^c \cap B^c$ b. $(A \cap B)^c = A^c \cup B^c$
a. $p \vee (p \wedge q) \equiv p$ b. $p \wedge (p \vee q) \equiv p$	a. $A \cup (A \cap B) = A$ b. $A \cap (A \cup B) = A$
a. $\sim \mathbf{t} \equiv \mathbf{c}$ b. $\sim \mathbf{c} \equiv \mathbf{t}$	a. $U^c = \emptyset$ b. $\emptyset^c = U$

If you let $\vee$ (or) correspond to $\cup$ (union), $\wedge$ (and) correspond to $\cap$ (intersection), $\mathbf{t}$ (a tautology) correspond to U (a universal set), $\mathbf{c}$ (a contradiction) correspond to $\emptyset$ (the empty set), and $\sim$ (negation) correspond to c (complementation), then you can see that the structure of the set of statement forms with operations $\vee$ and $\wedge$ is essentially identical to the structure of the set of subsets of a universal set with operations $\cup$ and $\cap$. In fact, both are special cases of the same general structure, known as a *Boolean algebra*. The essential idea of a Boolean algebra was introduced by the self-taught English mathematician/logician George Boole in 1847 in a book entitled *The Mathematical Analysis of Logic*. During the remainder of the nineteenth century, Boole and others amplified and clarified the concept until it reached the form in which we use it today.

In this section we show how to derive the various properties associated with a Boolean algebra from a set of five axioms.

Definition and Axioms for a Boolean Algebra

A **Boolean algebra** is a set B together with two operations, generally denoted $+$ and $\cdot$, such that for all a and b in B both $a + b$ and $a \cdot b$ are in B and the following axioms are assumed to hold:

1. *Commutative Laws*: For all a and b in B ,

$$(a) \ a + b = b + a \quad \text{and} \quad (b) \ a \cdot b = b \cdot a.$$

2. *Associative Laws*: For all a , b , and c in B ,

$$(a) \ (a + b) + c = a + (b + c) \quad \text{and} \quad (b) \ (a \cdot b) \cdot c = a \cdot (b \cdot c).$$

3. *Distributive Laws*: For all a , b , and c in B ,

$$(a) \ a + (b \cdot c) = (a + b) \cdot (a + c) \quad \text{and} \quad (b) \ a \cdot (b + c) = (a \cdot b) + (a \cdot c).$$

4. *Identity Laws*: There exist distinct elements 0 and 1 in B such that for each a in B ,

$$(a) \ a + 0 = a \quad \text{and} \quad (b) \ a \cdot 1 = a.$$

5. *Complement Laws*: For each a in B , there exists an element in B , denoted $\bar{a}$ and called the **complement** or **negation** of a , such that

$$(a) \ a + \bar{a} = 1 \quad \text{and} \quad (b) \ a \cdot \bar{a} = 0.$$

In any Boolean algebra, the complement of each element is unique, the quantities 0 and 1 are unique, and identities analogous to those in Theorem 2.1.1 and Theorem 6.2.2 can be deduced.

Theorem 6.4.1 Properties of a Boolean Algebra

Let B be any Boolean algebra.

1. *Uniqueness of the Complement Laws*: For all a and x in B , if $a + x = 1$ and $a \cdot x = 0$ then $x = \bar{a}$.

2. *Uniqueness of 0 and 1*: If there exists x in B such that $a + x = a$ for every a in B , then $x = 0$, and if there exists y in B such that $a \cdot y = a$ for every a in B , then $y = 1$.

3. *Double Complement Law*: For every $a \in B$, $\overline{(\bar{a})} = a$.

4. *Idempotent Laws*: For every $a \in B$,

$$(a) \ a + a = a \quad \text{and} \quad (b) \ a \cdot a = a.$$

5. *Universal Bound Laws*: For every $a \in B$,

$$(a) \ a + 1 = 1 \quad \text{and} \quad (b) \ a \cdot 0 = 0.$$

6. *De Morgan's Laws*: For all a and $b \in B$,

$$(a) \ \overline{a + b} = \bar{a} \cdot \bar{b} \quad \text{and} \quad (b) \ \overline{a \cdot b} = \bar{a} + \bar{b}.$$

7. *Absorption Laws*: For all a and $b \in B$,

$$(a) (a + b) \cdot a = a \quad \text{and} \quad (b) (a \cdot b) + a = a.$$

8. *Complements of 0 and 1*:

$$(a) \bar{0} = 1 \quad \text{and} \quad (b) \bar{1} = 0.$$

Proof:

Part 1: Uniqueness of the Complement Law

Suppose a and x are particular, but arbitrarily chosen, elements of B that satisfy the following hypothesis: $a + x = 1$ and $a \cdot x = 0$. Then

$$\begin{aligned} x &= x \cdot 1 && \text{because 1 is an identity for } \cdot \\ &= x \cdot (a + \bar{a}) && \text{by the complement law for } + \\ &= x \cdot a + x \cdot \bar{a} && \text{by the distributive law for } \cdot \text{ over } + \\ &= a \cdot x + x \cdot \bar{a} && \text{by the commutative law for } \cdot \\ &= 0 + x \cdot \bar{a} && \text{by hypothesis} \\ &= a \cdot \bar{a} + x \cdot \bar{a} && \text{by the complement law for } \cdot \\ &= (\bar{a} \cdot a) + (\bar{a} \cdot x) && \text{by the commutative law for } \cdot \\ &= \bar{a} \cdot (a + x) && \text{by the distributive law for } \cdot \text{ over } + \\ &= \bar{a} \cdot 1 && \text{by hypothesis} \\ &= \bar{a} && \text{because 1 is an identity for } \cdot. \end{aligned}$$

Proofs of the other parts of the theorem are discussed in the examples that follow and in the exercises.

You may notice that all parts of the definition of a Boolean algebra and most parts of Theorem 6.4.1 contain paired statements. For instance, the distributive laws state that for all a , b , and c in B ,

$$(a) a + (b \cdot c) = (a + b) \cdot (a + c) \quad \text{and} \quad (b) a \cdot (b + c) = (a \cdot b) + (a \cdot c),$$

and the identity laws state that for every a in B ,

$$(a) a + 0 = a \quad \text{and} \quad (b) a \cdot 1 = a.$$

Each of the paired statements can be obtained from the other by interchanging all the $+$ and $\cdot$ signs and interchanging 1 and 0. Such interchanges transform any Boolean identity into its **dual** identity. It can be proved that the dual of any Boolean identity is also an identity. This fact is often called the **duality principle** for a Boolean algebra.

Example 6.4.1

Proof of the Double Complement Law

Prove that for all elements a in a Boolean algebra B , $\overline{\bar{a}} = a$.

Solution Start by supposing that B is a Boolean algebra and a is any element of B . The basis for the proof is the uniqueness of the complement law: that each element in B has a unique complement, which satisfies certain equations with respect to it. So if a can be shown to satisfy those equations with respect to $\bar{a}$, then a must be the complement of $\bar{a}$.

Theorem 6.4.1(3) Double Complement Law

For every element a in a Boolean algebra B , $\overline{(\bar{a})} = a$.

Proof:

Suppose B is a Boolean algebra and a is any element of B . Then

$$\begin{aligned}\bar{a} + a &= a + \bar{a} && \text{by the commutative law for } + \\ &= 1 && \text{by the complement law for } 1\end{aligned}$$

and

$$\begin{aligned}\bar{a} \cdot a &= a \cdot \bar{a} && \text{by the commutative law for } \cdot \\ &= 0 && \text{by the complement law for } 0.\end{aligned}$$

Thus a satisfies the two equations with respect to $\bar{a}$ that are satisfied by the complement of $\bar{a}$. From the fact that the complement of a is unique, we conclude that $\overline{(\bar{a})} = a$.

Example 6.4.2

Proof of an Idempotent Law

Fill in the blanks in the following proof that for all elements a in a Boolean algebra B , $a + a = a$.

Proof:

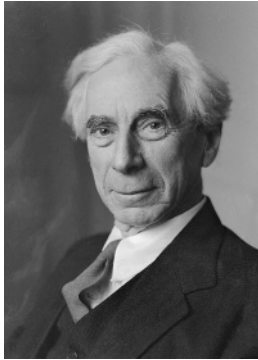
Suppose B is a Boolean algebra and a is any element of B . Then

$$\begin{aligned}a &= a + 0 && \text{(a)} \\ &= a + (a \cdot \bar{a}) && \text{(b)} \\ &= (a + a) \cdot (a + \bar{a}) && \text{(c)} \\ &= (a + a) \cdot 1 && \text{(d)} \\ &= a + a && \text{(e)}.\end{aligned}$$

Solution

- (a) because 0 is an identity for $+$
- (b) by the complement law for $\cdot$
- (c) by the distributive law for $+$ over $\cdot$
- (d) by the complement law for $+$
- (e) because 1 is an identity for $\cdot$

Russell's Paradox



Bertrand Russell
(1872–1970)

Bettmann/Getty Images

By the beginning of the twentieth century, abstract set theory had gained such wide acceptance that a number of mathematicians were working hard to show that all of mathematics could be built upon a foundation of set theory. In the midst of this activity, the English mathematician and philosopher Bertrand Russell discovered a “paradox” (really a genuine contradiction) that seemed to shake the very core of the foundation. The paradox assumes Cantor’s definition of set as “any collection into a whole of definite and separate objects of our intuition or our thought.”

Russell’s Paradox: Most sets are not elements of themselves. For instance, the set of all integers is not an integer and the set of all horses is not a horse. However, we can imagine the possibility of a set’s being an element of itself. For instance, the set of all abstract ideas might be considered an abstract idea. If we are allowed to use any description of a property as the defining property of a set, we can let S be the set of all sets that are not elements of themselves:

$$S = \{A \mid A \text{ is a set and } A \notin A\}.$$

Is S an element of itself?

The answer is both yes and no. For suppose $S \in S$. Then S satisfies the defining property for S , and hence $S \notin S$. This contradicts the supposition that $S \in S$ and shows that $S \notin S$. Next suppose $S \notin S$. Then S is a set such that $S \notin S$ and so S satisfies the defining property for S , which implies that $S \in S$. This contradicts the supposition that $S \notin S$ and shows that $S \in S$. Thus both $S \in S$ and $S \notin S$, which is impossible because a statement is either true or false but not both. To help explain his discovery to laypeople, Russell devised a puzzle, the barber puzzle, whose solution exhibits the same logic as his paradox.

Example 6.4.3

The Barber Puzzle

In a certain town there is a male barber who shaves all those men, and only those men, who do not shave themselves. *Question:* Does the barber shave himself?

Solution The answer is both yes and no. If the barber shaves himself, he is a member of the class of men who shave themselves. But no member of this class is shaved by the barber, and so the barber does *not* shave himself. On the other hand, if the barber does not shave himself, he belongs to the class of men who do not shave themselves. But the barber shaves every man in this class, so the barber *does* shave himself. ■

How can the answer be both yes and no? Surely any barber either does or does not shave himself. You might try to think of circumstances that would make the paradox disappear. For instance, maybe the barber happens to have no beard and never shaves. But a condition of the puzzle is that the barber is a man who shaves *all* those men who do not shave themselves. If he does not shave, then he does not shave himself, in which case he is shaved by the barber and the contradiction is as present as ever. Other attempts at resolving the paradox by considering details of the barber’s situation are similarly doomed to failure.

So let’s accept the fact that the paradox has no easy resolution and see where that thought leads. Since the barber both shaves himself and doesn’t shave himself, the sentence “The barber shaves himself” is both true and false. Yet the sentence arose in a natural way from a description of a situation. If the situation actually existed, then the sentence would have to be either true or false but not both. Thus we are forced to conclude that the situation described in the puzzle simply cannot exist in the world as we know it.

In a similar way, the conclusion to be drawn from Russell's paradox itself is that the object S is not a set. Because if it actually were a set, in the sense of satisfying the general properties of sets that we have been assuming, then it either would be an element of itself or not.

In the years following Russell's discovery, several ways were found to define the basic concepts of set theory so as to avoid his contradiction. The way used in this text requires that, except for the power set whose existence is guaranteed by an axiom, whenever a set is defined using a predicate as a defining property, the stipulation must also be made that the set is a subset of a known set. This method does not allow us to talk about "the set of all sets that are not elements of themselves." We can speak only of "the set of all sets that are subsets of some known set and that are not elements of themselves." When this restriction is made, Russell's paradox ceases to be contradictory. Here is what happens:

Let U be a universal set and suppose that all sets under discussion are subsets of U . Let

$$S = \{A \mid A \subseteq U \text{ and } A \notin A\}.$$

In Russell's paradox, both implications

$$S \in S \rightarrow S \notin S \quad \text{and} \quad S \notin S \rightarrow S \in S$$

are proved, and the contradictory conclusion

$$\text{both } S \in S \quad \text{and} \quad S \notin S$$

is therefore deduced. In the situation in which all sets under discussion are subsets of U , the implication $S \in S \rightarrow S \notin S$ is proved in almost the same way as it is for Russell's paradox: (Suppose $S \in S$. Then by definition of S , $S \subseteq U$ and $S \notin S$. In particular, $S \notin S$.) On the other hand, from the supposition that $S \notin S$ we can only deduce that the statement " $S \subseteq U$ and $S \notin S$ " is false. By one of De Morgan's laws, this means that " $S \not\subseteq U$ or $S \in S$." Since $S \in S$ would contradict the supposition that $S \notin S$, we eliminate it and conclude that $S \not\subseteq U$. In other words, the only conclusion we can draw is that the seeming "definition" of S is faulty—in other words, S is not a set in U .

Russell's discovery had a profound impact on mathematics because even though his contradiction could be made to disappear by more careful definitions, its existence caused people to wonder whether other contradictions remained. In 1931 Kurt Gödel showed that it is not possible to prove, in a mathematically rigorous way, that mathematics is free of contradictions. You might think that Gödel's result would have caused mathematicians to give up their work in despair, but that has not happened. On the contrary, there has been more mathematical activity since 1931 than in any other period in history.



Pictorial Press Ltd./Alamy Stock Photo

Kurt Gödel
(1906–1978)

The Halting Problem

Well before the actual construction of an electronic computer, Alan M. Turing (1912–1954) deduced a profound theorem about how such computers would have to work. The argument he used is similar to that in Russell's paradox. It is also related to those used by Gödel to prove his theorem and by Cantor to prove that it is impossible to write all the real numbers in an infinitely long list, even given an infinitely long period of time (see Section 7.4).

If you have some experience programming computers, you know how badly an infinite loop can tie up a computer system. It would be useful to be able to preprocess a program and its data set by running it through a checking program that determines whether execution of the given program with the given data set would result in an infinite loop. Can an algorithm for such a program be written? In other words, can an algorithm be written that will accept any algorithm X and any data set D as input and will then print

“halts” or “loops forever” to indicate whether X terminates in a finite number of steps or loops forever when run with data set D ? In the 1930s, Turing proved that the answer to this question is no.

Theorem 6.4.2

There is no computer algorithm that will accept any algorithm X and data set D as input and then will output “halts” or “loops forever” to indicate whether or not X terminates in a finite number of steps when X is run with data set D .

Proof (by contradiction):

Suppose there is an algorithm, CheckHalt , such that if an algorithm X and a data set D are input, then

$\text{CheckHalt}(X, D)$ prints

“halts” if X terminates in a finite number of steps
when run with data set D

or

“loops forever” if X does not terminate in a finite number of
steps when run with data set D .

[To show that no algorithm such as CheckHalt can exist, we will deduce a contradiction.]

Observe that the sequence of characters making up an algorithm X can be regarded as a data set itself. Thus it is possible to consider running CheckHalt with input (X, X) . Define a new algorithm, Test , as follows: For any input algorithm X ,

$\text{Test}(X)$

loops forever if $\text{CheckHalt}(X, X)$ prints “halts”

or

stops if $\text{CheckHalt}(X, X)$ prints “loops forever”.

Now run algorithm Test with input Test . If $\text{Test}(\text{Test})$ terminates after a finite number of steps, then the value of $\text{CheckHalt}(\text{Test}, \text{Test})$ is “halts” and so $\text{Test}(\text{Test})$ loops forever.

On the other hand, if $\text{Test}(\text{Test})$ does not terminate after a finite number of steps, then $\text{CheckHalt}(\text{Test}, \text{Test})$ prints “loops forever” and so $\text{Test}(\text{Test})$ terminates.

The two paragraphs above show that $\text{Test}(\text{Test})$ loops forever and also that it terminates. This is a contradiction. But the existence of Test follows logically from the supposition of the existence of an algorithm CheckHalt that can check any algorithm and data set for termination. *[Hence the supposition must be false, and there is no such algorithm.]*

The axioms introduced into set theory to avoid Russell's paradox are not entirely adequate to deal with the full range of recursively defined objects in computer algorithms. One response has been to develop an extension of set theory that includes new objects called hypersets. In addition, the kinds of semantic issues raised by the barber paradox are related to problems involved in processing natural language by computers.

TEST YOURSELF

- In the comparison between the structure of the set of statement forms and the set of subsets of a universal set, the *or* operation $\vee$ corresponds to _____, the *and* operation $\wedge$ corresponds to _____, a tautology **t** corresponds to _____, a contradiction **c** corresponds to _____, and the negation operation, denoted $\sim$, corresponds to _____.
- The operations of $+$ and $\cdot$ in a Boolean algebra are generalizations of the operations of _____ and _____ in the set of all statement forms in a given finite number of variables and the operations of _____ and _____ in the set of all subsets of a given set.
- Russell showed that the following proposed “set definition” could not actually define a set: _____.

EXERCISE SET 6.4

In 1–3 assume that B is a Boolean algebra with operations $+$ and $\cdot$. Give the reasons needed to fill in the blanks in the proofs using only the axioms for a Boolean algebra.

1. *Idempotent law for $\cdot$* : For every a in B , $a \cdot a = a$.

Proof: Let a be any element of B . Then

$$\begin{array}{ll} a = a \cdot 1 & \text{(a)} \\ = a \cdot (a + \bar{a}) & \text{(b)} \\ = (a \cdot a) + (a \cdot \bar{a}) & \text{(c)} \\ = (a \cdot a) + 0 & \text{(d)} \\ = a \cdot a & \text{(e)} \end{array}$$

2. *Universal bound law for $+$* : For every a in B , $a + 1 = 1$.

Proof: Let a be any element of B . Then

$$\begin{array}{ll} a + 1 = a + (a + \bar{a}) & \text{(a)} \\ = (a + a) + \bar{a} & \text{(b)} \\ = a + \bar{a} & \text{by Example 6.4.2} \\ = 1 & \text{(c)} \end{array}$$

3. *Absorption law for $\cdot$ over $+$* : For all a and b in B , $(a + b) \cdot a = a$.

Proof: Let a be any element of B . Then

$$\begin{array}{ll} (a + b) \cdot a = a \cdot (a + b) & \text{(a)} \\ = a \cdot a + a \cdot b & \text{(b)} \\ = a + a \cdot b & \text{by exercise 1} \\ = a \cdot 1 + a \cdot b & \text{(c)} \\ = a \cdot (1 + b) & \text{(d)} \\ = a \cdot (b + 1) & \text{(e)} \\ = a \cdot 1 & \text{by exercise 2} \\ = a & \text{(f)} \end{array}$$

In 4–10 assume that B is a Boolean algebra with operations $+$ and $\cdot$. Prove each statement using only the axioms

for a Boolean algebra and statements proved in the text or in lower-numbered exercises.

- Universal bound for 0*: For every a in B , $a \cdot 0 = 0$.
- Complements of 0 and 1*:
a. $\bar{0} = 1$ b. $\bar{1} = 0$
- Uniqueness of 0*: There is only one element of B that is an identity for $+$.
- Uniqueness of 1*: There is only one element of B that is an identity for $\cdot$.
- De Morgan's law for $\cdot$* : For all a and b in B , $\overline{a \cdot b} = \bar{a} + \bar{b}$. (Hint: Prove that $(a \cdot b) + (\bar{a} + \bar{b}) = 1$ and that $(a \cdot b) + (\bar{a} + \bar{b}) = 0$, and use the fact that $a \cdot b$ has a unique complement.)
- De Morgan's law for $+$* : For all a and b in B , $\overline{a + b} = \bar{a} \cdot \bar{b}$.
- Cancellation law*: For all x , y , and z in B , if $x + y = x + z$ and $x \cdot y = x \cdot z$, then $y = z$.
- Let $S = \{0, 1\}$, and define operations $+$ and $\cdot$ on S by the following tables:

$+$	0	1	$\cdot$	0	1
0	0	1	0	0	0
1	1	1	1	0	1

- Show that the elements of S satisfy the following properties:
 - the commutative law for $+$
 - the commutative law for $\cdot$
 - the associative law for $+$
 - the associative law for $\cdot$
- the distributive law for $+$ over $\cdot$
- the distributive law for $\cdot$ over $+$

In this chapter we discuss the mathematics of relations defined on sets, focusing on ways to represent relations and exploring various properties they may have. The concept of equivalence relation is introduced in Section 8.3 and applied in Section 8.4 to modular arithmetic and cryptography. Partial order relations are discussed in Section 8.5, and an application is given showing how to use these relations to help coordinate and guide the flow of individual tasks that must be performed to accomplish a complex, large-scale project.

8.1 Relations on Sets

Strange as it may sound, the power of mathematics rests on its evasion of all unnecessary thought and on its wonderful saving of mental operations. —Ernst Mach, 1838–1916

A more formal way to refer to the kind of relation defined in Section 1.3 is to call it a **binary relation** because it is a subset of a Cartesian product of two sets. At the end of this section we define an *n*-ary relation to be a subset of a Cartesian product of *n* sets, where *n* is any integer greater than or equal to two. Such a relation is the fundamental structure used in relational databases. However, because we focus on binary relations in this text, when we use the term *relation* by itself, we will mean binary relation.

Example 8.1.1

The Less-than Relation for Real Numbers

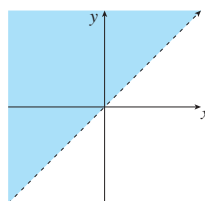
Define a relation *L* from **R** to **R** as follows: For all real numbers *x* and *y*,

$$x L y \Leftrightarrow x < y.$$

- a. Is $57 L 53$? b. Is $(-17) L (-14)$? c. Is $143 L 143$? d. Is $(-35) L 1$?
e. Draw the graph of *L* as a subset of the Cartesian plane **R** $\times$ **R**.

Solution

- a. No, $57 > 53$. b. Yes, $-17 < -14$. c. No, $143 = 143$. d. Yes, $-35 < 1$.
e. For each value of *x*, all the points (*x*, *y*) with $y > x$ are on the graph. So the graph consists of all the points above the line $x = y$.



Example 8.1.2 The Congruence Modulo 2 Relation

Define a relation E from $\mathbf{Z}$ to $\mathbf{Z}$ as follows: For every $(m, n) \in \mathbf{Z} \times \mathbf{Z}$,

$$m E n \iff m - n \text{ is even.}$$

- Is $4 E 0$? Is $2 E 6$? Is $3 E (-3)$? Is $5 E 2$?
- List five integers that are related by E to 1.
- Prove that if n is any odd integer, then $n E 1$.

Solution

- Yes, $4 E 0$ because $4 - 0 = 4$ and 4 is even.
Yes, $2 E 6$ because $2 - 6 = -4$ and -4 is even.
Yes, $3 E (-3)$ because $3 - (-3) = 6$ and 6 is even.
No, $5 \not E 2$ because $5 - 2 = 3$ and 3 is not even.

- There are many such lists. One is

$$\begin{aligned} 1 & \text{ because } 1 - 1 = 0 \text{ is even.} \\ 3 & \text{ because } 3 - 1 = 2 \text{ is even.} \\ 5 & \text{ because } 5 - 1 = 4 \text{ is even.} \\ -1 & \text{ because } -1 - 1 = -2 \text{ is even.} \\ -3 & \text{ because } -3 - 1 = -4 \text{ is even.} \end{aligned}$$

- Proof:** Suppose n is any odd integer. Then $n = 2k + 1$ for some integer k . Now by definition of E , $n E 1$ if, and only if, $n - 1$ is even. But by substitution,

$$n - 1 = (2k + 1) - 1 = 2k,$$

and since k is an integer, $2k$ is even. Hence $n E 1$ [as was to be shown].

It can be shown (see exercise 2 at the end of this section) that integers m and n are related by E if, and only if, $m \bmod 2 = n \bmod 2$ (that is, both are even or both are odd). When this occurs m and n are said to be **congruent modulo 2**. ■

Example 8.1.3 A Relation on a Power Set

Let $X = \{a, b, c\}$. Then $\mathcal{P}(X) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$. Define a relation $\mathbf{S}$ from $\mathcal{P}(X)$ to $\mathcal{P}(X)$ as follows: For all sets A and B in $\mathcal{P}(X)$ (that is, for all subsets A and B of X),

$$A \mathbf{S} B \iff A \text{ has at least as many elements as } B.$$

- Is $\{a, b\} \mathbf{S} \{b, c\}$? b. Is $\{a\} \mathbf{S} \emptyset$? c. Is $\{b, c\} \mathbf{S} \{a, b, c\}$? d. Is $\{c\} \mathbf{S} \{a\}$?

Solution

- Yes, both sets have two elements.
- Yes, $\{a\}$ has one element and $\emptyset$ has zero elements, and $1 \geq 0$.
- No, $\{b, c\}$ has two elements and $\{a, b, c\}$ has three elements and $2 < 3$.
- Yes, both sets have one element. ■

The Inverse of a Relation

If R is a relation from A to B , then a relation R^{-1} from B to A can be defined by interchanging the elements of all the ordered pairs of R .

Definition

Let R be a relation from A to B . Define the inverse relation R^{-1} from B to A as follows:

$$R^{-1} = \{(y, x) \in B \times A \mid (x, y) \in R\}.$$

This definition can be written operationally as follows:

$$\text{For all } x \in A \text{ and } y \in B, \quad (y, x) \in R^{-1} \iff (x, y) \in R.$$

Example 8.1.4 The Inverse of a Finite Relation

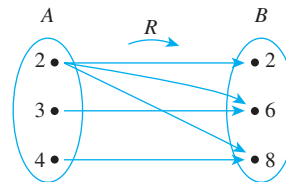
Let $A = \{2, 3, 4\}$ and $B = \{2, 6, 8\}$, and let R be the “divides” relation from A to B : For every ordered pair $(x, y) \in A \times B$,

$$x R y \iff x \mid y \quad x \text{ divides } y.$$

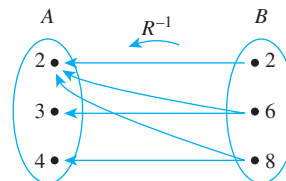
- State explicitly which ordered pairs are in R and R^{-1} , and draw arrow diagrams for R and R^{-1} .
- Describe R^{-1} in words.

Solution

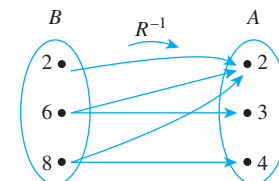
- $R = \{(2, 2), (2, 6), (2, 8), (3, 6), (4, 8)\}$
 $R^{-1} = \{(2, 2), (6, 2), (8, 2), (6, 3), (8, 4)\}$



To draw the arrow diagram for R^{-1} , you can copy the arrow diagram for R but reverse the directions of the arrows.



Or you can redraw the diagram so that B is on the left.



- R^{-1} can be described in words as follows: For every ordered pair $(y, x) \in B \times A$,

$$y R^{-1} x \iff y \text{ is a multiple of } x.$$

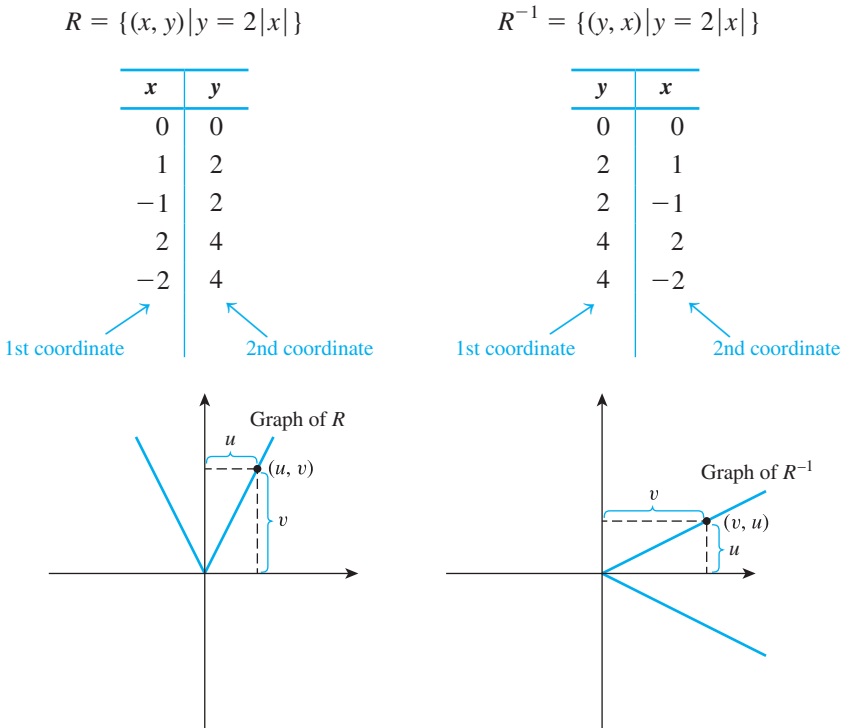
Example 8.1.5 The Inverse of an Infinite Relation

Define a relation R from $\mathbf{R}$ to $\mathbf{R}$ as follows: For every ordered pair $(x, y) \in \mathbf{R} \times \mathbf{R}$,

$$x R y \iff y = 2|x|.$$

Draw the graphs of R and R^{-1} in the Cartesian plane. Is R^{-1} a function?

Solution A point (v, u) is on the graph of R^{-1} if, and only if, (u, v) is on the graph of R . Note that if $x \geq 0$, then the graph of $y = 2|x| = 2x$ is a straight line with slope 2. And if $x < 0$, then the graph of $y = 2|x| = 2(-x) = -2x$ is a straight line with slope -2 . Some sample values are tabulated and the graphs are shown below.

 $R^{-1} = \{(y, x) | y = 2|x|\}$

R^{-1} is not a function because, for instance, both $(2, 1)$ and $(2, -1)$ are in R^{-1} . ■

Directed Graph of a Relation

In the remaining sections of this chapter, we discuss important properties of relations that are defined from a set to itself.

Note Be careful to distinguish clearly between a relation and the set on which it is defined.

Definition

A relation on a set A is a relation from A to A .

When a relation R is defined on a set A , the arrow diagram of the relation can be modified so that it becomes a **directed graph**. Instead of representing A as two separate sets of points, represent A only once, and draw an arrow from each point of A to each related point. As with an ordinary arrow diagram,

For all points x and y in A ,
 there is an arrow from x to $y \iff x R y \iff (x, y) \in R$.

If a point is related to itself, a loop is drawn that extends out from the point and goes back to it.

Example 8.1.6 Directed Graph of a Relation

Note Read “For every $x, y \in A$ ” as “For every x and y in A .”

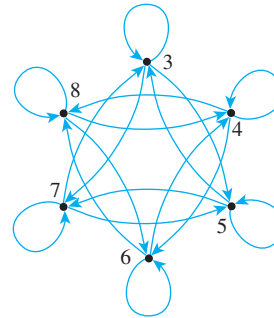
Let $A = \{3, 4, 5, 6, 7, 8\}$ and define a relation R on A as follows: For every $x, y \in A$,

$$x R y \iff 2 \mid (x - y).$$

Draw the directed graph of R .

Solution Note that $3 R 3$ because $3 - 3 = 0$ and $2 \mid 0$ since $0 = 2 \cdot 0$. Thus there is a loop from 3 to itself. Similarly, there is a loop from 4 to itself, from 5 to itself, and so forth, since the difference of each integer with itself is 0, and $2 \mid 0$.

Note also that $3 R 5$ because $3 - 5 = -2 = 2 \cdot (-1)$. And $5 R 3$ because $5 - 3 = 2 = 2 \cdot 1$. Hence there is an arrow from 3 to 5 and also an arrow from 5 to 3. The other arrows in the directed graph, as shown below, are obtained by similar reasoning.



N-ary Relations and Relational Databases

A special group of relations, called n -ary relations, form the mathematical foundation for relational database theory. Just as a binary relation is a subset of a Cartesian product of two sets, an n -ary relation is a subset of a Cartesian product of n sets.

Definition

Given sets $A_1, A_2, \dots, A_n$, an **n -ary relation** R on $A_1 \times A_2 \times \dots \times A_n$ is a subset of $A_1 \times A_2 \times \dots \times A_n$. The special cases of 2-ary, 3-ary, and 4-ary relations are called **binary**, **ternary**, and **quaternary relations**, respectively.

Example 8.1.7 A Simple Database

The following is a radically simplified version of a database that might be used in a hospital. Let A_1 be a set of positive integers, A_2 a set of alphabetic character strings, A_3 a set of numeric character strings, and A_4 a set of alphabetic character strings. Define a quaternary relation R on $A_1 \times A_2 \times A_3 \times A_4$ as follows:

$$(a_1, a_2, a_3, a_4) \in R \iff \text{a patient with patient ID number } a_1, \text{ named } a_2, \text{ was admitted on date } a_3, \text{ with primary diagnosis } a_4.$$

At a particular hospital, this relation might contain the following 4-tuples:

- (011985, John Schmidt, 020719, asthma)
- (574329, Tak Kurosawa, 011419, pneumonia)
- (466581, Mary Lazars, 010319, appendicitis)
- (008352, Joan Kaplan, 112419, gastritis)
- (011985, John Schmidt, 021719, pneumonia)
- (244388, Sarah Wu, 010319, broken leg)
- (778400, Jamal Baskers, 122719, appendicitis)

In discussions of relational databases, the n -tuples are normally thought of as being written in tables. Each row of the table corresponds to one n -tuple, and the header for each column gives the descriptive attribute for the elements in the column.

Operations within a database allow the data to be manipulated in many different ways. For example, in the database language SQL, if the above database is denoted S , the result of the query

```
SELECT Patient_ID#, Name FROM S WHERE
Admission_Date = 010319
```

would be a list of the ID numbers and names of all patients admitted on 01-03-19:

- 466581 Mary Lazars
- 244388 Sarah Wu

This is obtained by taking the intersection of the set $A_1 \times A_2 \times \{010319\} \times A_4$ with the database and then projecting onto the first two coordinates. (See exercise 25 of Section 7.1.) Similarly, SELECT can be used to obtain a list of all admission dates of a given patient. For John Schmidt this list is

- 02–07–19
- 02–17–19

Individual entries in a database can be added, deleted, or updated, and most databases can sort data entries in various ways. In addition, entire databases can be merged, and the entries common to two databases can be moved to a new database. ■

TEST YOURSELF

Answers to Test Yourself questions are located at the end of each section.

1. If R is a relation from A to B , $x \in A$, and $y \in B$, the notation $x R y$ means that ____.

2. If R is a relation from A to B , $x \in A$, and $y \in B$, the notation $x \not R y$ means that ____.

3. If R is a relation from A to B , $x \in A$, and $y \in B$, then $(y, x) \in R^{-1}$ if, and only if, ____.
4. A relation on a set A is a relation from ____ to ____.

5. If R is a relation on a set A , the directed graph of R has an arrow from x to y if, and only if, ____.

EXERCISE SET 8.1*

1. As in Example 8.1.2, the **congruence modulo 2** relation E is defined from $\mathbf{Z}$ to $\mathbf{Z}$ as follows: For every ordered pair $(m, n) \in \mathbf{Z} \times \mathbf{Z}$,

$$m E n \iff m - n \text{ is even.}$$

- a. Is $0 E 0$? Is $5 E 2$? Is $(6, 6) \in E$?
Is $(-1, 7) \in E$?
b. Prove that for any even integer n , $n E 0$.

- H 2. Prove that for all integers m and n , $m - n$ is even if, and only if, both m and n are even or both m and n are odd.

3. The **congruence modulo 3** relation, T , is defined from $\mathbf{Z}$ to $\mathbf{Z}$ as follows: For all integers m and n ,

$$m T n \iff 3 \mid (m - n).$$

- a. Is $10 T 1$? Is $1 T 10$? Is $(2, 2) \in T$?
Is $(8, 1) \in T$?
b. List five integers n such that $n T 0$.
c. List five integers n such that $n T 1$.
d. List five integers n such that $n T 2$.

- H e. Make and prove a conjecture about which integers are related by T to 0, which integers are related by T to 1, and which integers are related by T to 2.

4. Define a relation P on $\mathbf{Z}$ as follows: For every ordered pair $(m, n) \in \mathbf{Z} \times \mathbf{Z}$,

$$m P n \iff m \text{ and } n \text{ have a common prime factor.}$$

- a. Is $15 P 25$? b. Is $22 P 27$?
c. Is $0 P 5$? d. Is $8 P 8$?

5. Let $X = \{a, b, c\}$. Recall that $\mathcal{P}(X)$ is the power set of X . Define a relation $\mathbf{S}$ on $\mathcal{P}(X)$ as follows: For all sets A and B in $\mathcal{P}(X)$,

$$A \mathbf{S} B \iff A \text{ has the same number of elements as } B.$$

- a. Is $\{a, b\} \mathbf{S} \{b, c\}$? b. Is $\{a\} \mathbf{S} \{a, b\}$?
c. Is $\{c\} \mathbf{S} \{b\}$?

6. Let $X = \{a, b, c\}$. Define a relation $\mathbf{J}$ on $\mathcal{P}(X)$ as follows: For all sets A and B in $\mathcal{P}(X)$,

$$A \mathbf{J} B \iff A \cap B \neq \emptyset.$$

- a. Is $\{a\} \mathbf{J} \{c\}$? b. Is $\{a, b\} \mathbf{J} \{b, c\}$?
c. Is $\{a, b\} \mathbf{J} \{a, b, c\}$?

7. Define a relation R on $\mathbf{Z}$ as follows: For all integers m and n ,

$$m R n \iff 5 \mid (m^2 - n^2).$$

- a. Is $1 R (-9)$? b. Is $2 R 13$?
c. Is $2 R (-8)$? d. Is $(-8) R 2$?

8. Let A be the set of all strings of a 's and b 's of length 4. Define a relation R on A as follows: For every $s, t \in A$,

$$s R t \iff s \text{ has the same first two characters as } t.$$

- a. Is $abaa R abba$? b. Is $aabb R bbaa$?
c. Is $aaaa R aaab$? d. Is $baaa R abaa$?

9. Let A be the set of all strings of 0's, 1's, and 2's of length 4. Define a relation R on A as follows: For every $s, t \in A$,

$$s R t \iff \text{the sum of the characters in } s \text{ equals the sum of the characters in } t.$$

- a. Is $0121 R 2200$? b. Is $1011 R 2101$?
c. Is $2212 R 2121$? d. Is $1220 R 2111$?

10. Let $A = \{3, 4, 5\}$ and $B = \{4, 5, 6\}$ and let R be the "less than" relation. That is, for every ordered pair $(x, y) \in A \times B$,

$$x R y \iff x < y.$$

State explicitly which ordered pairs are in R and R^{-1} .

11. Let $A = \{3, 4, 5\}$ and $B = \{4, 5, 6\}$ and let S be the "divides" relation. That is, for every ordered pair $(x, y) \in A \times B$,

$$x S y \iff x \mid y.$$

State explicitly which ordered pairs are in S and S^{-1} .

12. a. Suppose a function $F: X \rightarrow Y$ is one-to-one but not onto. Is F^{-1} (the inverse relation for F) a function? Explain your answer.
b. Suppose a function $F: X \rightarrow Y$ is onto but not one-to-one. Is F^{-1} (the inverse relation for F) a function? Explain your answer.

*For exercises with blue numbers or letters, solutions are given in Appendix B. The symbol H indicates that only a hint or a partial solution is given. The symbol * signals that an exercise is more challenging than usual.

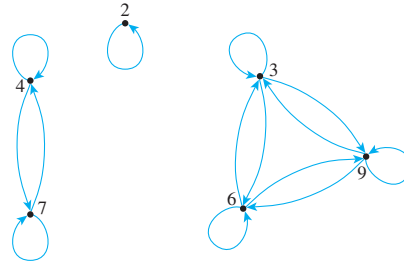
8.2 Reflexivity, Symmetry, and Transitivity

Mathematics is the tool specially suited for dealing with abstract concepts of any kind and there is no limit to its power in this field. —P. A. M. Dirac, 1902–1984

Let $A = \{2, 3, 4, 6, 7, 9\}$ and define a relation R on A as follows: For every $x, y \in A$,

$$x R y \iff 3 \mid (x - y).$$

Then $2 R 2$ because $2 - 2 = 0$, and $3 \mid 0$. Similarly, $3 R 3$, $4 R 4$, $6 R 6$, $7 R 7$, and $9 R 9$. Also $6 R 3$ because $6 - 3 = 3$, and $3 \mid 3$. And $3 R 6$ because $3 - 6 = -(6 - 3) = -3$, and $3 \mid (-3)$. Similarly, $3 R 9$, $9 R 3$, $6 R 9$, $9 R 6$, $4 R 7$, and $7 R 4$. Thus the directed graph for R has the appearance shown below.



This graph has three important properties:

1. Each point of the graph has an arrow looping around from it and going back to it.
2. In each case where there is an arrow going from one point to a second, there is an arrow going from the second point back to the first.
3. In each case where there is an arrow going from one point to a second and from the second point to a third, there is an arrow going from the first point to the third. That is, there are no “incomplete directed triangles” in the graph.

Properties (1), (2), and (3) correspond to properties of general relations called *reflexivity*, *symmetry*, and *transitivity*.



Caution! The definition of symmetric does not say that x is related to y by R ; rather, it states only that **if it happens** that x is related to y , then y must be related to x .

Definition

Let R be a relation on a set A .

1. R is **reflexive** if, and only if, for every $x \in A$, $x R x$.
2. R is **symmetric** if, and only if, for every $x, y \in A$, if $x R y$ then $y R x$.
3. R is **transitive** if, and only if, for every $x, y, z \in A$, if $x R y$ and $y R z$ then $x R z$.

Because of the equivalence of the expressions $x R y$ and $(x, y) \in R$ for every x and y in A , the reflexive, symmetric, and transitive properties can also be written as follows:

1. R is reflexive $\iff$ for every x in A , $(x, x) \in R$.
2. R is symmetric $\iff$ for every x and y in A , **if** $(x, y) \in R$ then $(y, x) \in R$.
3. R is transitive $\iff$ for every x, y , and z in A , **if** $(x, y) \in R$ and $(y, z) \in R$ then $(x, z) \in R$.



Caution! The “first,” “second,” and “third” elements in the informal versions need not all be distinct. This is a disadvantage of informality: It may mask nuances that a formal definition makes clear.

In informal terms, properties (1)–(3) say the following:

1. **Reflexive:** Each element is related to itself.
2. **Symmetric:** If any one element is related to any other element, then the second element is related to the first.
3. **Transitive:** If any one element is related to a second and that second element is related to a third, then the first element is related to the third.

Note that the definitions of reflexivity, symmetry, and transitivity are universal statements. This means that to prove a relation has one of the properties, you use either the method of exhaustion or the method of generalizing from the generic particular.

Now consider what it means for a relation *not* to have one of the properties defined previously. Recall that the negation of a universal statement is existential. Hence if R is a relation on a set A , then

1. R is **not reflexive** $\Leftrightarrow$ there is an element x in A such that $x \not R x$ [that is, such that $(x, x) \notin R$].
2. R is **not symmetric** $\Leftrightarrow$ there are elements x and y in A such that $x R y$ but $y \not R x$ [that is, such that $(x, y) \in R$ but $(y, x) \notin R$].
3. R is **not transitive** $\Leftrightarrow$ there are elements x , y , and z in A such that $x R y$ and $y R z$ but $x \not R z$ [that is, such that $(x, y) \in R$ and $(y, z) \in R$ but $(x, z) \notin R$].

It follows that you can show that a relation does *not* have one of the properties by finding a counterexample.

Example 8.2.1 Properties of Relations on Finite Sets

Let $A = \{0, 1, 2, 3\}$ and define relations R , S , and T on A as follows:

$$R = \{(0, 0), (0, 1), (0, 3), (1, 0), (1, 1), (2, 2), (3, 0), (3, 3)\},$$

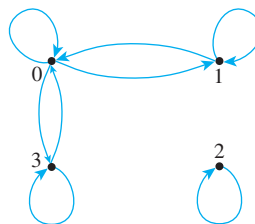
$$S = \{(0, 0), (0, 2), (0, 3), (2, 3)\},$$

$$T = \{(0, 1), (2, 3)\}.$$

- a. Is R reflexive? symmetric? transitive?
- b. Is S reflexive? symmetric? transitive?
- c. Is T reflexive? symmetric? transitive?

Solution

- a. The directed graph of R has the appearance shown below.

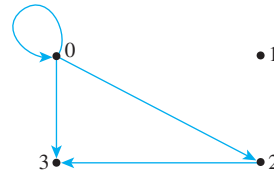


R is reflexive: There is a loop at each point of the directed graph. This means that each element of A is related to itself, so R is reflexive.

R is symmetric: In each case where there is an arrow going from one point of the graph to a second, there is an arrow going from the second point back to the first. This means that whenever one element of A is related by R to a second, then the second is related to the first. Hence R is symmetric.

R is not transitive: There is an arrow going from 1 to 0 and an arrow going from 0 to 3, but there is no arrow going from 1 to 3. This means that there are elements of A —0, 1, and 3—such that $1 R 0$ and $0 R 3$ but $1 \not R 3$. Hence R is not transitive.

b. The directed graph of S has the appearance shown below.



S is not reflexive: There is no loop at 1, for example. Thus $(1, 1) \notin S$, and so S is not reflexive.

S is not symmetric: There is an arrow from 0 to 2 but not from 2 to 0. Hence $(0, 2) \in S$ but $(2, 0) \notin S$, and so S is not symmetric.

S is transitive: There are three cases for which there is an arrow going from one point of the graph to a second and from the second point to a third. In particular, there are arrows going from 0 to 2 and from 2 to 3; there are arrows going from 0 to 0 and from 0 to 2; and there are arrows going from 0 to 0 and from 0 to 3. In each case there is an arrow going from the first point to the third. (Note again that the “first,” “second,” and “third” points need not be distinct.) This means that whenever $(x, y) \in S$ and $(y, z) \in S$, then $(x, z) \in S$, for every $x, y, z \in \{0, 1, 2, 3\}$, and so S is transitive.

c. The directed graph of T has the appearance shown below.



T is not reflexive: There is no loop at 0, for example. Thus $(0, 0) \notin T$, so T is not reflexive.

T is not symmetric: There is an arrow from 0 to 1 but not from 1 to 0. Thus $(0, 1) \in T$ but $(1, 0) \notin T$, and so T is not symmetric.

T is transitive: The transitivity condition is vacuously true for T . To see this, observe that the transitivity condition says that

$$\text{For every } x, y, z \in A, \text{ if } (x, y) \in T \text{ and } (y, z) \in T \text{ then } (x, z) \in T.$$

The only way for this to be false would be for there to exist elements of A that make the hypothesis true and the conclusion false. That is, there would have to be elements x, y , and z in A such that

$$(x, y) \in T \quad \text{and} \quad (y, z) \in T \quad \text{and} \quad (x, z) \notin T.$$

In other words, there would have to be two ordered pairs in T that have the potential to “link up” by having the *second* element of one pair be the *first* element of the other pair. But the only elements in T are $(0, 1)$ and $(2, 3)$, and these do not have the potential to link up. Hence the hypothesis is never true. It follows that it is impossible for T *not* to be transitive, and thus T is transitive. ■

Note T is transitive by default because it is *not* not transitive!

When a relation R is defined on a finite set A , it is possible to write computer algorithms to check whether R is reflexive, symmetric, and transitive. One way to do this is to represent A as a one-dimensional array, $(a[1], a[2], \dots, a[n])$ and use a modification of the algorithm of exercise 38 in Section 6.1 to check whether an ordered pair in $A \times A$ is in R . Checking whether R is reflexive can be done with a loop that examines each element $a[i]$ of A in turn. If, for some i , $(a[i], a[i]) \notin R$, then R is not reflexive. Otherwise, R is reflexive. Checking for symmetry can be done with a nested loop that examines each pair $(a[i], a[j])$ of $A \times A$ in turn. If, for some i and j , $(a[i], a[j]) \in R$ and $(a[j], a[i]) \notin R$, then R is not symmetric. Otherwise, R is symmetric. Checking whether R is transitive can be done with a triply nested loop that examines each triple $(a[i], a[j], a[k])$ of $A \times A \times A$ in turn. If, for some triple, $(a[i], a[j]) \in R$, $(a[j], a[k]) \in R$, and $(a[i], a[k]) \notin R$, then R is not transitive. Otherwise, R is transitive. In the exercises for this section, you are asked to formalize these algorithms.

Properties of Relations on Infinite Sets

Suppose a relation R is defined on an infinite set A . To prove the relation is reflexive, symmetric, or transitive, first write down what is to be proved. For instance, for symmetry you need to prove that

$$\forall x, y \in A, \text{ if } x R y \text{ then } y R x.$$

Then use the definitions of A and R to rewrite the statement for the particular case in question. For instance, for the “equality” relation on the set of real numbers, the rewritten statement is

$$\forall x, y \in \mathbf{R}, \text{ if } x = y \text{ then } y = x.$$

Sometimes the truth of the rewritten statement will be immediately obvious (as it is here). At other times you will need to prove it using the method of generalizing from the generic particular. We give examples of both cases in this section. We begin with the relation of equality, one of the simplest and yet most important relations.

Example 8.2.2 Properties of Equality

Define a relation R on $\mathbf{R}$ as follows: For all real numbers x and y ,

$$x R y \iff x = y.$$

- a. Is R reflexive? b. Is R symmetric? c. Is R transitive?

Solution

- a. **R is reflexive:** R is reflexive if, and only if, the following statement is true:

$$\text{For every } x \in \mathbf{R}, \quad x R x.$$

Since $x R x$ just means that $x = x$, this is the same as saying

$$\text{For every } x \in \mathbf{R}, \quad x = x.$$

But this statement is certainly true; every real number is equal to itself.

- b. **R is symmetric:** R is symmetric if, and only if, the following statement is true:

$$\text{For every } x, y \in \mathbf{R}, \quad \text{if } x R y \text{ then } y R x.$$

By definition of R , $x R y$ means that $x = y$ and $y R x$ means that $y = x$. Hence R is symmetric if, and only if,

For every $x, y \in \mathbf{R}$, **if** $x = y$ then $y = x$.

But this statement is certainly true; if one number is equal to a second, then the second is equal to the first.

c. **R is transitive:** R is transitive if, and only if, the following statement is true:

For every $x, y, z \in \mathbf{R}$, **if** $x R y$ and $y R z$ then $x R z$.

By definition of R , $x R y$ means that $x = y$, $y R z$ means that $y = z$, and $x R z$ means that $x = z$. Hence R is transitive if, and only if, the following statement is true:

For every $x, y, z \in \mathbf{R}$, **if** $x = y$ and $y = z$ then $x = z$.

But this statement is certainly true: If one real number equals a second and the second equals a third, then the first equals the third. ■

Example 8.2.3 Properties of “Less Than”

Define a relation R on $\mathbf{R}$ as follows: For all real numbers x and y ,

$$x R y \iff x < y.$$

- a. Is R reflexive? b. Is R symmetric? c. Is R transitive?

Solution

- a. **R is not reflexive:** R is reflexive if, and only if, $\forall x \in \mathbf{R}$, $x R x$. By definition of R , this means that $\forall x \in \mathbf{R}$, $x < x$. But this is false: $\exists x \in \mathbf{R}$ such that $x \not< x$. As a counterexample, let $x = 0$ and note that $0 \not< 0$. Hence R is not reflexive.
- b. **R is not symmetric:** R is symmetric if, and only if, $\forall x, y \in \mathbf{R}$, if $x R y$ then $y R x$. By definition of R , this means that $\forall x, y \in \mathbf{R}$, if $x < y$ then $y < x$. But this is false: $\exists x, y \in \mathbf{R}$ such that $x < y$ and $y \not< x$. As a counterexample, let $x = 0$ and $y = 1$ and note that $0 < 1$ but $1 \not< 0$. Hence R is not symmetric.
- c. **R is transitive:** R is transitive if, and only if, $\forall x, y, z \in \mathbf{R}$, if $x R y$ and $y R z$ then $x R z$. By definition of R , this means that $\forall x, y, z \in \mathbf{R}$, if $x < y$ and $y < z$, then $x < z$. But this statement is true by the transitive law of order for real numbers (Appendix A, T18). Hence R is transitive. ■

Sometimes a property is “universally false” in the sense that it is false for *every* element of its domain. It follows immediately, of course, that the property is false for each particular element of the domain and hence counterexamples abound. In such a case, it may seem more natural to prove the universal falseness of the property rather than to give a single counterexample. In the example above, for instance, you might find it natural to answer (a) and (b) as follows:

Alternative Answer to (a): R is not reflexive because $x \not< x$ for every real number x (by the trichotomy law—Appendix A, T17).

Alternative Answer to (b): R is not symmetric because for all real numbers x and y in A , if $x < y$ then $y \not< x$ (by the trichotomy law).

Example 8.2.4 Properties of Congruence Modulo 3

Define a relation T on $\mathbf{Z}$ (the set of all integers) as follows: For all integers m and n ,

$$m T n \iff 3 \mid (m - n).$$

This relation is called **congruence modulo 3**.

- a. Is T reflexive? b. Is T symmetric? c. Is T transitive?

Solution

- a. **T is reflexive:** To show that T is reflexive, it is necessary to show that

$$\text{For every } m \in \mathbf{Z}, \quad m T m.$$

By definition of T , this means that

$$\text{For every } m \in \mathbf{Z}, \quad 3 \mid (m - m),$$

which is true because $m - m = 0$ and $3 \mid 0$ (since $0 = 3 \cdot 0$). Hence T is reflexive. This reasoning is formalized in the following proof.

Proof of Reflexivity: Suppose m is a particular but arbitrarily chosen integer. [We must show that $m T m$.] Now $m - m = 0$. But $3 \mid 0$ since $0 = 3 \cdot 0$. Hence $3 \mid (m - m)$. Thus, by definition of T , $m T m$ [as was to be shown].

- b. **T is symmetric:** To show that T is symmetric, it is necessary to show that

$$\text{For every } m, n \in \mathbf{Z}, \quad \text{if } m T n \text{ then } n T m.$$

By definition of T this means that

$$\text{For every } m, n \in \mathbf{Z}, \quad \text{if } 3 \mid (m - n) \text{ then } 3 \mid (n - m).$$

Is this true? Suppose m and n are particular but arbitrarily chosen integers such that $3 \mid (m - n)$. Must it follow that $3 \mid (n - m)$? [In other words, can we find an integer so that $n - m = 3 \cdot (\text{that integer})$?] By definition of “divides,” since

$$3 \mid (m - n),$$

then

$$m - n = 3k \quad \text{for some integer } k.$$

The crucial observation is that $n - m = -(m - n)$. Hence, you can multiply both sides of this equation by -1 to obtain

$$-(m - n) = -3k,$$

which is equivalent to

$$n - m = 3(-k).$$

[Thus we have found an integer, $-k$, so that $n - m = 3 \cdot (\text{that integer})$.]

Since $-k$ is an integer, this equation shows that

$$3 \mid (n - m).$$

It follows that T is symmetric.

The reasoning above is formalized in the following proof.

Proof of Symmetry: Suppose m and n are particular but arbitrarily chosen integers that satisfy the condition $m T n$. [We must show that $n T m$.] By definition of T , since $m T n$ then $3 \mid (m - n)$. By definition of “divides,” this means that $m - n = 3k$, for some integer k . Multiplying both sides by -1 gives $n - m = 3(-k)$. Since $-k$ is an integer, this equation shows that $3 \mid (n - m)$. Hence, by definition of T , $n T m$ [as was to be shown].

- c. **T is transitive:** To show that T is transitive, it is necessary to show that

For every $m, n, p \in \mathbf{Z}$, **if** $m T n$ and $n T p$ then $m T p$.

By definition of T this means that

For every $m, n \in \mathbf{Z}$, **if** $3 \mid (m - n)$ and $3 \mid (n - p)$ then $3 \mid (m - p)$.

Is this true? Suppose m, n , and p are particular but arbitrarily chosen integers such that $3 \mid (m - n)$ and $3 \mid (n - p)$. Must it follow that $3 \mid (m - p)$? [In other words, can we find an integer so that $m - p = 3 \cdot (\text{that integer})$?] By definition of “divides,” since

$$3 \mid (m - n) \quad \text{and} \quad 3 \mid (n - p),$$

then

$$m - n = 3r \quad \text{for some integer } r,$$

and

$$n - p = 3s \quad \text{for some integer } s.$$

The crucial observation is that $(m - n) + (n - p) = m - p$. Add these two equations together to obtain

$$(m - n) + (n - p) = 3r + 3s,$$

which is equivalent to

$$m - p = 3(r + s).$$

[Thus we have found an integer so that $m - p = 3 \cdot (\text{that integer})$.] Since r and s are integers, $r + s$ is an integer. So this equation shows that

$$3 \mid (m - p).$$

It follows that T is transitive.

The reasoning above is formalized in the following proof.

Proof of Transitivity: Suppose m, n , and p are particular but arbitrarily chosen integers that satisfy the condition $m T n$ and $n T p$. [We must show that $m T p$.] By definition of T , since $m T n$ and $n T p$, then $3 \mid (m - n)$ and $3 \mid (n - p)$. By definition of “divides,” this means that $m - n = 3r$ and $n - p = 3s$, for some integers r and s . Adding the two equations gives $(m - n) + (n - p) = 3r + 3s$, and simplifying gives that $m - p = 3(r + s)$. Since $r + s$ is an integer, this equation shows that $3 \mid (m - p)$. Hence, by definition of T , $m T p$ [as was to be shown].

The Transitive Closure of a Relation

Generally speaking, a relation fails to be transitive because it fails to contain certain ordered pairs. For example, if $(1, 3)$ and $(3, 4)$ are in a relation R , then the pair $(1, 4)$ *must* be in R if R is to be transitive. To obtain a transitive relation from one that is not transitive, it is necessary to add ordered pairs. Roughly speaking, the relation obtained by adding the least number of ordered pairs to ensure transitivity is called the *transitive closure* of the relation. More precisely, the transitive closure of a relation is the smallest transitive relation that contains the relation.

Definition

Let A be a set and R a relation on A . The **transitive closure** of R is the relation R^t on A that satisfies the following three properties:

1. R^t is transitive.
2. $R \subseteq R^t$.
3. If S is any other transitive relation that contains R , then $R^t \subseteq S$.

Example 8.2.5 Transitive Closure of a Relation

Let $A = \{0, 1, 2, 3\}$ and consider the relation R defined on A as follows:

$$R = \{(0, 1), (1, 2), (2, 3)\}.$$

Find the transitive closure of R .

Solution Every ordered pair in R is in R^t , so

$$\{(0, 1), (1, 2), (2, 3)\} \subseteq R^t.$$

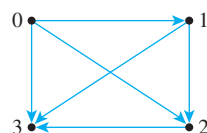
Thus the directed graph of R contains the arrows shown below.



Since there are arrows going from 0 to 1 and from 1 to 2, R^t must have an arrow going from 0 to 2. Hence $(0, 2) \in R^t$. Then $(0, 2) \in R^t$ and $(2, 3) \in R^t$, so since R^t is transitive, $(0, 3) \in R^t$. Also, since $(1, 2) \in R^t$ and $(2, 3) \in R^t$, then $(1, 3) \in R^t$. Thus R^t contains at least the following ordered pairs:

$$\{(0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (2, 3)\}.$$

But this relation *is* transitive; hence it equals R^t . The directed graph of R^t is shown below.



TEST YOURSELF

- For a relation R on a set A to be reflexive means that _____.
- For a relation R on a set A to be symmetric means that _____.
- For a relation R on a set A to be transitive means that _____.
- To show that a relation R on an infinite set A is reflexive, you suppose that _____ and you show that _____.
- To show that a relation R on an infinite set A is symmetric, you suppose that _____ and you show that _____.
- To show that a relation R on an infinite set A is transitive, you suppose that _____ and you show that _____.
- To show that a relation R on a set A is not reflexive, you _____.
- To show that a relation R on a set A is not symmetric, you _____.
- To show that a relation R on a set A is not transitive, you _____.
- Given a relation R on a set A , the transitive closure of R is the relation R' on A that satisfies the following three properties: _____, _____, and _____.

EXERCISE SET 8.2

In 1–8, a number of relations are defined on the set $A = \{0, 1, 2, 3\}$. For each relation:

- Draw the directed graph.
- Determine whether the relation is reflexive.
- Determine whether the relation is symmetric.
- Determine whether the relation is transitive.

Give a counterexample in each case in which the relation does not satisfy one of the properties.

- $R_1 = \{(0, 0), (0, 1), (0, 3), (1, 1), (1, 0), (2, 3), (3, 3)\}$
- $R_2 = \{(0, 0), (0, 1), (1, 1), (1, 2), (2, 2), (2, 3)\}$
- $R_3 = \{(2, 3), (3, 2)\}$
- $R_4 = \{(1, 2), (2, 1), (1, 3), (3, 1)\}$
- $R_5 = \{(0, 0), (0, 1), (0, 2), (1, 2)\}$
- $R_6 = \{(0, 1), (0, 2)\}$
- $R_7 = \{(0, 3), (2, 3)\}$
- $R_8 = \{(0, 0), (1, 1)\}$

In 9–33, determine whether the given relation is reflexive, symmetric, transitive, or none of these. Justify your answers.

- R is the “greater than or equal to” relation on the set of real numbers: For every $x, y \in \mathbf{R}$, $x R y \Leftrightarrow x \geq y$.
- C is the circle relation on the set of real numbers: For every $x, y \in \mathbf{R}$, $x C y \Leftrightarrow x^2 + y^2 = 1$.
- D is the relation defined on $\mathbf{R}$ as follows: For every $x, y \in \mathbf{R}$, $x D y \Leftrightarrow xy \geq 0$.
- E is the congruence modulo 4 relation on $\mathbf{Z}$: For every $m, n \in \mathbf{Z}$, $m E n \Leftrightarrow 4 \mid (m - n)$.
- F is the congruence modulo 5 relation on $\mathbf{Z}$: For every $m, n \in \mathbf{Z}$, $m F n \Leftrightarrow 5 \mid (m - n)$.
- O is the relation defined on $\mathbf{Z}$ as follows: For every $m, n \in \mathbf{Z}$, $m O n \Leftrightarrow m - n$ is odd.
- D is the “divides” relation on $\mathbf{Z}^+$: For all positive integers m and n , $m D n \Leftrightarrow m \mid n$.
- A is the “absolute value” relation on $\mathbf{R}$: For all real numbers x and y , $x A y \Leftrightarrow |x| = |y|$.
- Recall that a prime number is an integer that is greater than 1 and has no positive integer divisors other than 1 and itself. (In particular, 1 is not prime.) A relation P is defined on $\mathbf{Z}$ as follows: For every $m, n \in \mathbf{Z}$, $m P n \Leftrightarrow \exists$ a prime number p such that $p \mid m$ and $p \mid n$.
- Define a relation Q on $\mathbf{R}$ as follows: For all real numbers x and y , $x Q y \Leftrightarrow x - y$ is rational.
- Define a relation I on $\mathbf{R}$ as follows: For all real numbers x and y , $x I y \Leftrightarrow x - y$ is irrational.
- Let $X = \{a, b, c\}$ and $\mathcal{P}(X)$ be the power set of X (the set of all subsets of X). A relation $\mathbf{E}$ is defined on $\mathcal{P}(X)$ as follows: For every $A, B \in \mathcal{P}(X)$, $A \mathbf{E} B \Leftrightarrow$ the number of elements in A equals the number of elements in B .

53. Let $T = \{(0, 2), (1, 0), (2, 3), (3, 1)\}$. Find T^t , the transitive closure of T .
54. Write a computer algorithm to test whether a relation R defined on a finite set A is reflexive, where
- $$A = \{a[1], a[2], \dots, a[n]\}.$$
55. Write a computer algorithm to test whether a relation R defined on a finite set A is symmetric, where
- $$A = \{a[1], a[2], \dots, a[n]\}.$$
56. Write a computer algorithm to test whether a relation R defined on a finite set A is transitive, where
- $$A = \{a[1], a[2], \dots, a[n]\}.$$

ANSWERS FOR TEST YOURSELF

1. for every x in A , $x R x$ 2. for every x and y in A , if $x R y$ then $y R x$ 3. for every x, y , and z in A , if $x R y$ and $y R z$ then $x R z$ 4. x is any element of A ; $x R x$ 5. x and y are any elements of A such that $x R y$; $y R x$ 6. x, y , and z are any elements of A such that $x R y$ and $y R z$; $x R z$ 7. show that there is an element x in A such that $x \not R x$ 8. show that there are elements x and y in A such that $x R y$ but $y \not R x$ 9. show that there are elements x, y , and z in A such that $x R y$ and $y R z$ but $x \not R z$ 10. R' is transitive; $R \subseteq R'$; if S is any other transitive relation that contains R , then $R' \subseteq S$

8.3 Equivalence Relations

“You are sad” the Knight said in an anxious tone: “let me sing you a song to comfort you.”

... The name of the song is called ‘Haddocks’ Eyes.’”

“Oh, that’s the name of the song, is it?” Alice said, trying to feel interested.

“No, you don’t understand,” the Knight said, looking a little vexed. “That’s what the name is called. The name really is ‘The Aged Aged Man.’”

“Then I ought to have said ‘That’s what the song is called?’” Alice corrected herself.

“No, you oughtn’t: that’s quite another thing! The song is called ‘Ways and Means’: but that’s only what it’s called, you know!”

“Well, what is the song, then?” said Alice, who was by this time completely bewildered.

“I was coming to that,” the Knight said. “The song really is ‘A-sitting on a Gate’: and the tune’s my own invention.”

—Lewis Carroll, *Through the Looking Glass*, 1872

You know from your early study of fractions that each fraction has many equivalent forms. For example,

$$\frac{1}{2}, \frac{2}{4}, \frac{3}{6}, \frac{-1}{-2}, \frac{-3}{-6}, \frac{15}{30}, \dots, \text{ and so on,}$$

are all different ways to represent the same number. They may look different; they may be called different names; but they are all equal. The idea of grouping together things that “look different but are really the same” is the central idea of equivalence relations.

The Relation Induced by a Partition

A **partition** of a set A is a finite or infinite collection of nonempty, mutually disjoint subsets whose union is A . The diagram of Figure 8.3.1 illustrates a partition of a set A by subsets $A_1, A_2, \dots, A_6$.

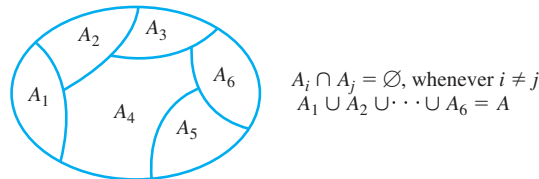


FIGURE 8.3.1 A Partition of a Set

Definition

Given a partition of a set A , the **relation induced by the partition**, R , is defined on A as follows: For every $x, y \in A$,

$$x R y \iff \text{there is a subset } A_i \text{ of the partition such that both } x \text{ and } y \text{ are in } A_i.$$

Example 8.3.1 Relation Induced by a Partition

Let $A = \{0, 1, 2, 3, 4\}$ and consider the following partition of A :

$$\{0, 3, 4\}, \{1\}, \{2\}.$$

Find the relation R induced by this partition.

Solution Since $\{0, 3, 4\}$ is a subset of the partition,

$$\begin{aligned} 0 R 3 & \text{ because both 0 and 3 are in } \{0, 3, 4\} \\ 3 R 0 & \text{ because both 3 and 0 are in } \{0, 3, 4\} \\ 0 R 4 & \text{ because both 0 and 4 are in } \{0, 3, 4\} \\ 4 R 0 & \text{ because both 4 and 0 are in } \{0, 3, 4\} \\ 3 R 4 & \text{ because both 3 and 4 are in } \{0, 3, 4\} \end{aligned}$$

and

$$4 R 3 \text{ because both 4 and 3 are in } \{0, 3, 4\}.$$

Also,

$$\begin{aligned} 0 R 0 & \text{ because both 0 and 0 are in } \{0, 3, 4\} \\ 3 R 3 & \text{ because both 3 and 3 are in } \{0, 3, 4\} \end{aligned}$$

and

$$4 R 4 \text{ because both 4 and 4 are in } \{0, 3, 4\}.$$

Since $\{1\}$ is a subset of the partition,

$$1 R 1 \text{ because both 1 and 1 are in } \{1\},$$

and since $\{2\}$ is a subset of the partition,

$$2 R 2 \text{ because both 2 and 2 are in } \{2\}.$$

Note These statements may seem strange, but, after all, they are not false!

Hence

$$R = \{(0, 0), (0, 3), (0, 4), (1, 1), (2, 2), (3, 0), (3, 3), (3, 4), (4, 0), (4, 3), (4, 4)\}.$$

The fact is that a relation induced by a partition of a set satisfies all three properties studied in Section 8.2: reflexivity, symmetry, and transitivity.

Theorem 8.3.1

Let A be a set with a partition and let R be the relation induced by the partition. Then R is reflexive, symmetric, and transitive.

Proof: Suppose A is a set with a partition. In order to simplify notation, we assume that the partition consists of only a finite number of sets. The proof for an infinite partition is identical except for notation. Denote the partition subsets by

$$A_1, A_2, \dots, A_n.$$

Then $A_i \cap A_j = \emptyset$ whenever $i \neq j$, and $A_1 \cup A_2 \cup \dots \cup A_n = A$. The relation R induced by the partition is defined as follows: For every $x, y \in A$,

$$x R y \iff \text{there is a set } A_i \text{ of the partition} \\ \text{such that } x \in A_i \text{ and } y \in A_i.$$

[Idea for the proof of reflexivity: For R to be reflexive means that each element of A is related by R to itself. But by definition of R , for an element x to be related to itself means that x is in the same subset of the partition as itself. Well, if x is in some subset of the partition, then it is certainly in the same subset as itself. And x is in some subset of the partition because the union of the subsets of the partition is all of A . This reasoning is formalized as follows.]

Proof that R is reflexive: Suppose $x \in A$. Since $A_1, A_2, \dots, A_n$ is a partition of A , it follows that $x \in A_i$ for some i , and so the statement

$$\text{there is a set } A_i \text{ of the partition such that } x \in A_i \text{ and } x \in A_i$$

is true. Thus, by definition of R , $x R x$.

[Idea for the proof of symmetry: For R to be symmetric means that any time one element is related to a second, then the second is related to the first. Now for one element x to be related to a second element y means that x and y are in the same subset of the partition. But if this is the case, then y is in the same subset of the partition as x , so y is related to x by definition of R . This reasoning is formalized as follows.]

Proof that R is symmetric: Suppose x and y are elements of A such that $x R y$. Then

$$\text{there is a subset } A_i \text{ of the partition such that } x \in A_i \text{ and } y \in A_i$$

by definition of R . It follows that the statement

$$\text{there is a subset } A_i \text{ of the partition such that } y \in A_i \text{ and } x \in A_i$$

is also true. Hence, by definition of R , $y R x$.

(continued on page 508)

Note The deduction that $x \in A_i$ and $x \in A_i$ is based on the logical equivalence of the statement forms p and $p \wedge p$.

Note The deduction that $y \in A_i$ and $x \in A_i$ is based on the logical equivalence of the statement forms $p \wedge q$ and $q \wedge p$.

[Idea for the proof of transitivity: For R to be transitive means that any time one element of A is related by R to a second and that second is related to a third, then the first element is related to the third. But for one element to be related to another means that there is a subset of the partition that contains both. So suppose x , y , and z are elements such that x is in the same subset as y and y is in the same subset as z . Must x be in the same subset as z ? Yes, because the subsets of the partition are mutually disjoint. Since the subset that contains x and y has an element in common with the subset that contains y and z (namely, y), the two subsets are equal. But this means that x , y , and z are all in the same subset, and so, in particular, x and z are in the same subset. Hence x is related by R to z . This reasoning is formalized as follows.]

Proof that R is transitive: Suppose x , y , and z are in A and $x R y$ and $y R z$. By definition of R , there are subsets A_i and A_j of the partition such that

$$x \text{ and } y \text{ are in } A_i \quad \text{and} \quad y \text{ and } z \text{ are in } A_j.$$

Suppose $A_i \neq A_j$. [We will deduce a contradiction.] Then $A_i \cap A_j = \emptyset$ since $\{A_1, A_2, A_3, \dots, A_n\}$ is a partition of A . But y is in A_i and y is in A_j also. Hence $A_i \cap A_j \neq \emptyset$. [This contradicts the statement that $A_i \cap A_j = \emptyset$.] Thus $A_i = A_j$. It follows that x , y , and z are all in A_i , and so, in particular,

$$x \text{ and } z \text{ are in } A_i.$$

Thus $x R z$ by definition of R .

Definition of an Equivalence Relation

A relation on a set that satisfies the three properties of reflexivity, symmetry, and transitivity is called an *equivalence relation*.

Definition

Let A be a set and R a relation on A . R is an **equivalence relation** if, and only if, R is reflexive, symmetric, and transitive.

Thus, according to Theorem 8.3.1, the relation induced by a partition is an equivalence relation. A variety of additional examples of equivalence relations are given below and in the exercises.

Example 8.3.2 An Equivalence Relation on a Set of Subsets

Let X be the set of all nonempty subsets of $\{1, 2, 3\}$. Then

$$X = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

Define a relation $\mathbf{R}$ on X as follows: For every A and B in X ,

$$A \mathbf{R} B \Leftrightarrow \text{the least element of } A \text{ equals the least element of } B.$$

Prove that $\mathbf{R}$ is an equivalence relation on X .

Solution

$\mathbf{R}$ is reflexive: Suppose A is a nonempty subset of $\{1, 2, 3\}$. [We must show that $A \mathbf{R} A$.] It is true to say that the least element of A equals the least element of A . Thus, by definition of $\mathbf{R}$, $A \mathbf{R} A$.

R is symmetric: Suppose A and B are nonempty subsets of $\{1, 2, 3\}$ and $A \mathbf{R} B$. [We must show that $B \mathbf{R} A$.] Since $A \mathbf{R} B$, the least element of A equals the least element of B . But this implies that the least element of B equals the least element of A , and so, by definition of $\mathbf{R}$, $B \mathbf{R} A$.

R is transitive: Suppose A , B , and C are nonempty subsets of $\{1, 2, 3\}$, $A \mathbf{R} B$, and $B \mathbf{R} C$. [We must show that $A \mathbf{R} C$.] Since $A \mathbf{R} B$, the least element of A equals the least element of B and since $B \mathbf{R} C$, the least element of B equals the least element of C . Thus the least element of A equals the least element of C , and so, by definition of $\mathbf{R}$, $A \mathbf{R} C$. ■

Example 8.3.3 Equivalence of Digital Logic Circuits Is an Equivalence Relation

Let S be the set of all digital logic circuits with a fixed number n of inputs. Define a relation $\mathbf{E}$ on S as follows: For all circuits C_1 and C_2 in S ,

$$C_1 \mathbf{E} C_2 \iff C_1 \text{ has the same input/output table as } C_2.$$

If $C_1 \mathbf{E} C_2$, then circuit C_1 is said to be *equivalent* to circuit C_2 . Prove that $\mathbf{E}$ is an equivalence relation on S .

Solution

E is reflexive: Suppose C is a digital logic circuit in S . [We must show that $C \mathbf{E} C$.] Certainly C has the same input/output table as itself. Thus, by definition of $\mathbf{E}$, $C \mathbf{E} C$ [as was to be shown].

E is symmetric: Suppose C_1 and C_2 are digital logic circuits in S such that $C_1 \mathbf{E} C_2$. [We must show that $C_2 \mathbf{E} C_1$.] By definition of $\mathbf{E}$, since $C_1 \mathbf{E} C_2$, then C_1 has the same input/output table as C_2 . It follows that C_2 has the same input/output table as C_1 . Hence, by definition of $\mathbf{E}$, $C_2 \mathbf{E} C_1$ [as was to be shown].

E is transitive: Suppose C_1 , C_2 , and C_3 are digital logic circuits in S such that $C_1 \mathbf{E} C_2$ and $C_2 \mathbf{E} C_3$. [We must show that $C_1 \mathbf{E} C_3$.] By definition of $\mathbf{E}$, since $C_1 \mathbf{E} C_2$ and $C_2 \mathbf{E} C_3$, then

C_1 has the same input/output table as C_2

and

C_2 has the same input/output table as C_3 .

It follows that

C_1 has the same input/output table as C_3 .

Hence, by definition of $\mathbf{E}$, $C_1 \mathbf{E} C_3$ [as was to be shown].

Since $\mathbf{E}$ is reflexive, symmetric, and transitive, $\mathbf{E}$ is an equivalence relation on S . ■

Certain implementations of computer languages do not place a limit on the allowable length of an identifier. This permits a programmer to be as precise as necessary in naming variables without having to worry about exceeding length limitations. However, compilers for such languages often ignore all but some specified number of initial characters: As far as the compiler is concerned, two identifiers are the same if they have the same initial characters, even though they may look different to a human reader of the program. For example, to a compiler that ignores all but the first eight characters of an identifier, the following identifiers would be the same:

NumberOfScrews NumberOfBolts.

Obviously, in using such a language, the programmer has to be sure to avoid giving two distinct identifiers the same first eight characters. When a compiler lumps identifiers together in this way, it sets up an equivalence relation on the set of all possible identifiers in the language. Such a relation is described in the next example.

Example 8.3.4 A Relation on a Set of Identifiers

Let L be the set of all allowable identifiers in a certain computer language, and define a relation R on L as follows: For all strings s and t in L ,

$$s R t \iff \text{the first eight characters of } s \text{ equal the first eight characters of } t.$$

Prove that R is an equivalence relation on L .

Solution

R is reflexive: Let $s \in L$. [We must show that $s R s$.] Clearly s has the same first eight characters as itself. Thus, by definition of R , $s R s$ [as was to be shown].

R is symmetric: Let s and t be in L and suppose that $s R t$. [We must show that $t R s$.] By definition of R , since $s R t$, the first eight characters of s equal the first eight characters of t . It follows that the first eight characters of t equal the first eight characters of s , and so, by definition of R , $t R s$ [as was to be shown].

R is transitive: Let s , t , and u be in L and suppose that $s R t$ and $t R u$. [We must show that $s R u$.] By definition of R , since $s R t$ and $t R u$, the first eight characters of s equal the first eight characters of t , and the first eight characters of t equal the first eight characters of u . Hence the first eight characters of s equal the first eight characters of u . Hence, by definition of R , $s R u$ [as was to be shown].

Since R is reflexive, symmetric, and transitive, R is an equivalence relation on L . ■

Equivalence Classes of an Equivalence Relation

Suppose there is an equivalence relation on a certain set. If a is any particular element of the set, then one can ask, “What is the subset of all elements that are related to a ?” This subset is called the *equivalence class* of a .

Note Be careful to distinguish among the following: (1) a relation on a set, (2) the (underlying) set itself, and (3) the equivalence class for an element of the (underlying) set.

Definition

Suppose A is a set and R is an equivalence relation on A . For each element a in A , the **equivalence class of a** , denoted $[a]$ and called the **class of a** for short, is the set of all elements x in A such that x is related to a by R .

In symbols:

$$[a] = \{x \in A \mid x R a\}$$

The procedural version of this definition is

$$\text{for every } x \in A, \quad x \in [a] \iff x R a.$$

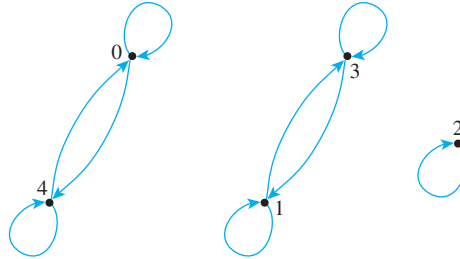
When several equivalence relations on a set are under discussion, the notation $[a]_R$ may be used to denote the equivalence class of a for the relation R .

Example 8.3.5 Equivalence Classes of a Relation Given as a Set of Ordered Pairs

Let $A = \{0, 1, 2, 3, 4\}$ and define a relation R on A as follows:

$$R = \{(0, 0), (0, 4), (1, 1), (1, 3), (2, 2), (3, 1), (3, 3), (4, 0), (4, 4)\}.$$

The directed graph for R is as shown below. As can be seen by inspection, R is an equivalence relation on A . Find the distinct equivalence classes of R .



Solution First find the equivalence class of every element of A .

$$[0] = \{x \in A \mid x R 0\} = \{0, 4\}$$

$$[1] = \{x \in A \mid x R 1\} = \{1, 3\}$$

$$[2] = \{x \in A \mid x R 2\} = \{2\}$$

$$[3] = \{x \in A \mid x R 3\} = \{1, 3\}$$

$$[4] = \{x \in A \mid x R 4\} = \{0, 4\}$$

Note that $[0] = [4]$ and $[1] = [3]$. Thus the *distinct* equivalence classes of the relation are

$$\{0, 4\}, \{1, 3\}, \text{ and } \{2\}.$$

When a problem asks you to find the *distinct* equivalence classes of an equivalence relation, you will generally solve the problem in two steps. In the first step you either explicitly construct (as in Example 8.3.5) or imagine constructing (as in infinite cases) the equivalence class for each element of the domain A of the relation. Usually several of the classes will contain exactly the same elements, so in the second step you must take a careful look at the classes to determine which are the same. You then indicate the distinct equivalence classes by describing them without duplication.

Example 8.3.6 Equivalence Classes of a Relation on a Set of Subsets

In Example 8.3.2 it was shown that the relation $\mathbf{R}$ was an equivalence relation, where for nonempty subsets A and B of $\{1, 2, 3\}$ to be related by $\mathbf{R}$ means that they have the same least element. Describe the distinct equivalence classes of $\mathbf{R}$.

Solution The equivalence class of $\{1\}$ is the set of all the nonempty subsets of $\{1, 2, 3\}$ whose least element is 1. Thus

$$[\{1\}] = \{\{1\}, \{1, 2\}, \{1, 3\}, \{1, 2, 3\}\}.$$

The equivalence class of $\{2\}$ is the set of all the nonempty subsets of $\{1, 2, 3\}$ whose least element is 2. Thus

$$[\{2\}] = \{\{2\}, \{2, 3\}\}.$$

The equivalence class of $\{3\}$ is the set of all the nonempty subsets of $\{1, 2, 3\}$ whose least element is 3. There is only one such set: $\{3\}$ itself. Thus

$$[\{3\}] = \{\{3\}\}.$$

Since all the nonempty subsets of $\{1, 2, 3\}$ are in one of the equivalence classes, this is a complete listing. Moreover, these classes are all distinct. ■

Example 8.3.7 Equivalence Classes of Identifiers

In Example 8.3.4 it was shown that the relation R of having the same first eight characters is an equivalence relation on the set L of allowable identifiers in a computer language. Describe the distinct equivalence classes of R .

Solution By definition of R , two strings in L are related by R if, and only if, they have the same first eight characters. Given any string s in L ,

$$\begin{aligned} [s] &= \{t \in L \mid t R s\} \\ &= \{t \in L \mid \text{the first eight characters of } t \text{ equal the first eight characters of } s\}. \end{aligned}$$

Thus the distinct equivalence classes of R are sets of strings such that (1) each class consists entirely of strings all of which have the same first eight characters, and (2) any two distinct classes contain strings that differ somewhere in their first eight characters. ■

Example 8.3.8 Equivalence Classes of the Identity Relation

Let A be any set and define a relation R on A as follows: For every x and y in A ,

$$x R y \iff x = y.$$

Then R is an equivalence relation. [To prove this, just generalize the argument used in Example 8.2.2.] Describe the distinct equivalence classes of R .

Solution Given any a in A , the class of a is

$$[a] = \{x \in A \mid x R a\}.$$

Now by definition of R , $a R x$ if, and only if, $a = x$. So

$$\begin{aligned} [a] &= \{x \in A \mid x = a\} \\ &= \{a\} \end{aligned} \quad \text{since the only element of } A \text{ that equals } a \text{ is } a.$$

Hence, given any a in A ,

$$[a] = \{a\},$$

and if $x \neq a$ then $\{x\} \neq \{a\}$. Consequently, all the classes of all the elements of A are distinct, and the distinct equivalence classes of R are all the single-element subsets of A . ■

In each of Examples 8.3.5, 8.3.6, 8.3.7, and 8.3.8, the set of distinct equivalence classes of the relation consists of mutually disjoint subsets whose union is the entire domain A of the relation. This means that the set of equivalence classes of the relation forms a partition of the domain A . In fact, it is always the case that the equivalence classes of an equivalence relation partition the domain of the relation into a union of mutually disjoint subsets. We establish the truth of this statement in stages, first proving two lemmas and then proving the main theorem.

The first lemma says that if two elements of A are related by an equivalence relation R , then their equivalence classes are the same.

Lemma 8.3.2

Suppose A is a set, R is an equivalence relation on A , and a and b are elements of A . If $a R b$, then $[a] = [b]$.

This lemma says that if a certain condition is satisfied, then $[a] = [b]$. Now $[a]$ and $[b]$ are *sets*, and two sets are equal if, and only if, each is a subset of the other. Hence the proof of the lemma consists of two parts: first, a proof that $[a] \subseteq [b]$ and second, a proof that $[b] \subseteq [a]$. To show each subset relation, it is necessary to show that every element in the left-hand set is an element of the right-hand set.

Proof of Lemma 8.3.2:

Let A be a set, let R be an equivalence relation on A , and suppose

a and b are elements of A such that $a R b$.

[We must show that $[a] = [b]$.]

Proof that $[a] \subseteq [b]$: Let $x \in [a]$. *[We must show that $x \in [b]$.]* Since

$$x \in [a],$$

then

$$x R a$$

by definition of class. But

$$a R b$$

by hypothesis. Thus, by transitivity of R ,

$$x R b.$$

Hence

$$x \in [b]$$

by definition of class. *[This is what was to be shown.]*

Proof that $[b] \subseteq [a]$: Let $x \in [b]$. *[We must show that $x \in [a]$.]* Since

$$x \in [b]$$

then

$$x R b$$

by definition of class. Now

$$a R b$$

by hypothesis. Thus, since R is symmetric,

$$b R a$$

also. Then, since R is transitive and $x R b$ and $b R a$,

$$x R a.$$

Hence,

$$x \in [a]$$

by definition of class. *[This is what was to be shown.]*

Since $[a] \subseteq [b]$ and $[b] \subseteq [a]$, it follows that $[a] = [b]$ by definition of set equality.

The second lemma says that any two equivalence classes of an equivalence relation are either mutually disjoint or identical.

Lemma 8.3.3

If A is a set, R is an equivalence relation on A , and a and b are elements of A , then
either $[a] \cap [b] = \emptyset$ or $[a] = [b]$.

The statement of Lemma 8.3.3 has the form

if p then $(q \text{ or } r)$,

where p is the statement “ A is a set, R is an equivalence relation on A , and a and b are elements of A ,” q is the statement “ $[a] \cap [b] = \emptyset$,” and r is the statement “ $[a] = [b]$.” To prove the lemma, we will prove the logically equivalent statement

if $(p \text{ and not } q)$ then r .

That is, we will prove the following:

If A is a set, R is an equivalence relation on A , a and b are elements of A , and $[a] \cap [b] \neq \emptyset$, then $[a] = [b]$.

Note You can always prove a statement of the form “if p then $(q \text{ or } r)$ ” by proving one of the logically equivalent statements: “if $(p \text{ and not } q)$ then r ” or “if $(p \text{ and not } r)$ then q .”*

Proof of Lemma 8.3.3:

Suppose A is a set, R is an equivalence relation on A , a and b are elements of A , and

$$[a] \cap [b] \neq \emptyset.$$

[We must show that $[a] = [b]$.] Since $[a] \cap [b] \neq \emptyset$, there exists an element x in A such that $x \in [a] \cap [b]$. By definition of intersection,

$$x \in [a] \quad \text{and} \quad x \in [b],$$

and so

$$x R a \quad \text{and} \quad x R b$$

by definition of class. Since R is symmetric [being an equivalence relation] and $x R a$, then $a R x$. But R is also transitive [since it is an equivalence relation], and so, since $a R x$ and $x R b$,

$$a R b.$$

Now a and b satisfy the hypothesis of Lemma 8.3.2. Hence, by that lemma,

$$[a] = [b]$$

[as was to be shown].

Theorem 8.3.4 The Partition Induced by an Equivalence Relation

If A is a set and R is an equivalence relation on A , then the distinct equivalence classes of R form a partition of A ; that is, the union of the equivalence classes is all of A , and the intersection of any two distinct classes is empty.

*See exercise 14 in Section 2.2.

The proof of Theorem 8.3.4 is divided into two parts: first, a proof that A is the union of the equivalence classes of R and second, a proof that the intersection of any two distinct equivalence classes is empty. The proof of the first part follows from the fact that the relation is reflexive. The proof of the second part follows from Lemma 8.3.3.

Proof of Theorem 8.3.4:

Suppose A is a set and R is an equivalence relation on A . For notational simplicity, we assume that R has only a finite number of distinct equivalence classes, which we denote

$$A_1, A_2, \dots, A_n,$$

where n is a positive integer. (When the number of classes is infinite, the proof is identical except for notation.)

Proof that $A = A_1 \cup A_2 \cup \dots \cup A_n$: [We must show that $A \subseteq A_1 \cup A_2 \cup \dots \cup A_n$ and that $A_1 \cup A_2 \cup \dots \cup A_n \subseteq A$.]

To show that $A \subseteq A_1 \cup A_2 \cup \dots \cup A_n$, suppose x is any element of A . [We must show that $x \in A_1 \cup A_2 \cup \dots \cup A_n$.] By reflexivity of R , $x R x$. And this implies that $x \in [x]$ by definition of class. Since x is in *some* equivalence class, it must be in one of the distinct equivalence classes $A_1, A_2, \dots$, or A_n . Thus $x \in A_i$ for some index i , and hence $x \in A_1 \cup A_2 \cup \dots \cup A_n$ by definition of union [as was to be shown].

To show that $A_1 \cup A_2 \cup \dots \cup A_n \subseteq A$, suppose $x \in A_1 \cup A_2 \cup \dots \cup A_n$. [We must show that $x \in A$.] Then $x \in A_i$ for some $i = 1, 2, \dots, n$, by definition of union. Now each A_i is an equivalence class of R , and equivalence classes are subsets of A . Hence $A_i \subseteq A$ and so $x \in A$ [as was to be shown].

Since $A \subseteq A_1 \cup A_2 \cup \dots \cup A_n$ and $A_1 \cup A_2 \cup \dots \cup A_n \subseteq A$, then by definition of set equality, $A = A_1 \cup A_2 \cup \dots \cup A_n$.

Proof that the distinct classes of R are mutually disjoint: Suppose that A_i and A_j are any two distinct equivalence classes of R . [We must show that A_i and A_j are disjoint.] Since A_i and A_j are distinct, then $A_i \neq A_j$. And since A_i and A_j are equivalence classes of R , there must exist elements a and b in A such that $A_i = [a]$ and $A_j = [b]$. By Lemma 8.3.3,

$$\text{either } [a] \cap [b] = \emptyset \quad \text{or} \quad [a] = [b].$$

Now $[a] \neq [b]$ because $A_i \neq A_j$, and hence $[a] \cap [b] = \emptyset$. Thus $A_i \cap A_j = \emptyset$, and so A_i and A_j are disjoint [as was to be shown].

Example 8.3.9

Equivalence Classes of Digital Logic Circuits

In Example 8.3.3 it was shown that the relation of equivalence among circuits is an equivalence relation. Let S be the set of all digital logic circuits with exactly two inputs and one output. The binary relation $\mathbf{E}$ is defined on S as follows: For every C_1 and C_2 in S ,

$$C_1 \mathbf{E} C_2 \iff C_1 \text{ has the same input/output table as } C_2.$$

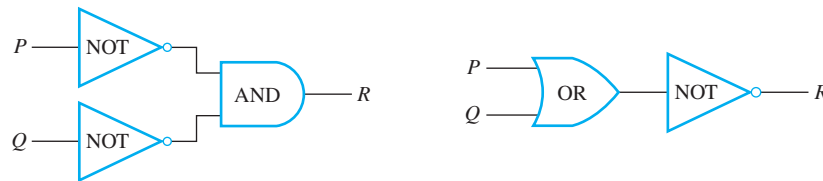
Describe the equivalence classes of this relation. How many distinct equivalence classes are there? Find two different circuits that are in one of the classes.

Solution Given a circuit C , the equivalence class of C is the set of all circuits with two input signals and one output signal that have the same input/output table as C . Now each input/output table has exactly four rows, corresponding to the four possible combinations of inputs: 11, 10, 01, and 00. A typical input/output table is the following:

Input		Output
P	Q	R
1	1	0
1	0	0
0	1	0
0	0	1

There are exactly as many such tables as there are binary strings of length 4. The reason is that distinct input/output tables can be formed by changing the pattern of the four 0's and 1's in the output column, and there are as many ways to do that as there are strings of four 0's and 1's. And since the number of binary strings of length 4 is $2^4 = 16$, there are 16 distinct input/output tables.

This implies that there are exactly 16 equivalence classes of circuits, one for each distinct input/output table. However, there are infinitely many circuits that give rise to each table. For instance, two circuits for the previous input/output table are shown below.



Congruence Modulo n

Example 8.2.4 showed that the relation of congruence modulo 3 is reflexive, symmetric, and transitive. Therefore, it is an equivalence relation.

Example 8.3.10 Equivalence Classes of Congruence Modulo 3

Let R be the relation of congruence modulo 3 on the set $\mathbf{Z}$ of all integers. That is, for all integers m and n ,

$$m R n \iff 3 \mid (m - n).$$

Describe the distinct equivalence classes of R .

Solution For each integer a ,

$$\begin{aligned} [a] &= \{x \in \mathbf{Z} \mid x R a\} \\ &= \{x \in \mathbf{Z} \mid 3 \mid (x - a)\} \\ &= \{x \in \mathbf{Z} \mid x - a = 3k, \text{ for some integer } k\}. \end{aligned}$$

Therefore,

$$[a] = \{x \in \mathbf{Z} \mid x = 3k + a, \text{ for some integer } k\}.$$

In particular,

$$\begin{aligned}
 [0] &= \{x \in \mathbf{Z} \mid x = 3k + 0, \text{ for some integer } k\} \\
 &= \{x \in \mathbf{Z} \mid x = 3k, \text{ for some integer } k\} \\
 &= \{\dots -9, -6, -3, 0, 3, 6, 9, \dots\}, \\
 [1] &= \{x \in \mathbf{Z} \mid x = 3k + 1, \text{ for some integer } k\} \\
 &= \{\dots -8, -5, -2, 1, 4, 7, 10, \dots\}, \\
 [2] &= \{x \in \mathbf{Z} \mid x = 3k + 2, \text{ for some integer } k\} \\
 &= \{\dots -7, -4, -1, 2, 5, 8, 11, \dots\}.
 \end{aligned}$$

Now since $3 R 0$, then by Lemma 8.3.2,

$$[3] = [0].$$

More generally, by the same reasoning,

$$[0] = [3] = [-3] = [6] = [-6] = \dots, \text{ and so on.}$$

Similarly,

$$[1] = [4] = [-2] = [7] = [-5] = \dots, \text{ and so on.}$$

And

$$[2] = [5] = [-1] = [8] = [-4] = \dots, \text{ and so on.}$$

Notice that every integer is in class $[0]$, $[1]$, or $[2]$. Hence the distinct equivalence classes are

$$\begin{aligned}
 &\{x \in \mathbf{Z} \mid x = 3k, \text{ for some integer } k\}, \\
 &\{x \in \mathbf{Z} \mid x = 3k + 1, \text{ for some integer } k\}, \quad \text{and} \\
 &\{x \in \mathbf{Z} \mid x = 3k + 2, \text{ for some integer } k\}.
 \end{aligned}$$

In words, the three classes of congruence modulo 3 are (1) the set of all integers that are divisible by 3, (2) the set of all integers that leave a remainder of 1 when divided by 3, and (3) the set of all integers that leave a remainder of 2 when divided by 3. ■

Example 8.3.10 illustrates a very important property of equivalence classes, namely that an equivalence class may have many different names. In Example 8.3.10, for instance, the class of 0, $[0]$, may also be *called* the class of 3, $[3]$, or the class of -6 , $[-6]$. But what the class *is* is the set of all integers that are divisible by 3:

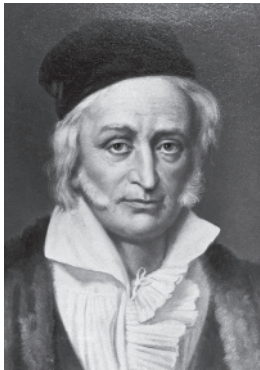
$$\{x \in \mathbf{Z} \mid x = 3k, \text{ for some integers } k\}.$$

(The quote at the beginning of this section refers in a humorous way to the philosophically interesting distinction between what things are *called* and what they *are*.)

Definition

Suppose R is an equivalence relation on a set A and S is an equivalence class of R . A **representative** of the class S is any element a such that $[a] = S$.

In exercises 36–41 at the end of this section, you are asked to show, in effect, that if a is any element of an equivalence class S , then $S = [a]$. Hence *any* element of an equivalence class is a representative of that class.



Carl Friedrich Gauss
(1777–1855)

Bettmann/Getty Images

The following notation is used frequently when referring to congruence relations. It was introduced by Carl Friedrich Gauss in the first chapter of his book *Disquisitiones Arithmeticae*. This work, which was published when Gauss was only 24, laid the foundation for modern number theory.

Definition

Let m and n be integers and let d be a positive integer. We say that **m is congruent to n modulo d** and write

$$m \equiv n \pmod{d}$$

if, and only if,

$$d \mid (m - n).$$

Symbolically:

$$m \equiv n \pmod{d} \iff d \mid (m - n).$$

Exercise 17(b) at the end of this section asks you to show that $m \equiv n \pmod{d}$ if, and only if, $m \bmod d = n \bmod d$, where m, n , and d are integers and d is positive.

Example 8.3.11 **Evaluating Congruences**

Determine which of the following congruences are true and which are false.

- a. $12 \equiv 7 \pmod{5}$ b. $6 \equiv -8 \pmod{4}$ c. $3 \equiv 3 \pmod{7}$

Solution

- a. True. $12 - 7 = 5 = 5 \cdot 1$. Hence $5 \mid (12 - 7)$, and so $12 \equiv 7 \pmod{5}$.
b. False. $6 - (-8) = 14$, and $4 \nmid 14$ because $14 \neq 4 \cdot k$ for any integer k . Consequently, $6 \not\equiv -8 \pmod{4}$.
c. True. $3 - 3 = 0 = 7 \cdot 0$. Hence $7 \mid (3 - 3)$, and so $3 \equiv 3 \pmod{7}$. ■

The discussion of binary arithmetic in Section 2.5 used a simplified computer model, which used 8-bit two’s complements and binary arithmetic to represent and compute with the 256 integers from -128 to $127 - 1$. To find $78 + (-46)$, the following scheme was shown:

	0	1	0	0	1	1	1	0	↔ 78
+	1	1	0	1	0	0	1	0	↔ -46
1	0	0	1	0	0	0	0	0	

Now the number corresponding to 10010000_2 is 288, which is too large to be the 8-bit binary representation of a number. But the directions said to discard the 1 in the 2^9 th position. When this is done, the result is 00100000_2 , which corresponds to 32 and is the correct answer for the problem.

To see why this method works, observe that the decimal forms of the two’s complements for 78 and -46 are 78 and $2^8 - 46$, respectively. In addition,

$$78 + (2^8 - 46) = 2^8 + (78 - 46) \equiv (78 - 46) \pmod{2^8}.$$

Thus reducing the number $78 + (2^8 - 46)$ modulo 2^8 gives the correct result. And doing so is equivalent to dropping the 1 in the 2^9 th position of the computation’s result because the 2^9 th position holds the place for the number 2^8 when integers are represented in binary notation.

A Definition for Rational Numbers

For a moment, forget what you know about fractional arithmetic and look at the numbers

$$\frac{1}{3} \quad \text{and} \quad \frac{2}{6}$$

as *symbols*. Considered as symbolic expressions, these *appear* quite different. In fact, if they were written as ordered pairs

$$(1, 3) \quad \text{and} \quad (2, 6)$$

they would *be* different. The fact that we regard them as “the same” is a specific instance of our general agreement to regard any two numbers

$$\frac{a}{b} \quad \text{and} \quad \frac{c}{d}$$

as equal provided the *cross products* are equal; in other words, if, and only if, $ad = bc$. This can be formalized as follows, using the language of equivalence relations.

Example 8.3.12

Rational Numbers Are Really Equivalence Classes

Let A be the set of all ordered pairs of integers for which the second element of the pair is nonzero. Symbolically:

$$A = \mathbf{Z} \times (\mathbf{Z} - \{0\}).$$

Define a relation R on A as follows: For all pairs (a, b) and (c, d) in A ,

$$(a, b) R (c, d) \iff ad = bc.$$

The fact is that R is an equivalence relation.

- Prove that R is transitive. (Proofs that R is reflexive and symmetric are left to exercise 42 at the end of the section.)
- Describe the distinct equivalence classes of R .

Solution

- [We must show that for all ordered pairs (a, b) , (c, d) , and (e, f) in A , if $(a, b) R (c, d)$ and $(c, d) R (e, f)$, then $(a, b) R (e, f)$.] Suppose (a, b) , (c, d) , and (e, f) are particular but arbitrarily chosen elements of A such that $(a, b) R (c, d)$ and $(c, d) R (e, f)$. [We must show that $(a, b) R (e, f)$.] By definition of R ,

$$(1) \quad ad = bc \quad \text{and} \quad (2) \quad cf = de.$$

Since the second elements of all ordered pairs in A are nonzero, $b \neq 0$, $d \neq 0$, and $f \neq 0$. Multiply both sides of equation (1) by f and both sides of equation (2) by b to obtain

$$(1') \quad adf = bcf \quad \text{and} \quad (2') \quad bcf = bde.$$

Because both equal bcf ,

$$adf = bde,$$

and, since $d \neq 0$, it follows from the cancellation law for multiplication (T7 in Appendix A) that

$$af = be.$$

Hence, by definition of R , $(a, b) R (e, f)$ [as was to be shown].

- b. There is one equivalence class for each distinct rational number. Each equivalence class consists of all ordered pairs (a, b) that, if written as fractions a/b , would equal each other. The reason is that the condition for two rational numbers to be equal is the same as the condition for two ordered pairs to be related. For instance, the class of $(1, 2)$ is

$$[(1, 2)] = \{(1, 2), (-1, -2), (2, 4), (-2, -4), (3, 6), (-3, -6), \dots\}$$

since $\frac{1}{2} = \frac{-1}{-2} = \frac{2}{4} = \frac{-2}{-4} = \frac{3}{6} = \frac{-3}{-6}$ and so forth. ■

It is possible to expand the result of Example 8.3.12 to define operations of addition and multiplication on the equivalence classes of R that satisfy all the same properties as the addition and multiplication of rational numbers. (See exercise 43.) It follows that the rational numbers can be defined as equivalence classes of ordered pairs of integers. Similarly (see exercise 44), it can be shown that all integers, negative and zero included, can be defined as equivalence classes of ordered pairs of positive integers. In the late nineteenth century, F. L. G. Frege and Giuseppe Peano showed that the positive integers can be defined entirely in terms of sets. And just a little earlier, Richard Dedekind (1848–1916) showed that all real numbers can be defined as sets of rational numbers. Taken together, these results show that the set of real numbers can be defined using logic and set theory alone.

TEST YOURSELF

- For a relation on a set to be an equivalence relation, it must be _____.
- The notation $m \equiv n \pmod{d}$ is read “_____” and means that _____.
- Given an equivalence relation R on a set A and given an element a in A , the equivalence class of a is denoted _____ and is defined to be _____.
- If A is a set, R is an equivalence relation on A , and a and b are elements of A , then either $[a] = [b]$ or _____.
- If A is a set and R is an equivalence relation on A , then the distinct equivalence classes of R form _____.
- Let $A = \mathbf{Z} \times (\mathbf{Z} - \{0\})$, and define a relation R on A by specifying that for every (a, b) and (c, d) in A , $(a, b) R (c, d)$ if, and only if, $ad = bc$. Then there is exactly one equivalence class of R for each _____.

EXERCISE SET 8.3

- Suppose that $S = \{a, b, c, d, e\}$ and R is a relation on S such that $a R b$, $b R c$, and $d R e$. List all of the following that must be true if R is (a) reflexive (but not symmetric or transitive), (b) symmetric (but not reflexive or transitive), (c) transitive (but not reflexive or symmetric), and (d) an equivalence relation.
 $c R b$ $c R c$ $a R c$ $b R a$
 $a R d$ $e R a$ $e R d$ $c R a$
- Each of the following partitions of $\{0, 1, 2, 3, 4\}$ induces a relation R on $\{0, 1, 2, 3, 4\}$. In each case, find the ordered pairs in R .
a. $\{0, 2\}, \{1\}, \{3, 4\}$ **b.** $\{0\}, \{1, 3, 4\}, \{2\}$
c. $\{0\}, \{1, 2, 3, 4\}$

In each of 3–6, the relation R is an equivalence relation on A . As in Example 8.3.5, first find the specified equivalence classes. Then state the number of distinct equivalence classes for R and list them.

- $A = \{0, 1, 2, 3, 4\}$
 $R = \{(0, 0), (0, 4), (1, 1), (1, 3), (2, 2), (3, 1), (3, 3), (4, 0), (4, 4)\}$
 equivalence classes: $[0], [1], [2], [3]$
- $A = \{a, b, c, d\}$
 $R = \{(a, a), (b, b), (b, d), (c, c), (d, b), (d, d)\}$
 equivalence classes: $[a], [b], [c], [d]$

that n is an integer with two different prime factorizations: $n = p_1 p_2 \cdots p_t = q_1 q_2 \cdots q_u$. All the prime factors that appear on both sides can be cancelled (as many times as they appear on both sides) to arrive at the situation where $p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s$, $p_1 \leq p_2 \leq \cdots \leq p_r$, $q_1 \leq q_2 \leq \cdots \leq q_s$, and $p_i \neq q_j$ for any integers i and j . Then use part (a) to deduce a contradiction, and conclude that the prime factorization of n is unique except, possibly, for the order in which the prime factors are written.

42. According to Fermat's little theorem, if p is a prime number and a and p are relatively prime,

then $a^{p-1} \equiv 1 \pmod{p}$. Verify that this theorem gives correct results for the following:

- a. $a = 15$ and $p = 7$ b. $a = 8$ and $p = 11$

43. Fermat's little theorem can be used to show that a number is not prime by finding a number a relatively prime to p with the property that $a^{p-1} \not\equiv 1 \pmod{p}$. However, it cannot be used to show that a number *is* prime. Find an example to illustrate this fact. That is, find integers a and p such that a and p are relatively prime and $a^{p-1} \equiv 1 \pmod{p}$ but p is not prime.

ANSWERS FOR TEST YOURSELF

1. the letter that is three places in the alphabet to the right of the given letter, with X wrapped around to A , Y to B , and Z to C 2. $a \equiv b \pmod{n}$; $a = b + kn$ for some integer k ; a and b have the same nonnegative remainder when divided by n ; $a \bmod n = b \bmod n$ 3. $(c + d) \pmod{n}$; $(c - d) \pmod{n}$;

- $(cd) \pmod{n}$; $c^m \pmod{n}$ 4. sum of powers of 2
5. Euclidean 6. a linear combination of a and n
7. $C = M^e \bmod pq$; $M = C^d \bmod pq$; d is a positive inverse for e modulo $(p-1)(q-1)$ 8. $a \mid b$ 9. $a^{p-1} \equiv 1 \pmod{p}$
10. $M^{ed} \bmod pq$

8.5 Partial Order Relations

There is no branch of mathematics, however abstract, which may not some day be applied to phenomena of the real world. —Nicolai Ivanovitch Lobachevsky, 1792–1856

In order to obtain a degree in computer science at a certain university, a student must take a specified set of required courses, some of which must be completed before others can be started. Given the prerequisite structure of the program, one might ask what is the least number of school terms needed to fulfill the degree requirements, or what is the maximum number of courses that can be taken in the same term, or whether there is a sequence in which a part-time student can take the courses one per term. Later in this section, we will show how representing the prerequisite structure of the program as a partial order relation makes it relatively easy to answer such questions.

Antisymmetry

In Section 8.2 we defined three properties of relations: reflexivity, symmetry, and transitivity. A fourth property of relations is called *antisymmetry*. In terms of the arrow diagram of a relation, saying that a relation is antisymmetric is the same as saying that whenever there is an arrow going from one element to another *distinct* element, there is *not* an arrow going back from the second to the first.

Definition

Let R be a relation on a set A . R is **antisymmetric** if, and only if,
for every a and b in A , if $a R b$ and $b R a$ then $a = b$.

By taking the negation of the definition, you can see that a relation R is **not antisymmetric** if, and only if,

there are elements a and b in A such that $a R b$ and $b R a$ but $a \neq b$.

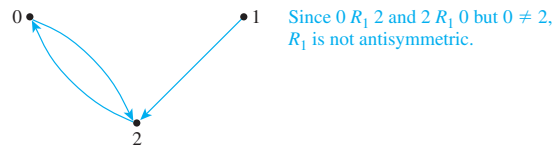
Example 8.5.1 Testing for Antisymmetry of Finite Relations

Let R_1 and R_2 be the relations on $\{0, 1, 2\}$ defined as follows: Draw the directed graphs for R_1 and R_2 and indicate which relations are antisymmetric.

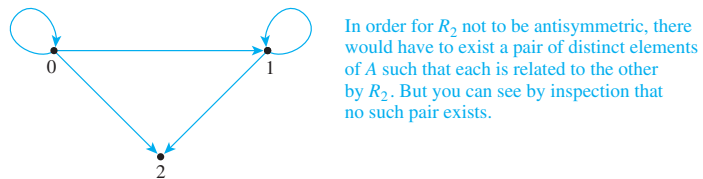
- a. $R_1 = \{(0, 2), (1, 2), (2, 0)\}$ b. $R_2 = \{(0, 0), (0, 1), (0, 2), (1, 1), (1, 2)\}$

Solution

- a. R_1 is not antisymmetric.



- b. R_2 is antisymmetric.



Example 8.5.2 Testing for Antisymmetry of “Divides” Relations

Let R_1 be the “divides” relation on the set of all positive integers, and let R_2 be the “divides” relation on the set of all integers.

$$\begin{array}{ll} \text{For every } a, b \in \mathbb{Z}^+, & a R_1 b \iff a \mid b. \\ \text{For every } a, b \in \mathbb{Z}, & a R_2 b \iff a \mid b. \end{array}$$

- a. Is R_1 antisymmetric? Prove or give a counterexample.
b. Is R_2 antisymmetric? Prove or give a counterexample.

Solution

- a. R_1 is antisymmetric.

Proof: Suppose a and b are positive integers such that $a R_1 b$ and $b R_1 a$. [We must show that $a = b$.] By definition of R_1 , $a \mid b$ and $b \mid a$. Thus, by definition of divides, there are integers k_1 and k_2 with $b = k_1 a$ and $a = k_2 b$. It follows that

$$b = k_1 a = k_1(k_2 b) = (k_1 k_2) b.$$

Dividing both sides by b gives

$$k_1 k_2 = 1.$$

Now since a and b are both integers, k_1 and k_2 are both positive integers also. And the only product of two positive integers that equals 1 is $1 \cdot 1$. Thus

$$k_1 = k_2 = 1$$

and so

$$a = k_2 b = 1 \cdot b = b$$

[as was to be shown].

b. R_2 is not antisymmetric.

Counterexample:

Let $a = 2$ and $b = -2$. Then $a \mid b$ [since $-2 = (-1) \cdot 2$] and $b \mid a$ [since $2 = (-1)(-2)$].

Hence $a R_2 b$ and $b R_2 a$ but $a \neq b$. ■

Example 8.5.2 illustrates the fact that a relation may be antisymmetric on a subset of a set but not antisymmetric on the set itself.

Partial Order Relations

A relation that is reflexive, antisymmetric, and transitive is called a *partial order*.

Definition

Let R be a relation defined on a set A . R is a **partial order relation** if, and only if, R is reflexive, antisymmetric, and transitive.

Two fundamental partial order relations are the “less than or equal to” relation on a set of real numbers and the “subset” relation on a set of sets. These can be thought of as models, or paradigms, for general partial order relations.

Example 8.5.3 The “Subset” Relation

Let $\mathcal{A}$ be any collection of sets and define the “subset” relation, $\subseteq$, on $\mathcal{A}$ as follows: For every $U, V \in \mathcal{A}$,

$$U \subseteq V \iff \text{for each } x, \text{ if } x \in U \text{ then } x \in V.$$

By an argument almost identical to that of the solution for exercise 23 of Section 8.2, $\subseteq$ is reflexive and transitive. Finish the proof that $\subseteq$ is a partial order relation by proving that $\subseteq$ is antisymmetric.

Solution For $\subseteq$ to be antisymmetric means that for all sets U and V in $\mathcal{A}$, if $U \subseteq V$ and $V \subseteq U$ then $U = V$. This is true by definition of equality of sets. ■

Example 8.5.4 A “Divides” Relation on a Set of Positive Integers

Let $\mid$ be the “divides” relation on a set A of positive integers. That is, for all a and b in A ,

$$a \mid b \iff b = ka \text{ for some integer } k.$$

Prove that $\mid$ is a partial order relation on A .

Solution

$|$ is reflexive: [We must show that for each $a \in A$, $a | a$.] Suppose $a \in A$. Then $a = 1 \cdot a$, so $a | a$ by definition of divisibility.

$|$ is antisymmetric: [We must show that for every $a, b \in A$, if $a | b$ and $b | a$ then $a = b$.] The proof of this is virtually identical to that of Example 8.5.2(a).

$|$ is transitive: To show transitivity means to show that for every $a, b, c \in A$, if $a | b$ and $b | c$ then $a | c$. But this is the transitivity of divisibility property, which was proved as Theorem 4.4.3.

Since $|$ is reflexive, antisymmetric, and transitive, $|$ is a partial order relation on A . ■

Example 8.5.5**The “Less Than or Equal to” Relation**

Let S be a set of real numbers and define the “less than or equal to” relation, $\leq$, on S as follows: For all real numbers x and y in S ,

$$x \leq y \iff x < y \text{ or } x = y.$$

Show that $\leq$ is a partial order relation.

Solution

$\leq$ is reflexive: For $\leq$ to be reflexive means that $x \leq x$ for every real number x in S . But $x \leq x$ means that $x < x$ or $x = x$, and $x = x$ is always true.

$\leq$ is antisymmetric: For $\leq$ to be antisymmetric means that for all real numbers x and y in S , if $x \leq y$ and $y \leq x$ then $x = y$. This follows immediately from the definition of $\leq$ and the trichotomy property (see Appendix A, T17), which says that given any real numbers x and y , exactly one of the following holds: $x < y$ or $x = y$ or $x > y$.

$\leq$ is transitive: For $\leq$ to be transitive means that for all real numbers x, y , and z in S if $x \leq y$ and $y \leq z$ then $x \leq z$. This follows from the definition of $\leq$ and the transitivity property of order (see Appendix A, T18), which says that given any real numbers x, y , and z , if $x < y$ and $y < z$ then $x < z$.

Because $\leq$ is reflexive, antisymmetric, and transitive, it is a partial order relation. ■

Notation

Because of the special paradigmatic role played by the $\leq$ relation in the study of partial order relations, the symbol $\preceq$ is often used to refer to a general partial order relation, and the notation $x \preceq y$ is read “ x is less than or equal to y ” or “ y is greater than or equal to x .”

Lexicographic Order

To figure out which of two words comes first in an English dictionary, you compare their letters one by one from left to right. If all letters have been the same to a certain point and one word runs out of letters, that word comes first in the dictionary. For example, *play* comes before *playhouse*. If all letters up to a certain point are the same and the next letters differ, then the word whose next letter is located earlier in the alphabet comes first in the dictionary. For instance, *playhouse* comes before *playmate*.

More generally, if A is any set with a partial order relation, then a *dictionary* or *lexicographic* order can be defined on a set of strings over A as indicated in the following theorem.

Theorem 8.5.1 Lexicographic Order

Let A be a set with a partial order relation R , and let S be a set of strings over A . Define a relation $\preceq$ on S as follows:

Let s and t be any strings in S of lengths m and n , respectively, where m and n are positive integers, and let s_m and t_m be the characters in the m th position for s and t , respectively.

1. If $m \leq n$ and the first m characters of s and t are the same, then $s \preceq t$.
2. If the first $m - 1$ characters in s and t are the same, $s_m R t_m$, and $s_m \neq t_m$, then $s \preceq t$.
3. If λ is the null string then $\lambda \preceq s$.

If no strings are related by $\preceq$ other than by these three conditions, then $\preceq$ is a partial order relation on S .

The proof of Theorem 8.5.1 is technical but straightforward. It is left for the exercises.

Definition

The partial order relation of Theorem 8.5.1 is called the **lexicographic order for S** that corresponds to the partial order R on A .

Example 8.5.6**Testing Strings for Lexicographic Order**

Let $A = \{x, y\}$ and let R be the following partial order relation on A :

$$R = \{(x, x), (x, y), (y, y)\}.$$

Let S be the set of all strings over A , and denote by the lexicographic order for S that corresponds to R .

- a. Is $x \preceq x$? Is $x \preceq xx$? Is $yx \preceq yxy$?
- b. Is $xxxxyy \preceq xy$?
- c. Is $x \preceq y$?
- d. Is $\lambda \preceq xy$?
- e. Is $xyy \preceq xyx$?

Solution

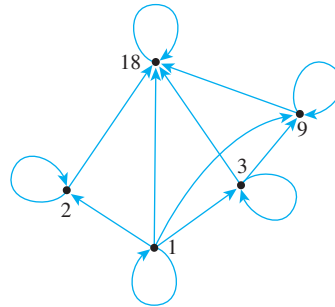
- a. Yes in all three cases, by property (1) of the definition of $\preceq$.
- b. Yes in all cases, by property (2) of the definition of $\preceq$.
- c. Yes in all cases, by property (2) of the definition of $\preceq$. In this case $m - 1 = 0$, and the statement that the first zero characters of x and y are the same is true by default.
- d. Yes by property (3) of the definition of $\preceq$.
- e. No because y is not related to x by R . ■

Hasse Diagrams

Let $A = \{1, 2, 3, 9, 18\}$ and consider the “divides” relation on A : For every $a, b \in A$,

$$a \mid b \iff b = ka \text{ for some integer } k.$$

The directed graph of this relation has the following appearance:

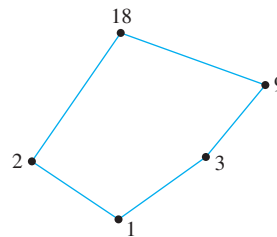


Note that there is a loop at every vertex, all other arrows point in the same direction (upward), and any time there is an arrow from one point to a second and from the second point to a third, there is an arrow from the first point to the third. Given any partial order relation defined on a finite set, it is possible to draw the directed graph in such a way that all of these properties are satisfied. This makes it possible to associate a somewhat simpler graph, called a **Hasse diagram** (after Helmut Hasse, a twentieth-century German number theorist), with a partial order relation defined on a finite set. To obtain a Hasse diagram, proceed as follows:

Start with a directed graph of the relation, placing vertices on the page so that all arrows point upward. Then eliminate

1. the loops at all the vertices,
2. all arrows whose existence is implied by the transitive property, and
3. the direction indicators on the arrows.

For the relation given previously, the Hasse diagram is as follows:



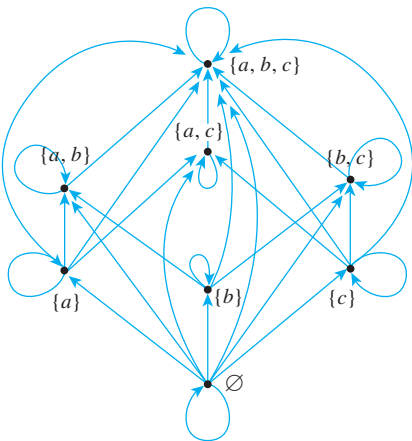
Example 8.5.7 Constructing a Hasse Diagram

Consider the “subset” relation, $\subseteq$, on the set $\mathcal{P}(\{a, b, c\})$. That is, for all sets U and V in $\mathcal{P}(\{a, b, c\})$,

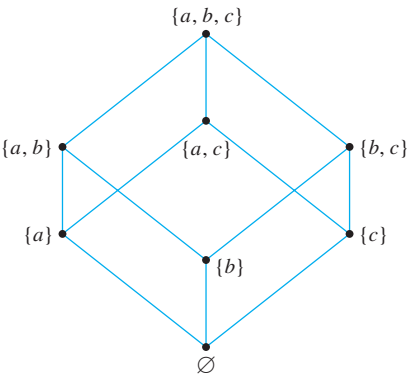
$$U \subseteq V \iff \forall x, \text{ if } x \in U \text{ then } x \in V.$$

Construct the Hasse diagram for this relation.

Solution Draw the directed graph of the relation in such a way that all arrows except loops point upward.



Then strip away all loops, unnecessary arrows, and direction indicators to obtain the Hasse diagram.

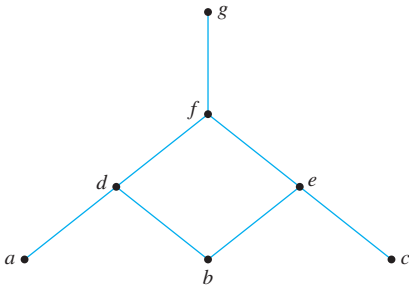


To recover the directed graph of a relation from the Hasse diagram, just reverse the instructions given previously, using the knowledge that the original directed graph was sketched so that all arrows pointed upward:

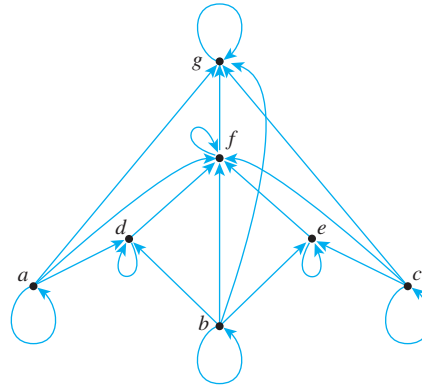
- 1. Reinsert the direction markers on the arrows making all arrows point upward.
- 2. Add loops at each vertex.
- 3. For each sequence of arrows from one point to a second and from that second point to a third, add an arrow from the first point to the third.

Example 8.5.8 Obtaining the Directed Graph of a Partial Order Relation from the Hasse Diagram of the Relation

A partial order relation R has the following Hasse diagram. Find the directed graph of R .



Solution



Partially and Totally Ordered Sets

Given any two real numbers x and y , either $x \leq y$ or $y \leq x$. In a situation like this, the elements x and y are said to be *comparable*. On the other hand, given two subsets A and B of $\{a, b, c\}$, it may be the case that neither $A \subseteq B$ nor $B \subseteq A$. For instance, let $A = \{a, b\}$ and $B = \{b, c\}$. Then $A \not\subseteq B$ and $B \not\subseteq A$. In such a case, A and B are said to be *noncomparable*.

Definition

Suppose $\preceq$ is a partial order relation on a set A . Elements a and b of A are said to be **comparable** if, and only if, either $a \preceq b$ or $b \preceq a$. Otherwise, a and b are called **noncomparable**.

When all the elements of a partial order relation are comparable, the relation is called a *total order*.

Definition

If R is a partial order relation on a set A , and for any two elements a and b in A either $a R b$ or $b R a$, then R is a **total order relation** on A .

Both the “less than or equal to” relation on sets of real numbers and the lexicographic order of the set of words in a dictionary are total order relations. Note that the Hasse diagram for a total order relation can be drawn as a single vertical “chain.”

Many important partial order relations have elements that are not comparable and are, therefore, not total order relations. For instance, the subset relation on $\mathcal{P}(\{a, b, c\})$ is not a total order relation because, as shown previously, the subsets $\{a, b\}$ and $\{a, c\}$ of $\{a, b, c\}$ are not comparable. In addition, a “divides” relation is not a total order relation unless the elements are all powers of a single integer. (See exercise 21 at the end of this section.)

A set A is called a **partially ordered set** (or **poset**) with respect to a relation $\preceq$ if, and only if, $\preceq$ is a partial order relation on A . For instance, the set of real numbers is a partially ordered set with respect to the “less than or equal to” relation $\leq$, and a set of sets is partially ordered with respect to the “subset” relation $\subseteq$. It is entirely straightforward to show that *any subset of a partially ordered set is partially ordered*. (See exercise 35 at the end of this

section.) This, of course, assumes the “same definition” for the relation on the subset as for the set as a whole. A set A is called a **totally ordered set** with respect to a relation $\preceq$ if, and only if, A is partially ordered with respect to $\preceq$ and $\preceq$ is a total order.

A set that is partially ordered but not totally ordered may have totally ordered subsets. Such subsets are called *chains*.

Definition

Let A be a set that is partially ordered with respect to a relation $\preceq$. A subset B of A is called a **chain** if, and only if, the elements in each pair of elements in B are comparable. In other words, $a \preceq b$ or $b \preceq a$ for every a and b in B . The **length of a chain** is one less than the number of elements in the chain.

Observe that if B is a chain in A , then B is a totally ordered set with respect to the “restriction” of $\preceq$ to B .

Example 8.5.9 A Chain of Subsets

The set $\mathcal{P}(\{a, b, c\})$ is partially ordered with respect to the subset relation. Find a chain of length 3 in $\mathcal{P}(\{a, b, c\})$.

Solution Since $\emptyset \subseteq \{a\} \subseteq \{a, b\} \subseteq \{a, b, c\}$, the set

$$S = \{\emptyset, \{a\}, \{a, b\}, \{a, b, c\}\}$$

is a chain of length 3 in $\mathcal{P}(\{a, b, c\})$. ■

In exercise 39 at the end of this section, you are asked to show that a set that is partially ordered with respect to a relation $\preceq$ is totally ordered with respect to $\preceq$ if, and only if, it is a chain.

A *maximal element* in a partially ordered set is an element that is greater than or equal to every element to which it is comparable. (There may be many elements to which it is not comparable.) A *greatest element* in a partially ordered set is an element that is greater than or equal to every element in the set (so it is comparable to every element in the set). Minimal and least elements are defined similarly.

Definition

Let a set A be partially ordered with respect to a relation $\preceq$.

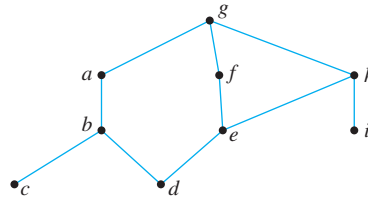
1. An element a in A is called a **maximal element of A** if, and only if, for each b in A , either $b \preceq a$ or b and a are not comparable.
2. An element a in A is called a **greatest element of A** if, and only if, for each b in A , $b \preceq a$.
3. An element a in A is called a **minimal element of A** if, and only if, for each b in A , either $a \preceq b$ or b and a are not comparable.
4. An element a in A is called a **least element of A** if, and only if, for each b in A , $a \preceq b$.

A greatest element is maximal, but a maximal element need not be a greatest element. However, every finite subset of a totally ordered set has both a least element and a greatest element. (See exercise 40 at the end of the section.) Similarly, a least element is minimal, but a minimal element need not be a least element. Furthermore, a set that is partially ordered with respect to a relation can have at most one greatest element and one least element

(see exercise 42 at the end of the section), but it may have more than one maximal or minimal element. The next example illustrates some of these facts.

Example 8.5.10**Maximal, Minimal, Greatest, and Least Elements**

Let $A = \{a, b, c, d, e, f, g, h, i\}$ have the partial ordering $\preceq$ defined by the following Hasse diagram. Find all maximal, minimal, greatest, and least elements of A .



Solution There is just one maximal element, g , which is also the greatest element. The minimal elements are c , d , and i , and there is no least element. ■

Topological Sorting

Is it possible to input the sets of $\mathcal{P}(\{a, b, c\})$ into a computer in a way that is *compatible* with the subset relation $\subseteq$ in the sense that if set U is a subset of set V , then U is input before V ? The answer, as it turns out, is yes. For instance, the following input order satisfies the given condition:

$$\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}.$$

Another input order that satisfies the condition is

$$\emptyset, \{a\}, \{b\}, \{a, b\}, \{c\}, \{b, c\}, \{a, c\}, \{a, b, c\}.$$

Definition

Given partial order relations $\preceq$ and $\preceq'$ on a set A , $\preceq'$ is **compatible** with $\preceq$ if, and only if, for every a and b in A , if $a \preceq b$ then $a \preceq' b$.

Given an arbitrary partial order relation $\preceq$ on a set A , is there a total order $\preceq'$ on A that is compatible with $\preceq$? If the set on which the partial order is defined is finite, then the answer is yes. A total order that is compatible with a given order is called a *topological sorting*.

Definition

Given partial order relations $\preceq$ and $\preceq'$ on a set A , $\preceq'$ is a **topological sorting** for $\preceq$, if, and only if, $\preceq'$ is a total order that is compatible with $\preceq$.

The construction of a topological sorting for a general finite partially ordered set is based on the fact that *any partially ordered set that is finite and nonempty has a minimal element*. (See exercise 41 at the end of the section.) To create a total order for a partially ordered set, simply pick any minimal element and make it number one. Then consider the set obtained when this element is removed. Since the new set is a subset of a partially ordered set, it is partially ordered. If it is empty, stop the process. If not, pick a minimal

element from it and call that element number two. Then consider the set obtained when this element also is removed. If this set is empty, stop the process. If not, pick a minimal element and call it number three. Continue in this way until all the elements of the set have been used up.

Here is a somewhat more formal version of the algorithm:

Constructing a Topological Sorting

Let $\preceq$ be a partial order relation on a nonempty finite set A . To construct a topological sorting:

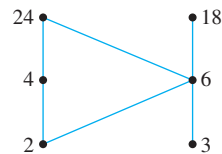
1. Pick any minimal element x in A . [Such an element exists since A is nonempty.]
2. Set $A' := A - \{x\}$.
3. Repeat steps a–c while $A' \neq \emptyset$.
 - a. Pick any minimal element y in A' .
 - b. Define $x \preceq' y$.
 - c. Set $A' := A' - \{y\}$ and $x := y$.

[Completion of steps 1–3 of this algorithm gives enough information to construct the Hasse diagram for the total ordering $\preceq$. We have already shown how to use the Hasse diagram to obtain a complete directed graph for a relation.]

Example 8.5.11

A Topological Sorting

Consider the set $A = \{2, 3, 4, 6, 18, 24\}$ ordered by the “divides” relation $|$. The Hasse diagram of this relation is the following:

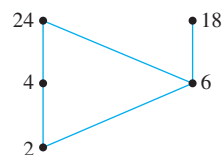


The ordinary “less than or equal to” relation $\leq$ on this set is a topological sorting for it since for positive integers a and b , if $a|b$ then $a \leq b$. Find another topological sorting for this set.

Solution The set has two minimal elements: 2 and 3. Either one may be chosen; suppose you pick 3. The beginning of the total order is

total order: 3.

Set $A' = A - \{3\}$. You can indicate this by removing 3 from the Hasse diagram as shown below.



Next choose a minimal element from $A' - \{3\}$. Only 2 is minimal, so you must pick it. The total order thus far is

total order: $3 \preceq 2$.

Set $A' = (A - \{3\}) - \{2\} = A - \{3, 2\}$. You can indicate this by removing 2 from the Hasse diagram, as is shown below.



Choose a minimal element from $A' - \{3, 2\}$. Again you have two choices: 4 and 6. Suppose you pick 6. The total order for the elements chosen thus far is

$$\text{total order: } 3 \preceq 2 \preceq 6.$$

You continue in this way until every element of A has been picked. One possible sequence of choices gives

$$\text{total order: } 3 \preceq 2 \preceq 6 \preceq 18 \preceq 4 \preceq 24.$$

You can verify that this order is compatible with the “divides” partial order by checking that for each pair of elements a and b in A such that $a|b$, then $a \preceq b$. Note that it is *not* the case that if $a \preceq b$ then $a|b$. ■

An Application

To return to the example that introduced this section, note that the following defines a partial order relation on the set of courses required for a university degree: For all required courses x and y ,

$$x \preceq y \iff x = y \text{ or } x \text{ is a prerequisite for } y.$$

If the Hasse diagram for the relation is drawn, then the questions raised at the beginning of this section can be answered easily. For instance, consider the Hasse diagram for the requirements at a particular university, which is shown in Figure 8.5.1.

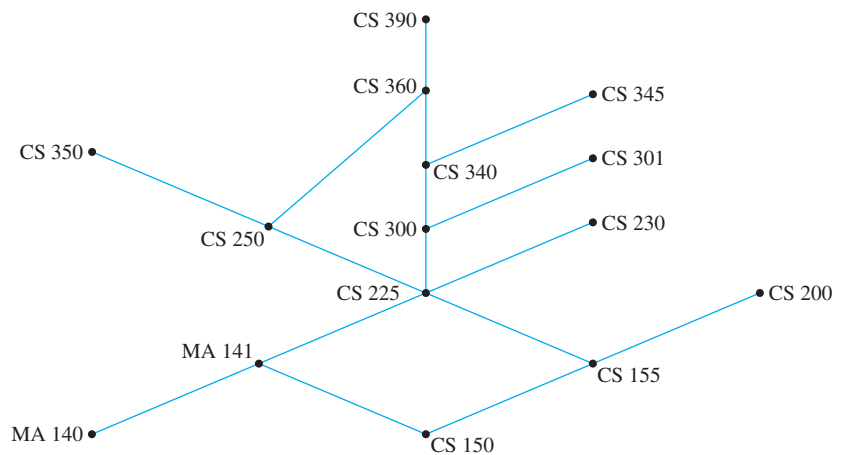


FIGURE 8.5.1

The minimum number of school terms needed to complete the requirements is the size of a longest chain, which is 7 (150, 155, 225, 300, 340, 360, 390, for example). The maximum number of courses that could be taken in the same term (assuming the university allows it) is the maximum number of noncomparable courses, which is 6 (350, 360, 345, 301, 230, 200, for example). A part-time student could take the courses in a sequence

determined by constructing a topological sorting for the set. (One such sorting is 140, 150, 141, 155, 200, 225, 230, 300, 250, 301, 340, 345, 350, 360, 390. There are many others.)

PERT and CPM

Two important and widely used applications of partial order relations are **PERT** (Program Evaluation and Review Technique) and **CPM** (Critical Path Method). These techniques were developed in the 1950s as planners came to grips with the complexities of scheduling the individual activities needed to complete very large projects, and although they are very similar, their developments were independent. PERT was developed by the U.S. Navy to help organize the construction of the Polaris submarine, and CPM was developed by the E. I. Du Pont de Nemours company for scheduling chemical plant maintenance. Here is a somewhat simplified example of the way the techniques work.

Example 8.5.12 A Job Scheduling Problem

At an automobile assembly plant, the job of assembling an automobile can be broken down into these tasks:

- 1. Build frame.
- 2. Install engine, power train components, gas tank.
- 3. Install brakes, wheels, tires.
- 4. Install dashboard, floor, seats.
- 5. Install electrical lines.
- 6. Install gas lines.
- 7. Install brake lines.
- 8. Attach body panels to frame.
- 9. Paint body.

Certain of these tasks can be carried out at the same time, whereas some cannot be started until other tasks are finished. Table 8.5.1 summarizes the order in which tasks can be performed and the time required to perform each task.

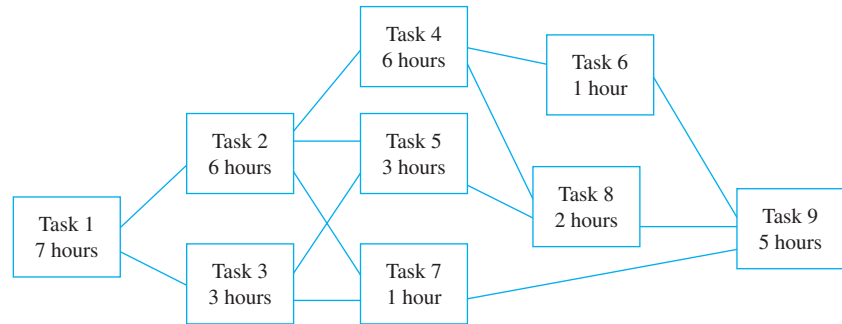
TABLE 8.5.1

Task	Immediately Preceding Tasks	Time Needed to Perform Task
1		7 hours
2	1	6 hours
3	1	3 hours
4	2	6 hours
5	2, 3	3 hours
6	4	1 hour
7	2, 3	1 hour
8	4, 5	2 hours
9	6, 7, 8	5 hours

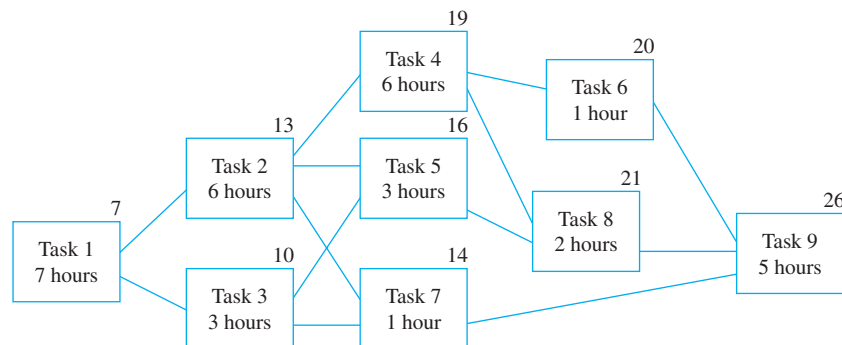
Let T be the set of all tasks, and consider the partial order relation $\preceq$ defined on T as follows: For all tasks x and y in T ,

$$x \preceq y \iff x = y \text{ or } x \text{ precedes } y.$$

If the Hasse diagram of this relation is turned sideways (as is customary in PERT and CPM analysis), it has the appearance shown below.



What is the minimum time required to assemble a car? You can determine this by working from left to right across the diagram, noting for each task (say, just above the box representing that task) the minimum time needed to complete that task starting from the beginning of the assembly process. For instance, you can put a 7 above the box for task 1 because task 1 requires 7 hours. Task 2 requires completion of task 1 (7 hours) plus 6 hours for itself, so the minimum time required to complete task 2, starting at the beginning of the assembly process, is $7 + 6 = 13$ hours. So you can put a 13 above the box for task 2. Similarly, you can put a 10 above the box for task 3 because $7 + 3 = 10$. Now consider what number you should write above the box for task 5. The minimum times to complete tasks 2 and 3, starting from the beginning of the assembly process, are 13 and 10 hours respectively. Since *both* tasks must be completed before task 5 can be started, the minimum time to complete task 5, starting from the beginning, is the time needed for task 5 itself (3 hours) plus the *maximum* of the times to complete tasks 2 and 3 (13 hours), and this equals $3 + 13 = 16$ hours. Thus you should place the number 16 above the box for task 5. The same reasoning leads you to place a 14 above the box for task 7. Similarly, you can place a 19 above the box for task 4, a 20 above the box for task 6, a 21 above the box for task 8, and a 26 above the box for task 9, as shown below.



This analysis shows that at least 26 hours are required to complete task 9 starting from the beginning of the assembly process. When task 9 is finished, the assembly is complete, so 26 hours is the minimum time needed to accomplish the whole process.

Note that the minimum time required to complete tasks 1, 2, 4, 8, and 9 in sequence is exactly 26 hours. This means that a delay in performing any one of these tasks causes a delay in the total time required for assembly of the car. For this reason, the path through tasks 1, 2, 4, 8, and 9 is called a **critical path**. ■

TEST YOURSELF

- For a relation R on a set A to be antisymmetric means that _____.
- To show that a relation R on an infinite set A is antisymmetric, you suppose that _____ and you show that _____.
- To show that a relation R on a set A is not antisymmetric, you _____.
- To construct a Hasse diagram for a partial order relation, you start with a directed graph of the relation in which all arrows point upward and you eliminate _____, _____, and _____.
- If A is a set that is partially ordered with respect to a relation $\preceq$ and if a and b are elements of A , we say that a and b are comparable if, and only if, _____ or _____.
- A relation $\preceq$ on a set A is a total order if, and only if, _____.
- If A is a set that is partially ordered with respect to a relation $\preceq$, and if B is a subset of A , then B is a chain if, and only if, for all a and b in B , _____.
- Let A be a set that is partially ordered with respect to a relation $\preceq$, and let a be an element of A .
 - a is maximal if, and only if, _____.
 - a is a greatest element of A if, and only if, _____.
 - a is minimal if, and only if, _____.
 - a is a least element of A if, and only if, _____.
- Given a set A that is partially ordered with respect to a relation $\preceq$, the relation $\preceq'$ is a topological sorting for $\preceq$, if, and only if, $\preceq'$ is a _____ and for all a and b in A if $a \preceq b$ then _____.
- PERT and CPM are used to produce efficient _____.

EXERCISE SET 8.5

- Each of the following is a relation on $\{0, 1, 2, 3\}$. Draw directed graphs for each relation, and indicate which relations are antisymmetric.
 - $R_1 = \{(0, 0), (0, 2), (1, 0), (1, 3), (2, 2), (3, 0), (3, 1)\}$
 - $R_2 = \{(0, 1), (0, 2), (1, 1), (1, 2), (1, 3), (2, 2), (3, 2)\}$
 - $R_3 = \{(0, 0), (0, 3), (1, 0), (1, 3), (2, 2), (3, 3), (3, 2)\}$
 - $R_4 = \{(0, 0), (1, 0), (1, 2), (1, 3), (2, 0), (2, 1), (3, 2), (3, 0)\}$
- Let P be the set of all people in the world and define a relation R on P as follows: For all people x and y ,

$$x R y \iff x \text{ is no older than } y.$$
 Is R antisymmetric? Prove or give a counterexample.
- Let S be the set of all strings of a 's and b 's. Define a relation R on S as follows: For every $s, t \in S$,

$$s R t \iff L(s) \leq L(t),$$
 where $L(x)$ denotes the length of a string x . Is R antisymmetric? Prove or give a counterexample.
- Let R be the “less than” relation on $\mathbf{R}$, the set of all real numbers: For every $x, y \in \mathbf{R}$,

$$x R y \iff x < y.$$
 Is R antisymmetric? Prove or give a counterexample.
- Let $\mathbf{R}$ be the set of all real numbers and define a relation R on $\mathbf{R} \times \mathbf{R}$ as follows: For every (a, b) and (c, d) in $\mathbf{R} \times \mathbf{R}$,

$$(a, b) R (c, d) \iff \text{either } a < c \text{ or both } a = c \text{ and } b \leq d.$$
 Is R a partial order relation? Prove or give a counterexample.
- Let P be the set of all people who have ever lived and define a relation R on P as follows: For every $r, s \in P$,

$$r R s \iff r \text{ is an ancestor of } s \text{ or } r = s.$$
 Is R a partial order relation? Prove or give a counterexample.

Lecture 6: two topics not in Epp

Composition of relations · Linearization of partial orders

Epp’s Chapter 8 defines relations, their inverses and directed graphs (§8.1), but never composes two relations—composition appears only for *functions*, in §7.3, which is outside the midterm scope. Likewise, Epp §8.5 treats topological sorting through its own “compatible total order” construction, and states neither the lecture’s definition of a *linearization* nor Kahn’s algorithm. Lecture 6 states that part of its content is taken from Dr Wong Tin Lok’s notes; these are those parts. Everything below is typeset from the Lecture 6 slides, in the lecture’s notation.

S.1 Composition of Relations

Lecture 6, §6.1.4

Definition: Composition of Relations

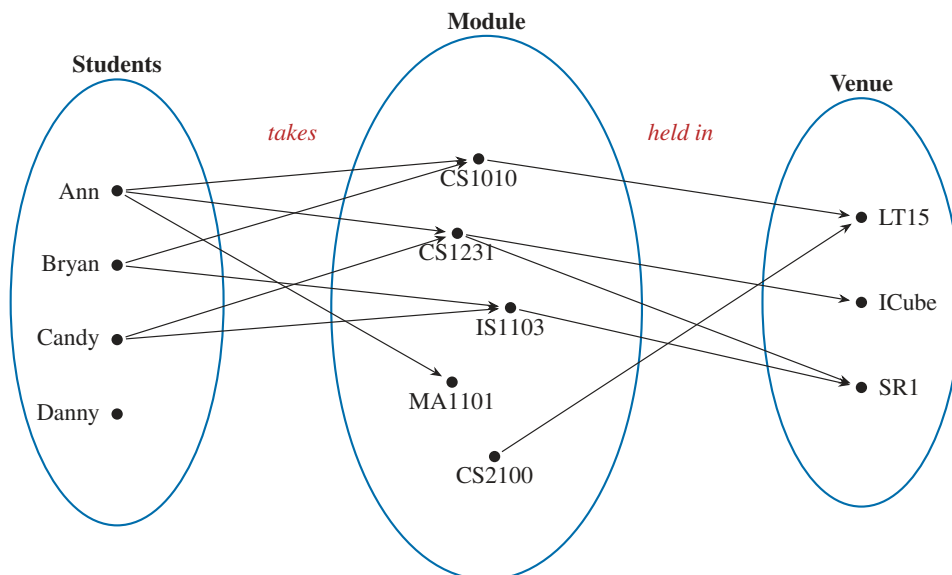
Let A , B and C be sets. Let $R \subseteq A \times B$ be a relation. Let $S \subseteq B \times C$ be a relation. The **composition of R with S** , denoted $S \circ R$, is the relation from A to C such that

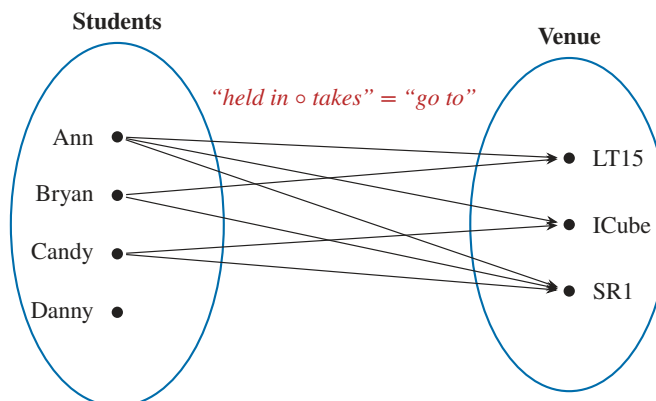
$$\forall x \in A, \forall z \in C \left(x (S \circ R) z \iff \exists y \in B (x R y \wedge y S z) \right).$$

In other words, $x \in A$ and $z \in C$ are “ $S \circ R$ ”-related if, and only if, there is a “path” from x to z via some intermediate element $y \in B$ in the arrow diagram.

Note the order: $S \circ R$ applies R first and then S , matching the convention for composition of functions.

Example





Reading the definition off the picture: Ann takes CS1010, CS1231 and MA1101; CS1010 is held in LT15 and CS1231 in both ICube and SR1, while MA1101 is held nowhere—so Ann “go to” LT15, ICube and SR1. Danny takes no module, so Danny is related to no venue. CS2100 is held in LT15 but is taken by nobody, so it contributes no pair to the composition.

Proposition: Composition is Associative

Let A, B, C, D be sets. Let $R \subseteq A \times B$, $S \subseteq B \times C$ and $T \subseteq C \times D$ be relations. Then

$$T \circ (S \circ R) = (T \circ S) \circ R = T \circ S \circ R.$$

Proposition: Inverse of Composition

Let A, B and C be sets. Let $R \subseteq A \times B$ and $S \subseteq B \times C$ be relations. Then

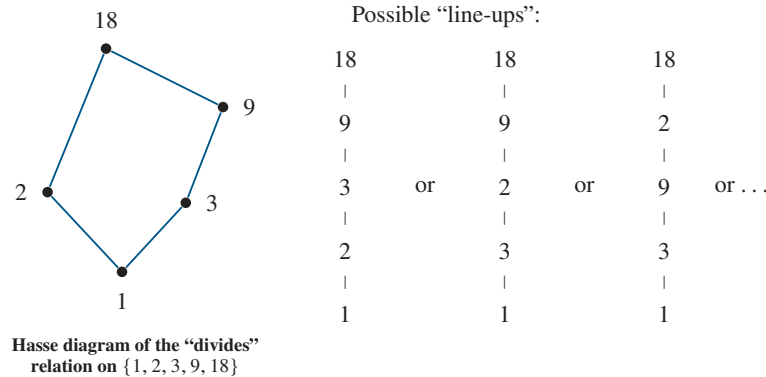
$$(S \circ R)^{-1} = R^{-1} \circ S^{-1}.$$

The lecture states both propositions without proof. Mind the reversal in the second one: it is the same swap you see for inverses of composed functions.

S.2 Linearization of Partial Orders

Lecture 6, §§6.4.6–6.4.7

Let $\preceq$ be a partial order relation on a set A . Suppose we view A as a set of tasks and $x \preceq y$ to mean x must be performed before or at the same time as y . How can we “line up” the tasks in A , assuming that we cannot perform two tasks simultaneously?



A “line-up” is called a **linearization** of $\preceq$. Such a line-up happens to be a **total order**.

Definition: Total Order Relation (Epp §8.5, restated in the lecture)

If R is a partial order relation on a set A , and for any two elements x and y in A , either $x R y$ or $y R x$, then R is a **total order relation** (or simply **total order**) on A . In other words, R is a total order if, and only if,

$$R \text{ is a partial order} \quad \text{and} \quad \forall x, y \in A \ (x R y \vee y R x).$$

Example. The divisibility relation $|$ on $\mathbb{Z}^+$ is a partial order but not a total order, because 3 and 5 are not comparable ($3 \nmid 5$ and $5 \nmid 3$). The relation $\leq$ on $\mathbb{Q}$ is a total order, because $\forall x, y \in \mathbb{Q} \ (x \leq y \vee y \leq x)$.

It follows that the Hasse diagram of a total order is one single line (chain). Hence the linearization of a total order is the total order itself. A linearization of a partial order can be seen as deriving one total order (among many possible total orders) from that partial order.

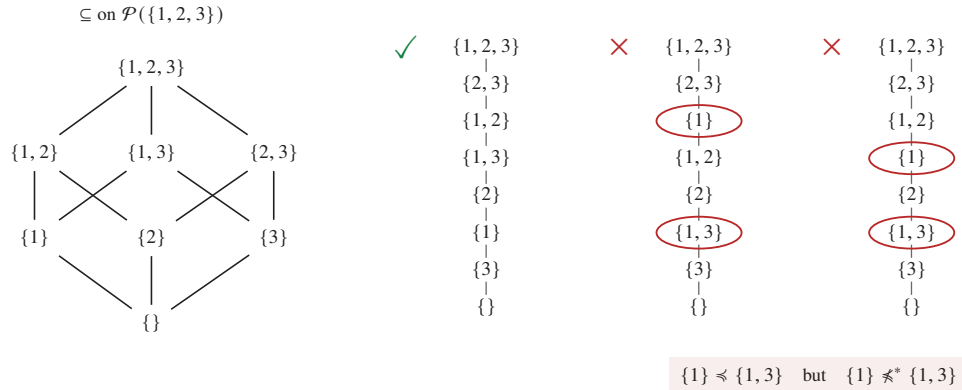
Definition: Linearization of a Partial Order

Let $\preceq$ be a partial order on a set A . A **linearization** of $\preceq$ is a total order $\preceq^*$ on A such that

$$\forall x, y \in A \ (x \preceq y \implies x \preceq^* y).$$

Example: the subset relation $\subseteq$ on $\mathcal{P}(\{1, 2, 3\})$

Which of the following is a linearization of $(\mathcal{P}(\{1, 2, 3\}), \subseteq)$?



Only the first chain is a linearization. In the other two, $\{1, 3\}$ is placed below $\{1\}$, so $\{1, 3\} \preceq^* \{1\}$ even though $\{1\} \subseteq \{1, 3\}$ — which breaks the requirement $x \preceq y \Rightarrow x \preceq^* y$. Chains are read bottom to top.

Kahn's Algorithm

Kahn's Algorithm (1962)

Input: A finite set A and a partial order $\preceq$ on A .

1. Set $A_0 := A$ and $i := 0$.
2. Repeat until $A_i = \emptyset$:
 - 2.1. find a minimal element c_i of A_i with respect to $\preceq$;
 - 2.2. set $A_{i+1} = A_i \setminus \{c_i\}$;
 - 2.3. set $i := i + 1$.

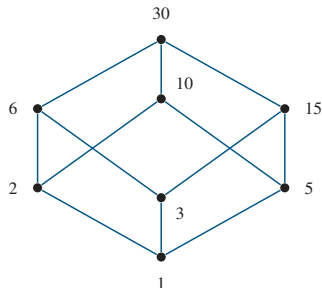
Output: A linearization $\preceq^*$ of $\preceq$ defined by setting, for all indices i, j ,

$$c_i \preceq^* c_j \iff i \leq j.$$

At step 2.1 there is in general a *choice* of minimal element; each sequence of choices yields a different linearization. That is exactly why a partial order usually has many linearizations.

Example: a run of Kahn's algorithm

On the set $\{d \in \mathbb{Z}^+ : d \mid 30\}$ partially ordered by the divisibility relation $\mid$:



$$A_0 := \{d \in \mathbb{Z}^+ : d \mid 30\}.$$

1 is the only minimal element of A_0 .
 2, 3, 5 are the minimal elements of A_1 .
 2, 5 are the minimal elements of A_2 .
 5, 6 are the minimal elements of A_3 .
 5 is the only minimal element of A_4 .
 10, 15 are the minimal elements of A_5 .
 10 is the only minimal element of A_6 .
 30 is the only minimal element of A_7 .
 $A_8 = \emptyset$, and so we stop.

Set $c_0 := 1$, $A_1 := A_0 \setminus \{1\}$.
 Set $c_1 := 3$, $A_2 := A_1 \setminus \{3\}$.
 Set $c_2 := 2$, $A_3 := A_2 \setminus \{2\}$.
 Set $c_3 := 6$, $A_4 := A_3 \setminus \{6\}$.
 Set $c_4 := 5$, $A_5 := A_4 \setminus \{5\}$.
 Set $c_5 := 15$, $A_6 := A_5 \setminus \{15\}$.
 Set $c_6 := 10$, $A_7 := A_6 \setminus \{10\}$.
 Set $c_7 := 30$, $A_8 := A_7 \setminus \{30\}$.

A linearization is obtained:

$$1 \preceq^* 3 \preceq^* 2 \preceq^* 6 \preceq^* 5 \preceq^* 15 \preceq^* 10 \preceq^* 30.$$

PROPERTIES OF THE REAL NUMBERS*

In this text we take the real numbers and their basic properties as our starting point. We give a core set of properties, called axioms, which the real numbers are assumed to satisfy, and we state some useful properties that can be deduced from these axioms.

We assume that there are two binary operations defined on the set of real numbers, called **addition** and **multiplication**, such that if a and b are any two real numbers, the **sum** of a and b , denoted $a + b$, and the **product** of a and b , denoted $a \cdot b$ or ab , are also real numbers. These operations satisfy properties F1–F6, which are called the **field axioms**.

F1. *Commutative Laws* For all real numbers a and b ,

$$a + b = b + a \quad \text{and} \quad ab = ba.$$

F2. *Associative Laws* For all real numbers a , b , and c ,

$$(a + b) + c = a + (b + c) \quad \text{and} \quad (ab)c = a(bc).$$

F3. *Distributive Laws* For all real numbers a , b , and c ,

$$a(b + c) = ab + ac \quad \text{and} \quad (b + c)a = ba + ca.$$

F4. *Existence of Identity Elements* There exist two distinct real numbers, denoted 0 and 1, such that for every real number a ,

$$0 + a = a + 0 = a \quad \text{and} \quad 1 \cdot a = a \cdot 1 = a.$$

F5. *Existence of Additive Inverses* For every real number a , there is a real number, denoted $-a$ and called the **additive inverse** of a , such that

$$a + (-a) = (-a) + a = 0.$$

F6. *Existence of Reciprocals* For every real number $a \neq 0$, there is a real number, denoted $1/a$ or a^{-1} , called the **reciprocal** of a , such that

$$a \cdot \left(\frac{1}{a}\right) = \left(\frac{1}{a}\right) \cdot a = 1.$$

All the usual algebraic properties of the real numbers that do not involve order can be derived from the field axioms. The most important are collected as theorems T1–T16 as follows. In all these theorems the symbols a , b , c , and d represent arbitrary real numbers.

T1. *Cancellation Law for Addition* If $a + b = a + c$, then $b = c$. (In particular, this shows that the number 0 of Axiom F4 is unique.)

T2. *Possibility of Subtraction* Given a and b , there is exactly one x such that $a + x = b$. This x is denoted by $b - a$. In particular, $0 - a$ is the additive inverse of a , $-a$.

*Adapted from Tom M. Apostol, *Calculus, Volume I* (New York: Blaisdell, 1961), pp. 13–19.

T3. $b - a = b + (-a)$.

T4. $-(-a) = a$.

T5. $a(b - c) = ab - ac$.

T6. $0 \cdot a = a \cdot 0 = 0$.

T7. *Cancellation Law for Multiplication* If $ab = ac$ and $a \neq 0$, then $b = c$. (In particular, this shows that the number 1 of Axiom F4 is unique.)

T8. *Possibility of Division* Given a and b with $a \neq 0$, there is exactly one x such that $ax = b$. This x is denoted by b/a and is called the **quotient** of b and a . In particular, $1/a$ is the reciprocal of a .

T9. If $a \neq 0$, then $b/a = b \cdot a^{-1}$.

T10. If $a \neq 0$, then $(a^{-1})^{-1} = a$.

T11. *Zero Product Property* If $ab = 0$, then $a = 0$ or $b = 0$.

T12. *Rule for Multiplication with Negative Signs*

$$(-a)b = a(-b) = -(ab), \quad (-a)(-b) = ab,$$

and

$$-\frac{a}{b} = \frac{-a}{b} = \frac{a}{-b}.$$

T13. *Equivalent Fractions Property*

$$\frac{a}{b} = \frac{ac}{bc}, \quad \text{if } b \neq 0 \text{ and } c \neq 0.$$

T14. *Rule for Addition of Fractions*

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \text{if } b \neq 0 \text{ and } d \neq 0.$$

T15. *Rule for Multiplication of Fractions*

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}, \quad \text{if } b \neq 0 \text{ and } d \neq 0.$$

T16. *Rule for Division of Fractions*

$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{ad}{bc}, \quad \text{if } b \neq 0, c \neq 0, \text{ and } d \neq 0.$$

The real numbers also satisfy the following axioms, called the **order axioms**. It is assumed that among all real numbers there are certain ones, called the **positive real numbers**, that satisfy properties Ord1–Ord3.

Ord1. For any real numbers a and b , if a and b are positive, so are $a + b$ and ab .

Ord2. For every real number $a \neq 0$, either a is positive or $-a$ is positive but not both.

Ord3. The number 0 is not positive.

The symbols $<$, $>$, $\leq$, and $\geq$, and negative numbers are defined in terms of positive numbers.

Definition	
Given real numbers a and b ,	
$a < b$ means $b + (-a)$ is positive.	$b > a$ means $a < b$.
$a \leq b$ means $a < b$ or $a = b$.	$b \geq a$ means $a \leq b$.
If $a < 0$, we say that a is negative .	If $a \geq 0$, we say that a is nonnegative .

From the order axioms Ord1–Ord3 and the above definition, all the usual rules for calculating with inequalities can be derived. The most important are collected as theorems T17–T27 as follows. In all these theorems the symbols a , b , c , and d represent arbitrary real numbers.

- T17. *Trichotomy Law* For arbitrary real numbers a and b , exactly one of the three relations $a < b$, $b < a$, or $a = b$ holds.
- T18. *Transitive Law* If $a < b$ and $b < c$, then $a < c$.
- T19. If $a < b$, then $a + c < b + c$.
- T20. If $a < b$ and $c > 0$, then $ac < bc$.
- T21. If $a \neq 0$, then $a^2 > 0$.
- T22. $1 > 0$.
- T23. If $a < b$ and $c < 0$, then $ac > bc$.
- T24. If $a < b$, then $-a > -b$. In particular, if $a < 0$, then $-a > 0$.
- T25. If $ab > 0$, then both a and b are positive or both are negative.
- T26. If $a < c$ and $b < d$, then $a + b < c + d$.
- T27. If $0 < a < c$ and $0 < b < d$, then $0 < ab < cd$.

One final axiom distinguishes the set of real numbers from the set of rational numbers. It is called the **least upper bound axiom**.

- LUB. Any nonempty set S of real numbers that is bounded above has a least upper bound. That is, if B is the set of all real numbers x such that $x \geq s$ for every s in S and if B has at least one element, then B has a smallest element. This element is called the **least upper bound** of S .

The least upper bound axiom holds for the set of real numbers but not for the set of rational numbers. For example, the set of all rational numbers that are less than $\sqrt{2}$ has upper bounds but not a least upper bound within the set of rational numbers.